

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 95 on 19/12/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTkwODU1MTRfNTM0Mw%3D%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

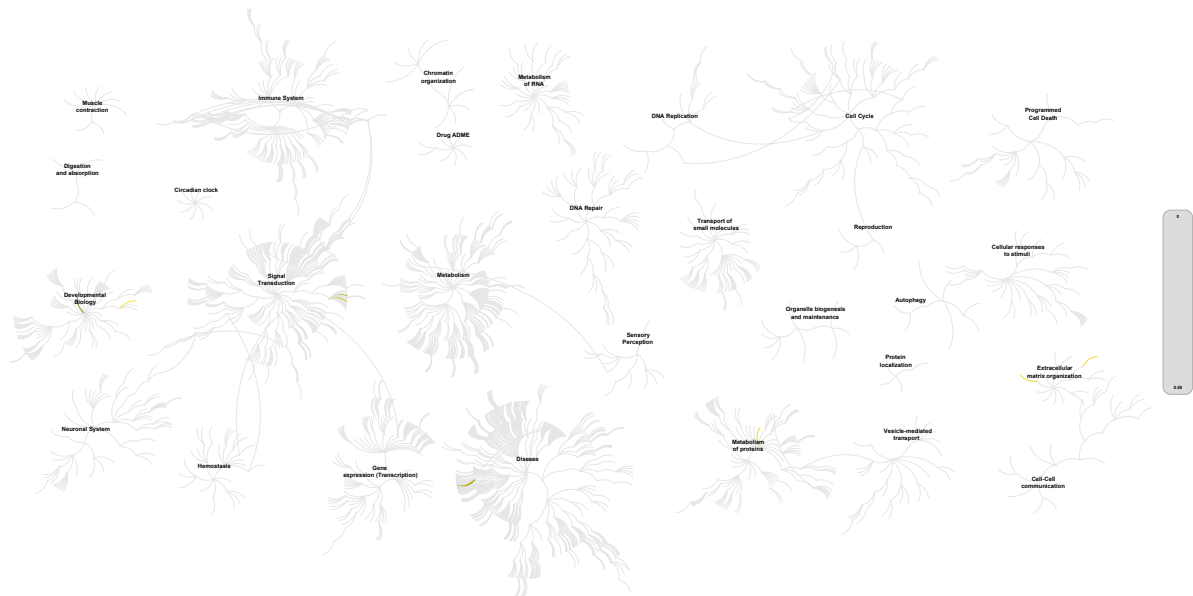
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 4180 out of 6241 identifiers in the sample were found in Reactome, where 2587 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTkwODU1MTRfNTM0Mw%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

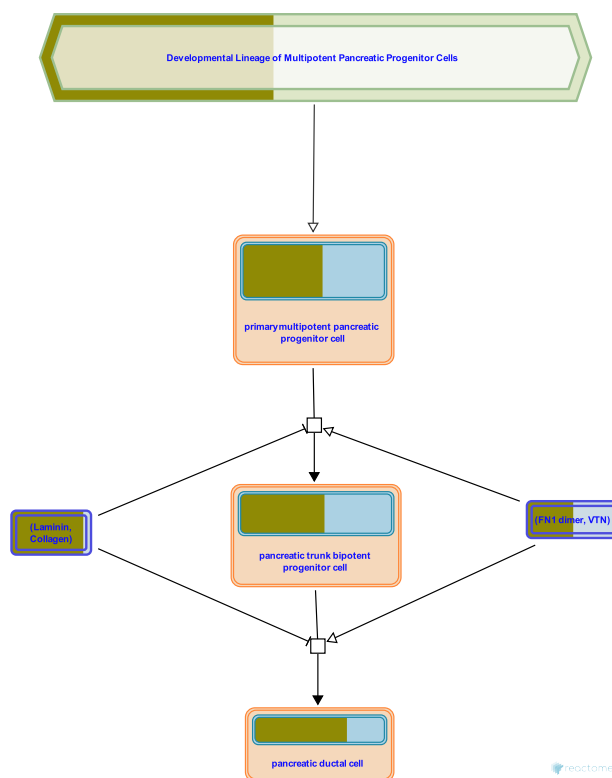
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Developmental Lineage of Pancreatic Ductal Cells	57 / 77	0.003	1.52e-04	0.421	2 / 2	1.26e-04
Collagen chain trimerization	33 / 44	0.002	0.003	1	23 / 28	0.002
Post-translational protein phosphorylation	67 / 109	0.004	0.005	1	1 / 1	6.30e-05
Molecules associated with elastic fibres	36 / 44	0.002	0.005	1	10 / 10	6.30e-04
RHOG GTPase cycle	48 / 78	0.003	0.014	1	6 / 6	3.78e-04
Defective B3GALT6 causes EDSP2 and SEMDJL1	16 / 21	8.46e-04	0.026	1	1 / 1	6.30e-05
Defective B4GALT7 causes EDS, progeroid type	16 / 21	8.46e-04	0.026	1	1 / 1	6.30e-05
RHOA GTPase cycle	83 / 154	0.006	0.036	1	5 / 6	3.78e-04
Defective B3GAT3 causes JDSSDHD	16 / 22	8.86e-04	0.037	1	1 / 1	6.30e-05
LGI-ADAM interactions	11 / 14	5.64e-04	0.048	1	5 / 5	3.15e-04
CDC42 GTPase cycle	84 / 159	0.006	0.05	1	5 / 6	3.78e-04
Defective EXT1 causes exostoses 1, TRPS2 and CHDS	12 / 16	6.44e-04	0.053	1	3 / 3	1.89e-04
Defective EXT2 causes exostoses 2	12 / 16	6.44e-04	0.053	1	3 / 3	1.89e-04
Interaction between L1 and Ankyrins	21 / 33	0.001	0.062	1	4 / 4	2.52e-04
RAC1 GTPase cycle	98 / 191	0.008	0.066	1	5 / 6	3.78e-04
RHOB GTPase cycle	42 / 75	0.003	0.069	1	6 / 6	3.78e-04
Loss of MECP2 binding ability to 5mC-DNA	5 / 5	2.01e-04	0.071	1	2 / 2	1.26e-04
RAC3 GTPase cycle	53 / 100	0.004	0.095	1	5 / 6	3.78e-04
Defective CHST14 causes EDS, musculocontractural type	7 / 9	3.62e-04	0.104	1	1 / 1	6.30e-05
RAC2 GTPase cycle	57 / 101	0.004	0.107	1	10 / 10	6.30e-04
Crosslinking of collagen fibrils	15 / 24	9.67e-04	0.112	1	13 / 13	8.19e-04
Assembly of collagen fibrils and other multimeric structures	59 / 87	0.004	0.116	1	24 / 26	0.002
Anchoring fibril formation	10 / 15	6.04e-04	0.128	1	2 / 4	2.52e-04
Signaling by RNF43 mutants	6 / 8	3.22e-04	0.142	1	1 / 1	6.30e-05
Defective VWF binding to collagen type I	3 / 3	1.21e-04	0.146	1	1 / 1	6.30e-05

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Developmental Lineage of Pancreatic Ductal Cells ([R-HSA-9925563](#))



Ductal epithelial cells line the pancreatic ducts in a single layer, changing the shape as ducts increase in size from cuboidal, through low columnar, to high columnar epithelium (Liggitt and Dintzis, “Pancreas”, pp. 241-250). Ductal epithelial cells may have cilia or microvilli on their apical surface (Liggitt and Dintzis, “Pancreas”, pp. 241-250). Besides ductal epithelial cells, main pancreatic ducts sometimes include goblet cells (Liggitt and Dintzis, “Pancreas”, pp. 241-250).

Ductal epithelial cells develop from bipotent pancreatic trunk progenitor cells, which also give rise to endocrine progenitors and are derived from the primary multipotent pancreatic progenitor cells (reviewed in Shih et al. 2013, Jennings et al. 2015).

References

- Liggitt D & Dintzis SM (2018). *Pancreas*, “Pancreas”, in “Comparative Anatomy and Histology. A Mouse, Rat, and Human Atlas.”, edited by Treuting, Dintzis and Montine. , pp. 241-250.
- Shih HP, Wang A & Sander M (2013). Pancreas organogenesis: from lineage determination to morphogenesis. *Annu Rev Cell Dev Biol*, 29, 81-105. [↗](#)
- Jennings RE, Berry AA, Strutt JP, Gerrard DT & Hanley NA (2015). Human pancreas development. *Development*, 142, 3126-37. [↗](#)

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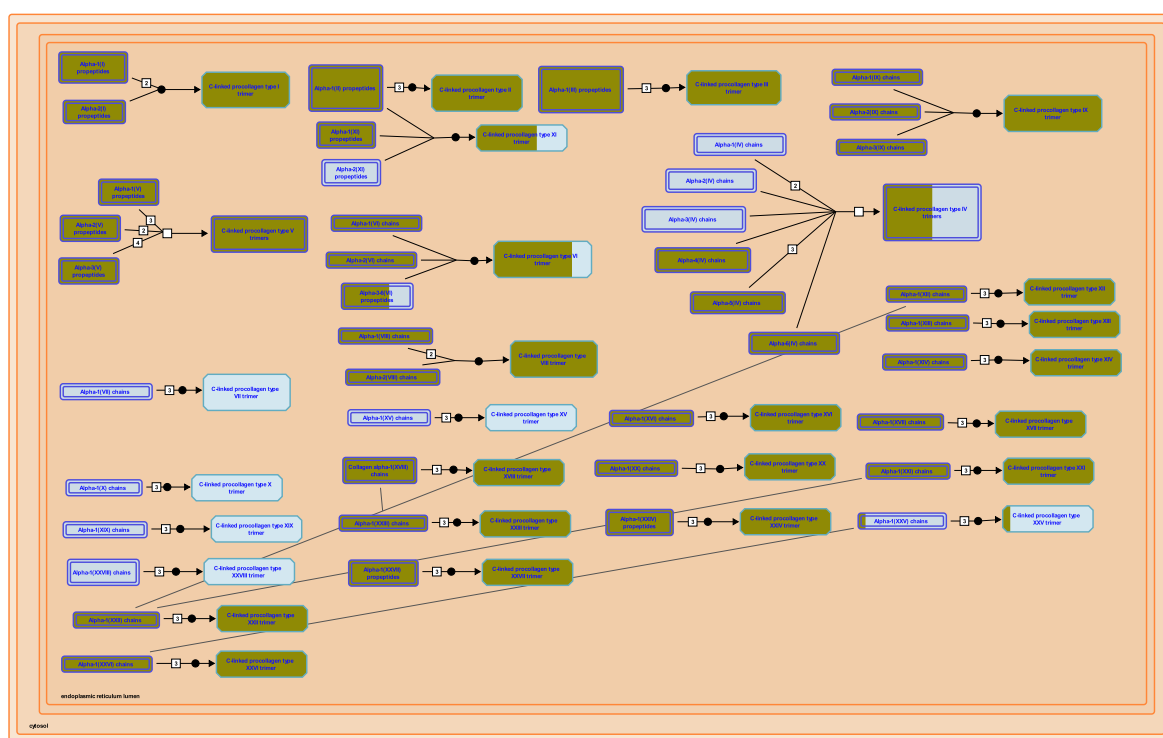
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2024-10-17	Authored	Orlic-Milacic M
2024-10-17	Created	Orlic-Milacic M
2025-02-21	Reviewed	Li NT
2025-02-25	Edited	Orlic-Milacic M
2025-03-05	Modified	Weiser JD

39 submitted entities found in this pathway, mapping to 57 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ANXA1	P04083	ANXA2	P07355	ANXA3	P12429
AQP1	P29972	CD74	P04233	CFTR	P13569
COL11A1	P12107	COL1A1	P02452	COL1A2	P08123
COL24A1	Q17RW2	COL27A1	Q8IZC6	COL2A1	P02458
COL3A1	P02461	COL5A1	P20908	COL5A2	P05997
COL5A3	P25940	ERICH5	Q6P6B1	FN1	P02751
FOXA2	Q9Y261	GATA4	P43694	KRT17	Q04695
KRT19	P08727	KRT7	P08729	LAMA1	P25391
LAMA2	P24043	LAMA3	Q16787	LAMA4	Q16363
LAMA5	O15230	LAMB1	P07942	LAMB2	P11047, P55268
LAMC2	Q13753	ONECUT1	Q9UBC0	SLC4A4	Q9Y6R1
SOX9	P48436	TFPI2	P48307	YAP1	P46937

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
ANXA1	ENST00000257497	ANXA2	ENST00000451270	ANXA3	ENST00000264908
ANXA4	ENST00000394295	AQP1	ENST00000311813	CD74	ENST00000009530
CFTR	ENST00000003084	ERICH5	ENST00000318528	FOXA2	ENST00000377115
GATA4	ENST00000335135	GATA6	ENST00000269216	HES1	ENST00000232424
KRT17	ENST00000311208	KRT19	ENST00000361566	KRT7	ENST00000331817
ONECUT1	ENST00000305901	SLC4A4	ENST00000264485	SOX9	ENST00000245479
TFPI2	ENST00000222543	YAP1	ENST00000282441		

2. Collagen chain trimerization ([R-HSA-8948216](#))



The C-propeptides of collagen propeptide chains are essential for the association of three peptide chains into a trimeric but non-helical procollagen. This initial binding event determines the composition of the trimer, brings the individual chains into the correct register and initiates formation of the triple helix at the C-terminus, which then proceeds towards the N-terminus in a zipper-like fashion (Engel & Prockop 1991). Most early refolding studies were performed with collagen type III, which contains a disulfide linkage at the C-terminus of its triple helix (Bächinger et al. 1978, Bruckner et al. 1978) that acts as a permanent linker even after removal of the non-collagenous domains.

Mutations within the C-propeptides further suggest that they are crucial for the correct interaction of the three polypeptide chains and for subsequent correct folding (refs. in Boudko et al. 2011).

References

Byers PH, Click EM, Harper E & Bornstein P (1975). Interchain disulfide bonds in procollagen are located in a large nontriple-helical COOH-terminal domain. *Proc Natl Acad Sci U S A*, 72, 3009-13

Bächinger HP, Brückner P, Timpl R & Engel J (1978). The role of cis-trans isomerization of peptide bonds in the coil leads to and comes from triple helix conversion of collagen. *Eur J Biochem*, 90, 605-13.

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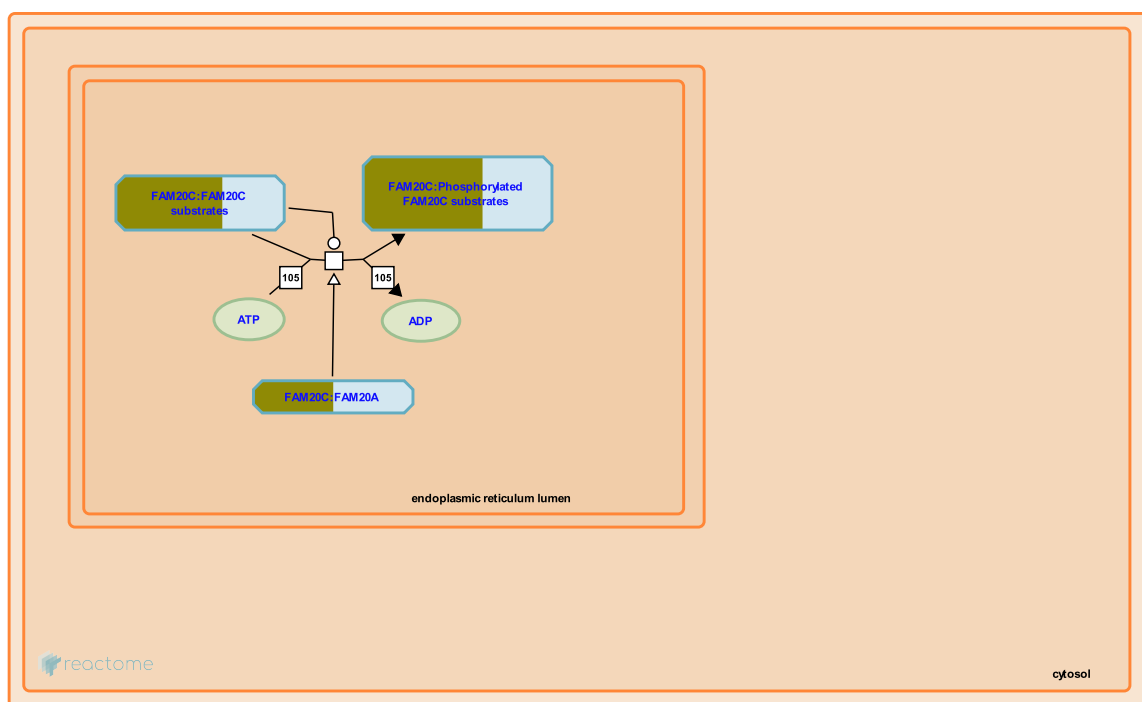
Date	Action	Author
2012-04-11	Authored	Jupe S
2012-05-24	Reviewed	Canty-Laird EG
2016-11-03	Edited	Jupe S

Date	Action	Author
2016-11-11	Created	Jupe S
2025-11-15	Modified	Weiser JD

31 submitted entities found in this pathway, mapping to 35 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
COL11A1	P12107	COL12A1	Q99715	COL13A1	Q5TAT6
COL14A1	Q05707	COL16A1	Q07092	COL18A1	P39060, P53420
COL1A1	P02452	COL1A2	P08123	COL20A1	Q9P218
COL21A1	Q96P44	COL22A1	Q8NFW1	COL23A1	Q86Y22
COL24A1	Q17RW2	COL26A1	Q86Y22, Q96A83	COL27A1	Q8IZC6
COL2A1	P02458	COL3A1	P02461	COL4A5	P29400
COL4A6	Q14031	COL5A1	P20908	COL5A2	P05997
COL5A3	P25940	COL6A1	P12109	COL6A2	P12110
COL6A3	A8TX70, P12111	COL8A1	P27658	COL8A2	P25067
COL9A1	P20849	COL9A2	Q14055, Q9UMD9	COL9A3	Q14050
EMID1	P39060				

3. Post-translational protein phosphorylation (R-HSA-8957275)



Secretory pathway kinases phosphorylate a diverse array of substrates involved in many physiological processes.

References

Sreelatha A, Kinch LN & Tagliabracci VS (2015). The secretory pathway kinases. *Biochim. Biophys. Acta*, 1854, 1687-93. [↗](#)

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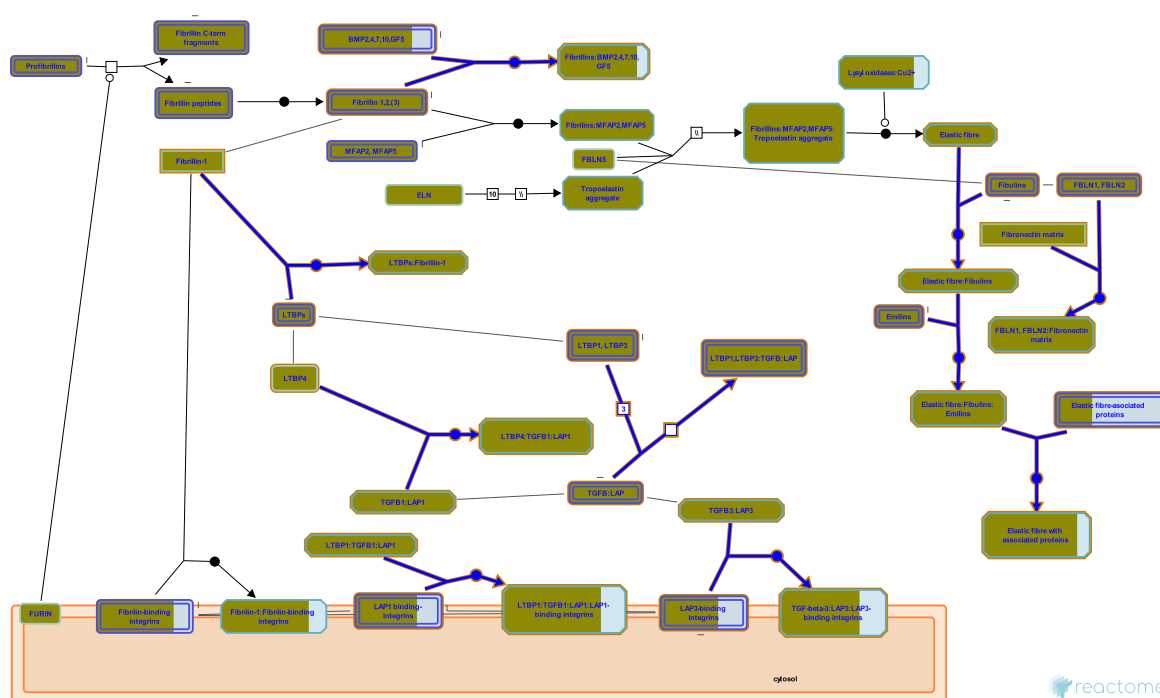
Date	Action	Author
2016-12-08	Authored	Jupe S
2017-01-23	Reviewed	Wiley SE
2017-01-24	Edited	Jupe S
2017-01-24	Created	Jupe S

70 submitted entities found in this pathway, mapping to 72 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AFP	P02771	ANO8	Q9HCE9	APLP2	Q06481
APOE	P02649	APOL1	O14791	APP	P05067
APPL2	Q06481	BMP4	P12644	C3	P01024
C4A	P0C0L4	C4B	P0C0L4	CABP1	Q15084
CALU	O43852	CCN1	O00622	CDH2	P19022
CHGB	P05060	CHRD1	Q9BU40	CKAP4	Q07065
CP	P00450	CSF1	P09603	CST3	P01034
DNAJC3	Q13217	FAM20C	Q8IXL6	FBN1	P35555
FN1	P02751	FSTL1	Q12841	FSTL3	O95633

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GAS6	Q14393	GOLM1	Q8NBJ4	GPC3	P51654
HRC	P23327	HSP90B1	P14625	IGFBP3	P17936
IGFBP4	P22692	IGFBP5	P24593	IGFBP7	Q16270
KTN1	Q86UP2	LAMB1	P07942	LAMB2	P11047, P55268
LGALS1	P09382	LTBP1	Q14766	MATN3	O15232
MBTPS1	Q14703	MELTF	P08582	MGAT4A	Q9UM21
MSLN	Q13421	MXRA8	Q9BRK3	NOTUM	Q6P988
NUCB1	Q02818	P4HB	P07237	PCSK9	Q8NBP7
PDIA6	Q15084	PEAR1	P11047	PROC	P04070
PRSS23	O95084	QSOX1	O00391	RCN1	Q15293
SCG2	O00255, P13521	SCG3	Q8WXD2	SDC2	P34741
SERPINA1	P01009	SHISA5	Q8N114	SPARC	Q14515
TF	P02787	TIMP1	P01033	TMEM132A	Q24JP5
TMEM132E	Q24JP5	VGF	O15240	VWA1	Q6PCB0
WFS1	O76024				

4. Molecules associated with elastic fibres ([R-HSA-2129379](#))



Proteins found associated with microfibrils include vitronectin (Dahlback et al. 1990), latent transforming growth factor beta-binding proteins (Kielty et al. 2002, Munger & Sheppard 2011), emilin (Bressan et al. 1993, Mongiat et al. 2000), members of the microfibrillar-associated proteins (MFAPs, Gibson et al. 1996), and fibulins (Roark et al. 1995, Yanagisawa et al. 2002). The significance of these interactions is not well understood but may help mediate elastin-fibrillin interactions during elastic fibre assembly.

Proteoglycans such as versican (Isogai et al. 2002), biglycan, and decorin (Reinboth et al. 2002) can interact with the microfibrils. They confer specific properties including hydration, impact absorption, molecular sieving, regulation of cellular activities, mediation of growth factor association, and release and transport within the extracellular matrix (Buczek-Thomas et al. 2002). In addition, glycosaminoglycans have been shown to interact with tropoelastin through its lysine side chains (Wu et al. 1999) regulating tropoelastin assembly (Tu and Weiss, 2008).

References

Wagenseil JE & Mecham RP (2007). New insights into elastic fiber assembly. Birth Defects Res. C Embryo Today, 81, 229-40. [🔗](#)

Kielty CM, Sherratt MJ & Shuttleworth CA (2002). Elastic fibres. *J Cell Sci*, 115, 2817-28. [🔗](#)

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Date	Action	Author
2012-02-21	Created	Jupe S
2012-04-30	Authored	Jupe S
2012-11-02	Reviewed	Muiznieks LD
2012-11-12	Edited	Jupe S
2025-11-15	Modified	Weiser ID

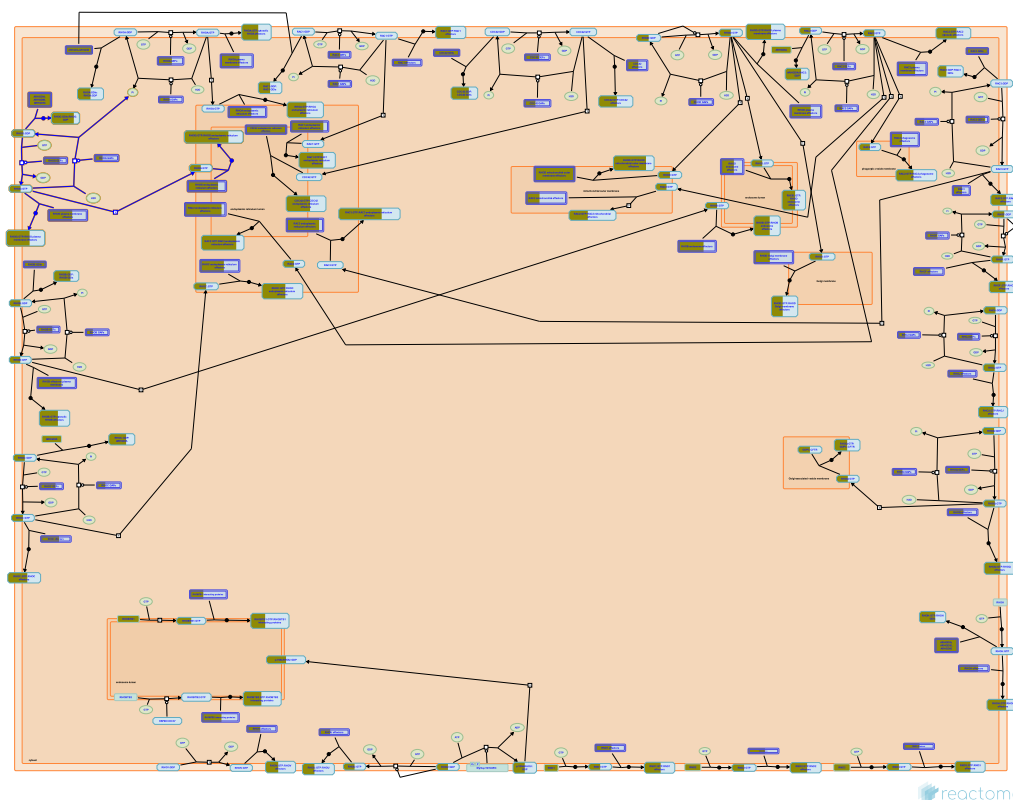
31 submitted entities found in this pathway, mapping to 35 Reactome entities

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BMP2	P12643	BMP4	P12644	BMP7	P18075
EFEMP1	Q12805	EFEMP2	O95967, Q12805	ELN	P15502
EMILIN1	Q9Y6C2	EMILIN2	Q9BXX0	EMILIN3	Q9NT22
FBLN1	P23142	FBLN2	P98095	FBLN5	Q12805, Q9UBX5
FBN1	P35555	FBN2	P35556, Q75N90	FN1	P02751
GDF5	P43026	ITGA8	P53708	ITGAV	P06756
ITGB1	P05556	ITGB3	P05106	ITGB5	P18084
LTBP1	Q14766	LTBP2	Q14767	LTBP3	Q14767, Q9NS15
LTBP4	Q8N2S1	MFAP2	P55001	MFAP4	P55083
MFAP5	Q13361	TGFB1	P01137	TGFB2	P61812
TGFB3	P10600				

Interactors found in this pathway (5)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CACNA1A	O00555	Q8N2S1	FBLN5	Q9UBX5	Q8N2S1
FBXO2	Q9UK22	Q8N2S1	NOTCH2	Q04721	Q8N2S1
ZNF74	Q16587	Q8N2S1			

5. RHOG GTPase cycle (R-HSA-9013408)



This pathway catalogues RHOG guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOG effectors. RHOG is a RAC-related RHO GTPase, ~70% identical to RAC1 (Vincent et al. 1992, de Curtis 2008). RHOG is broadly expressed in different tissue types. It regulates the cytoskeleton, acting either upstream of or in parallel to RAC1. RHOG regulates cell polarity, adhesion, migration and invasion, contributing to the formation of lamellipodia and invadopodia (Gauthier-Rouvière et al. 1998, Al-Koussa et al. 2020). The ortholog of RHOG is required for neuronal development in *C. elegans* (de Curtis 2008). RHOG is involved in VEGF signaling and angiogenesis (El Baba et al. 2020). RHOG cooperates with RAC1 and CDC42 in malignant cell transformation (Roux et al. 1997) and may contribute to invasiveness of cancer cells (Al-Koussa et al. 2020).

References

- de Curtis I (2008). Functions of Rac GTPases during neuronal development. *Dev. Neurosci.*, 30, 47-58. [🔗](#)
- El Baba N, Farran M, Khalil EA, Jaafar L, Fakhoury I & El-Sibai M (2020). The Role of Rho GTPases in VEGF Signaling in Cancer Cells. *Anal Cell Pathol (Amst)*, 2020, 2097214. [🔗](#)
- Al-Koussa H, Atat OE, Jaafar L, Tashjian H & El-Sibai M (2020). The Role of Rho GTPases in Motility and Invasion of Glioblastoma Cells. *Anal Cell Pathol (Amst)*, 2020, 9274016. [🔗](#)
- Vincent S, Jeanteur P & Fort P (1992). Growth-regulated expression of rhoG, a new member of the ras homolog gene family. *Mol Cell Biol*, 12, 3138-48. [🔗](#)
- Gauthier-Rouvière C, Vignal E, Mérianne M, Roux P, Montcourier P & Fort P (1998). RhoG GTPase controls a pathway that independently activates Rac1 and Cdc42Hs. *Mol Biol Cell*, 9, 1379-94. [🔗](#)

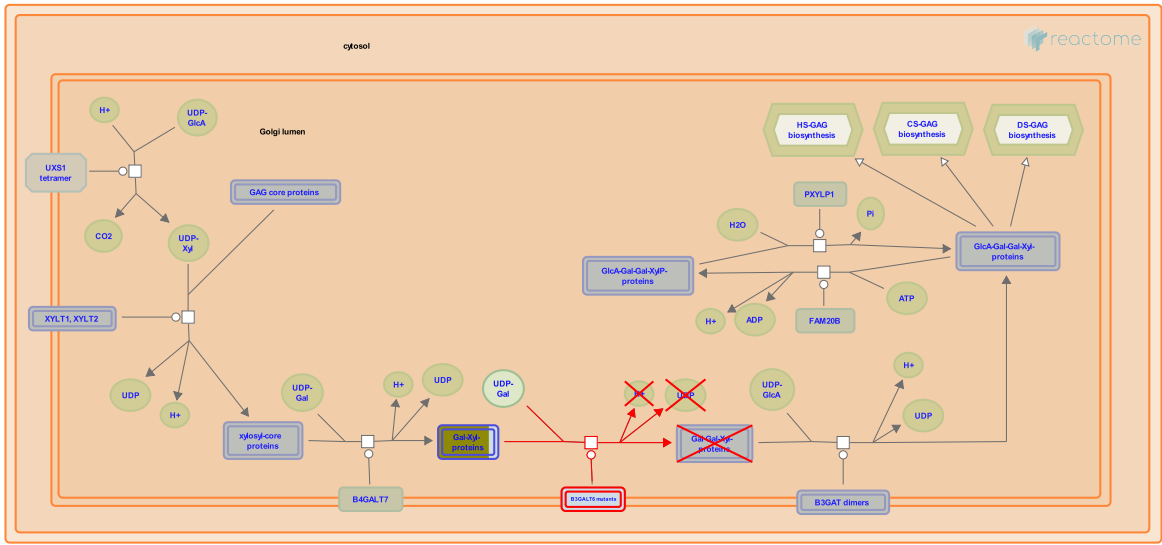
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Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

51 submitted entities found in this pathway, mapping to 51 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ANKLE2	Q86XL3	ARFGAP1	Q9NP61	ARFGAP3	Q9NP61
ARHGAP10	Q5T5U3	ARHGAP21	Q5T5U3	ARHGAP32	A7KAX9
ARHGAP33	A7KAX9	ARHGAP39	Q9C0H5	ARHGDIA	P52565
ARHGDIB	P52566	ARHGDIG	Q99819	ARHGEF5	Q12774
CAV1	Q03135	CDC42EP1	Q00587	DIAPH3	Q9NSV4
DOCK2	Q92608	DOCK3	Q8IZD9	DOCK4	Q8N110
DOCK5	Q9H7D0	DSG2	Q14126	ELMO2	Q96JJ3
EPHA2	P29317	ERBIN	Q96RT1	ESYT1	Q9BSJ8
ITGB1	P05556	ITSN1	Q15811	KALRN	O60229
KTN1	Q86UP2	LBR	Q14739	LEMD3	Q9Y2U8
LETM1	O95202	LMAN1	P49257	MCAM	P43121
MCF2	P10911	MCF2L	O15068	MPP7	Q5T2T1
NDUFS3	O75489	PAK4	O96013	PGRMC2	O15173
PLD1	Q13393	PLEKHG3	A1L390	PREX1	Q8TCU6
RHOG	P84095	TFRC	P02786	TRIO	O75962
VAMP3	Q15836	VANGL1	Q8TAA9	VAPB	O95292
VAV3	Q9UKW4	VRK2	Q86Y07	YKT6	O15498

6. Defective B3GALT6 causes EDSP2 and SEMDJL1 (R-HSA-4420332)



Diseases: Ehlers-Danlos syndrome, spondyloepimetaphyseal dysplasia.

The biosynthesis of dermatan sulfate/chondroitin sulfate and heparin/heparan sulfate glycosaminoglycans (GAGs) starts with the formation of a tetrasaccharide linker sequence attached to the core protein. Beta-1,3-galactosyltransferase 6 (B3GALT6) is one of the critical enzymes involved in the formation of this linker sequence. Defects in B3GALT6 causes Ehlers-Danlos syndrome progeroid type 2 (EDSP2; MIM:615349), a severe disorder resulting in a broad spectrum of skeletal, connective tissue and wound healing problems. Defects in B3GALT6 can also cause spondyloepimetaphyseal dysplasia with joint laxity type 1 (SEMDJL1; MIM:271640), characterised by spinal deformity and lax joints, especially of the hands and respiratory compromise resulting in early death (Nakajima et al. 2013, Malfait et al. 2013).

References

Malfait F, Kariminejad A, Van Damme T, Gauche C, Syx D, Merhi-Soussi F, ... De Paepe A (2013). Defective Initiation of Glycosaminoglycan Synthesis due to B3GALT6 Mutations Causes a Pleiotropic Ehlers-Danlos Syndrome-like Connective Tissue Disorder. *Am. J. Hum. Genet.* [\[link\]](#)

Nakajima M, Mizumoto S, Miyake N, Kogawa R, Iida A, Ito H, ... Ikegawa S (2013). Mutations in B3GALT6, which Encodes a Glycosaminoglycan Linker Region Enzyme, Cause a Spectrum of Skeletal and Connective Tissue Disorders. *Am. J. Hum. Genet.* [\[link\]](#)

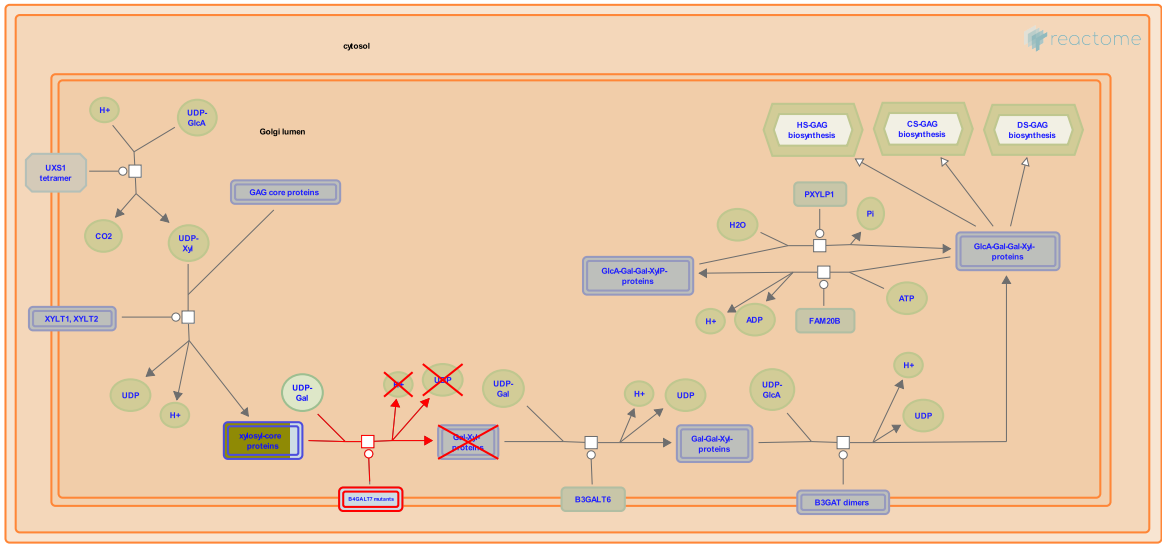
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Date	Action	Author
2013-09-03	Edited	Jassal B
2013-09-03	Authored	Jassal B
2013-09-03	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

16 submitted entities found in this pathway, mapping to 16 Reactome entities

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GPC2	Q8N158	GPC3	P51654	GPC4	O75487
GPC5	P78333	GPC6	Q9Y625	HSPG2	P98160
NCAN	O14594	SDC2	P34741	SDC3	O75056
SDC4	P31431				

7. Defective B4GALT7 causes EDS, progeroid type (R-HSA-3560783)



Diseases: Ehlers-Danlos syndrome.

Ehlers–Danlos syndrome (EDS) is a group of inherited connective tissue disorders, caused by a defect in the synthesis of collagen types I or III. Abnormal collagen renders connective tissues more elastic. The severity of the mutation can vary from mild to life-threatening. There is no cure and treatment is supportive, including close monitoring of the digestive, excretory and particularly the cardiovascular systems. Defective B4GALT7, a galactosyltransferase important in proteoglycan synthesis, causes the progeroid variant of EDS (MIM:130070). Features include an aged appearance, developmental delay, short stature, generalized osteopenia, defective wound healing, hypermobile joints, hypotonic muscles, and loose but elastic skin (Okajima et al. 1999).

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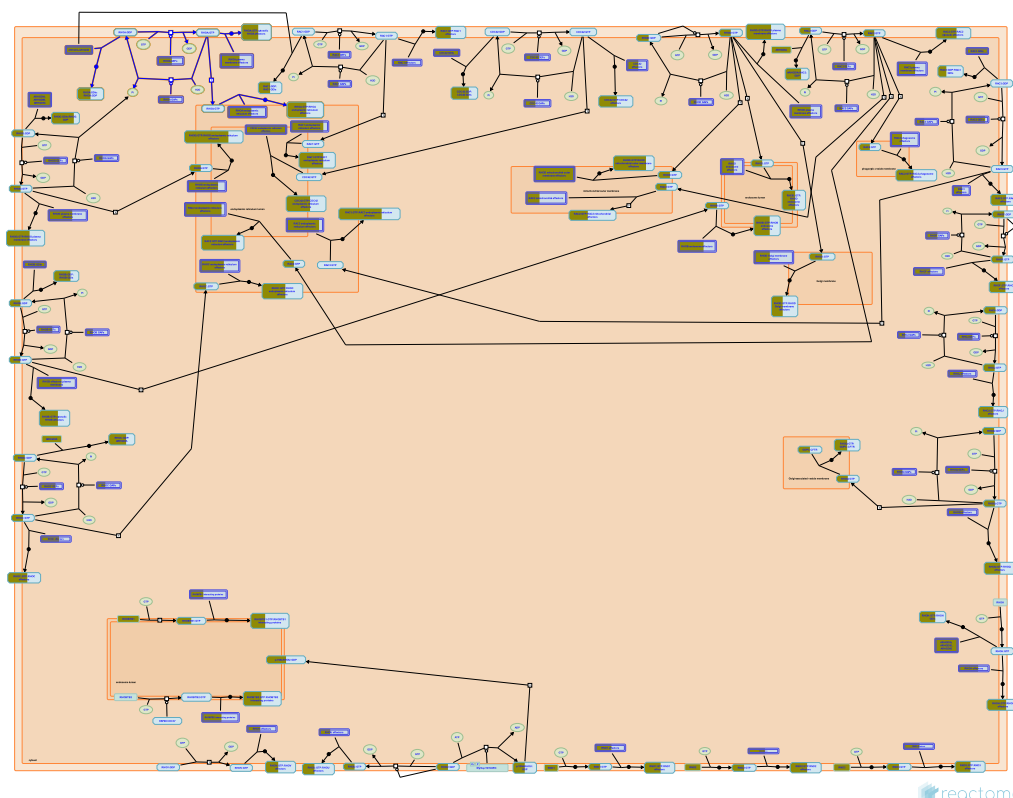
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2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

16 submitted entities found in this pathway, mapping to 16 Reactome entities

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GPC2	Q8N158	GPC3	P51654	GPC4	O75487

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GPC5	P78333	GPC6	Q9Y625	HSPG2	P98160
NCAN	O14594	SDC2	P34741	SDC3	O75056
SDC4	P31431				

8. RHOA GTPase cycle (R-HSA-8980692)



This pathway catalogues RHOA guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOA effectors. RHOA is one of the three best characterized RHO GTPases, the other two being RAC1 and CDC42, and is the founding member of the RHO GTPase family (Zhou and Zheng 2013). RHOA regulates the cytoskeleton and cell contractility (Lessey et al. 2012), thus playing a role in a number of cellular functions, such as adhesion, migration, survival, division, vesicle trafficking and gene expression (Zhou and Zheng 2013). RHOA-regulated processes are involved in mechanotransduction (Lessey et al. 2012; Marjoram et al. 2014), neuronal development (Zhou and Zheng 2013; Fujita and Yamashita 2014; Hu and Selzer 2017), immune system development (Zhou and Zheng 2013; Ricker et al. 2016; Bros et al. 2019), and cardiovascular regulation (Zhou and Zheng 2013; Cai et al. 2016; Shimokawa et al. 2016). RHOA mutations are frequently found in cancer (Kataoka and Ogawa 2016). Toxins of numerous pathogens target RHOA to hijack the host cytoskeleton (Jamilloux et al. 2018).

References

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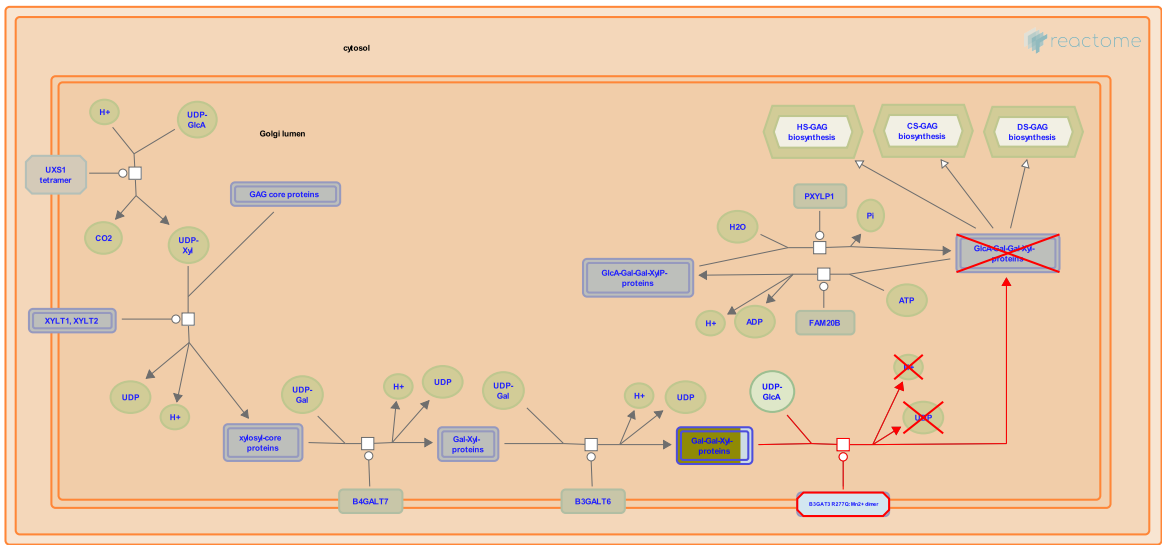
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2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M

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ARHGAP10	A1A4S6, Q5T5U3	ARHGAP18	Q8N392	ARHGAP19	Q14CB8
ARHGAP20	Q9P2F6	ARHGAP21	Q5T5U3	ARHGAP22	Q7Z5H3
ARHGAP23	Q9P227	ARHGAP28	Q9P2N2	ARHGAP29	Q52LW3
ARHGAP31	Q2M1Z3	ARHGAP32	A7KAX9	ARHGAP33	A7KAX9
ARHGAP39	Q9C0H5	ARHGAP42	A6NI28	ARHGAP6	O43182
ARHGAP8	P85298	ARHGDIA	P52565	ARHGDIB	P52566
ARHGEF1	Q92888	ARHGEF10	O15013	ARHGEF10L	Q9HCE6
ARHGEF11	O15085	ARHGEF17	Q96PE2	ARHGEF28	Q8N1W1
ARHGEF3	Q9NR81	ARHGEF5	Q12774	ARHGEF7	Q14155
C1QBP	Q07021	CAV1	Q03135	CAVIN1	Q6NZI2
CIT	O14578, O14578-3	DAAM1	Q9Y4D1	DDRCK1	Q96HY6
DIAPH3	Q9NSV4	DLC1	Q96QB1	DOCK2	Q92608
ERBIN	Q96RT1	FAM13A	O94988	FARP1	Q9Y4F1
FHOD3	Q8IVF7	FLOT1	O75955	FMNL3	Q8IVF7
IQGAP1	P46940	JUP	P14923	KALRN	O60229
KTN1	Q86UP2	LBR	Q14739	LMAN1	P49257
MCAM	P43121	MCF2	P10911	MCF2L	O15068
MID1	Q9NQW6	NET1	Q7Z628	NGEF	Q8N5V2
OBSCN	Q5VST9	PAK1	Q16512	PCDH7	O60245
PGRMC2	O15173	PKN3	Q6P5Z2	PLD1	Q13393
PLEKHG3	A1L390	PREX1	Q8TCU6	PREX2	Q70Z35
RHPN2	Q8IUC4	ROCK1	Q13464	SCFD1	Q8WVM8
SNAP23	O00161	STARD13	Q9Y3M8	STARD8	Q92502
STK10	O94804	STOM	P27105	TFRC	P02786
TIAM1	Q13009	TMEM87B	Q8NBN3	TRIO	O75962
VAMP3	Q15836	VANGL1	Q8TAA9	VAPB	O95292
VAV3	Q9UKW4	VMA22	Q96NT0	YKT6	O15498

9. Defective B3GAT3 causes JDSSDHD ([R-HSA-3560801](#))



Diseases: congenital heart defect, Larsen syndrome.

Galactosylgalactosylxylosylprotein 3-beta-glucuronosyltransferases1, 2 and 3 (B3GAT1-3) are involved in forming the linker tetrasaccharide present in heparan sulfate and chondroitin sulfate. Defects in B3GAT3 cause multiple joint dislocations, short stature, craniofacial dysmorphism, and congenital heart defects (JDSSDHD; MIM:245600). This is an autosomal recessive disease characterized by dysmorphic facies, elbow, hip and knee dislocations, clubfeet, short stature and cardiovascular defects (Steel & Kohl 1972, Bonaventure et al. 1992, Baasanjav et al. 2011). JDSSDHD has phenotypic similarities to Larsen syndrome (Larsen et al. 1950).

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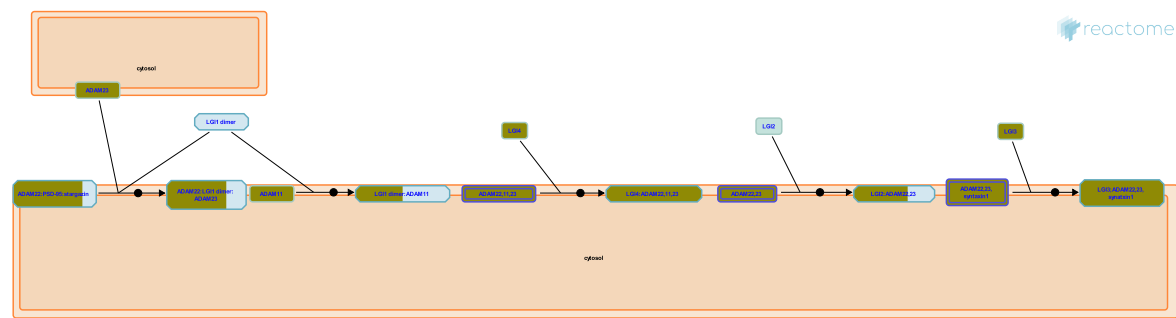
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2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

16 submitted entities found in this pathway, mapping to 16 Reactome entities

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GPC2	Q8N158	GPC3	P51654	GPC4	O75487
GPC5	P78333	GPC6	Q9Y625	HSPG2	P98160
NCAN	O14594	SDC2	P34741	SDC3	O75056
SDC4	P31431				

10. LGI-ADAM interactions (R-HSA-5682910)



Cellular compartments: extracellular region.

Synapse formation and maturation require multiple interactions between presynaptic and postsynaptic neurons. These interactions are mediated by a diverse set of synaptogenic proteins (Kegel et al. 2013, Siddiqui & Craig 2011). Initial synapse formation needs both the binding of secreted proteins to presynaptic and postsynaptic receptors, and the direct binding between presynaptic and postsynaptic transmembrane proteins. One class of molecules that plays an important role in cellular interactions in nervous system development and function is the leucine-rich glioma inactivated (LGI) protein family. These are secreted synaptogenic proteins consisting of an LRR (leucine-rich repeat) domain and a epilepsy-associated or EPTP (epitempin) domain (Gu et al. 2002). Both protein domains are generally involved in protein-protein interactions. Genetic and biochemical evidence suggests that the mechanism of action of LGI proteins involves binding to a subset of cell surface receptors belonging to the ADAM (a disintegrin and metalloproteinase) family, i.e. ADAM11, ADAM22 and ADAM23. These interactions play crucial role in the development and function of the vertebrate nervous system mainly mediating synaptic transmission and myelination (Kegel et al. 2013, Novak 2004, Seals & Courtneidge 2003).

References

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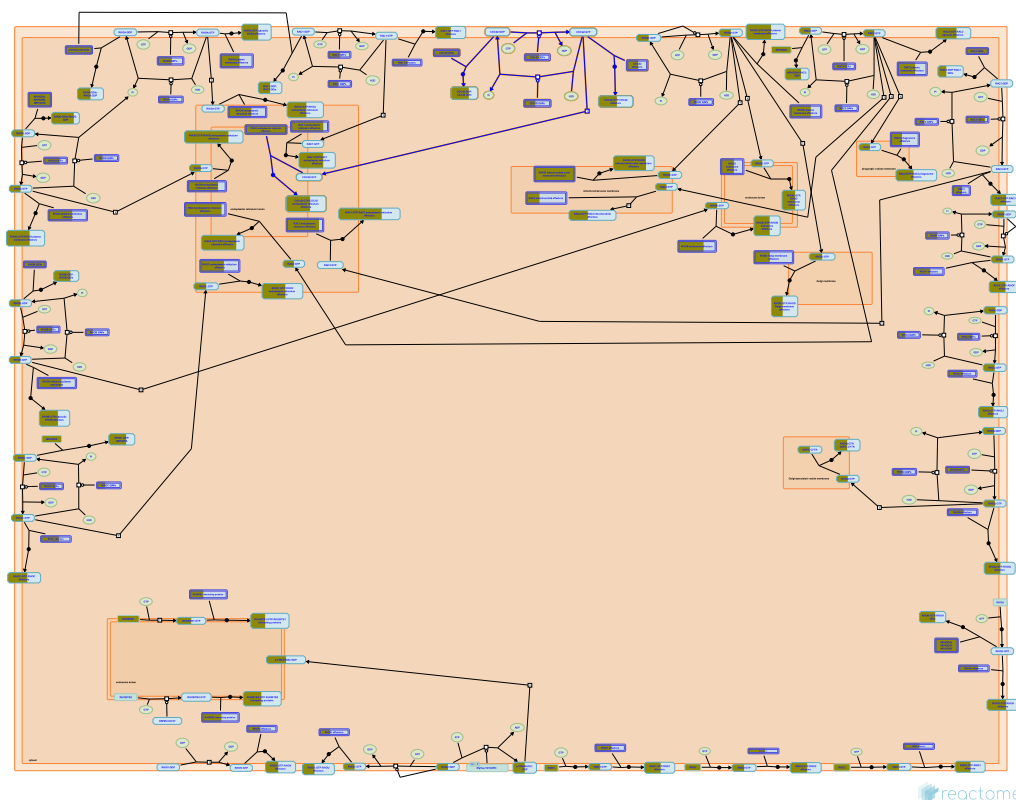
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2015-03-11	Created	Garapati P V
2015-04-20	Reviewed	Meijer D
2025-11-15	Modified	Weiser JD

11 submitted entities found in this pathway, mapping to 11 Reactome entities

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CACNG2	Q9Y698	CACNG3	O60359	CACNG8	Q8WXS5
DLG4	P78352	LGI3	Q8N145	LGI4	Q8N135
STX1A	Q16623	STX1B	P61266		

11. CDC42 GTPase cycle (R-HSA-9013148)



This pathway catalogues CDC42 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and CDC42 effectors. CDC42 is one of the three best characterized RHO GTPases, the other two being RHOA and RAC1. By regulating the cytoskeleton, CDC42 regulates cell polarity across different species, from yeast to humans (Pichaud et al. 2019, Woods and Lew 2019). CDC42 is an essential regulator of polarized morphogenesis in epithelial cells, where it coordinates formation of the apical membrane and lumen formation, as well as junction maturation (Pichaud et al. 2019). CDC42 plays a role in cell-to-cell adhesion and cell cycle regulation (Xiao et al. 2018). CDC42 takes part in the regulation of membrane trafficking. Dysfunction of several CDC42-specific GEFs has been shown to impair intracellular trafficking (Egorov and Polishchuk 2017). CDC42 participates in insulin synthesis and secretion and contributes to the pathogenesis of insulin resistance and diabetic nephropathy (Huang et al. 2019). CDC42 is often dysregulated in cancer because a number of GEFs and GEF activators that act upstream of RAC1 and CDC42 are known oncogenes (Aguilar et al. 2017; Maldonado et al. 2018; Zhang et al. 2019; Maldonado et al. 2020). CDC4 promotes cancer cell proliferation, survival, invasion, migration and metastasis (Xiao et al. 2018), especially under hyperglycemia (Huang et al. 2019).

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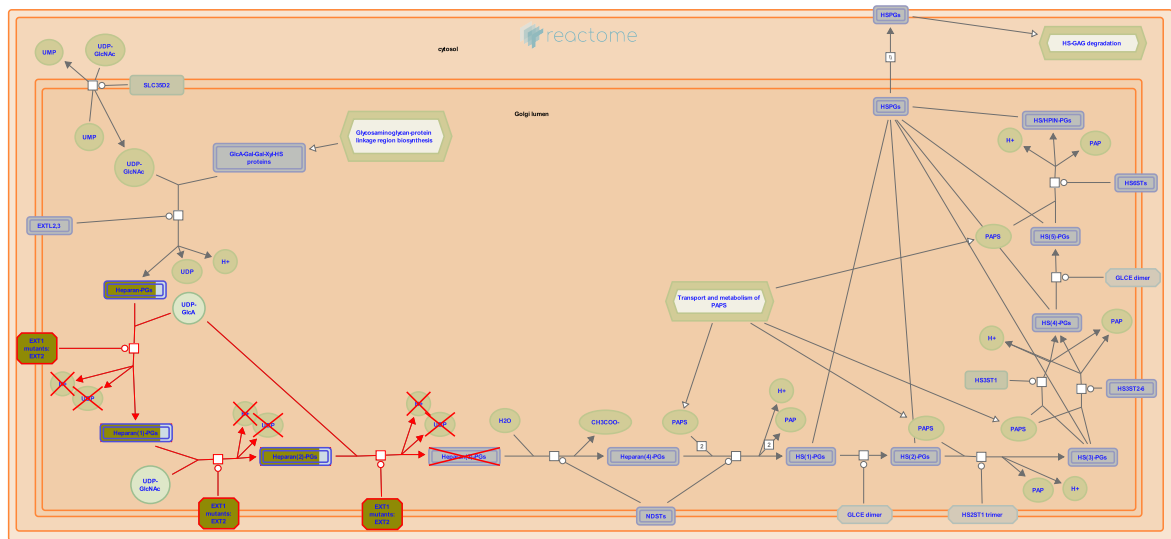
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2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

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ARHGAP20	Q9P2F6	ARHGAP21	Q5T5U3	ARHGAP22	Q7Z5H3
ARHGAP29	Q52LW3	ARHGAP31	Q2M1Z3	ARHGAP32	A7KAX9
ARHGAP33	A7KAX9, O14559	ARHGAP39	Q9C0H5	ARHGAP42	A6NI28
ARHGDIA	P52565	ARHGDIB	P52566	ARHGDIG	Q99819
ARHGEF10	O15013	ARHGEF11	O15085	ARHGEF5	Q12774
ARHGEF6	Q15052	ARHGEF7	Q14155	ARHGEF9	O43307
CAV1	Q03135	CDC42EP1	Q00587	CDC42EP3	Q9UKI2
CDC42EP4	Q9H3Q1	CDC42EP5	Q6NZY7	CHN1	P15882
CPNE8	Q86YQ8	DAAM1	Q9Y4D1	DIAPH3	Q9NSV4
DLC1	Q96QB1	DNMBP	Q6XZF7	DOCK10	Q96BY6
DOCK11	Q5JSL3	DOCK7	Q96N67	DOCK9	Q9BZ29
FARP1	Q9Y4F1	FGD1	P98174	FGD3	Q5JSP0
FGD4	Q96M96	FHOD3	Q8IVF7	FMNL2	Q96PY5
FMNL3	Q8IVF7	FNBPL	Q5T0N5	IQGAP1	P46940
ITSN1	Q15811	JUP	P14923	KCTD3	Q9Y597
KTN1	Q86UP2	LBR	Q14739	MCF2	P10911
MCF2L	O15068	NGEF	Q8N5V2	PAK1	Q13153
PAK3	O75914, Q13153	PAK4	O96013	PAK5	Q9NQU5, Q9P286
PAK6	Q9NQU5	PLD1	Q13393	PLEKHG3	A1L390
PLEKHG4B	Q96PX9	PREX1	Q8TCU6	PREX2	Q70Z35
RALBP1	Q15311	SNAP23	O00161	SPATA13	Q96N96
SRGAP2	O43295, O75044	STARD13	Q9Y3M8	STARD8	Q92502
STEAP3	Q658P3	STOM	P27105	SYDE1	Q6ZW31
TFRC	P02786	TIAM1	Q13009	TRIO	O75962, Q96PX9
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Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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YKT6	O15498				

12. Defective EXT1 causes exostoses 1, TRPS2 and CHDS ([R-HSA-3656253](#))



Diseases: hereditary multiple exostoses.

Heparan sulfate (HS) is involved in regulating various body functions during development, homeostasis and pathology including blood clotting, angiogenesis and metastasis of cancer cells. Exostosin 1 and 2 (EXT1 and 2) glycosyltransferases are required to form HS. They are able to transfer N-acetylglucosamine (GlcNAc) and glucuronate (GlcA) to HS during its synthesis. The functional form of these enzymes appears to be a complex of the two located on the Golgi membrane. Defects in either EXT1 or EXT2 can cause hereditary multiple exostoses 1 (Petersen 1989) and 2 (McGaughran et al. 1995) respectively (MIM:133700 and MIM:133701), autosomal dominant disorders characterized by multiple projections of bone capped by cartilage resulting in deformed legs, forearms and hands. Trichorhinophalangeal syndrome, type II (TRPS2 aka Langer-Giedion syndrome, LGS) is a disorder that combines the clinical features of trichorhinophalangeal syndrome type I (TRPS1, MIM:190350) and multiple exostoses type I, caused by mutations in the TRPS1 and EXT1 genes, respectively (Langer et al. 1984, Ludecke et al. 1995). Defects in EXT1 may also be responsible for chondrosarcoma (CHDS; MIM:215300) (Schajowicz & Bessone 1967, Hecht et al. 1995).

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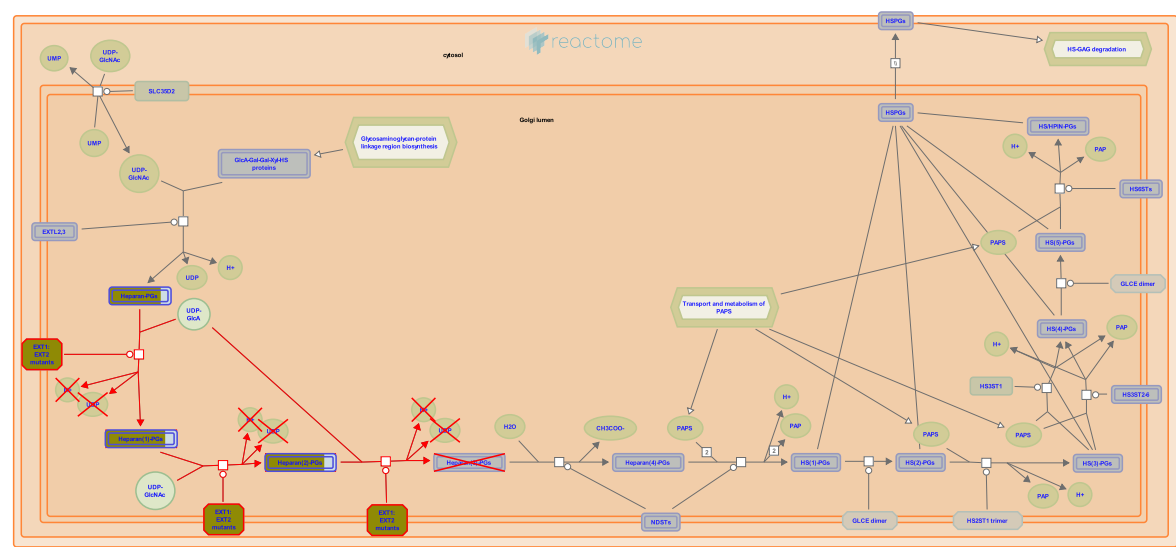
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2014-07-09	Reviewed	Spillmann D
2025-05-30	Reviewed	Matthews L
2025-06-02	Modified	Stephan R

13 submitted entities found in this pathway, mapping to 13 Reactome entities

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GPC4	O75487	GPC5	P78333	GPC6	Q9Y625
HSPG2	P98160	SDC2	P34741	SDC3	O75056
SDC4	P31431				

13. Defective EXT2 causes exostoses 2 (R-HSA-3656237)



Diseases: hereditary multiple exostoses.

Heparan sulfate (HS) is involved in regulating various body functions during development, homeostasis and pathology including blood clotting, angiogenesis and metastasis of cancer cells. Exostosin 1 and 2 (EXT1 and 2) glycosyltransferases are required to form HS. They are able to transfer N-acetylglucosamine (GlcNAc) and glucuronate (GlcA) to HS during its synthesis. The functional form of these enzymes appears to be a complex of the two located on the Golgi membrane. Defects in either EXT1 or EXT2 can cause hereditary multiple exostoses 1 (Petersen 1989) and 2 (McGaughran et al. 1995) respectively (MIM:133700 and MIM:133701), autosomal dominant disorders characterised by multiple projections of bone capped by cartilage resulting in deformed legs, forearms and hands.

References

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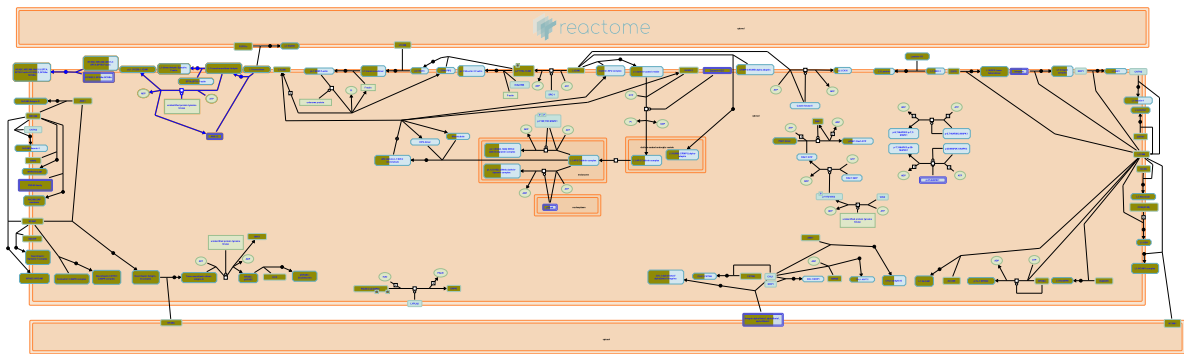
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2014-07-09	Reviewed	Spillmann D
2025-05-30	Reviewed	Matthews L
2025-06-02	Modified	Stephan R

13 submitted entities found in this pathway, mapping to 13 Reactome entities

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GPC4	O75487	GPC5	P78333	GPC6	Q9Y625
HSPG2	P98160	SDC2	P34741	SDC3	O75056
SDC4	P31431				

14. Interaction between L1 and Ankyrins (R-HSA-445095)



Ankyrins are a family of adaptor proteins that couple membrane proteins such as voltage gated Na⁺ channels and the Na⁺/K⁺ anion exchanger to the spectrin actin cytoskeleton. Ankyrins are encoded by three genes (ankyrin-G, -B and -R) of which ankyrin-G and -B are the major forms expressed in the developing nervous system. Ankyrins bind to the cytoplasmic domain of L1 CAMs and couple them and ion channel proteins, to the spectrin cytoskeleton. This binding enhances the homophilic adhesive activity of L1 and reduces its mobility within the plasma membrane. L1 interaction with ankyrin mediates branching and synaptogenesis of cortical inhibitory neurons.

References

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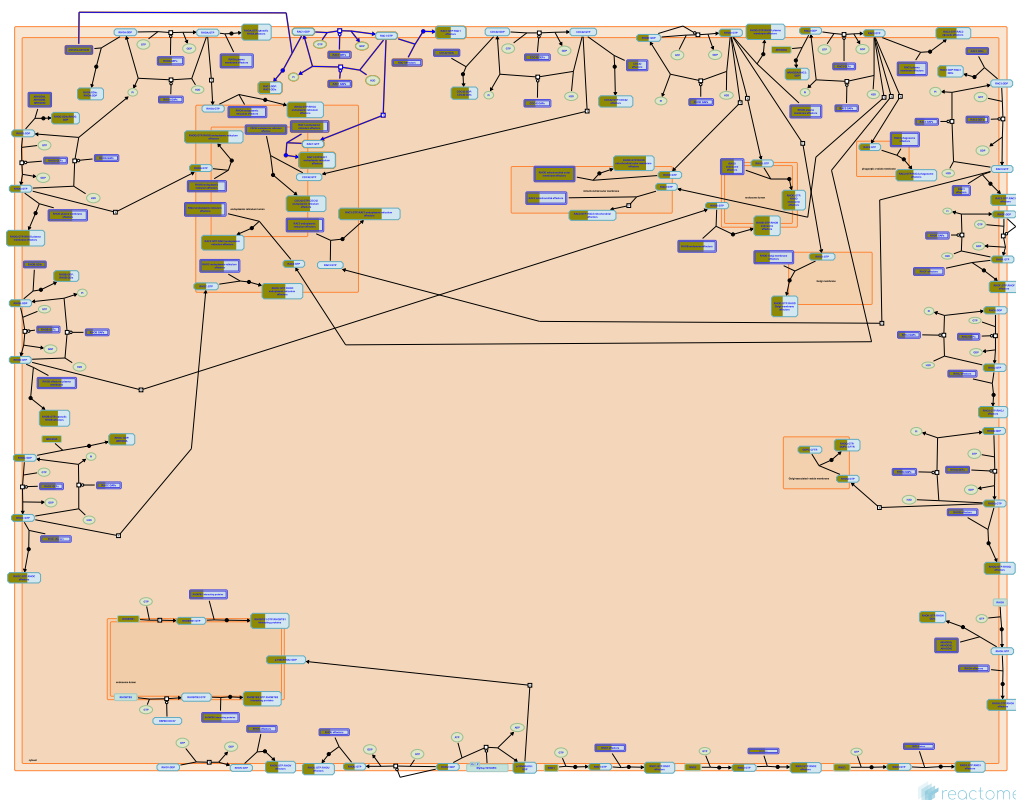
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2010-02-16	Reviewed	Maness PF
2025-11-15	Modified	Weiser JD

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NFASC	O94856	NRCAM	Q92823	SCN2A	Q99250
SCN2B	O60939	SCN3A	Q99250, Q9NY46	SCN3B	Q9NY72
SCN4A	P35499	SCN7A	Q01118	SCN8A	Q9UQD0

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SPTA1	P02549	SPTB	P11277	SPTBN1	Q01082
SPTBN2	O15020	SPTBN4	Q9H254	SPTBN5	Q9NRC6
TNKS2	P16157				

15. RAC1 GTPase cycle (R-HSA-9013149)



This pathway catalogues RAC1 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RAC1 effectors (reviewed by Payapilli and Malliri 2018). RAC1 is one of the three best characterized RHO GTPases, the other two being RHOA and CDC42. RAC1 regulates the cytoskeleton and the production of reactive oxygen species (ROS) (Acevedo and Gonzalez-Billault 2018) and is involved in cell adhesion and cell migration (Marei and Malliri 2017). RAC1 is involved in neuronal development (de Curtis et al. 2014). In neurons, RAC1 activity is regulated by synaptic activation and RAC1-mediated changes in actin cytoskeleton are implicated in dendritic spine morphogenesis, which plays a role in memory formation and learning (Tajeda-Simon 2015; Costa et al. 2020). RAC1 is involved in metabolic regulation of pancreatic islet β -cells and in diabetes pathophysiology (Kowluru 2017; Kowluru et al. 2020). RAC1-mediated activation of NOX2 contributes to mitochondrial damage and the development of retinopathy in patients with diabetes (Sahajpal et al. 2019). RAC1 is important for exercise and contraction-stimulated glucose uptake in skeletal muscles (Sylov et al. 2014). RAC1 plays an important role in the maintenance of intestinal barrier integrity under physiological conditions and during tissue repair after resolution of colitis. Toxins of many diarrhea-causing bacteria target RAC1 (Kotelevets and Chastre 2020). RAC1 is important for skin homeostasis and wound healing and is involved in the pathogenesis of psoriasis (Winge and Marinkovich 2019). RAC1 is essential to vascular homeostasis and chronically elevated RAC1 signaling contributes to vascular pathology (Marinkovic et al. 2015). RAC1 hyperactivation, mutation and copy-number gain are frequently observed in solid tumors (Zou et al. 2017; De et al. 2019; De et al. 2020; Cannon et al. 2020; Kotelevets and Chastre 2020).

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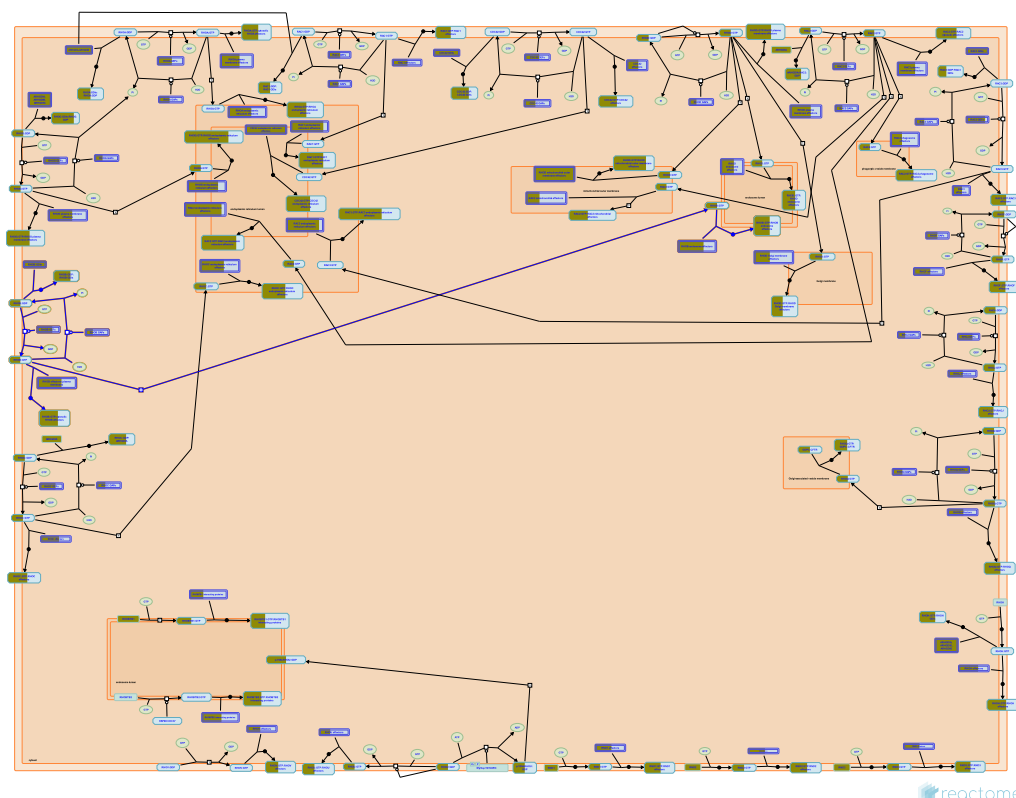
Date	Action	Author
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2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

95 submitted entities found in this pathway, mapping to 103 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABI1	Q8IZP0	ABI2	Q9NYB9	ARAP1	Q96P48
ARAP2	Q8WZ64	ARHGAP10	A1A4S6, Q5T5U3	ARHGAP12	Q8IWW6
ARHGAP20	Q9P2F6	ARHGAP21	Q5T5U3	ARHGAP22	Q7Z5H3
ARHGAP23	Q9P227	ARHGAP29	Q52LW3	ARHGAP31	Q2M1Z3
ARHGAP32	A7KAX9	ARHGAP33	A7KAX9, O14559	ARHGAP39	Q9C0H5
ARHGAP42	A6NI28	ARHGDIA	P52565	ARHGDIB	P52566
ARHGEF10	O15013	ARHGEF11	O15085	ARHGEF5	Q12774
ARHGEF6	Q15052	ARHGEF7	Q14155	BAIAP2L1	Q9UHR4
CAV1	Q03135	CDC42EP1	Q00587	CDC42EP4	Q9H3Q1
CHN1	P15882	CHN2	P52757	CIT	O14578, O14578-3
CYBA	P13498	CYFIP2	Q96F07	DIAPH3	Q9NSV4
DLC1	Q96QB1	DOCK10	Q96BY6	DOCK11	Q5JSL3
DOCK2	Q92608	DOCK3	Q8IZD9	DOCK4	Q8N110
DOCK5	Q9H7D0	DOCK7	Q96N67	DOCK9	Q9BZ29
EPHA2	P29317	ERBIN	Q96RT1	ESYT1	Q9BSJ8
FAM13A	O94988	FARP1	Q9Y4F1	FARP2	O94887
FERMT2	Q96AC1	IQGAP1	P46940	ITGB1	P05556
KALRN	O60229	KTN1	Q86UP2	LBR	Q14739
LEMD3	Q9Y2U8	MCAM	P43121	MCF2	P10911
MCF2L	O15068	MPP7	Q5T2T1	NCF2	P19878
NGEF	Q8N5V2	NHS	Q6T4R5	NOX1	Q9Y5S8
PAK1	Q13153, Q16512	PAK3	O75914, Q13153	PAK4	O96013
PAK5	Q9NQU5, Q9P286	PAK6	Q9NQU5	PIK3R3	Q92569
PLD1	O14939, Q13393	PLD2	O14939	PLEKHG3	A1L390
PREX1	Q8TCU6	PREX2	Q70Z35	RALBP1	Q15311
SNAP23	O00161	SOS2	Q07890	SPATA13	Q96N96

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SRGAP2	O43295, O75044	SWAP70	Q9UH65	SYDE2	Q5VT97
TAOK3	Q9H2K8	TFRC	P02786	TIAM1	Q13009
TRIO	O75962	VAMP3	Q15836	VANGL1	Q8TAA9
VAV3	Q9UKW4	VRK2	Q86Y07	WASF1	Q92558
WASF2	Q9Y6W5	WASF3	Q9UPY6	WIPF2	Q8TF74
WIPF3	A6NGB9	YKT6	O15498		

16. RHOB GTPase cycle (R-HSA-9013026)



This pathway catalogues RHOB guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOB effectors. RHOB belongs to the RHOA subfamily of RHO GTPases and shares 85% sequence identity with RHOA and RHOC. Posttranslational modifications of the unique C-terminus of RHOB regulate its subcellular localization. Similar to RHOA and RHOC, RHOB regulates the cytoskeleton and plays a role in cell migration. RHOB plays a role in regulation of membrane trafficking (Vega and Ridley 2018), cell proliferation (Vega and Ridley 2018), cell adhesion (Vega and Ridley 2018), DNA repair (Vega and Ridley 2018), and apoptosis (Prendergast 2001; Vega and Ridley 2018). RHOB-regulated cellular processes are important for the immune system function (Vega and Ridley 2018), pulmonary gas exchange (Wojciak-Stothard et al. 2012; Vega and Ridley 2018), and angiogenesis and vascular function (Vega and Ridley 2018). RHOB has been implicated both as a tumor suppressor and as an oncogene in cancer (Prendergast 2001; Huang and Prendergast 2006; Ju and Gilkes 2018; Calvayrac et al. 2018).

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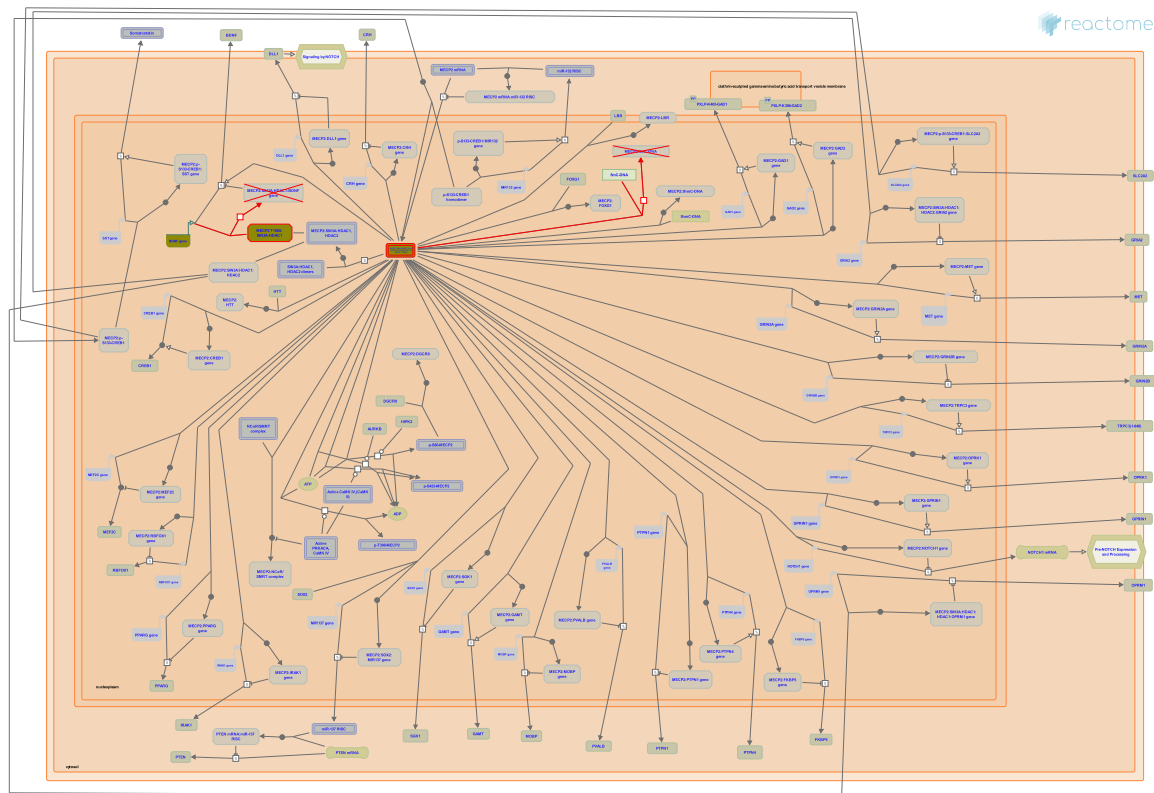
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Date	Action	Author
2017-07-21	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

43 submitted entities found in this pathway, mapping to 44 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP10	Q5T5U3	ARHGAP21	Q5T5U3	ARHGAP32	A7KAX9
ARHGAP33	A7KAX9	ARHGAP39	Q9C0H5	ARHGDIG	Q99819
ARHGEF1	Q92888	ARHGEF10	O15013	ARHGEF10L	Q9HCE6
ARHGEF11	O15085	ARHGEF17	Q96PE2	ARHGEF28	Q8N1W1
ARHGEF3	Q9NR81	ARHGEF5	Q12774	CAV1	Q03135
CAVIN1	Q6NZI2	CIT	O14578, O14578-3	DAAM1	Q9Y4D1
DIAPH3	Q9NSV4	DLC1	Q96QB1	ERBIN	Q96RT1
FLOT1	O75955	JUP	P14923	MCAM	P43121
MCF2	P10911	MCF2L	O15068	MID1	Q9NQW6
NET1	Q7Z628	PAK1	Q16512	PCDH7	O60245
PKN3	Q6P5Z2	PREX1	Q8TCU6	RHOB	P62745
RHPN2	Q8IUC4	ROCK1	Q13464	SNAP23	O00161
STARD13	Q9Y3M8	STARD8	Q92502	STK10	O94804
STOM	P27105	TFRC	P02786	VAMP3	Q15836
VANGL1	Q8TAA9				

17. Loss of MECP2 binding ability to 5mC-DNA ([R-HSA-9022538](#))



Cellular compartments: nucleoplasm.

Diseases: Rett syndrome.

Missense mutations in the methyl-CpG binding domain (MBD) of methyl-CpG-binding protein 2 (MECP2), spanning amino acids 90 to 162, negatively affect the binding ability of MECP2 to methylated DNA (Ghosh et al. 2008, Ho et al. 2008, Goffin et al. 2012, Mellen et al. 2012).

References

Ho KL, McNae IW, Schmiedeberg L, Klose RJ, Bird AP & Walkinshaw MD (2008). MeCP2 binding to DNA depends upon hydration at methyl-CpG. *Mol. Cell*, 29, 525-31. [🔗](#)

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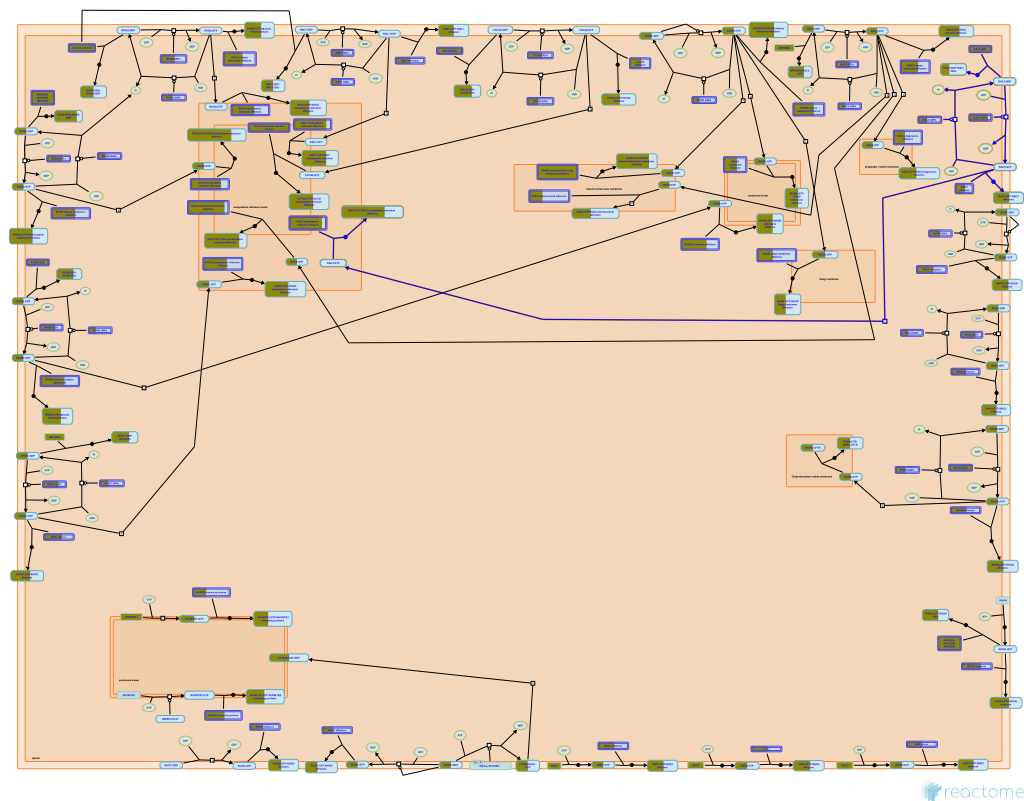
Date	Action	Author
2017-09-25	Created	Orlic-Milacic M
2017-10-02	Authored	Orlic-Milacic M

Date	Action	Author
2018-08-07	Reviewed	Christodoulou J, Krishnaraj R
2018-08-08	Edited	Orlic-Milacic M
2023-03-08	Modified	Matthews L

4 submitted entities found in this pathway, mapping to 5 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HDAC1	Q13547	MECP2	P51608-1, P51608-2	SIN3A	Q96ST3
Input	Ensembl Id				
BDNF	ENSG00000176697				

18. RAC3 GTPase cycle (R-HSA-9013423)



This pathway catalogues RAC3 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RAC3 effectors. RAC3 is highly similar to RAC1 (92% amino acid sequence identity), but it is expressed in fewer tissues than RAC1. RAC3 orthologues only exist in vertebrates (de Curtis 2019). RAC3 is highly expressed in neurons and plays an important role in neuronal and brain development (de Curtis 2014, de Curtis 2019). RAC3 mutations have been reported in patients with intellectual disability and brain malformations (de Curtis 2019). RAC3 is frequently overexpressed in cancer and contributes to proliferation, migration and invasiveness of cancer cells (de Curtis 2019).

References

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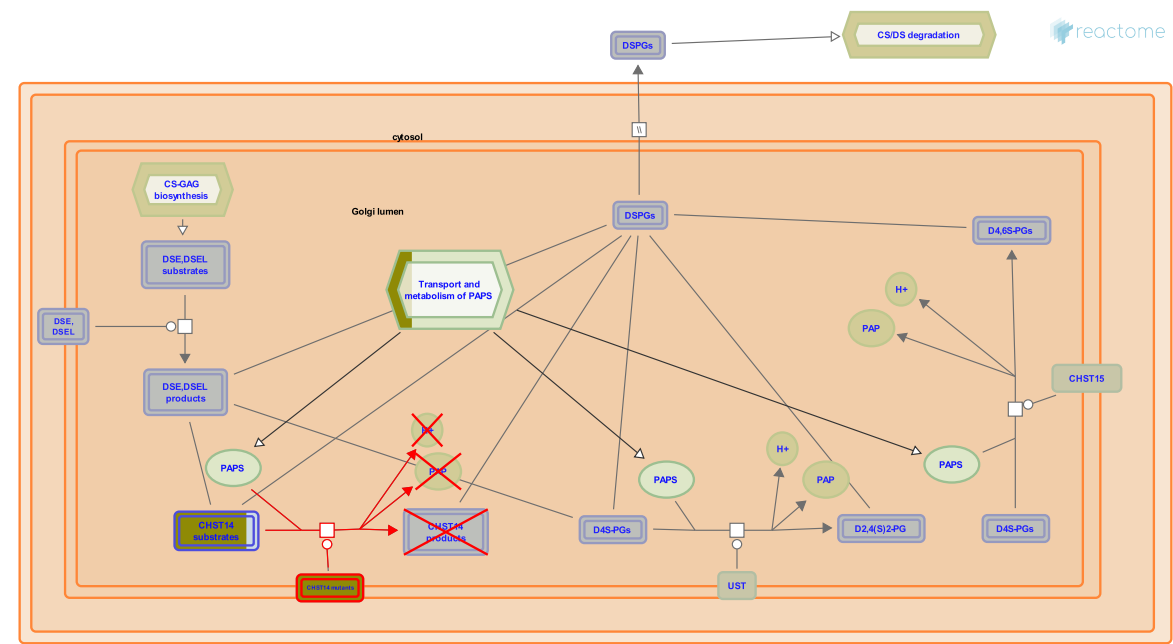
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Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

56 submitted entities found in this pathway, mapping to 56 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABI1	Q8IZP0	ABI2	Q9NYB9	ARAP2	Q8WZ64
ARHGAP10	Q5T5U3	ARHGAP21	Q5T5U3	ARHGAP32	A7KAX9
ARHGAP33	A7KAX9	ARHGAP39	Q9C0H5	ARHGAP42	A6NI28
ARHGAP6	O43182	ARHGDIB	P52566	BAIAP2L1	Q9UHR4
CAV1	Q03135	CDC42EP1	Q00587	CYBA	P13498
DIAPH3	Q9NSV4	DOCK10	Q96BY6	DSG2	Q14126
EPHA2	P29317	ERBIN	Q96RT1	ESYT1	Q9BSJ8
FERMT2	Q96AC1	IL32	P24001	ITGB1	P05556
LBR	Q14739	LEMD3	Q9Y2U8	LMAN1	P49257
MCAM	P43121	MCF2	P10911	MPP7	Q5T2T1
NCF2	P19878	NHS	Q6T4R5	NOX1	Q9Y5S8
OCRL	Q01968	PAK1	Q13153	PAK3	Q13153
PAK4	O96013	PGRMC2	O15173	PREX1	Q8TCU6
RAPGEF1	Q13905	SLITRK3	O94933	SLITRK5	O94991
SNAP23	O00161	SRGAP2	O75044	SWAP70	Q9UH65
SYDE1	Q6ZW31	TAOK3	Q9H2K8	TFRC	P02786
TIAM1	Q13009	TRIO	O75962	VAMP3	Q15836
VANGL1	Q8TAA9	VRK2	Q86Y07	WASF1	Q92558
WASF2	Q9Y6W5	YKT6	O15498		

19. Defective CHST14 causes EDS, musculocontractural type ([R-HSA-3595174](#))



Diseases: Ehlers-Danlos syndrome.

Carbohydrate sulfotransferase 14 (CHST14 also known as D4ST-1) mediates the transfer of sulfate to position 4 of further N-acetylgalactosamine (GalNAc) residues of dermatan sulfate (DS). Defects in CHST14 cause Ehlers-Danlos syndrome, musculocontractural type (MIM:601776). The Ehlers-Danlos syndromes (EDS) are a group of connective tissue disorders that share common features such as skin hyperextensibility, articular hypermobility and tissue fragility (Beighton et al. 1998). The musculocontractural form of EDS (MIM:601776) include distinctive characteristics such as craniofacial dysmorphism, congenital contractures of fingers and thumbs, clubfeet, severe kyphoscoliosis and muscular hypotonia (Malfait et al. 2010).

References

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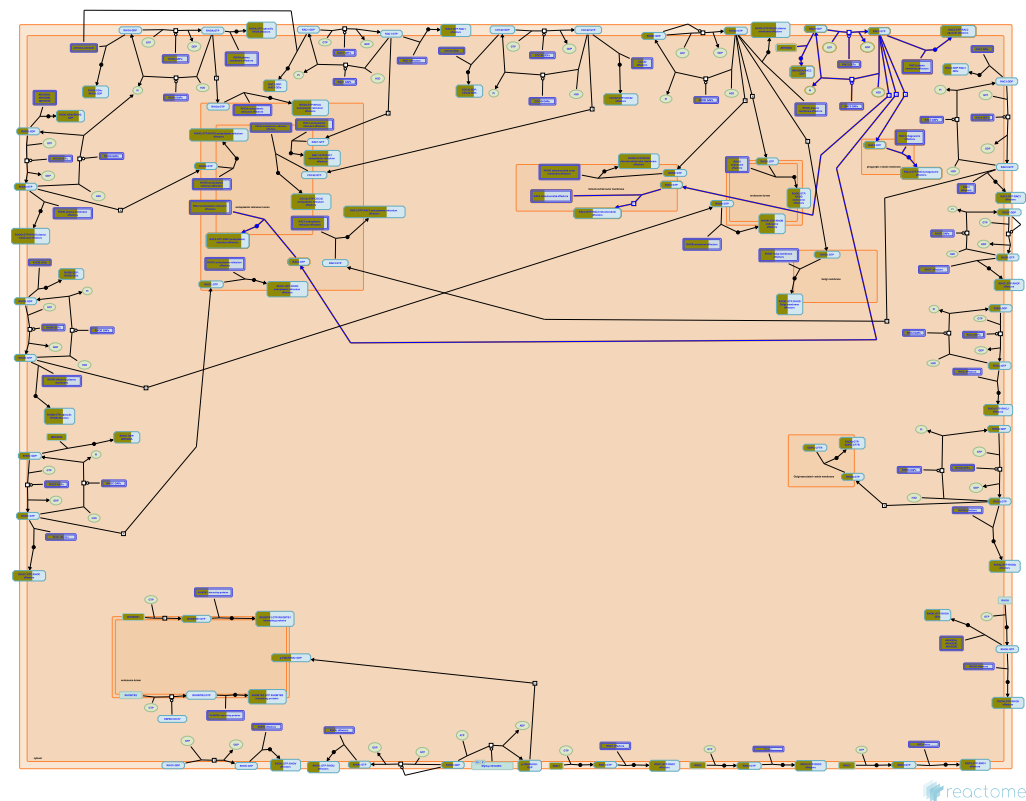
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2013-05-21	Edited	Jassal B
2013-05-21	Authored	Jassal B
2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

7 submitted entities found in this pathway, mapping to 7 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BCAN	Q96GW7	BGN	P21810	CHST14	Q8NCH0
CSPG4	Q6UVK1	CSPG5	O95196	DCN	P07585
NCAN	O14594				

20. RAC2 GTPase cycle (R-HSA-9013404)



This pathway catalogues RAC2 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RAC2 effectors. RAC2 is exclusively expressed in hematopoietic cells (Troeger and Williams 2013). RAC2 is a component of the phagocytic oxidase complex in neutrophils (Troeger and Williams 2013). RAC2 is required for adhesion and mobilization of hematopoietic stem cells and progenitor cells (Troeger and Williams 2013). RAC2 is also needed for adhesion, migration and degranulation of mast cells (Troeger and Williams 2013). Mutations in RAC2 have been found in a small number of patients with primary immunodeficiencies (Gu and Williams 2002; Troeger and Williams 2013; Lougaris et al. 2020).

References

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Gu Y & Williams DA (2002). RAC2 GTPase deficiency and myeloid cell dysfunction in human and mouse. *J. Pediatr. Hematol. Oncol.*, 24, 791-4. [🔗](#)

Edit history

Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M

Date	Action	Author
2025-11-15	Modified	Weiser JD

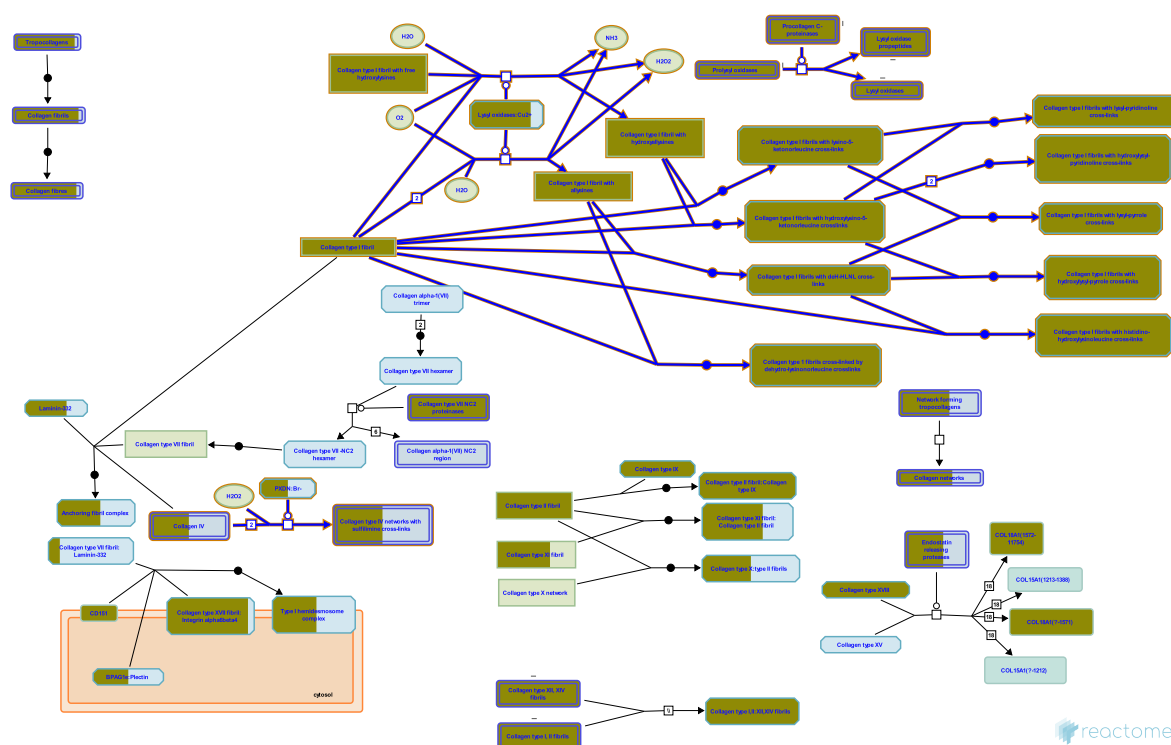
57 submitted entities found in this pathway, mapping to 57 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABI1	Q8IZP0	ABI2	Q9NYB9	ANKLE2	Q86XL3
ARHGAP10	Q5T5U3	ARHGAP21	Q5T5U3	ARHGAP32	A7KAX9
ARHGAP33	A7KAX9	ARHGAP39	Q9C0H5	ARHGAP42	A6NI28
ARHGDIA	P52565	BAIAP2L1	Q9UHR4	CAV1	Q03135
CDC42EP1	Q00587	CDC42EP4	Q9H3Q1	CYBA	P13498
DIAPH3	Q9NSV4	DOCK10	Q96BY6	DOCK2	Q92608
DOCK3	Q8IZD9	DOCK4	Q8N1I0	DSG2	Q14126
EPHA2	P29317	ERBIN	Q96RT1	ESYT1	Q9BSJ8
IQGAP1	P46940	ITGB1	P05556	LBR	Q14739
LEMD3	Q9Y2U8	LMAN1	P49257	MCAM	P43121
MCF2	P10911	MPP7	Q5T2T1	MTX1	Q13505
NCF2	P19878	NHS	Q6T4R5	PAK1	Q13153
PAK3	Q13153	PAK4	O96013	PGRMC2	O15173
PIK3R3	Q92569	PLD1	O14939	PLD2	O14939
PREX1	Q8TCU6	RAC2	P15153	SLITRK5	O94991
SWAP70	Q9UH65	SYDE1	Q6ZW31	TAOK3	Q9H2K8
TFRC	P02786	TIAM1	Q13009	TRIO	O75962
VAMP3	Q15836	VANGL1	Q8TAA9	VAPB	O95292
VAV3	Q9UKW4	VRK2	Q86Y07	WASF2	Q9Y6W5

Interactors found in this pathway (4)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EZR	P15311	P52565	RHOC	P08134	P52565
RHOG	P84095	P52565	TNFRSF19	Q9NS68	P52565

21. Crosslinking of collagen fibrils (R-HSA-2243919)



After removal of the N- and C-procollagen propeptides, fibrillar collagen molecules aggregate into microfibrillar arrays, stabilized by covalent intermolecular cross-links. These depend on the oxidative deamination of specific lysine or hydroxylysine residues in the telopeptide region by lysyl oxidase (LOX) with the subsequent spontaneous formation of covalent intermolecular cross-links (Pinnell & Martin 1968, Siegel et al. 1970, 1974, Maki 2009, Nishioka et al. 2012). Hydroxylysine is formed intracellularly by lysine hydroxylases (LH). There are different forms of LH responsible for hydroxylation of helical and telopeptide lysines (Royce & Barnes 1985, Knott et al. 1997, Takaluoma et al. 2007, Myllyla 2007). The chemistry of the cross-links formed depends on whether lysines or hydroxylysines are present in the telopeptides (Barnes et al. 1974), which depends on the proportion of collagen lysines post-translationally converted to hydroxylysine by LH. The lysine pathway predominates in adult skin, cornea and sclera while the hydroxylysine pathway occurs primarily in bone, cartilage, ligament, tendons, embryonic skin and most connective tissues (Eyre 1987, Eyre & Wu 2005, Eyre et al. 2008). Oxidative deamination of lysine or hydroxylysine residues by LOX generates the allysine and hydroxyallysine aldehydes respectively. These can spontaneously react with either another aldehyde to form an aldol condensation product (intramolecular cross-link), or with an unmodified lysine or hydroxylysine residue to form intermolecular cross-links.

The pathway of cross-linking is regulated primarily by the hydroxylation pattern of telopeptide and triple-helix domain lysine residues. When lysine residues are the source of aldehydes formed by lysyl oxidase the allysine cross-linking pathway leads to the formation of aldimine cross-links (Eyre & Wu 2005). These are stable at physiological conditions but readily cleaved at acid pH or elevated temperature. When hydroxylysine residues are the source of aldehydes formed by lysyl oxidase the hydroxyallysine cross-linking pathway leads to the formation of more stable ketoimine cross-links.

Telopeptide lysine residues can be converted by LOX to allysine, which can react with a helical hydroxylysine residue forming the lysine aldehyde aldimine cross-link dehydro hydroxylysino norleucine (deHHLNL) (Bailey & Peach 1968, Eyre et al. 2008). If the telopeptide residue is hydroxylysine, the hydroxyallysine formed by LOX can react with a helical hydroxylysine forming the Schiff base, which spontaneously undergoes an Amadori rearrangement resulting in the ketoimine cross link hydroxylysino 5 ketonorleucine (HLKNL). This stable cross-link is formed in tissues where telopeptide residues are predominantly hydroxylated, such as foetal bone and cartilage, accounting for the relative insolubility of collagen from these tissues (Bailey et al. 1998). In bone, telopeptide hydroxyallysines can react with the epsilon-amino group of a helical lysine (Robins & Bailey 1975). The resulting Schiff base undergoes Amadori rearrangement to form lysino-hydroxynorleucine (LHNL). An alternative mechanism of maturation of ketoimine cross-links has been reported in cartilage leading to the formation of arginoline (Eyre et al. 2010).

These divalent crosslinks greatly diminish as connective tissues mature, due to further spontaneous reactions (Bailey & Shimokomaki 1971, Robins & Bailey 1973) with neighbouring peptides that result in tri- and tetrafunctional cross-links. In mature tissues collagen cross-links are predominantly trivalent. The most common are pyridinoline or 3-hydroxypyridinium cross-links, namely hydroxylysyl-pyridinoline (HL-Pyr) and lysyl-pyridinoline (L-Pyr) cross-links (Eyre 1987, Ogawa et al. 1982, Fujimoto et al. 1978). HL-Pyr is formed from three hydroxylysine residues, HLKNL plus a further hydroxyallysine. It predominates in highly hydroxylated collagens such as type II collagen in cartilage. L-Pyr is formed from two hydroxylysines and a lysine, LKNL plus a further hydroxyallysine, found mostly in calcified tissues (Bailey et al. 1998). Trivalent collagen cross-links can also form as pyrroles, either Lysyl-Pyrrole (L-Pyrrole) or hydroxylysyl-pyrrole (HL-Pyrrole), respectively formed when LKNL or HLKNL react with allysine (Scott et al. 1981, Kuypers et al. 1992). A further three-way crosslink can form when DeH-HLNL reacts with histidine to form histidino-hydroxylysino norleucine (HHL), found in skin and cornea (Yamauchi et al. 1987, 1996). This can react with an additional lysine to form the tetrafunctional cross-link histidinohydroxymerodesmosine (Reiser et al. 1992, Yamauchi et al. 1996).

Another mechanism which could be involved in the cross-linking of collagen IV networks is the sulfilimine bond (Vanacore et al. 2009), catalyzed by peroxidasin, an enzyme found in basement membrane (Bhave 2012).

To improve clarity inter-chain cross-linking is represented here for Collagen type I only. Although the formation of each type of cross-link is represented here as an independent event, the partial and random nature of lysine hydroxylation and subsequent lysyl oxidation means that any combination of these cross-linking events could occur within the same collagen fibril .

References

Bailey AJ, Paul RG & Knott L (1998). Mechanisms of maturation and ageing of collagen. *Mech Ageing Dev*, 106, 1-56. [🔗](#)

Edit history

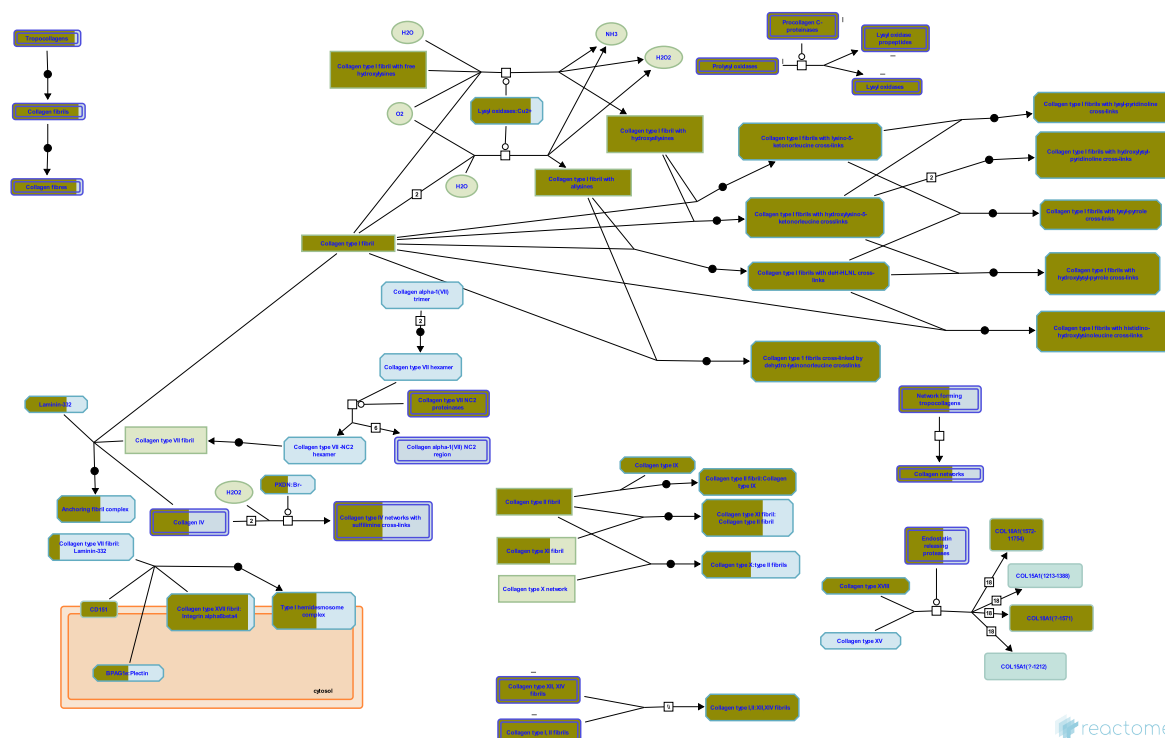
Date	Action	Author
2012-04-30	Authored	Jupe S
2012-05-09	Created	Jupe S
2012-10-08	Reviewed	Kalamajski S, Raleigh S

Date	Action	Author
2012-11-12	Edited	Jupe S
2012-11-19	Reviewed	Ricard-Blum S
2025-11-15	Modified	Weiser JD

15 submitted entities found in this pathway, mapping to 16 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BMP1	P13497	COL18A1	P53420	COL1A1	P02452
COL1A2	P08123	COL4A5	P29400	COL4A6	Q14031
LOX	P28300	LOXL1	P58215, Q08397	LOXL2	Q9Y4K0
LOXL3	P58215	LOXL4	Q96JB6	PCOLCE	Q15113
PXDN	Q92626	TLL1	O43897	TLL2	Q9Y6L7

22. Assembly of collagen fibrils and other multimeric structures (R-HSA-2022090)



Collagen trimers in triple-helical form, referred to as procollagen or collagen molecules, are exported from the ER and trafficked through the Golgi network before secretion into the extracellular space. For fibrillar collagens namely types I, II, III, V, XI, XXIV and XXVII (Gordon & Hahn 2010, Ricard-Blum 2011) secretion is concomitant with processing of the N and C terminal collagen propeptides. These processed molecules are known as tropocollagens, considered to be the units of higher order collagen structures. They form within the extracellular space via a process that can proceed spontaneously, but in the cellular environment is regulated by many collagen binding proteins such as the FACIT (Fibril Associated Collagens with Interrupted Triple helices) family collagens and Small Leucine-Rich Proteoglycans (SLRPs). The architecture formed ultimately depends on the collagen subtype and the cellular conditions. Structures include the well-known fibrils and fibres formed by the major structural collagens type I and II plus several different types of supra-molecular assembly (Bruckner 2010). The mechanical and physical properties of tissues depend on the spatial arrangement and composition of these collagen-containing structures (Kadler et al. 1996, Shoulders & Raines 2009, Birk & Bruckner 2011).

Fibrillar collagen structures are frequently heterotypic, composed of a major collagen type in association with smaller amounts of other types, e.g. type I collagen fibrils are associated with types III and V, while type II fibrils frequently contain types IX and XI (Wess 2005). Fibres composed exclusively of a single collagen type probably do not exist, as type I and II fibrils require collagens V and XI respectively as nucleators (Kadler et al. 2008, Wenstrup et al. 2011). Much of the structural understanding of collagen fibrils has been obtained with fibril-forming collagens, particularly type I, but some central features are believed to apply to at least the other fibrillar collagen subtypes (Wess 2005). Fibril diameter and length varies considerably, depending on the tissue and collagen types (Fang et al. 2012). The reasons for this are poorly understood (Wess 2005).

Some tissues such as skin have fibres that are approximately the same diameter while others such as tendon or cartilage have a bimodal distribution of thick and thin fibrils. Mature type I collagen fibrils in tendon are up to 1 μm in length, with a diameter of approx. 500 nm. An individual fibrillar collagen triple helix is less than 1.5 nm in diameter and around 300 nm long; collagen molecules must assemble to give rise to the higher-order fibril structure, a process known as fibrillogenesis, prevented by the presence of C-terminal propeptides (Kadler et al. 1987). In electron micrographs, fibrils have a banded appearance, due to regular gaps where fewer collagen molecules overlap, which occur because the fibrils are aligned in a quarter-stagger arrangement (Hodge & Petruska 1963). Collagen microfibrils are believed to have a quasi-hexagonal unit cell, with tropocollagen arranged to form supertwisted, right-handed microfibrils that interdigitate with neighbouring microfibrils, leading to a spiral-like structure for the mature collagen fibril (Orgel et al. 2006, Holmes & Kadler 2006).

Neighbouring tropocollagen monomers interact with each other and are cross-linked covalently by lysyl oxidase (Orgel et al. 2000, Maki 2006). Mature collagen fibrils are stabilized by lysyl oxidase-mediated cross-links. Hydroxylysyl pyridinoline and lysyl pyridinoline cross-links form between (hydroxy) lysine and hydroxylysine residues in bone and cartilage (Eyre et al. 1984). Arginoline cross-links can form in cartilage (Eyre et al. 2010); mature bovine articular cartilage contains roughly equimolar amounts of arginoline and hydroxylysyl pyridinoline based on peptide yields. Mature collagen fibrils in skin are stabilized by the lysyl oxidase-mediated cross-link histidinohydroxylysinonorleucine (Yamauch et al. 1987). Due to the quarter-staggered arrangement of collagen molecules in a fibril, telopeptides most often interact with the triple helix of a neighbouring collagen molecule in the fibril, except for collagen molecules in register staggered by 4D from another collagen molecule. Fibril aggregation *in vitro* can be unipolar or bipolar, influenced by temperature and levels of C-proteinase, suggesting a role for the N- and C- propeptides in regulation of the aggregation process (Kadler et al. 1996). *In vivo*, collagen molecules at the fibril surface may retain their N-propeptides, suggesting that this may limit further accretion, or alternatively represents a transient stage in a model whereby fibrils grow in diameter through a cycle of deposition, cleavage and further deposition (Chapman 1989).

In vivo, fibrils are often composed from more than one type of collagen. Type III collagen is found associated with type I collagen in dermal fibrils, with the collagen III on the periphery, suggesting a regulatory role (Fleischmajer et al. 1990). Type V collagen associates with type I collagen fibrils, where it may limit fibril diameter (Birk et al. 1990, White et al. 1997). Type IX associates with the surface of narrow diameter collagen II fibrils in cartilage and the cornea (Wu et al. 1992, Eyre et al. 2004). Highly specific patterns of crosslinking sites suggest that collagen IX functions in interfibrillar networking (Wess 2005). Type XII and XIV collagens are localized near the surface of banded collagen I fibrils (Nishiyama et al. 1994). Certain fibril-associated collagens with interrupted triple helices (FACITs) associate with the surface of collagen fibrils, where they may serve to limit fibril fusion and thereby regulate fibril diameter (Gordon & Hahn 2010). Collagen XV, a member of the multiplexin family, is almost exclusively associated with the fibrillar collagen network, in very close proximity to the basement membrane. In human tissues collagen XV is seen linking banded collagen fibers subjacent to the basement membrane (Amenta et al. 2005). Type XIV collagen, SLRPs and discoidin domain receptors also regulate fibrillogenesis (Ansorge et al. 2009, Kalamajski et al. 2010, Flynn et al. 2010).

Collagen IX is cross-linked to the surface of collagen type II fibrils (Eyre et al. 1987). Type XII and XIV collagens are found in association with type I (Walchli et al. 1994) and type II (Watt et al. 1992, Eyre 2002) fibrils in cartilage. They are thought to associate non-covalently via their COL1/NC1 domains (Watt et al. 1992, Eyre 2002).

Some non-fibrillar collagens form supramolecular assemblies that are distinct from typical fibrils. Collagen VII forms anchoring fibrils, composed of antiparallel dimers that connect the dermis to the epidermis (Bruckner-Tuderman 2009). During fibrillogenesis, the nascent type VII procollagen molecules dimerize in an antiparallel manner. The C-propeptides are then removed by Bone morphogenetic protein 1 (Rattenholl et al. 2002) and the processed antiparallel dimers aggregate laterally. Collagens VIII and X form hexagonal networks and collagen VI forms beaded filament (Gordon & Hahn 2010, Ricard-Blum et al. 2011).

References

Kadler KE, Holmes DF, Trotter JA & Chapman JA (1996). Collagen fibril formation. *Biochem J*, 316, 1-11. [🔗](#)

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Edit history

Date	Action	Author
2011-08-05	Authored	Jupe S
2011-11-25	Created	Jupe S
2012-10-08	Reviewed	Kalamajski S, Raleigh S
2012-11-12	Edited	Jupe S
2012-11-19	Reviewed	Ricard-Blum S
2025-11-15	Modified	Weiser JD

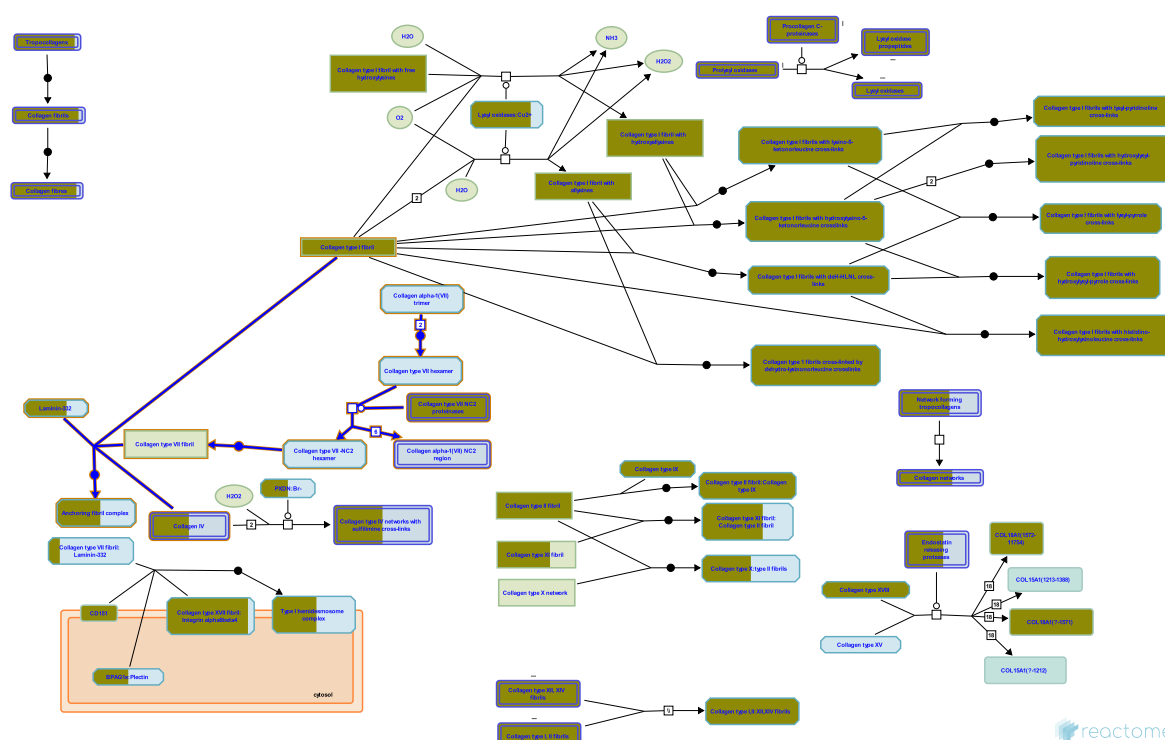
44 submitted entities found in this pathway, mapping to 50 Reactome entities

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COL27A1	Q8IZC6	COL2A1	P02458	COL3A1	P02461
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COL6A2	P12110	COL6A3	A8TX70, P12111	COL8A1	P27658
COL8A2	P25067	COL9A1	P20849	COL9A2	Q14055, Q9UMD9
COL9A3	Q14050	CTSB	P07858	CTSL	O60911, P07711
CTSS	P25774	CTSV	O60911, P07711	EMID1	P39060
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LOX	P28300	LOXL1	P58215, Q08397	LOXL2	Q9Y4K0
LOXL3	P58215	LOXL4	Q96JB6	MMP10	P08254
PCOLCE	Q15113	PLEC	Q15149	PXDN	Q92626
TLL1	O43897	TLL2	Q9Y6L7		

Interactors found in this pathway (14)

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CD9	P21926	P48509	GRAMD1C	Q8IYS0	P48509
ITGA3	P26006	P48509	ITGB1	P05556	P48509
ITGB4	P16144	P48509	LGALS1	P09382	P48509
MMP2	P08253	P39060	P4HA2	O15460	P39060
PLOD3	O60568	P39060	TSPAN14	Q8NG11	P48509
TSPAN18	Q8WUV1	P48509	TSPAN5	P62079	P48509

23. Anchoring fibril formation (R-HSA-2214320)



Cellular compartments: extracellular region.

Collagen VII forms anchoring fibrils, composed of antiparallel dimers that connect the dermis to the epidermis (Bruckner-Tuderman 2009, Has & Kern 2010). During fibrillogenesis, the nascent type VII procollagen molecules dimerize in an antiparallel manner. The C-propeptide is then removed by Bone morphogenetic protein 1 (Rattenholl et al. 2002) and the processed antiparallel dimers laterally aggregate (Villone et al. 2008, Gordon & Hahn 2010).

References

Chung HJ & Uitto J (2010). Type VII collagen: the anchoring fibril protein at fault in dystrophic epidermolysis bullosa. *Dermatol Clin*, 28, 93-105. [🔗](#)

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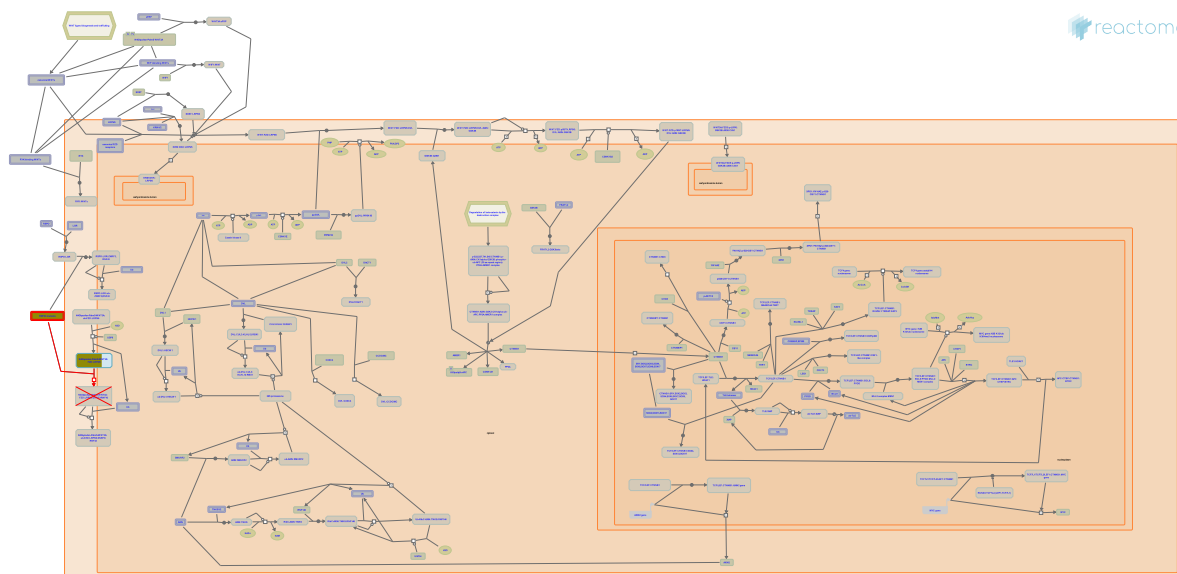
Date	Action	Author
2012-04-30	Authored	Jupe S
2012-04-30	Created	Jupe S
2012-10-08	Reviewed	Kalamajski S, Raleigh S
2012-11-12	Edited	Jupe S
2012-11-19	Reviewed	Ricard-Blum S
2025-11-15	Modified	Weiser JD

10 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BMP1	P13497	COL18A1	P53420	COL1A1	P02452

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
COL1A2	P08123	COL4A5	P29400	COL4A6	Q14031
LAMA3	Q16787	LAMC2	Q13753	TLL1	O43897
TLL2	Q9Y6L7				

24. Signaling by RNF43 mutants (R-HSA-5340588)



Cellular compartments: plasma membrane.

Diseases: cancer.

RNF43 and related protein ZNRF3 are E3 ubiquitin ligases that negatively regulate WNT signaling by downregulating FZD receptors at the cell surface (Mukai et al, 2010; Hao et al, 2012). Frameshift loss-of-function mutations in RNF43 that enhance WNT signaling have been identified in pancreatic and colorectal cancers; the proliferation of these cells is dependent on the presence of secreted WNT, as their growth is abrogated by treatment of cells with the Porcupine inhibitor LGK974 (Koo et al, 2012; Jiang et al, 2013).

References

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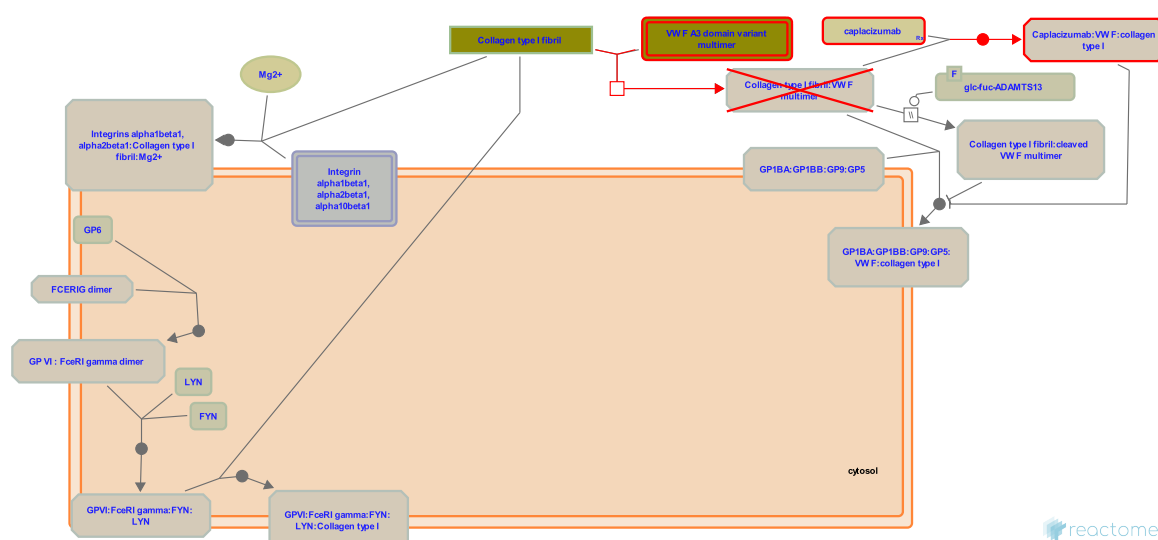
Date	Action	Author
2014-03-11	Authored	Rothfels K
2014-03-20	Created	Rothfels K
2014-04-03	Edited	Matthews L

Date	Action	Author
2014-05-12	Reviewed	Salahshor S
2014-05-22	Reviewed	Woodgett J
2023-10-12	Modified	Weiser JD

6 submitted entities found in this pathway, mapping to 6 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
FZD4	Q9ULV1	FZD5	Q13467	FZD6	O60353
LRP6	O75581	RNF43	Q68DV7	WNT3A	P56704

25. Defective VWF binding to collagen type I ([R-HSA-9845622](#))



Diseases: blood platelet disease.

Upon vascular injury, circulating von Willebrand factor (VWF) binds to exposed vascular collagen. This Reactome event shows defective binding of VWF to collagen type I caused by loss-of-function mutations in the A3 domain of VWF found in patients with von Willebrand disease (VWD) type 2M, which is characterized by defects in platelet adhesion and/or collagen binding with normal or sub-normal VWF multimer distribution.

References

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- Shida Y, Rydz N, Stegner D, Brown C, Mewburn J, Sponagle K, ... Lillicrap D (2014). Analysis of the role of von Willebrand factor, platelet glycoprotein VI-, and $\alpha 2\beta 1$ -mediated collagen binding in thrombus formation. *Blood*, 124, 1799-807. [↗](#)
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Edit history

Date	Action	Author
2023-01-07	Authored	Shamovsky V
2023-10-06	Created	Shamovsky V
2023-11-06	Reviewed	Gao R
2023-11-07	Modified	Shamovsky V
2023-11-07	Edited	Shamovsky V

3 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
COL1A1	P02452	COL1A2	P08123	VWF	P04275

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

4180 of the submitted entities were found, mapping to 5076 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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ABCA9	Q8IUA7, Q8WWZ4	ABCB1	P21439	ABCB10	Q9NRK6
ABCB11	O95342	ABCB5	O75027	ABCB9	Q9NP78
ABCC1	P33527	ABCC10	Q5T3U5	ABCC3	O15438
ABCC4	O15439	ABCC5	O15439, O15440	ABCC8	O60706, Q09428
ABCC9	O60706	ABCD2	P33897	ABCD3	P28288
ABCD4	O14678	ABCE1	P61221	ABCF2	Q9UG63
ABCG1	P45844	ABCG4	P45844, Q9H172	ABHD10	Q9NUJ1
ABHD12	Q8N2K0	ABHD14B	Q96IU4	ABHD17C	Q6PCB6
ABHD3	Q8WU67	ABHD4	Q8TB40	ABI1	Q8IZP0
ABI2	Q9NYB9	ABLIM1	O14639	ABLIM2	Q6H8Q1
ABLIM3	O94929	ABO	P16442	ACACA	Q13085
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ACAT1	P24752	ACBD3	Q9H3P7	ACCS	Q93063
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ACP6	Q9NPH0	ACSF2	Q96CM8	ACSL1	P33121, Q9ULC5
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ACSS1	Q9NUB1	ACTA2	P62736	ACTG2	P63267
ACTL6B	O94805	ACTN1	P12814	ACTN2	P35609
ACTN4	O43707	ACTR2	P61160	ACTR3	P61158
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ADAM22	Q9P0K1	ADAM23	O75077	ADAM33	Q9BZ11
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ADAMTSL3	P82987	ADAMTSL4	Q6UY14	ADAMTSL5	Q6ZMM2
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AK7	Q96M32	AKAP12	Q02952	AKAP5	P24588
AKR1B1	P15121	AKR1C1	Q04828	AKR1C2	P52895
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ALMS1	Q8TCU4	ALOX15	P16050	ALOX5	P09917
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AMT	P48728	ANAPC1	Q9H1A4	ANAPC5	Q9UJX4
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APBB1	O00213	APC	P25054	APC2	Q9UJX6
APEX1	P27695	APH1B	Q8WW43	APLNR	P35414
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ARPC2	O15144	ARPC5	O15511	ARPP19	P56211
ARRB1	P49407	ARRB2	P32121	ARSA	O43681
ARSB	P15848	ARSF	P54793	ARSG	Q96EG1
ARSI	Q5FYB1	ARSJ	Q5FYB0	ARSK	Q6UWY0
ARSL	P51690	ART3	Q13508	ASAH1	Q13510
ASAP1	Q9ULH1	ASB3	A6NK59, Q9Y575	ASCC2	Q9H1I8
ASGR2	P07307	ASH2L	Q9UBL3	ASIC2	Q16515
ASIC4	Q96FT7	ASNS	P08243	ASPG	Q86U10
ASPH	Q12797	ASPN	Q9BXN1	ASS1	P00966
ASXL1	Q8IXJ9	ATAD1	Q8NBU5	ATAD3A	Q5T9A4
ATAD3B	Q5T9A4	ATAT1	Q5SQI0	ATF1	P18846
ATF3	P18847	ATF5	Q9Y2D1	ATF7IP	Q6VMQ6
ATG5	Q9H1Y0	ATIC	P31939	ATL2	Q8NHH9
ATM	Q13315	ATP10A	O60312	ATP10B	O94823
ATP10D	Q9P241	ATP11B	Q9Y2G3	ATP1A1	P05023
ATP1A2	P50993, Q13733	ATP1A3	P13637	ATP1B3	P54709
ATP2A1	O14983	ATP2B2	P23634, Q01814	ATP2B3	Q16720
ATP2B4	P23634	ATP5MC2	P05496, P48201, Q06055	ATP6AP2	O75787
ATP6V0A1	Q93050	ATP6V0A4	Q9HBG4	ATP6V0E1	O15342
ATP6V1A	P38606	ATP6V1G2	O95670	ATP6V1H	Q9UI12
ATP7B	P35670	ATP8A2	Q9NTI2	ATP8B1	O43520
ATP8B2	P98198	ATP8B3	O60423	ATP9A	O75110
ATXN3	P54252, Q9H3M9	AUTS2	Q8WXX7	AVEN	Q9NQS1
AXIN1	O15169	AXIN2	Q9Y2T1	AXL	P30530
AZIN2	Q96A70	B2M	P61769	B3GALNT1	O75752
B3GALNT2	Q8NCR0	B3GALT1	Q9Y5Z6	B3GALT2	O43825
B3GAT2	Q9NPZ5	B3GLCT	Q6Y288	B3GNT2	Q9NY97
B3GNT5	Q9BYG0	B3GNT7	Q8NFL0	B3GNT8	Q67FW5, Q7Z7M8
B3GNT9	Q6UX72	B4GALNT1	Q00973	B4GALT1	P15291
B4GALT4	O60513	BAG2	O95816	BAG3	O95817
BAG6	P46379	BAIAP2L1	Q9UHR4	BAMBI	Q13145
BASP1	P80723	BAX	Q07812	BAZ2A	Q9UIF9
BBOX1	O75936	BCAN	Q96GW7	BCAR1	P56945
BCDIN3D	Q7Z5W3	BCL2A1	Q16548	BCL2L1	Q07817
BCL2L14	Q9BZR8	BCL6	P41182	BCL7A	Q4VC05
BCL9	O00512, Q9Y3Y4	BCL9L	Q86UU0	BCO2	Q9BYV7
BDKRB1	P46663	BDKRB2	P30411, P46663	BDNF	P23560
BECN1	Q14457	BEGAIN	Q9BUH8	BEST3	Q8N1M1
BET1	O15155	BET1L	Q9NYM9	BGN	P21810
BHLHE40	O14503	BID	P55957	BIN1	O00422

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BLMH	Q13867	BLTP3B	A0JNW5	BLVRA	P53004
BMP1	P13497	BMP2	P12643	BMP2K	Q2M2I8
BMP4	P12644	BMP7	P18075	BMPR1B	O00238
BMPR2	Q13873	BMS1	Q14692	BNIP3L	O60238
BOC	Q9BWV1	BOD1	Q8NFC6, Q96IK1	BPHL	Q86WA6
BPNT2	Q9NX62	BRAF	P15056	BRCA1	P38398
BRWD3	Q9NSI6	BSG	P35613	BSN	Q9UPA5
BST1	Q10588	BST2	Q10589	BTBD1	Q9H0C5
BTC	P35070	BTD	P43251	BTF3	O00478, P78410
BTG1	P62324	BTG2	P78543	BTN2A2	Q8WVV5
BTN3A2	P78410	BTNL8	Q6UX41	BTNL9	Q6UXG8
BTRC	Q9Y297	BYSL	Q13895	C1GALT1	Q9NS00
C1GALT1C1	Q96EU7	C1QBP	Q07021	C1R	P00738
C1S	P09871	C2CD3	Q4AC94	C2CD5	Q86YS7
C3	P01024	C4A	P0C0L4	C4B	P0C0L4
C4BPB	P20851	C6	P13671	C7	P10643
CA12	O43570	CA13	Q8N1Q1	CA14	Q9ULX7
CA3	P07451	CA9	Q16790	CABLES2	Q9BTV7
CABP1	Q15084	CACNA1A	O00555	CACNA1B	Q00975
CACNA1C	Q13936	CACNA1D	Q01668	CACNA1E	Q15878
CACNA1G	O43497	CACNA1I	Q9P0X4	CACNA2D2	Q9NY47
CACNA2D3	Q8IZS8	CACNB1	Q02641	CACNB4	O00305
CACNG2	Q9Y698	CACNG3	O60359	CACNG7	P62955
CACNG8	Q8WXS5	CADM1	Q9BY67	CADM2	Q8N3J6
CALB1	P05937	CALCA	P01258, P06881, P10092	CALCRL	Q16602
CALD1	Q05682	CALM1	P0DP23	CALM2	P0DP24
CALM3	P0DP23	CALR	P27797	CALU	O43852
CALY	Q92560	CAMK2A	Q9UQM7	CAMK2B	Q13554
CAMK4	Q16566	CAP1	Q01518	CAPG	Q9BPX3
CAPN1	P07384	CAPN14	A8MX76	CAPN2	P17655
CAPN6	Q9Y6Q1	CAPNS1	P04632, Q96L46	CAPZA2	P47755
CARD11	Q9BXL7	CARD8	Q9C000	CARNMT1	Q8N4J0
CARNS1	A5YM72	CASP1	P29466	CASP2	P42575
CASP4	P49662	CASP6	Q9BZM1	CASP7	P55210
CASP8	Q14790	CASP8AP2	Q9UKL3	CAST	O15446
CASTOR2	A6NHX0	CATSPERG	Q6ZRH7	CAV1	Q03135
CAV2	P51636	CAVIN1	Q6NZI2	CBFA2T2	O43439
CBFB	Q13951	CBLB	Q13191	CBX2	Q14781
CBX3	Q13185	CBX4	O00257	CBX5	P45973
CBX8	Q9HC52	CC2D1B	Q13415	CCDC59	Q9P031
CCDC8	Q9H0W5	CCDC88A	Q3V6T2	CCDC88C	Q9P219
CCKAR	P32238	CCL2	P13500	CCN1	O00622
CCN2	P29279	CCNA1	P78396	CCNC	P24863
CCND1	P24385	CCND2	P30279	CCND3	P30281
CCNE2	O96020	CCNG2	Q16589	CCNT2	O60583
CCP110	O43303	CCR7	P32248	CCS	O14618
CCT5	P48643	CCT6A	P40227, Q92526	CD109	Q6YHK3
CD151	P48509	CD177	Q8N6Q3	CD2	P06729

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CD200	P41217	CD22	P20273	CD24	P25063
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CD320	Q9NPF0	CD36	P16671	CD38	P28907
CD4	P01730	CD44	P16070	CD46	P15529
CD47	Q08722	CD55	P08174	CD58	P19256
CD59	P13987	CD6	Q86VB7	CD74	P04233
CD81	P60033	CD8A	P01732	CD9	P21926
CD93	Q9NPY3	CD96	P40200	CDA	P32320
CDC14A	Q9UNH5	CDC16	Q13042	CDC25A	P30304
CDC25B	P30304	CDC25C	P30307	CDC27	Q15054
CDC42EP1	Q00587	CDC42EP3	Q9UKI2	CDC42EP4	Q9H3Q1
CDC42EP5	Q6NZY7	CDC7	O00311	CDC73	Q6P1J9
CDCA5	Q96FF9	CDH1	P12830	CDH10	Q9Y6N8
CDH11	P55287, P55287-2	CDH12	P55289	CDH13	P55290
CDH18	Q13634	CDH19	Q9H159	CDH2	P19022
CDH24	Q86UP0	CDH3	P55291	CDH4	P55283
CDH5	P33151	CDH6	P55285, Q9ULB4	CDH7	Q9ULB5
CDH8	P55286	CDH9	Q9ULB4	CDK5R1	Q15078
CDK5R2	Q13319	CDK7	P50613	CDK8	P49336
CDKN1B	P46527	CDKN2A	Q8N726	CDKN2B	P42771
CDKN2D	P55273	CDON	Q4KMG0	CDS2	O95674
CDT1	Q9H211	CEACAM21	P31997	CEBPB	P17676
CEBPD	P49716	CEL	P19835	CENPH	Q9H3R5
CEP135	Q66GS9	CEP192	Q8TEP8	CEP20	O75665
CEP41	Q9BYV8	CEP57	Q86XR8	CEP70	Q8NHQ1
CEP78	Q5JTW2	CEP83	Q9Y592	CEP97	Q8IW35
CERK	Q8TCT0	CERS2	Q96G23	CERS6	Q6ZMG9
CES1	P23141	CES2	P23141	CETN2	P41208
CFB	P00751	CFH	P08603, Q92496	CFHR1	Q03591, Q92496
CFI	P05156	CFTR	P13569	CGB8	P0DN86
CGN	Q9P2M7	CHAT	P28329	CHCHD10	Q8WYQ3
CHCHD5	Q9BSY4	CHCHD7	Q9BUK0	CHD3	Q12873
CHD4	Q14839	CHD5	O00258	CHD8	Q9HCK8
CHD9	Q3L8U1	CHGA	P10645	CHGB	P05060
CHM	P24386, P26374	CHML	P26374	CHMP2B	Q9UQN3
CHMP6	Q96FZ7	CHN1	P15882	CHN2	P52757
CHPF2	Q9P2E5	CHPT1	Q8WUD6	CHRD1	Q9BU40
CHRM3	P20309	CHRM4	P08173	CHRNA1	P02708
CHRNA3	P32297	CHRNA4	P43681	CHRNA2	P17787
CHRNA4	P30926	CHST1	O43916	CHST14	Q8NCH0
CHST15	Q7LFX5	CHST2	Q9Y4C5	CHST9	Q7LIS5-2
CHSY3	Q70JA7, Q9P2E5	CHTOP	Q9Y3Y2	CHUK	O15111
CIB2	O75838	CIITA	P33076	CISH	Q9NSE2
CIT	O14578, O14578-3	CITED2	Q99967	CKAP4	Q07065
CKB	P12277	CKMT1A	P12532	CKMT1B	P12532
CLASP1	Q7Z460	CLASP2	O75122	CLCF1	Q9UBD9
CLCN2	P51788	CLCN3	P51790	CLCN4	P51793
CLCN5	P51795	CLCN6	P51797	CLCNKB	P51800, P51801
CLDN1	O95832	CLDN11	O75508	CLDN12	P56749

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CLDN2	P57739	CLDN6	P56747	CLEC4E	Q9ULY5
CLIC5	Q9NZA1	CLINT1	Q14677	CLIP1	P30622
CLIP3	Q96DZ5	CLTCL1	P53675	CLVS1	Q8IUQ0
CLVS2	Q5SYC1	CMKLR1	Q99788	CMPK1	P30085
CNDP2	Q96KP4	CNGB1	Q14028	CNIH1	O95406
CNIH3	Q8TBE1	CNKS2	Q8WXI2	CNN1	P51911
CNN2	Q99439	CNOT11	Q9UKZ1	CNOT9	Q92600
CNR1	P21554	CNTFR	P26992	CNTN1	Q12860
CNTN3	Q9P232	CNTN6	Q9UQ52	CNTNAP1	P78357
CNTRL	Q7Z7A1	COA4	Q9NYJ1	COASY	Q13057
COG4	Q9H9E3	COG6	Q9Y2V7	COG8	Q96MW5
COL11A1	P12107	COL12A1	Q99715	COL13A1	Q5TAT6
COL14A1	Q05707	COL16A1	Q07092	COL18A1	P39060, P53420
COL1A1	P02452	COL1A2	P08123	COL20A1	Q9P218
COL21A1	Q96P44	COL22A1	Q8NFW1	COL23A1	Q86Y22
COL24A1	Q17RW2	COL26A1	Q86Y22, Q96A83	COL27A1	Q8IZC6
COL2A1	P02458	COL3A1	P02461	COL4A5	P29400
COL4A6	Q14031	COL5A1	P20908	COL5A2	P05997
COL5A3	P25940	COL6A1	P12109	COL6A2	P12110
COL6A3	A8TX70, P12111	COL8A1	P27658	COL8A2	P25067
COL9A1	P20849	COL9A2	Q14055, Q9UMD9	COL9A3	Q14050
COLEC12	Q5KU26	COLGALT2	Q8IYK4	COMMMD8	Q9NX08
COMP	P49747	COMT	P21964	COPA	P53621
COPB1	P53618	COPB2	P35606	COPG1	Q9Y678
COPG2	Q9UBF2	COPS5	Q92905	COPZ2	Q9P299
COQ10B	Q9H8M1	COQ2	Q96H96	CORO1A	P31146
COTL1	Q14019	COX5B	P10606	COX6A1	P12074
COX6B2	Q6YFQ2	COX7A2L	O14548	CP	P00450
CPA1	P15085	CPA3	P15088	CPB1	P15086
CPNE7	Q9UBL6	CPNE8	Q86YQ8	CPPED1	Q9BRF8
CPSF2	Q9P2I0	CPT1A	P50416	CRABP1	P29762
CRABP2	P29373	CREB3L1	Q96BA8	CREB3L2	Q70SY1
CREB3L4	O43889, Q8TEY5	CREG1	O75629	CREM	Q03060-6
CRH	P06850	CRISPLD2	Q9H0B8	CRLF1	O75462
CRLS1	Q9UJA2	CRMP1	Q14194	CRTAP	O75718
CRTC3	Q6UUV7	CRY1	Q16526	CRY2	Q49AN0
CRYAB	P02511	CRYM	Q14894	CSF1	P09603
CSF2RB	P32927	CSF3R	Q99062	CSGALNACT2	Q8N6G5
CSPG4	Q6UVK1	CSPG5	O95196	CSRP1	P21291
CST3	P01034	CTDNEP1	O95476	CTDP1	Q9Y5B0
CTDSP2	O14595	CTF1	P33240	CTH	P32929
CTNNA1	P35221	CTNBNIP1	Q9NSA3	CTSA	P10619
CTSB	P07858	CTSC	P53634	CTSD	P07339
CTSH	P09668	CTSL	O60911, P07711	CTSS	P25774
CTSV	O60911, P07711	CTSZ	Q9UBR2	CTTN	Q14247
CTU2	Q2VPK5	CUBN	O60494	CUL1	Q13616
CUL4A	Q13619	CUL4B	Q13620	CUL7	Q14999
CUX1	Q13948	CX3CL1	P78423	CXCL1	P09341
CXCL10	P02778	CXCL11	O14625	CXCL12	P48061

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CXCL5	P42830, P80162	CXCL6	P80162	CXXC1	Q9P0U4
CXXC4	Q9H2H0	CXXC5	Q7LFL8	CYB5R2	Q6BCY4
CYB5RL	Q6IPT4	CYBA	P13498	CYBRD1	Q53TN4
CYCS	P99999	CYFIP2	Q96F07	CYGB	Q8WWM9
CYP1B1	Q16678	CYP21A2	P08686	CYP26A1	O43174
CYP26B1	Q9NR63	CYP26C1	Q6V0L0	CYP27A1	Q02318
CYP2C8	P10632	CYP2E1	P05181	CYP2R1	Q6VVX0
CYP2S1	Q96SQ9	CYP2U1	Q7Z449	CYP2W1	Q8TAV3
CYP39A1	Q9NYL5	CYSTM1	Q9H1C7	CYTH3	O43739
DAAM1	Q9Y4D1	DAB1	O75553	DAB2	P98082
DACT1	Q9NYF0	DAG1	Q14118	DAGLA	Q9Y4D2
DAP	P46527	DAPK1	P53355	DAPK3	O43293
DAPP1	Q9UN19	DARS1	P14868	DBN1	Q16643
DBP	Q10586	DCAF11	Q8TEB1	DCAF4	Q8WV16
DCAF5	Q96JK2	DCAF6	Q58WW2	DCAF7	P61962
DCC	P43146	DCLRE1A	Q6PJP8	DCN	P07585
DCTD	P32321	DCTN3	O75935	DCTN4	Q9UJW0
DCTN5	Q9BTE1	DCTN6	O00399	DCUN1D1	Q96GG9
DCUN1D3	Q8IWE4	DCX	O43602	DDAH1	O94760
DDB2	Q92466	DDC	P20711	DDIT3	P35638
DDIT4	Q9NX09	DDOST	P39656	DDR1	Q08345
DDR2	Q16832	DDRGK1	Q96HY6	DDX11	Q96FC9
DDX21	Q9NR30	DDX3X	O00571	DDX5	P17844
DEGS1	O15121	DEGS2	Q6QHC5	DENND1B	Q6P3S1
DENND3	A2RUS2	DENND5B	Q6ZUT9	DEPDC5	O75140
DES	P17661	DFFA	O00273	DGKA	P23743, P52824
DGKB	Q9Y6T7	DGKD	Q16760	DGKI	O75912
DGKK	Q5KSL6	DHDDS	Q86SQ9	DHRS3	O75911
DHRS9	Q9BPW9	DHTKD1	Q96HY7	DHX16	O60231
DHX37	Q8IY37	DIAPH2	O60879-3	DIAPH3	Q9NSV4
DIMT1	Q9UNQ2	DIS3	Q9Y2L1	DISP2	A7MBM2
DKC1	O60832	DKK2	Q9UBU2	DLC1	Q96QB1
DLG1	Q12959, Q92796	DLG2	Q15700	DLG3	Q92796
DLG4	P78352	DLGAP1	O14490	DLGAP2	Q9P1A6
DLGAP3	O95886	DLK1	P80370	DLL1	O00548
DLX5	P56178	DMAP1	Q9NPF5	DMKN	Q6E0U4
DMTN	Q08495	DNAJB11	Q9UBS4	DNAJB6	O75190
DNAJB9	Q9UBS3	DNAJC10	Q8IXB1	DNAJC11	Q9NVH1
DNAJC2	Q99543	DNAJC3	Q13217	DNAJC6	O75061
DNASE1L1	P49184	DNASE2	O00115	DNER	Q8NFT8
DNM1	Q05193	DNM2	P50570	DNM3	Q9UQ16
DNMBP	Q6XZF7	DNMT3B	Q9UBC3	DNPH1	O43598
DOCK10	Q96BY6	DOCK11	Q5JSL3	DOCK2	Q92608
DOCK3	Q8IZD9	DOCK4	Q8N1I0	DOCK5	Q9H7D0
DOCK7	Q96N67	DOCK9	Q9BZ29	DOHH	Q9BU89
DOK3	Q7L591	DOK5	Q9P104	DOT1L	Q8TEK3
DPEP1	P16444	DPF1	Q92782	DPF2	Q92785
DPH2	Q9BQC3	DPP3	Q9NY33	DPP4	P27487
DPPA4	Q7L190	DPYD	Q12882	DPYSL2	Q16555

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DPYSL3	Q14195	DPYSL4	O14531	DPYSL5	Q9BPU6
DRD2	P14416	DRP2	Q13474	DSC1	Q08554
DSC2	Q02487	DSC3	Q08554	DSCAM	O60469
DSCAML1	Q8TD84	DSE	Q9UL01	DSEL	Q8IZU8
DSG2	Q14126	DSG4	Q86SJ6	DSN1	Q9H410
DSP	P15924	DTL	Q9NZJ0	DTNA	Q9Y4J8
DTNB	O60941	DTNBP1	Q96EV8	DTX1	Q86Y01
DTX3L	Q8TDB6	DTX4	Q9Y2E6	DUSP1	P28562
DUSP16	Q9BY84	DUSP3	P51452	DUSP4	Q13115
DUSP6	Q16828	DUSP8	Q13202	DVL3	Q92997
DYNLL1	P63167	DYNLL2	Q96FJ2	DYNLT1	P63172
DYNLT5	Q8N7M0	DYRK3	Q92630	DYSF	Q8IUM7
DZIP1	Q86YF9	DZIP3	Q86Y13	E2F2	Q14209
E2F4	Q16254	E2F5	Q15329	E2F8	A0AVK6
EARS2	Q5JPH6	EBF1	Q9UH73	EBI3	Q14213
EBNA1BP2	Q99848	ECH1	Q13011	ECI2	O75521-2
EDA2R	Q9HAV5	EDC3	Q96F86	EDEM2	Q9BV94
EDIL3	O43854	EDN1	P05305	EDN2	P20800
EDNRA	P25101	EDNRB	P24530	EEA1	Q15075
EEF1A1	P68104	EEF1A2	Q05639	EEF1B2	P24534
EEF1D	P29692	EEF2	P13639	EEF2K	O00418
EFEMP1	Q12805	EFEMP2	O95967, Q12805	EFHD2	Q96C19
EFNA1	P20827	EFNA5	P52803	EFNB1	P98172
EFTUD2	Q15029	EGF	P01133	EGFR	P00533, P04626
EGLN3	Q9H6Z9	EGR1	P18146	EGR2	P11161
EHD1	Q9H4M9	EHD2	Q9NZN3, Q9NZN4	EHHADH	Q08426
EHMT2	Q96KQ7	EIF2A	P05198	EIF2AK1	Q9BQI3
EIF2AK2	P19525	EIF2AK4	Q9P2K8	EIF2S1	P05198
EIF3A	Q14152	EIF3B	P55884	EIF3D	O15371
EIF3G	O75821	EIF3J	O75822	EIF3M	Q7L2H7
EIF4EBP1	Q13541	EIF4G1	Q04637	EIF4H	Q15056
EIF5A2	Q9GZV4	EIF5B	O60841	ELAC2	Q9BQ52
ELAVL2	Q12926	ELF1	O43921	ELL	P55199
ELL2	O00472	ELMO1	Q92556	ELMO2	Q96JJ3
ELN	P15502	ELOVL1	Q9BW60	ELOVL2	Q9NXB9, Q9NYP7
ELOVL4	Q9GZR5	ELOVL6	Q9H5J4	ELOVL7	A1L3X0
EME1	A4GXA9, Q96AY2	EMID1	P39060	EMILIN1	Q9Y6C2
EMILIN2	Q9BXX0	EMILIN3	Q9NT22	EMX2	Q04743
ENAH	Q8N8S7	ENO2	P09104	ENO4	A6NNW6
ENPP1	P22413	ENPP2	Q13822	ENTPD2	Q9Y5L3
ENTPD3	O75355	ENTPD4	Q9Y227	ENTPD7	Q9NQZ7
EPAS1	Q99814	EPB41	O43491, P11171	EPB41L1	Q9H4G0
EPB41L2	O43491	EPB41L3	Q9Y2J2	EPHA1	P21709
EPHA10	Q5JZY3	EPHA2	P29317	EPHA3	P29320
EPHA4	P54764	EPHA5	P54756	EPHA6	Q9UF33
EPHA7	Q15375	EPHA8	P29322	EPHB1	P54762
EPHB2	P29323	EPHB3	P54753	EPHB4	P54760
EPHB6	O15197	EPHX1	P07099	EPN2	O95208
EPS15	P42566	EPS8L2	Q9H6S3	EPSTI1	Q96J88

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ERAP1	Q9NZ08	ERAP2	Q6P179	ERBB3	P21860-1
ERBIN	Q96RT1	ERCC2	P18074	ERF	P50548
ERG	Q12809	ERI1	Q8IV48	ERICH5	Q6P6B1
ERLEC1	Q96DZ1	ERO1A	Q96HE7	ERP44	Q9BS26
ESAM	Q96AP7	ESPN	B1AK53	ESRP2	Q9H6T0
ESYT1	Q9BSJ8	ESYT2	A0FGR8, A0FGR9	ESYT3	A0FGR9
ETF1	P62495	ETFA	P13804	ETFB	P38117
ETNK2	Q9NVF9	ETNPPL	Q8TBG4	ETS1	P14921
ETS2	P15036	ETV4	P43268	ETV6	P41212
EVC	P57679	EVC2	Q86UK5	EVL	Q9UI08
EXO1	Q9UQ84	EXOC2	Q96KP1	EXT1	Q16394
EXT2	Q93063	EXTL3	O43909	EYA1	Q99502
EYA3	Q99504	EYA4	O95677	EZH1	Q92800
EZH2	Q15910, Q92800	EZR	P15311	F11	P03951
F11R	Q9Y624	F2RL1	P25116	F2RL2	O00254
F3	P13726	FAAH2	Q6GMR7	FABP4	P15090
FABP5	Q01469	FABP7	O15540	FADD	Q13158
FADS1	O60427	FADS2	O95864	FAM131B	Q86XD5
FAM13A	O94988	FAM20B	O75063	FAM20C	Q8IXL6
FAM3C	Q92520	FAM53B	Q5JPI9	FAM83B	Q5T0W9
FAM91A1	Q658Y4	FANCA	O15360	FANCD2	Q9BXW9
FANCE	Q9HB96	FANCG	O15287	FANCI	Q9NVI1
FANCM	Q8IYD8	FAP	P27487	FAR2	Q96K12
FARP1	Q9Y4F1	FARP2	O94887	FAS	P25445
FASN	P49327	FAU	P62861	FBLIM1	Q8WUP2
FBLN1	P23142	FBLN2	P98095	FBLN5	Q12805, Q9UBX5
FBN1	P35555	FBN2	P35556, Q75N90	FBXL14	Q8N1E6
FBXL16	Q8N461	FBXL17	Q9UF56	FBXL19	Q6PCT2
FBXL20	Q96IG2	FBXL3	Q9UKT7	FBXL4	Q9UKA2
FBXL7	Q9UJT9	FBXL8	Q96CD0	FBXO10	Q9UK96
FBXO11	Q86XK2	FBXO17	Q96EF6	FBXO2	Q96EF6, Q9H4M3, Q9UK22
FBXO22	Q8NEZ5	FBXO27	Q8NI29, Q96EF6	FBXO30	Q8TB52
FBXO32	Q969P5	FBXO41	Q8TF61	FBXO44	Q9NRD1
FBXO6	Q9NRD1	FBXO7	Q9Y3I1	FBXW8	Q8N3Y1
FBXW9	Q5XUX1	FCHO1	O14526	FECH	P22830
FEM1B	Q9UK73	FERMT2	Q96AC1	FES	P07332
FGD1	P98174	FGD3	Q5JSP0	FGD4	Q96M96
FGF10	O15520	FGF12	P61328	FGF13	Q92913
FGF14	Q92915	FGF17	O60258-1	FGF18	O76093
FGF19	O95750	FGF2	P09038	FGF5	P12034-1
FGF9	P31371	FGFBP1	Q14512	FGFBP3	Q8TAT2
FGFR1	P11362-1, P11362-19	FGFR1OP2	Q9NVK5	FGFRL1	Q8N441
FHL2	Q14192	FHOD3	Q8IVF7	FIGNL1	Q6PIW4
FIRRM	Q9NSG2	FKBP14	Q9NWM8	FKBP1B	P68106
FKBP5	Q13451	FKBP9	O95302	FLCN	Q8NFG4
FLI1	Q01543	FLNA	P21333	FLNB	O75369
FLNC	Q14315	FLOT1	O75955	FLRT1	Q9NZU1
FLRT2	O43155	FLRT3	Q9NZU0	FLT1	P17948

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
FLT4	P35916	FLVCR1	Q9Y5Y0-1, Q9Y5Y0-2	FMN1	Q969G6
FMNL2	Q96PY5	FMNL3	Q8IVF7	FMO1	Q01740, Q99518
FMOD	Q06828	FN1	P02751	FNBP1L	Q5T0N5
FNIP2	Q9P278	FOS	P01100	FOSB	P53539
FOSL1	P15407	FOXA2	Q9Y261	FOXC1	Q12948
FOXC2	Q99958	FOXF1	Q12946	FOXK1	P85037
FOXL2	P58012	FOXO1	Q12778	FOXO4	P98177
FOXO6	A8MYZ6	FOXP1	Q9H334-8	FOXP3	Q9BZS1
FOXP4	Q9BZS1	FPGS	Q05932-1, Q05932-2	FRAT1	Q92837
FRAT2	O75474	FRMPD3	Q5JV73	FRS3	O43559
FSTL1	Q12841	FSTL3	O95633	FTCD	O95954
FTSJ3	Q8IY81	FUCA1	P04066	FUOM	A2VDF0
FURIN	P09958	FUT4	P22083, Q11128	FUT8	Q9BYC5
FUT9	Q9Y231	FXYD5	Q14802	FXYD6	Q9H0Q3
FYN	P06241	FYTTD1	Q96QD9	FZD1	Q9UP38
FZD10	Q9ULW2	FZD3	O00144, Q9NPG1	FZD4	Q9ULV1
FZD5	Q13467	FZD6	O60353	FZD7	O75084
FZD9	O00144	G3BP1	Q13283	G3BP2	Q9UN86
GAB1	Q13480	GAB2	Q9UQC2	GABARAPL2	P60520
GABBR1	Q9UBS5	GABBR2	O75899	GABRA2	P47869
GABRA3	P34903	GABRA5	P31644	GABRB3	P28472
GABRG2	P18507	GABRG3	Q99928	GABRR1	P24046
GAD1	Q05329, Q99259	GAD2	Q05329	GADD45A	P24522
GADD45GIP1	Q8TAE8	GADL1	Q6ZQY3	GALK1	P51570
GALNS	P34059	GALNT1	Q10472	GALNT10	Q86SR1
GALNT13	Q8IUC8	GALNT14	Q96FL9	GALNT17	Q49A17, Q6IS24
GALNT18	Q6P9A2	GALNT2	Q10471	GALNT5	Q7Z7M9
GALNT6	Q14435	GALNT7	Q86SF2	GALNT9	Q9HCQ5
GALNTL6	Q49A17	GALT	P07902	GANAB	Q14697
GAP43	P17677	GAR1	Q9NY12	GAS1	P54826
GAS2	O43903	GAS6	Q14393	GATA4	P43694
GATA5	Q9BWX5	GATA6	Q92908	GATAD2B	Q8WXI9
GBP1	P32455, Q9H0R5	GBP2	P32456, Q9H0R5	GBP3	Q9H0R5
GBP5	Q96PP8	GBX2	P52951	GCA	P28676
GCFC2	P16383	GCHFR	P30047	GCLC	P48506
GCLM	P48507	GCN1	Q92616	GCNT1	Q02742
GCNT4	Q9P109	GDAP1	Q8TB36	GDE1	Q9NZC3
GDF5	P43026	GDI1	P50395	GDPD1	Q8N9F7
GEM	P55040	GEMIN4	P57678	GFAP	P14136
GFPT1	Q06210	GFPT2	O94808	GFRA1	P56159
GFUS	Q13630	GGCX	P38435	GGH	Q92820
GGT5	P36269	GGT7	Q9UJ14	GHR	P15509
GINS2	Q9Y248	GIPC1	O14908	GIPR	P48546
GJA1	P17302	GJA3	Q9Y6H8	GJA4	P35212
GJB2	P29033	GJC1	P36383	GJD2	Q9UKL4
GLB1	P16278, Q6UWU2, Q8NCI6	GLB1L2	Q8IW92, Q8NCI6	GLDC	P23378
GLI1	P08151	GLI2	P10070	GLI3	P10071
GLIPR1	P48060	GLP1R	P43220	GLRA1	P23415

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GLRA2	P23416	GLRX	P35754	GLRX5	Q86SX6
GLS	O94925	GLS2	O94925, Q9UI32	GLUL	P15104
GM2A	P17900	GMD5	O60547	GMPPA	Q96IJ6
GMPPB	Q9Y5P6	GMPR	P36959	GNA14	O95837, P29992
GNAI2	P04899	GNAL	P38405	GNAO1	P09471
GNAQ	P50148	GNB3	P16520	GNB4	Q9HAV0
GNG11	P61952	GNG12	Q9UBI6	GNG2	P59768
GNG3	P63215	GNG4	P50150	GNG7	O60262
GNL3	Q9BVP2	GPNPNT1	Q96EK6	GNS	P15586
GOLGA2	Q08379	GOLGA3	Q08378	GOLGA4	Q13439
GOLGA5	Q8TBA6	GOLIM4	O00461	GOLM1	Q8NBJ4
GORASP1	Q9BQQ3	GORASP2	Q9H8Y8	GOSR2	O14653
GPAM	Q9HCL2	GPAT3	Q53EU6	GPATCH1	Q9BRR8
GPC2	Q8N158	GPC3	P51654	GPC4	O75487
GPC5	P78333	GPC6	Q9Y625	GPCPD1	Q9NPB8
GPD1L	Q8N335	GPFR1	Q99527	GPHN	Q9NQX3
GPNMB	Q14956	GPR143	P51810	GPR17	Q13304
GPR176	Q14439	GPR27	Q9NS67	GPR37	O15354
GPR39	O43194	GPR55	Q9Y2T6	GPRIN1	Q7Z2K8
GPX3	O75715, P22352	GPX7	Q96SL4	GPX8	Q8TED1
GRB10	Q13322	GRB7	Q14451	GREB1	Q4ZG55
GREM1	O60565	GREM2	Q9H772	GRHL1	Q9NZI5
GRIA1	P42261	GRIA2	P42262	GRIA4	P48058
GRIK3	Q13003	GRIK5	Q16478	GRIN1	Q05586
GRIN2A	Q12879	GRIN2B	Q13224	GRIN3A	Q8TCU5
GRIN3B	O60391	GRIP2	Q9C0E4	GRK1	Q15835
GRK2	P25098	GRK3	P35626	GRM2	Q14416
GRM3	Q14832	GRM4	Q14833	GRM5	P41594
GRN	P28799	GRPEL1	Q9HAV7	GRSF1	Q12849
GSDMD	P57764	GSDME	O60443	GSK3B	P49841
GSN	P06396	GSPT1	P15170, Q8IYD1	GSPT2	Q8IYD1
GSR	P00390-2	GSTA4	O15217, P08263	GSTK1	Q9Y2Q3
GSTT2B	P0CG30	GTF2A2	P52657	GTF2E2	P29084
GTF2H1	P32780	GTF2H2	Q13888	GUCY1A1	Q02108
GUCY1A2	P33402	GUCY1B1	Q02153	GXYLT2	A0PJZ3
GYG2	O15488	GYPC	P04921	H1-2	P16403
H2AC6	Q93077	H2AZ2	Q71UI9	H2BC21	P06899, Q16778
H2BC9	P58876, Q93079, Q99877	H3-3A	P84243	HACD1	B0YJ81, Q6Y1H2
HACE1	Q8IYU2	HADHB	P55084	HAGH	Q16775-2
HAND1	O96004	HAND2	P61296	HAP1	P27695
HAPLN1	P10915	HAS1	Q92839	HBEGF	Q99075
HCCS	P53701	HCN2	Q9UL51	HCN3	Q9P1Z3
HCN4	Q9Y3Q4	HCRTR2	O43614	HDAC1	Q13547
HDAC5	Q9UQL6	HDAC7	Q8WUI4, Q9UKV0	HDAC9	Q9UKV0
HDGF	P51858	HDHD3	P13716	HDLBP	Q00341
HEATR1	Q9H583	HEBP1	Q9NRV9	HEBP2	Q9Y5Z4
HECTD1	Q9ULT8	HECTD2	Q5U5R9	HECTD3	Q5T447
HECW1	Q76N89	HECW2	Q9P2P5	HEPH	Q9BQS7
HERC1	Q15751	HERC5	Q9UII4	HES1	Q14469

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HEXB	P07686	HEYL	Q9NQ87	HFE	Q30201
HGD	Q93099	HGF	P14210	HGSNAT	Q68CP4
HHEX	Q03014	HHIP	Q96QV1	HIBADH	P31937
HIC1	Q14526	HIF1AN	Q9NWT6	HINT2	Q9BX68
HIP1	O00291	HIP1R	O75146	HIPK1	Q86Z02
HIPK2	Q9H2X6	HIVEP3	Q5T1R4	HK2	P52789
HLA-A	P04439	HLA-B	P01889	HLA-C	P10321
HLA-DPB1	P04440	HLA-DRA	P01903	HLA-DRB1	P01911
HLA-E	P13747	HLA-H	P01893	HLTF	Q14527
HM13	Q8TCT9	HMBS	P08397	HMCN1	Q96DU3
HMGA2	P52926	HMGB2	P26583	HMGCLL1	Q8TB92
HMGCR	P04035	HMGCS1	P54868	HNMT	P50135
HNRNPA1	P09651	HNRNPDL	O14979	HNRNPF	P52597
HNRNPH1	P31943, P55795	HOGA1	Q86XE5	HOMER2	Q9NSB8
HOMER3	Q9NSC5	HOXA1	P49639	HOXA11	P31270
HOXA2	O43364	HOXA3	O43365	HOXA4	Q00056
HOXA6	P31267	HOXB3	P14651	HOXB4	P17483
HOXC11	O43248	HOXC4	P09017	HOXC8	P31273
HOXD11	P31277	HOXD4	P09016	HPDL	Q96IR7
HPS1	Q92902	HPS4	Q9NQG7	HPSE	Q9Y251
HPX	P02790	HRC	P23327	HRH1	P35367
HS2ST1	Q7LGA3	HS3ST1	O14792	HS3ST2	Q9Y278
HS3ST3B1	Q9Y662, Q9Y663	HS6ST1	O60243	HS6ST2	Q96MM7
HS6ST3	Q8IZP7	HSD11B1L	P28845	HSD17B12	Q53GQ0
HSD17B2	P37059	HSD3B1	P14060, P26439	HSD3B7	Q9H2F3
HSP90B1	P14625	HSPA4L	O95757	HSPA5	P11021
HSPA8	P11142	HSPB8	Q9UJY1	HSPG2	P98160
HSPH1	Q92598	HTR1A	P08908	HTR1D	P28221
HTR2A	P28223	HTRA1	Q92743	HTT	P31645
ICAM1	P05362	ICE1	Q9Y2F5	ID1	P41134
ID4	P47928	IDE	P14735	IDH2	P48735
IDH3A	P50213	IER3	P46695	IFI16	Q16666
IFI27L2	P09912	IFI35	P80217	IFI44	Q8TCB0-1
IFI44L	Q53G44	IFIH1	Q9BYX4	IFIT1	P09914
IFITM3	Q01628	IFNAR1	P17181	IFNAR2	P48551-2
IFNGR1	P15260	IFNGR2	Q13651	IFNLR1	Q8IU57
IFT22	Q9H7X7	IFT46	Q9NQC8	IFT52	Q9Y366
IFT57	Q9NWB7	IGF1	P05019	IGF1R	P08069
IGF2	P01344	IGF2BP1	Q9NZI8	IGF2BP3	O00425
IGFBP3	P17936	IGFBP4	P22692	IGFBP5	P24593
IGFBP6	P24592	IGFBP7	Q16270	IGSF3	Q93033
IKBIP	Q70UQ0	IKZF1	Q13422-1	IL10RA	Q13651
IL11	P20809	IL12A	P29459	IL13RA1	P78552
IL13RA2	Q14627	IL15	P40933	IL16	Q14005
IL17RA	Q96F46	IL17RD	Q8NFM7	IL18	Q14116, Q14116-2
IL18R1	Q13478	IL1B	P01584	IL1R1	Q9HB29
IL1RAP	Q9NPH3-1	IL20RA	Q9UHF4	IL31RA	Q8NII7
IL32	P24001	IL33	O95760	IL34	Q6ZMJ4
IL41I	Q96RQ9	IL6R	P08887, P08887-2	IL6ST	P40189

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IL7R	P16871	ILF3	Q12906	IMP4	Q96G21
IMPA1	P29218	ING4	Q9UNL4	INHBA	P08476
INHBB	P09529	INHBE	P58166	INMT	O95050
INO80D	Q53TQ3	INPP5D	Q92835	INPP5F	Q9Y2H2
INPPL1	O15357	INSM1	Q01101	INSR	P06213
INTS13	Q9NVM9	INTS4	Q96HW7	INTS6	Q9UL03
IP6K2	Q9UHH9	IP6K3	Q96PC2	IPMK	Q8NFU5
IQGAP1	P46940	IQSEC3	Q9UPP2	IRAK1	P51617
IRF1	P10914	IRF7	Q92985	IRS1	P35568
IRS2	Q9Y4H2	IRX2	Q9BZI1	ISCA1	Q9BUE6
ISCU	Q9H1K1-1	ISLR	O14498	ISM1	Q9NSE4
ISYNA1	Q9NPH2	ITGA1	P56199	ITGA11	Q9UKX5
ITGA2	P17301	ITGA3	P26006	ITGA4	P13612
ITGA5	P08648	ITGA7	Q13683	ITGA8	P53708
ITGAV	P06756	ITGB1	P05556	ITGB3	P05106
ITGB3BP	Q13352	ITGB4	P16144	ITGB5	P18084
ITGB7	P26010	ITGBL1	O95965	ITIH3	Q06033
ITK	Q08881	ITLN1	Q8WWA0	ITM2B	Q9Y287
ITPK1	Q13572	ITPKB	P27987	ITPR1	Q14643
ITPR2	Q14571	ITPR3	Q14573	ITSN1	Q15811
ITSN2	Q9NZM3	JADE1	Q6IE81	JADE2	Q9NQC1
JAG2	Q9Y219	JAK1	P23458	JARID2	Q92833
JMJD4	Q9H9V9	JOSD1	Q15040	JOSD2	Q8TAC2
JUN	P05412	JUNB	P17275	JUP	P14923
KALRN	O60229	KANK1	Q14678	KANSL3	Q9P2N6
KAT2B	Q92831	KAT6B	Q8WYB5	KAT7	O95251
KAZN	Q674X7	KBTBD6	Q86V97	KBTBD7	Q8WVZ9
KCNA2	P16389	KCNA3	P22001	KCNA6	P17658
KCNB1	Q14721	KCNB2	Q92953	KCNC1	P48547
KCNC3	Q14003	KCNC4	Q03721	KCND1	Q9NSA2
KCNE1	P15382	KCNE2	Q9Y6J6	KCNE4	Q8WWG9
KCNE5	Q9UJ90	KCNF1	Q9H3M0	KCNG1	Q9UIX4
KCNH2	Q12809	KCNH5	Q8NCM2	KCNH8	Q96L42
KCNIP1	Q9NZI2	KCNIP2	Q9NS61	KCNIP4	Q6PIL6
KCNJ10	P78508	KCNJ15	Q99712	KCNJ16	Q9NPI9
KCNJ5	P48544	KCNJ8	Q15842	KCNJ9	Q92806
KCNK1	O00180, Q9Y2U2	KCNK10	P57789	KCNK13	Q9HB14
KCNK6	Q9Y257	KCNMA1	Q12791	KCNMB2	Q9NPA1, Q9Y691
KCNN1	Q92952	KCNN3	Q9UGI6	KCNQ2	O43526
KCNQ3	O43525	KCNQ4	P56696	KCNS1	Q96KK3
KCNS2	Q9ULS6	KCTD13	Q8WZ19	KCTD3	Q9Y597
KDELR1	P24390	KDELR2	P24390, P33947	KDELR3	O43731
KDM2A	Q9Y2K7	KDM2B	Q8NHM5	KDM3B	Q7LBC6
KDM4A	O75164	KDM4B	O94953	KDM5B	Q9UGL1
KDM6A	O15550	KDM6B	O15054	KDR	P35968
KDSR	Q06136	KHDRBS2	Q5VWX1	KHDRBS3	O75525
KIAA1549	Q9HCM3	KIAA1671	Q9C0C2	KICS2	Q96MD2
KIF13A	Q9H1H9	KIF15	Q9NS87	KIF1A	Q12756
KIF1B	O60333	KIF1C	O43896	KIF21A	Q7Z4S6

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KIF21B	O75037	KIF22	Q14807	KIF24	Q5T7B8
KIF26B	Q2KJY2	KIF2A	O00139, Q8N4N8	KIF3A	Q9Y496
KIF3B	O15066	KIF3C	O14782	KIF5A	Q12840
KIF5C	Q6PKD1	KIF7	Q2M1P5	KIF9	Q9HAQ2
KIFC2	Q96AC6	KIFC3	Q9BVG8-2	KIRREL1	Q96J84
KIRREL3	Q8IZU9	KISS1	Q15726	KITLG	P21583-1
KLC1	Q07866	KLC3	Q6P597	KLF16	Q9BXXK1
KLF4	O43474	KLF5	Q13887	KLHL12	Q53G59
KLHL13	Q9P2J3, Q9P2N7	KLHL20	Q9Y2M5	KLHL21	Q9UJP4
KLHL25	Q9H0H3	KLHL3	Q9UH77	KLHL42	Q9P2K6
KLHL9	Q9P2J3	KLK5	Q9Y337	KLK6	P06870
KMT2A	Q03164	KMT5B	Q4FZB7, Q86Y97	KNTC1	P50748
KPNA3	O00505	KPNA4	O00629	KRBOX4	Q5JUW0
KREMEN2	Q8NCW0	KRR1	Q13601	KRT10	P13645
KRT13	P13646	KRT17	Q04695	KRT18	P05783
KRT19	P08727	KRT4	P04259, P19013	KRT7	P08729
KRT75	O95678	KRT8	P13647	KRT80	P13647
KRT81	P78385, Q14533	KRTAP2-3	P0C7H8	KSR2	Q6VAB6, Q8WXI2
KTN1	Q86UP2	L1CAM	P32004	L3MBTL1	Q9Y468
LAG3	P18627	LAMA1	P25391	LAMA2	P24043
LAMA3	Q16787	LAMA4	Q16363	LAMA5	O15230
LAMB1	P07942	LAMB2	P11047, P55268	LAMC2	Q13753
LAMP1	P11279	LAMP2	P13473	LAMP3	P34810
LAP3	Q13867	LARGE1	O95461	LARGE2	Q8N3Y3
LARP1	Q6PKG0	LATS2	Q9NRM7	LBR	Q14739
LCAT	P04180	LCLAT1	Q6UWP7	LCN9	Q8WX39
LDLRAP1	Q5SW96	LEF1	Q9UJU2	LEFTY2	O00292
LEMD3	Q9Y2U8	LETM1	O95202	LGALS1	P09382
LGALS3BP	Q08380	LGI3	Q8N145	LGI4	Q8N135
LGMN	Q99538	LGR4	Q9BXB1	LGR5	O75473
LGR6	Q9HBX8	LHFPL2	Q6ZUX7	LHX2	P50458
LHX9	Q9NQ69	LIF	P15018	LIFR	P42702
LIG1	P18858	LIG3	P49916	LIMA1	Q8TDZ2
LIMS1	P48059	LIMS2	Q7Z4I7	LIN28A	Q9H9Z2
LIN7B	Q9HAP6	LINGO1	Q96FE5	LIPH	Q8WWY8
LMAN1	P49257	LMCD1	Q9NZU5	LMNA	P02545
LMO7	Q8WWI1	LMOD1	P29536	LNK1	Q8TBB1
LONRF1	Q17RB8	LOX	P28300	LOXL1	P58215, Q08397
LOXL2	Q9Y4K0	LOXL3	P58215	LOXL4	Q96JB6
LPAR1	Q92633	LPAR2	Q9HBW0	LPAR4	Q99677
LPAR5	Q9H1C0	LPAR6	P43657	LPCAT1	Q8NF37
LPCAT2	Q7L5N7	LPCAT3	Q6P1A2	LPGAT1	Q92604
LPIN1	Q14693	LPIN2	Q92539	LPIN3	Q9BQK8
LPL	P06858	LRAT	O95237	LRFN3	Q9BTN0
LRIG1	Q96JA1	LRP1	Q07954	LRP10	O75096
LRP2	P98164	LRP4	O75096	LRP6	O75581
LRP8	Q14114	LRRC1	Q9BTT6	LRRC41	Q15345
LRRC49	Q8IUZ0	LRRC4B	Q9NT99	LRRC7	Q96NW7
LRRC8B	Q6P9F7	LRRC8C	Q8TDW0	LRRC8D	Q7L1W4

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
LRRK8E	Q6NSJ5	LRRK1	Q96J92	LRRK2	Q5S007
LRRN2	O43300	LRRTM2	O43300	LRRTM3	Q86VH5
LSAMP	Q13449	LSM11	P83369	LSR	Q86X29
LTB4R	Q15722, Q9NPC1	LTBP1	Q14766	LTBP2	Q14767
LTBP3	Q14767, Q9NS15	LTBP4	Q8N2S1	LTBR	P36941
LTV1	Q96GA3	LUM	P51884	LY6E	Q16553
LY6H	O94772	LYL1	P12980	LYN	P07948
LYPD1	Q8N2G4	LYPD6B	Q8NI32	LYPLA1	O75608
LZTFL1	Q9NQ48	MAD2L2	Q9UI95	MADD	Q8WXG6-3
MAF	O75444	MAFK	O60675	MAGED2	Q9UNF1
MAGI2	Q86UL8	MAGT1	Q9H0U3	MALSU1	Q96EH3
MALT1	Q9UDY8	MAN1A1	P33908	MAN1A2	O60476
MAN2A1	Q16706	MAN2B1	O00754	MAN2B2	Q9Y2E5
MANF	P55145	MAP2	P50579	MAP2K3	P46734
MAP2K4	P45985	MAP2K5	Q13163	MAP2K6	P52564
MAP3K1	Q13233	MAP3K14	Q99558	MAP3K3	Q99759
MAP3K5	Q99683	MAP3K8	P41279	MAPK10	P53779
MAPK11	Q15759	MAPK14	Q16539	MAPK8	P45983
MAPKAP1	Q9BPZ7	MAPRE3	Q9UPY8	MAPT	P10636
MASP1	P48740	MAST1	Q7Z460	MAT2A	P31153
MATK	P42679	MATN3	O15232	MBD1	Q9UIS9
MBD2	Q9UBB5	MBD5	Q9P267	MBOAT1	Q6ZNC8
MBP	P13727	MBTPS1	Q14703	MCAM	P43121
MCAT	Q8IVS2	MCCC2	Q9HCC0	MCF2	P10911
MCF2L	O15068	MCFD2	Q8NI22	MCHR1	Q99705
MCHR2	Q969V1	MCL1	Q07820	MCM10	Q7L590
MCM2	P33993, P49736	MCM7	P33993	MCM8	Q9UJA3
MCOLN2	Q8IZK6	MCOLN3	Q8TDD5	MCU	Q8NE86
MCUB	Q9NWR8	MDGA1	Q8NFP4	MDGA2	Q7Z553
MDK	P21741	MDM2	Q00987	MDM4	O15151
ME1	P48163	ME2	P15923	MEAF6	Q9HAF1
MECOM	Q03112	MECP2	P51608-1, P51608-2	MED10	Q9BTT4
MED13L	Q71F56	MED15	Q96RN5	MED21	Q13503
MEF2C	Q06413	MEIS1	O00470	MELTF	P08582
MET	P08581	METAP2	P50579	METTL1	Q9UBP6
METTL21A	Q8WXB1	MFAP2	P55001	MFAP4	P55083
MFAP5	Q13361	MFN1	Q8IWA4	MFNG	O00587
MFSD2A	Q8NA29	MGAM	O43451	MGAT1	P26572
MGAT2	Q10469	MGAT3	Q09327	MGAT4A	Q9UM21
MGAT4C	Q9UBM8	MGAT5	Q09328	MGAT5B	Q3V5L5
MGST1	P10620	MGST2	Q99735	MIA2	Q96PC5
MICA	Q29983	MICAL1	Q8TDZ2	MICB	Q29980
MICU2	Q8IYU8	MID1	Q9NQW6	MINPP1	Q9UNW1
MITF	O75030-9	MKKS	Q9NPJ1	MKLN1	Q9UL63
MKRN1	Q9UHC7	MLH1	P40692	MLLT1	Q03111
MLLT3	P42568	MLPH	Q9BV36	MLXIP	Q9NP71
MLYCD	O95822-2	MMADHC	Q9H3L0	MME	P39900
MMP10	P08254	MMP14	P50281	MMP15	P51511
MMP17	Q9ULZ9	MMP19	Q99542	MMP2	P08253

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
MMRN1	Q13201	MMS19	Q96T76	MNDA	P41218
MOB1A	Q9H8S9	MOB1B	Q7L9L4	MOCOS	Q96EN8
MOCSS2	O96007, O96033	MOGS	Q13724	MORF4L2	Q15014
MOV10	Q9HCE1	MOV10L1	Q9BXT6	MPG	P29372
MPHOSPH6	Q99547	MPP2	Q08050	MPP7	Q5T2T1
MPST	P25325	MPZ	P25189	MR1	Q8N490-2
MRC2	Q9UBG0	MRM1	Q6IN84	MRPL10	Q7Z7H8
MRPL14	Q6P1L8	MRPL16	Q9NX20	MRPL19	P49406
MRPL3	P09001	MRPL32	Q6P1L8, Q9BYC8	MRPL33	O75394
MRPL4	Q9BYD3	MRPL54	Q6P161	MRPS10	P82664
MRPS12	O15235	MRPS15	P82914	MRPS18B	Q9Y676
MRPS34	P82930	MRPS35	P82673, Q9Y2Q9	MRS2	Q9HD23
MSH2	P43246	MSH5	O43196	MSI2	Q96DH6
MSL1	Q68DK7	MSLN	Q13421	MSN	P26038
MSRA	Q9UJ68	MSRB2	Q9Y3D2	MSRB3	Q8IXL7
MSX1	P28360	MSX2	P35548	MT-ATP6	P00846
MT-CO1	P00395	MT-CO2	P00403	MT-CO3	P00414
MT-ND2	P03891	MT-ND3	P03897	MT-ND4	P03905
MT-ND4L	P03901	MT-ND6	P03923	MT1E	P04732
MT3	P25713	MTARC1	Q5VT66	MTBP	Q96DY7
MTERF1	Q99551	MTF2	Q9Y483	MTHFD1	P11586
MTHFD1L	Q6UB35	MTHFR	P42898	MTIF3	Q9H2K0
MTMR3	Q96QG7	MTMR4	Q9NYA4	MTMR7	Q9Y216
MTMR8	Q96QG7	MTMR9	Q96QG7	MTNR1A	P48039
MTPN	Q96Q27	MTR	Q99707	MTTP	P55157
MTX1	Q13505	MUC13	Q9H3R2	MUC16	Q8WXI7
MUC4	Q99102	MUTYH	Q9UIF7-3, Q9UIF7-6	MVB12B	Q9H7P6
MVP	Q14764	MX1	P20591	MXRA8	Q9BRK3
MYBBP1A	Q9BQG0	MYBPC2	Q14324	MYC	P01106
MYCN	P04198	MYD88	Q99836	MYDGF	Q969H8
MYH10	P35580	MYH15	P13533	MYH9	P35579
MYL10	Q9BUA6	MYL12A	P19105	MYL12B	O14950
MYL7	Q01449	MYL9	P24844	MYLIP	Q8WY64
MYO19	Q96H55	MYO1C	O00159	MYO3A	Q8NEV4
MYO5A	Q9Y4I1	MYO5B	Q9ULV0	MYO6	Q9UM54
MYOCD	Q8IZQ8	MYRIP	Q8NFW9	MYT1	Q99640
MZT2A	Q6P582	NAA35	Q5VZE5	NAAA	Q02083
NAALAD2	Q9Y3Q0	NAB2	Q15742	NABP1	Q96AH0
NADK	O95544	NADSYN1	Q6IA69	NAF1	Q15025
NAGLU	P54802	NALCN	Q8IZF0	NAMPT	P43490
NANP	Q8TBE9	NANS	Q9NR45	NAPB	Q9H115
NAPRT	Q6XQN6	NAPSA	O96009	NARF	Q8WVD3
NAT8L	Q8N9F0	NAXE	Q8NCW5	NBAS	A2RRP1
NBEA	Q8NFP9	NBEAL2	Q6ZNJ1	NBN	O60934
NCALD	P61601	NCAM1	P13591	NCAN	O14594
NCAPH2	Q6IBW4	NCF2	P19878	NCL	P19338
NCOA2	Q15596	NCOA6	Q14686	NCOR2	Q9Y618
NCR3LG1	Q68D85	NDEL1	Q9GZM8	NDN	Q99608
NDST1	P52848	NDST4	Q9H3R1	NDUFA13	Q9P0J0

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
NDUFA6	P56556	NDUFAB3	Q9BU61	NDUFB11	Q9NX14
NDUFS3	O75489	NDUFS5	O43920	NDUFV3	P56181
NECAP1	Q8NC96	NECAP2	Q9NVZ3	NECTIN1	Q15223
NECTIN2	Q92692	NEFL	P07196	NEGR1	Q7Z3B1
NEK11	Q8NG66	NEK3	Q9UHY1	NEK6	Q9HC98
NEK7	Q8TDX7	NELFA	Q9H3P2	NELL2	Q99435
NEMF	O60524	NET1	Q7Z628	NEURL1	O76050
NEURL1B	A8MQ27	NEURL3	A8MQ27	NEUROG3	Q9Y4Z2
NFASC	O94856	NFATC1	O95644	NFATC2	Q12968, Q13469
NFE2L2	Q16236	NFIA	Q12857	NFIB	O00712
NFIC	P08651	NFIX	Q14938	NFKB1	P19838
NFKB2	Q00653	NFKBIA	P25963	NFYB	P25208
NGB	Q9NPG2	NGEF	Q8N5V2	NGF	P01138
NGFR	P08138	NGLY1	Q96IV0	NGRN	Q9NPE2
NHS	Q6T4R5	NID1	P14543	NID2	Q14112
NINL	Q9Y2I6	NIP7	Q9Y221	NIPA2	Q8N8Q9
NIPAL1	Q6NVV3	NIPAL2	Q9H841	NIPAL3	Q6P499
NIPAL4	Q0D2K0	NIPBL	Q6KC79	NIPSNAP2	O75323
NIT2	Q9NQR4	NKX2-2	O95096	NKX3-2	P78367
NLGN1	Q8N2Q7	NLGN2	Q8NFX4	NLGN3	Q9NZ94
NLGN4X	Q8N0W4	NLK	Q9UBE8	NLN	Q9BYT8
NLRC5	Q86WI3	NLRP1	Q9C000	NLRX1	Q86UT6
NMB	Q14956	NME4	O00746	NMI	Q13287
NMNAT2	Q9BZQ4	NMT1	P30419	NMU	P48645
NNMT	P40261	NNT	Q13423	NOB1	Q9ULX3
NOCT	Q9UK39	NOD1	Q9Y239	NOG	Q13253
NOL6	Q9H6R4	NOP14	P78316	NOP56	O00567
NOS3	P29474	NOTCH1	P46531	NOTCH2	Q04721
NOTCH4	Q99466	NOTUM	Q6P988	NOX1	Q9Y5S8
NOX4	Q9NPH5	NPAS2	Q99743	NPC2	P61916
NPHP1	O15259	NPLOC4	Q8TAT6	NPM1	P06748
NPM2	Q86SE8	NPNT	Q6UXI9	NPPA	P01160
NPPC	P23582	NPR1	P16066	NPR2	P20594
NPRL2	Q8WTW4	NPW	Q8N729	NPY1R	P25929
NR2F1	P10589	NR2F2	P24468	NR2F6	P10588
NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9	NR3C2	P08235-1, P08235-2, P08235-3, P08235-4	NR4A2	P43354
NR4A3	Q92570-1, Q92570-2	NR5A2	O00482-1, O00482-2, O00482-3	NR6A1	Q15406-1, Q15406-2, Q15406-3, Q15406-4, Q15406-5
NRBP1	Q9UHY1	NRCAM	Q92823	NRG2	O14511
NRG3	P56975	NRG4	Q8WWG1	NRGN	Q92686
NRN1	Q9NPD7	NRP2	O60462	NRTN	Q99748
NRXN1	P58400, Q9ULB1	NRXN2	P58401, Q9P2S2	NRXN3	Q9HDB5, Q9Y4C0
NSD2	O96028	NSD3	Q9BZ95	NSF	P46459
NSMAF	Q92636	NSRP1	Q9H0G5	NSUN2	Q08J23
NSUN6	Q8TEA1	NT5C1A	Q9BXI3	NT5M	Q8TCD5, Q9NPB1
NTF4	P34130	NTN1	O95631	NTN4	Q9HB63
NTNG1	Q9Y2I2	NTRK2	Q16620	NTRK3	Q16288
NUAK1	O60285	NUCB1	Q02818	NUDC	Q9Y266

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
NUDT15	Q9NV35	NUDT16	Q96DE0	NUDT18	Q6ZVK8-1
NUDT2	P50583	NUDT3	O95989	NUMA1	Q14980
NUP210	Q8TEM1	NUP214	P35658, Q96HA1	NUP93	Q8N1F7
NUP98	P52948-3, P52948-4, P52948-5	NUS1	Q96E22	OAS3	Q9Y6K5
OAT	P04181	OBSCN	Q5VST9	OCA2	Q04671
OCLN	Q16625	OCRL	Q01968	ODC1	Q96A70
OGFOD1	Q8N543	OGG1	O15527	OGN	P20774
OLR1	P78380	OMG	P23515	ONECUT1	Q9UBC0
ONECUT3	O60422	OPLAH	O14841	OPN3	Q9H1Y3
OPRK1	P41145	OPTN	Q96CV9	OR10G4	Q8NGN3
OR51E1	Q8TCB6	ORA12	Q96SN7	ORC1	Q13415
ORC2	Q13416	ORC4	O43929	ORMDL2	Q53FV1
OS9	Q13438	OSBPL10	Q9BXB5	OSBPL11	Q9BXB4
OSBPL5	Q9H0X9	OSGEPL1	Q9H4B0	OSR1	Q8TAX0
OST4	P0C6T2	OSTC	Q9NRP0	OSTF1	Q92882
OSTM1	Q86WC4	OTOF	Q9HC10	OTUB2	Q96DC9
OTUD3	Q5T2D3	OTUD7A	Q8TE49	OTUD7B	Q6GQQ9
OXTR	P30559	P2RX4	Q99571	P2RX6	O15547
P2RY6	Q15077	P3H1	Q32P28	P3H2	Q8IVL5
P3H3	Q8IVL6	P4HA1	P13674	P4HA2	O15460
P4HA3	Q7Z4N8	P4HB	P07237	PABPC1	P11940
PABPN1	Q86U42	PACSIN3	Q9UKS6	PADI1	Q9ULC6
PADI2	Q9Y2J8	PAFAH1B3	Q15102	PAFAH2	Q99487
PAG1	Q9NWX8	PAIP1	Q9H074	PAK1	Q16512
PAK3	O75914, Q13153	PAK4	O96013	PAK5	Q9NQU5, Q9P286
PAK6	Q9NQU5	PALS1	Q8N3R9	PAN2	Q9HBH5
PANK1	Q8TE04	PANX2	Q96RD6	PAPOLG	Q9BWT3
PAPPA	Q13219	PAPSS2	O95340	PARD3	Q8TEW0
PARD6B	Q9BYG5	PARP1	P09874	PARP10	Q53GL7
PARP14	Q460N5	PARP4	Q9UKK3	PARP9	Q8IXQ6
PARS2	Q7L3T8	PARVA	Q9NVD7	PARVB	Q9HBI1
PAX2	Q02962	PAX3	P23760	PAX6	P26367
PAX7	P23759	PAX8	Q06710	PBX2	Q9Y4H4
PCBD1	P61457	PCBP4	P57723	PCCA	P05165
PCDH15	Q96QU1	PCDH19	Q8TAB3	PCDH7	O60245
PCF11	O94913	PCGF5	Q86SE9	PCLO	Q9Y6V0
PCM1	Q15154	PCOLCE	Q15113	PCSK1	P29120
PCSK2	P16519	PCSK5	Q92824	PCSK6	P29122
PCSK9	Q8NBP7	PCYT1B	Q9Y5K3	PDCD1LG2	Q9BQ51
PDCD6IP	Q8WUM4	PDE10A	Q9Y233	PDE1A	P54750
PDE1B	Q01064	PDE1C	Q01064	PDE2A	O00408
PDE3A	Q14432	PDE3B	Q13370	PDE4A	P27815
PDE4B	Q07343-3	PDE4D	Q08499	PDE5A	O76074
PDE6D	O43924	PDE7A	Q13946	PDGFA	P04085-1, P04085-2
PDGFB	P01127	PDGFC	Q9NRA1-1	PDGFD	Q9GZP0
PDGFRA	P16234	PDGFRB	P09619	PDHB	P11177
PDHX	O00330	PDIA2	Q13087	PDIA3	P30101
PDIA5	Q14554	PDIA6	Q15084	PDK1	O15530
PDK3	Q15120	PDK4	Q16654	PDLIM7	Q9NR12

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PDP1	Q9P0J1	PDPN	Q86YL7	PDS5A	Q29RF7
PDYN	P01213	PEA15	Q15121	PEAR1	P11047
PECAM1	P16284	PECR	Q9BY49	PELO	Q9BRX2
PERP	Q96FX8	PES1	O00541	PEX1	O43933
PEX11A	O75192	PEX11B	O96011	PEX16	Q9Y5Y5
PEX2	P28328	PEX3	P56589	PFDN1	O60925
PFKFB1	P16118	PFKFB3	Q16875	PFKFB4	Q16877
PFKP	Q01813	PGAM5	Q96HS1-1	PGAP1	Q75T13
PGD	P52209	PGF	P49763	PGK1	P00558
PGM1	P36871	PGM2L1	Q6PCE3	PGM3	O95394
PGR	P06401	PGRMC1	O00264	PGRMC2	O15173
PHACTR2	O75167	PHETA2	Q9H4I9	PHF1	O43189
PHF10	Q8WUB8	PHF21A	Q96BD5	PHIP	Q8WWQ0
PHLDA1	Q8WV24	PHLPP1	O60346	PHYH	O14832
PHYKPL	Q8IUZ5	PI4K2B	Q8TCG2	PI4KA	P42356
PIANP	Q8IYJ0	PICALM	Q13492	PICK1	Q9NRD5
PIEZO1	Q92508	PIF1	Q9H611	PIGK	Q92643
PIGN	O95427	PIGS	Q96S52	PIGU	Q9H490
PIGW	Q7Z7B1	PIK3C2A	O00443	PIK3C2B	O00750
PIK3C2G	O75747	PIK3CB	P42338	PIK3CD	O00329
PIK3R3	Q92569	PIM1	P11309	PIP4K2B	P78356
PIP4K2C	Q8TBX8	PIP4P1	Q86T03	PIP5K1A	Q99755
PIP5K1B	O14986	PIPOX	Q9P0Z9	PITPNB	P48739
PITPNM1	O00562	PITPNM3	Q9BZ71	PITRM1	Q5JRX3
PITX2	Q99697	PJA1	Q8NG27	PJA2	O43164
PKD1	P98161	PKD2	Q13563	PKN3	Q6P5Z2
PKP2	Q99959	PKP3	Q9Y446	PKP4	Q99569
PLA2G12A	Q9BZM1	PLA2G2A	P14555, Q9UNK4	PLA2G4A	P47712
PLA2R1	Q13018	PLAAT3	P53816	PLAAT4	P53816
PLAAT5	Q96KN8	PLAGL1	Q9UM63	PLAT	P00750
PLAU	P00749	PLAUR	Q03405	PLB1	Q3MJ16
PLBD1	Q6P4A8	PLCB3	Q01970	PLCD1	P51178
PLCD3	Q8N3E9	PLCD4	Q9BRC7	PLCH1	Q4KWH8
PLD1	Q13393	PLD2	O14939	PLD3	Q8IV08
PLD6	Q8N2A8	PLEC	Q15149	PLEKHA2	Q9HB19
PLEKHA6	Q9Y2H5	PLEKHA8	Q96JA3	PLEKHG3	A1L390
PLEKHG4B	Q96PX9	PLIN3	Q8IUC6	PLK2	Q9NYY3
PLN	P26678	PLOD1	Q02809	PLOD2	O00469
PLOD3	O60568	PLPP4	Q5VZY2	PLPPR1	Q8TBJ4
PLPPR2	Q96GM1	PLPPR3	Q6T4P5	PLPPR4	Q7Z2D5
PLPPR5	Q32ZL2	PLS1	Q14651	PLXNA1	Q9UIW2
PLXNA4	Q9HCM2	PLXNB1	O43157	PLXNB3	O43157
PLXNC1	O60486	PLXND1	Q9Y4D7	PMEPA1	Q969W9
PML	P29590	PMM1	Q92871	PMM2	O15305
PMP22	Q9NR77	PMPCA	Q10713	PNKD	Q8N490-2
PNMT	P11086	PNO1	Q9NRX1	PNP	P00491
PNPT1	Q8TCS8	PODXL2	Q9NZ53	POFUT1	Q9H488
POFUT2	Q9Y2G5	POLA1	P09884	POLD2	P49005
POLE	Q07864	POLE2	P56282	POLI	Q9UNA4

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
POLR1A	O95602	POLR1C	O15160	POLR1D	P0DPB5
POLR1E	O15160	POLR1F	Q3B726	POLR1G	O15446
POLR2A	P24928	POLR2D	O15514	POLR2G	P62487
POLR3A	O14802	POLR3D	P05423	POLR3E	Q9NVU0
POLR3G	O15318, Q9BT43	POLR3H	Q9Y535	POMK	Q9H5K3
PON2	Q15165	POP1	Q9UKB1	POR	P16435
POU2F1	P14859	POU2F2	P09086	POU3F1	Q03052
POU3F2	P20265	POU3F3	P20264	POU4F1	Q01851
POU4F2	Q12837	PPARA	Q07869	PPFIA1	Q13136
PPFIA2	O75334	PPFIA3	O75145	PPFIA4	O75335
PPFIBP1	Q86W92	PPIB	P23284	PPIF	P62937
PPIL3	Q9H2H8	PPIL6	Q8IXY8	PPL	P23434
PPM1D	O15297	PPM1E	Q8WY54	PPM1K	Q8N3J5
PPM1L	Q5SGD2	PPP1CB	P62140	PPP1R12A	O14974
PPP1R12B	O60237	PPP1R13L	Q8WUF5	PPP1R14A	Q96A00
PPP1R3C	Q9UQK1	PPP2CB	P62714	PPP2R1A	P30153
PPP2R2A	P63151	PPP2R5A	Q15172	PPP2R5B	Q15173
PPP3R1	P63098	PPP4R2	Q9NY27	PPRC1	Q5VV67
PRAG1	Q86YV5	PRCP	P42785	PRDM1	O75626
PRDM4	Q9UKN5	PRDM8	Q9NQV8	PRDX2	P32119
PRDX3	P30048	PRDX4	Q13162	PRDX6	P30041
PRELID1	Q9Y255	PRELID3A	Q96N28	PREX1	Q8TCU6
PREX2	Q70Z35	PRICKLE1	Q96MT3	PRIM1	P49642
PRKAA1	Q13131	PRKAA2	P54646	PRKAB1	Q9Y478
PRKACB	P22694	PRKAG2	Q9UGJ0	PRKCA	P17252
PRKCB	P05771	PRKCD	Q05655	PRKCE	Q02156
PRKCI	P41743	PRKD1	Q15139	PRKD3	O94806
PRKG1	Q13976-1	PRMT3	O60678	PRMT7	Q9NVM4
PRNP	P04156	PROC	P04070	PROCR	Q9UNN8
PRODH	O43272, Q9UF12	PROKR1	Q8TCW9	PROS1	P07225
PRP4K	Q13523	PRPF3	O43395	PRPF38A	Q8NAV1
PRPF40A	O75400	PRPF6	O94906	PRR29	Q99943
PRR3	Q9NQS3	PRR5	P85299	PRSS23	O95084
PRSS8	Q16651	PRX	Q9BXM0, Q9BXM0-1	PSAP	Q8IWL1
PSIP1	O75475	PSMB5	P28074	PSMB8	P28062
PSMB9	P28065	PSMD5	Q16401	PSMD6	Q15008
PSME1	Q06323	PSME3	Q06323	PTBP1	P26599
PTGER2	P35408, P43116	PTGER3	P43115	PTGER4	P35408
PTGES2	Q9H7Z7	PTGFR	P43088	PTGIR	P43119
PTGIS	Q16647	PTGS1	P23219	PTGS2	P35354
PTK2	Q05397	PTK7	P42679	PTN	P21246
PTPMT1	Q8WUK0	PTPN11	Q06124	PTPN13	Q12923
PTPN14	Q15678	PTPN2	P17706	PTPN20	Q4JDL3
PTPN5	P54829	PTPN6	P29350	PTPRA	P18433
PTPRD	P23468	PTPRF	P10586	PTPRJ	Q12913
PTPRK	Q15262	PTPRN2	Q92932	PTPRO	Q16827
PTPRS	Q13332	PTPRU	Q92729	PTPRZ1	P23471
PTX3	P26022	PUS1	Q9Y606-1	PUS7	Q96PZ0
PVR	P16234	PXDN	Q92626	PXMP4	Q9Y6I8

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PXN	P11678	PXYLP1	Q8TE99	PYCARD	Q9C000
PYCR1	P32322	PYCR3	Q53H96	PYGL	P06737
PYGM	P11217	PYGO2	Q9BRQ0	QDPR	P09417
QPCT	Q16769	QPRT	Q15274	QRFPR	Q96P65
QSOX1	O00391	QTRT2	Q9H974	RAB11A	P62491
RAB11B	Q15907	RAB11FIP2	Q7L804	RAB12	Q6IQ22
RAB15	P59190	RAB17	Q9H0T7	RAB1A	P62820
RAB1B	Q9H0U4	RAB20	Q9NX57	RAB23	Q9ULC3
RAB27A	P51159	RAB27B	O00194	RAB29	O14966
RAB2A	P61019	RAB32	Q13637	RAB33A	Q14088
RAB34	Q9BZG1	RAB38	P57729	RAB39A	Q14964
RAB39B	Q96DA2	RAB3A	P20336	RAB3C	Q96E17
RAB3D	O95716	RAB3GAP2	Q9H2M9	RAB40C	Q96S21
RAB42	Q8N4Z0	RAB4A	P20338	RAB6B	Q9NRW1
RAB8B	Q92930	RAB9B	Q9NP90	RABEPK	Q7Z6M1
RABGAP1	Q9Y3P9	RABGGTB	P53611	RAC2	P15153
RAD23A	P54725	RAD23B	P54727	RALB	P11234
RALBP1	Q15311	RALGDS	Q12967	RAMP1	O60894
RAMP2	O60895	RAMP3	O60896	RANBP9	Q96S59
RAP1A	P62834	RAP1B	P61224	RAP1GAP	P47736
RAP1GAP2	Q684P5	RAP2A	P61225	RAP2C	Q9Y3L5
RAPGEF1	Q13905	RAPGEF2	Q9Y4G8	RAPGEF3	O95398
RAPGEF4	Q8WZA2	RAPGEF6	Q9Y4G8	RARA	P10276
RARB	P10826	RARG	P13631	RARS1	P54136
RASA2	Q15283	RASA4B	O43374	RASAL2	Q9UJF2
RASGEF1A	Q8N9B8	RASGRF1	Q13972	RASGRP1	O95267
RASGRP3	Q8IV61	RASGRP4	Q8TDF6	RBBP4	Q09028
RBBP8	Q99708	RBFOX1	Q9NWB1	RBFOX2	O43251
RBFOX3	A6NFN3	RBKS	Q9H477	RBL1	P28749
RBM10	P52756, P98175	RBM17	Q96I25	RBM4B	P52272
RBM7	Q9Y580	RBM8A	Q9Y5S9	RBMX	P38159
RBMX2	Q9Y388	RBP1	P09455	RBSN	Q9H1K0
RCC1L	Q96I51	RCL1	Q9Y2P8	RCN1	Q15293
RCOR1	Q9UKL0	RCVRN	P35243	RDH10	Q8IZV5
RDH14	Q9HBH5	RDH5	Q92781	REC8	O95072
RECK	O95980	REEP1	Q9H902	REL	Q00653
RELA	Q04206	RELB	Q01201	RENBP	P51606
REPS1	Q96D71	REST	Q13127	RET	P07949
RETSAT	Q6NUM9	REV3L	O60673	REXO2	Q9Y3B8
RFC5	P40937, P40938	RFK	Q969G6	RGCC	Q9H4X1
RGMA	Q96B86	RGMB	Q6NW40	RGR	P47804
RGS10	O43665	RGS17	Q9UGC6	RGS19	P49795
RGS2	Q9UGC6	RGS20	O76081	RGS4	P49798
RGS5	O15539	RGS6	P49758	RGS7	P49802
RHBDF2	Q6PJF5	RHEB	Q15382	RHOB	P62745
RHOBTB1	O94844	RHOC	P08134	RHOD	O00212
RHOF	Q9HBH0	RHOG	P84095	RHOJ	Q9H4E5
RHOQ	P17081	RHOU	Q7L0Q8	RHPN2	Q8IUC4
RIC1	Q4ADV7	RIMKLA	Q8IXN7	RIMKLB	Q9ULI2

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RIMS1	Q86UR5	RIN2	Q8WYP3	RIN3	Q8TB24
RINT1	Q6NUQ1	RIOK1	Q9BRS2	RIOK3	O14730
RIOX2	Q8IUF8	RIPK1	Q13546	RIPK2	O43353
RIPOR2	Q9Y4F9	RIPPLY2	Q5TAB7	RLN1	P04090
RMI1	Q9H9A7	RMI2	Q96E14	RNASEH1	O60930
RND1	Q92730	RND2	P52198	RND3	P61587
RNF103	O00237	RNF125	Q96EQ8	RNF130	Q86XS8
RNF135	Q8IUD6	RNF139	Q8WU17	RNF144A	P50876
RNF144B	Q7Z419	RNF146	Q9NTX7	RNF152	Q8N8N0
RNF182	Q8N6D2	RNF185	Q96GF1	RNF19A	Q9NV58
RNF20	Q5VTR2	RNF213	Q63HN8	RNF216	Q9NWF9
RNF217	Q8TC41	RNF41	Q9H4P4	RNF43	Q68DV7
RNF5	Q96GF1, Q99942	RNF6	Q9Y252	RNF7	Q9UBF6
RNH1	O60930	RNMT	O43148	ROBO1	Q9Y6N7
ROBO2	Q9HCK4	ROBO3	Q96MS0-1	ROCK1	Q13464
ROR1	Q01973	ROR2	Q01974	RORA	P35398
RORB	Q92753	RPA2	P15927	RPE	Q2QD12, Q96AT9
RPE65	Q16518	RPL12	P30050	RPL22L1	Q6P5R6
RPL23A	P62750	RPL36A	P83881, Q969Q0	RPL8	P62424, P62917
RPLP1	P05386	RPLP2	P05387	RPN1	Q13200
RPN2	Q99460	RPP38	P78345	RPRD2	Q5VT52
RPS13	P62277	RPS16	P62249	RPS2	P08865, P15880, P46782
RPS27A	P62979, P62987	RPS27L	Q71UM5	RPS6KA2	Q15349
RPS6KA4	O75676	RPS6KA5	O75582	RPS6KB1	P23443
RPS6KB2	Q9UBS0	RRAGA	Q5VZM2, Q7L523	RRAS	P10301
RRAS2	P62070	RRBP1	Q9P2E9	RRM2B	Q7LG56
RRP1	P05386	RSF1	Q16629	RSPO1	Q2MKA7
RSPO2	Q6UXX9	RSPO3	Q9BXY4	RSPO4	Q2IOM5
RSU1	Q15404	RTN3	O95197	RTN4RL1	Q86UN2
RTN4RL2	Q86UN3	RUNX1	Q01196	RUNX2	Q13950-1, Q13950-2
RUVBL1	Q9Y265	RWDD1	Q9H446	RWDD2B	Q9Y3V2
RXYLT1	Q9Y2B1	RYK	P34925	RYR1	P21817
RYR3	Q15413	S100A10	P60903	S100A11	P31949
S100B	P04271	S1PR1	P21453	S1PR2	O95136
S1PR3	Q99500	S1PR5	Q9H228	SALL1	Q9NSC2
SALL4	Q9UJQ4	SAMD10	Q96KM6	SAMHD1	Q9Y3Z3
SAP130	Q15393	SAP30	O75446, Q9HAJ7	SAP30BP	Q9UHR5
SARDH	Q9UL12	SARM1	Q6SZW1-1	SAT2	Q96QD8
SAV1	Q9H4B6	SBSPON	Q8IVN8	SCAI	Q8N9R8
SCAP	Q12770	SCARA5	P21757, Q6ZMJ2	SCARB1	Q8WTV0-2
SCARB2	Q14108	SCD	O00767	SCD5	Q86SK9
SCFD1	Q8WVM8	SCG2	O00255, P13521	SCG3	Q8WXD2
SCLT1	Q96NL6	SCN2A	Q99250	SCN2B	O60939
SCN3A	Q99250, Q9NY46	SCN3B	Q9NY72	SCN4A	P35499
SCN7A	Q01118	SCN8A	Q9UQD0	SCNN1A	P37088
SCNN1G	P51170	SCO1	O75880	SCTR	P47872
SCUBE1	Q8IWY4	SCUBE2	Q9NQ36	SCUBE3	Q8IX30
SDC2	P34741	SDC3	O75056	SDC4	P31431
SDCBP	O00560	SDHB	P21912	SDHD	O14521

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SDK2	Q58EX2	SDSL	Q96GA7	SEC11C	Q9BY50
SEC13	P55735	SEC22B	O75396	SEC23A	Q15436
SEC23IP	Q9Y6Y8	SEC24A	O95486	SEC24B	O95487
SEC24D	O94855	SEC31A	O94979	SEC61A1	P61619
SEC61A2	Q9H9S3	SEC61B	P60468	SEH1L	Q96EE3-1, Q96EE3-2
SELENOS	Q9BQE4	SEMA3E	O15041	SEMA4A	Q9H3S1
SEMA6A	Q9H2E6	SEMA6D	Q8NFY4	SENP1	Q9P0U3
SEPSECS	Q9HD40	SEPTIN4	O43236-6	SEPTIN7	Q16181
SERINC1	Q9NRX5	SERINC2	Q96SA4	SERINC5	Q86VE9
SERP1	Q9Y6X1	SERPINA1	P01009	SERPINA5	P05154
SERPINB1	P30740	SERPINB2	P05120	SERPINB6	P35237
SERPINB8	P50452	SERPINE1	P05121	SERPING1	P05155
SERPINH1	P50454	SESN2	P58004	SETD7	Q8WTS6
SETD9	Q8NE22	SETDB1	Q15047	SF1	Q13285
SF3A2	P03971	SFI1	A8K8P3	SFRP2	Q96HF1
SGCB	Q16585	SGCD	Q92629	SGCG	Q13326
SGIP1	O14526, Q9BQI5	SGK1	O00141	SGK3	Q96BR1
SGMS1	Q86VZ5	SGMS2	Q8NHU3	SGPP1	Q9BX95
SGPP2	Q8IWX5	SGSM2	Q9GZT4	SH2B2	O14492
SH3BP4	Q9P0V3	SH3D19	Q5HYK7	SH3GL1	Q99961
SH3GL2	Q99962	SH3RF1	Q7Z6J0	SHANK1	Q9Y566
SHANK2	Q9UPX8	SHARPIN	Q9H0F6	SHB	Q15464
SHC2	P98077	SHC3	Q92529	SHH	Q15465
SHISA5	Q8N114	SHPRH	Q149N8	SHQ1	Q6PI26
SIGIRR	Q6IA17	SIGMAR1	Q99720	SIN3A	Q96ST3
SIN3B	O75182	SIRPA	P78324	SIRT6	Q8N6T7
SIX1	Q15475	SIX2	Q9NPC8	SKAP2	O75563
SKI	Q5VTY9	SKIC2	Q15477	SKIC3	Q6PGP7
SKIL	P12757	SKP2	Q13309	SLBP	Q14493
SLC12A4	Q9UP95	SLC12A5	Q9H2X9	SLC12A7	Q9Y666
SLC13A5	Q86YT5	SLC15A3	Q8IY34	SLC16A1	P53985
SLC16A10	Q8TF71	SLC16A2	P36021	SLC16A3	O15427
SLC17A5	Q9NRA2	SLC17A6	Q9P2U8	SLC17A8	Q8NDX2
SLC18A3	Q16572	SLC19A1	P41440	SLC1A3	P43003
SLC1A6	P48664	SLC20A2	Q08357	SLC22A15	Q8IZD6
SLC22A17	Q8WUG5	SLC22A3	O75751	SLC22A5	O76082
SLC24A2	Q9UI40	SLC24A3	Q9HC58	SLC25A14	O95258
SLC25A16	P16260	SLC25A19	Q9HC21	SLC25A20	O43772
SLC25A22	Q9H936	SLC25A27	O95847	SLC25A28	Q96A46
SLC25A42	Q86VD7	SLC26A11	Q86WA9	SLC26A7	Q8TE54
SLC26A9	Q7LBE3	SLC27A1	Q6PCB7	SLC27A3	Q5K4L6
SLC27A6	Q9Y2P4	SLC29A2	Q14542	SLC2A11	Q9BYW1
SLC30A1	Q9Y6M5	SLC30A10	Q6XR72	SLC30A5	Q8TAD4
SLC31A1	O15431	SLC32A1	Q9H598	SLC34A2	O95436
SLC35A2	P78381	SLC35B3	Q9H1N7	SLC35B4	Q969S0
SLC35C1	Q96A29	SLC37A4	O43826	SLC38A2	Q96QD8
SLC38A3	Q99624	SLC38A4	Q969I6	SLC39A14	Q15043
SLC39A7	Q92504	SLC39A8	Q9C0K1	SLC40A1	Q9NP59
SLC41A2	Q96JW4	SLC43A2	Q8N370	SLC44A1	Q8WWI5

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SLC44A2	Q8IWA5	SLC44A3	Q8N4M1	SLC44A5	Q8NCS7
SLC46A1	Q96NT5	SLC47A1	Q96FL8	SLC4A10	Q6U841
SLC4A2	P04920	SLC4A4	Q9Y6R1	SLC4A8	Q2Y0W8
SLC4A9	Q96Q91	SLC50A1	Q9BRV3	SLC52A3	Q9NQ40
SLC5A12	Q1EHB4	SLC6A1	P30531	SLC6A11	P48066
SLC6A13	Q9NSD5	SLC6A5	P23975	SLC6A6	P31641
SLC6A7	Q99884	SLC6A8	P48029	SLC7A2	P52569-1, P52569-2
SLC7A3	Q8WY07	SLC7A5	Q01650	SLC7A6	Q92536
SLC7A7	Q9UM01	SLC7A8	Q9UHI5	SLC8A1	P32418
SLC8A3	P57103	SLC8B1	Q6J4K2	SLC9A2	Q9UBY0
SLC9A5	Q14940	SLC9A6	Q92581	SLC9A7	Q96T83
SLC9B2	Q4ZJI4, Q86UD5	SLCO1A2	P46721	SLCO1C1	Q9NYB5
SLCO2A1	Q92959	SLCO2B1	O94956	SLCO4A1	Q96BD0
SLCO4C1	Q6ZQN7	SLIT1	O75093	SLIT2	O94813
SLIT3	O75094	SLITRK1	Q96PX8	SLITRK2	Q9H156
SLITRK3	O94933	SLITRK4	Q8IW52	SLITRK5	O94991
SLITRK6	Q9H5Y7	SLPI	P03973	SLX4	Q8IY92
SMAD2	Q15796	SMAD3	P84022	SMAD6	O43541
SMAD7	O15105	SMAD9	O15198	SMAP2	Q9H0E9
SMARCC2	Q8TAQ2	SMARCD1	Q96GM5	SMARCD2	Q92925
SMC5	Q8IY18	SMG5	Q9UPR3	SMG6	Q86US8
SMG7	Q92540	SMPD1	P17405	SMPD2	O60906
SMPD3	Q9NY59	SMYD3	Q9H7B4, Q9NRG4	SNAI1	O95863
SNAP23	O00161	SNAP25	P60880	SNAP91	O60641
SNAPC4	Q5SXM2	SNCA	P37840	SNCAIP	Q9Y6H5, Q9Y6H5-2
SNCB	Q16143	SND1	Q7KZF4	SNRNP200	O75643
SNRPD3	P62318	SNTA1	Q13424	SNTB1	Q13884
SNTB2	Q13425	SNX18	Q96RF0	SNX3	O60493
SNX5	Q9Y5X3	SNX6	Q9Y5X3	SNX9	Q9Y5X1
SOCS1	O15524	SOCS3	O14543	SOCS5	O75159
SOCS6	O14544	SOCS7	Q8WXH5	SORBS1	Q9BX66
SORBS3	O60504	SORL1	Q92673	SORT1	Q99523
SOS2	Q07890	SOX1	O00570	SOX10	P56693
SOX13	Q9UN79	SOX2	P48431	SOX21	P56693
SOX3	P41225	SOX6	P35712	SOX9	P48436
SP100	P23497	SP3	Q9Y2H2	SPAG9	O60271
SPARC	Q14515	SPATA13	Q96N96	SPATA2	Q9UM82
SPC24	Q8NBT2	SPCS3	P61009	SPDL1	Q96EA4
SPHK1	Q9NYA1	SPINT1	O43278	SPINT2	O43291
SPNS2	Q8IVW8	SPOCK3	Q9BQ16	SPON1	Q9HCB6
SPOPL	Q6IQ16	SPPL2A	Q8TCT8	SPR	P35270
SPRED1	Q7Z699	SPRED2	Q7Z698	SPRED3	Q2MJR0
SPRR2E	P22531	SPRR2F	Q96RM1	SPRTN	Q9H040
SPRY1	O43609	SPRY2	O43597	SPSB1	Q96BD6
SPSB4	Q96A44, Q99619	SPTA1	P02549	SPTB	P11277
SPTBN1	Q01082	SPTBN2	O15020	SPTBN4	Q9H254
SPTBN5	Q9NRC6	SPTLC1	O15269	SPTLC3	Q9NUV7
SPX	Q15427	SQSTM1	Q13501	SRD5A1	P18405
SREBF1	P36956-3	SREBF2	Q12772	SRF	P11831

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SRGAP2	O43295, O75044	SRI	P30626	SRM	P19623
SRP54	Q05519	SRP68	Q9UHB9	SRP72	O76094
SRPRA	P08240	SRPRB	Q9Y5M8	SRR	Q9GZT4
SRRM1	Q8IYB3	SRSF11	Q05519	SRSF2	Q01130
SRSF3	P84103	SS18	Q15532	SSB	P05455
SSC5D	A1L4H1	SSPN	Q14714	SSR1	Q8TAQ2
SSR3	Q6STE5	SSR4	P51571	SST	P61278
SSTR1	P30872	SSTR2	P30874	SSU72	Q9NP77
ST3GAL1	Q11201	ST3GAL6	Q9Y274	ST6GALNAC1	Q9NSC7
ST6GALNAC2	Q9UJ37	ST6GALNAC4	Q9H4F1	ST8SIA1	Q92185
ST8SIA3	O43173	ST8SIA4	Q92187	STAMBP	O95630
STAMBPL1	Q96FJ0	STAP2	Q9UGK3	STAR	P25092
STARD13	Q9Y3M8	STARD3	Q14849	STARD3NL	O95772
STARD8	Q92502	STAT1	P42224	STAT2	P52630
STAT3	P40763	STAT5A	P42229, P51692	STAT5B	P51692
STAT6	P42226	STEAP3	Q658P3	STING1	Q86WV6
STK10	O94804	STK26	Q9P289	STK3	Q13188
STK38	Q15208	STMN2	Q93045	STOM	P27105
STON1	Q9Y6Q2	STRA6	Q9BX79	STRADB	Q9C0K7
STT3A	P46977	STT3B	Q8TCJ2	STX16	O14662
STX1A	Q16623	STX1B	P61266	STX3	Q13277
STX4	Q12846	STXBP1	P61764	STXBP2	Q15833
SUCLG2	Q96I99	SUDS3	Q9H7L9	SUFU	Q9UMX1
SULT1B1	O75897	SULT1C2	O75897	SULT1E1	P49888
SULT4A1	Q9BR01	SUMF1	Q8NBK3	SUMF2	Q92526
SUMO3	P55854	SUN1	O94901	SUOX	P51687
SUPT3H	O75486	SUPT5H	O00267	SUPT7L	O94864
SURF4	O15260	SV2A	Q7L0J3	SV2B	Q7L1I2
SVBP	Q8N300	SVIP	Q8NHG7	SWAP70	Q9UH65
SYDE1	Q6ZW31	SYDE2	Q5VT97	SYN1	Q13884
SYN2	Q9NY99	SYN3	O14994	SYNE2	Q8WXH0
SYNGR1	O43759	SYNJ1	O43426	SYNJ2	O15056
SYP	P08247	SYT1	P21579, Q8NBV8	SYT10	Q6XYQ8
SYT11	Q9BT88	SYT12	Q8IV01	SYT2	Q8N9I0
SYT4	Q9BT88	SYT5	O00445	SYT7	O43581
SYT9	Q86SS6	SYTL1	Q8IYJ3	TAB2	Q9NYJ8
TACR3	P29371	TADA1	Q96BN2	TADA2A	O75478
TAF13	Q15543	TAF1A	Q15573	TAF1D	Q9H5J8
TAF5	Q15542	TAF5L	O75529	TAF9	Q16594
TAGLN2	P37802	TALDO1	P37837	TANK	Q92844
TAOK3	Q9H2K8	TAP1	Q03518	TAP2	Q03519
TAPBP	O15533	TAS2R4	Q9NYW5	TASP1	Q9H6P5
TAX1BP1	Q86VP1	TBC1D10C	Q8IV04	TBC1D13	Q9NVG8
TBC1D16	Q8TBP0	TBC1D2	Q9BYX2	TBC1D24	Q9ULP9
TBC1D4	O60343	TBC1D8B	Q0IIM8	TBCD	Q9BTW9
TBK1	Q9UHD2	TBRG4	Q969Z0	TBX2	Q13207
TBX3	O15119	TBXA2R	P21731, P21731-2, P21731-3	TCAP	O15273
TCF12	Q99081	TCF7	P36402	TCHH	Q07283
TCIRG1	Q13488	TCN2	P20062	TDG	Q13569

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TDRKH	Q9Y2W6	TEAD3	Q99594	TEAD4	Q15561, Q15562
TECTA	O75443	TEK	Q02763	TEP1	P60484
TERF2IP	Q9NYB0	TES	Q9NZU5	TET1	Q8NFM7
TET3	O43151	TEX264	Q9Y6I9	TF	P02787
TFAP2B	Q92481	TFAP2C	Q92754	TFB2M	Q9H5Q4
TFDP1	Q14186	TFDP2	Q14188	TFE3	P19532
TFEB	P19484	TFF3	Q07654	TFG	Q92734
TFPI2	P48307	TFR2	Q9UP52	TFRC	P02786
TGFB1	P01137	TGFB2	P61812	TGFB3	P10600
TGFB1	Q15582	TGFB2	P37173	TGIF1	Q15583
TGIF2	Q9GZN2	TGM5	O43548	THBD	P07204
THBS1	P07996	THBS2	P35442	THBS3	P49746
THOC2	Q8NI27	THOC6	Q86W42	THPO	P40225
THRB	P10828	THSD7A	Q9UPZ6	THSD7B	Q9C014
TIAL1	Q01085	TIAM1	Q13009	TICAM1	Q8IUC6
TIFA	P38919	TIGAR	Q9NQ88	TIMM17A	Q99595
TIMM23	O14925	TIMM23B	O14925	TIMMDC1	Q9NPL8
TIMP1	P01033	TIMP2	P16035	TIMP3	P35625
TJP1	Q07157	TK1	P04183	TLE2	Q04725
TLE3	Q04726-3	TLE4	Q04727	TLL1	O43897
TLL2	Q9Y6L7	TLN1	Q9Y490	TLR2	O60603
TLR3	O15455	TLR6	Q9Y2C9	TMBIM1	Q969X1
TMED10	P49755	TMED2	Q15363	TMED3	Q9Y3Q3
TMED5	Q9Y3A6	TMED7	Q9Y3B3	TMED9	Q9BVK6
TMEM132A	Q24JP5	TMEM132E	Q24JP5	TMEM258	P61165
TMEM30A	Q9NV96	TMEM59	Q9BXS4	TMEM63A	O94886
TMEM86A	Q8N661	TMEM87B	Q8NBN3	TMF1	P82094
TMOD2	Q9NZR1	TNFAIP1	Q13829	TNFAIP3	P21580
TNFAIP8	O95379	TNFRSF10A	O00220	TNFRSF10D	Q9UBN6
TNFRSF11B	O00300	TNFRSF12A	Q9NP84	TNFRSF13C	Q96RJ3
TNFRSF19	Q9HAV5	TNFRSF1A	P19438	TNFRSF21	O75509
TNFRSF8	P28908	TNFRSF9	Q07011	TNFSF13	O75888
TNFSF14	O43557	TNFSF15	O95150	TNFSF4	P23510
TNFSF9	P41273	TNIP2	Q8NFZ5	TNIP3	Q96KP6
TNKS	O95271, Q9H2K2	TNKS1BP1	Q9C0C2	TNKS2	P16157
TNNC1	P19429, P63316	TNNI1	P19237	TNNT1	P13805
TNNT2	P45379	TNR	Q92752	TNRC6B	Q9UPQ9
TNRC6C	Q9HCJ0	TNS3	Q68CZ2	TOP2B	Q02880
TOP3A	Q13472	TOR1A	O14656	TOR1AIP1	Q5JTV8
TOR4A	Q9NXH8	TP53	P04637	TP53BP1	Q12888
TP53RK	Q96S44	TP73	O15350	TPCN1	Q9ULQ1
TPD52	Q9Y2Q9	TPD52L1	Q16890	TPH1	P17752
TPK1	Q9H3S4	TPM1	P09493	TPM2	P07951
TPM4	P67936	TPMT	P51580	TPST1	O60507
TPST2	O60704	TRADD	Q15628	TRAF3	Q13114
TRAF7	Q6Q0C0	TRAIP	Q9BWF2	TRAK1	Q9UPV9
TRAM1	Q15629	TRAPPC2	P0DI81	TRAPPC2L	Q9UL33
TRAPPC3	O43617	TRAPPC4	Q9Y296	TRAPPC5	Q8IUR0
TRAPPC6A	O75865	TRAPPC8	Q9Y2L5	TRAPPC9	Q96Q05

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TRDMT1	O14717	TRDN	Q13061	TRH	P20396
TRIM14	Q14142	TRIM2	Q9C040	TRIM21	P19474
TRIM22	Q8IYM9	TRIM24	O15164	TRIM25	Q14258
TRIM3	P53365	TRIM36	Q9NQ86	TRIM38	O00635
TRIM41	Q8WV44	TRIM47	Q8WV44	TRIM48	Q8IWZ4
TRIM5	Q9C035	TRIM56	Q9BRZ2	TRIM6	Q9C030
TRIM71	Q2Q1W2	TRIM9	Q9C026	TRIO	O75962
TRIP10	Q15642	TRMT1	O75648	TRMT6	Q9UJA5
TRMT61A	Q96FX7	TRMT9B	Q9P272	TRNT1	Q96Q11
TRPC3	Q13507	TRPC4	Q9UBN4	TRPC4AP	Q8TEL6
TRPC5	Q9UL62	TRPM3	Q9HCF6	TRPM4	Q8TD43
TRPM7	Q96QT4	TRPM8	Q7Z2W7	TRPV1	Q8NER1
TRPV3	Q8NET8	TRPV4	Q9HBA0	TRUB2	O95900
TSC1	Q92574	TSHR	P16473	TSPAN14	Q8NG11
TSPAN5	P62079	TSPAN7	P41732	TSP0	P30536
TSR1	Q2NL82	TST	Q15477	TTBK2	Q6IQ55
TTC19	Q6DKK2	TTC9	Q9UIM3	TTL	Q8NG68
TTL1	O95922	TTL11	Q8NHH1	TTL12	Q14166
TTYH1	Q9BSA4, Q9H313-3	TTYH2	Q9BSA4	TTYH3	Q9C0H2
TUBA1A	Q71U36	TUBA1C	P68363, Q9BQE3	TUBA3D	P0DPH8
TUBB2A	Q13885, Q9BVA1	TUBB2B	Q13885, Q9BVA1	TUBB4A	P04350
TUBB4B	P04350, P68371	TUBB6	Q9BUF5	TUBGCP6	Q96RT7
TULP3	O75386	TUSC3	Q13454	TUT4	Q5TAX3
TWF1	Q12792	TWIST1	Q15672	TXK	P42681
TXNDC11	Q6PKC3	TXNDC5	Q8NBS9	TXNIP	Q9H3M7
TXNRD1	Q16881	TXNRD2	Q9NNW7	TYK2	P29597
TYMP	P19971	TYRP1	P17643	TYW3	Q6IPR3
U2AF1	Q01081	UACA	Q9BZF9	UAP1	Q16222
UBA1	P22314	UBA3	Q8TBC4	UBA6	A0AVT1
UBA7	P22314, P41226	UBAP1	Q9NZ09	UBB	P0CG47, P62979, P62987
UBE2A	P49459	UBE2D3	P61077	UBE2E1	P51965
UBE2E2	Q96LR5	UBE2E3	Q969T4	UBE2G1	P62253
UBE2G2	P60604	UBE2H	P62256	UBE2J1	Q9Y385
UBE2K	P61086	UBE2L6	O14933	UBE2O	Q9C0C9
UBE2Q1	Q7Z7E8	UBE2Q2	Q8WVN8	UBE2Z	Q9H832
UBE3B	Q7Z3V4	UBE3C	Q15386	UBL4A	P11441
UBQLN1	Q9UMX0	UBR4	Q5T4S7	UBXN7	O94888
UCN	P55089	UCN2	Q96RP3	UCP3	P55916
UGCG	Q16739	UGDH	O60701	UGGT2	Q9NYU1
UGP2	Q16851	UGT2B4	P06133	UGT8	Q16880
UHKM1	Q8TAS1	UIMC1	Q96RL1	ULK3	Q6PHR2
UNC13B	O14795	UNC5A	Q6ZN44	UNC5B	Q8IZJ1
UNC5D	Q6UXZ4	UNC79	Q9P2D8	UNC80	Q8N2C7
UNC93B1	Q9H1C4	UPF3B	Q9BZI7	UQCC2	Q9BRT2
UROD	P06132	USO1	O60763	USP11	P51784
USP14	P54578	USP15	Q9Y4E8	USP18	Q9UMW8
USP20	Q9Y2K6	USP22	Q9UPT9	USP25	Q9UHP3
USP3	Q9Y6I4	USP30	Q70CQ3	USP33	Q8TEY7
USP34	Q70CQ2	USP37	Q86T82	USP43	Q70EL4

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
USP44	Q70CQ1, Q9H0E7	USP45	Q70EL2	USP48	Q86UV5
USP49	Q70CQ1, Q9H0E7	USP6NL	Q92738	USP8	P40818
USP9X	Q93008	UST	Q9Y2C2	UTP14A	Q9BVJ6
UTP15	Q8TED0	UTP20	O75691	UTP25	Q68CQ4
UTP3	Q9NQZ2	UTRN	P46939	UTS2R	Q9UKP6
UXS1	Q8NBZ7	VAMP2	P63027	VAMP3	Q15836
VAMP8	Q9BV40	VANGL1	Q8TAA9	VANGL2	Q9ULK5
VAPB	O95292	VARS1	P26640	VARS2	P26640
VASH1	Q7L8A9	VASP	Q2MJR0	VAV3	Q9UKW4
VCAM1	P19320	VCL	P18206	VCP	P55072
VDR	P11473	VEGFA	P15692, P49765	VEGFB	P49765
VEGFD	O43915	VGf	O15240	VIPR2	P41587
VKORC1L1	Q8N0U8	VMA22	Q96NT0	VPS26A	O75436
VPS33B	Q9H267	VPS41	P49754	VPS45	Q9NRW7
VRK2	Q86Y07	VWA1	Q6PCB0	VWF	P04275
WARS2	Q9UGM6	WASF1	Q92558	WASF2	Q9Y6W5
WASF3	Q9UPY6	WDCP	Q9H6R7	WDR1	O75083
WDR11	Q9BZH6	WDR3	Q9UNX4	WDR33	Q9C0J8
WDR43	Q15061	WDR45B	Q5MNZ6	WDR46	O15213
WDR48	Q8TAF3	WDR82	Q6UXN9	WDR91	A4D1P6
WDTC1	Q8N5D0	WFDC2	Q14508	WFS1	O76024
WIF1	Q9Y5W5	WIPF2	Q8TF74	WIPF3	A6NGB9
WIPI1	Q5MNZ9	WLS	Q5T9L3	WNK3	Q9BYP7
WNK4	Q96J92	WNT11	O96014	WNT16	Q9UBV4
WNT2	O00755, P09544	WNT2B	Q93097	WNT3	P56703
WNT3A	P56704	WNT4	O14905, P56705	WNT5A	P41221
WNT5B	Q9H1J7	WNT7A	O00755	WNT7B	P56706
WNT9A	O14904	WNT9B	O14905	WT1	P19544
WTAP	Q15007	WTIP	A6NIX2, Q96IF1	WWC1	Q8IX03
WWTR1	Q9GZV5	XDH	Q06278	XPNPEP1	O43895
XPO1	O14980	XRCC1	P18887	XRCC2	O43543
XRN2	Q9H0D6	XYLT2	Q9H1B5	YAP1	P46937
YARS2	Q9Y2Z4	YIF1A	O95070	YIPF6	Q96EC8
YKT6	O15498	YME1L1	Q96TA2	YRDC	Q86U90
YTHDC1	Q96MU7	ZBTB16	Q05516	ZC3H12A	Q5D1E8
ZC3H15	Q8WU90	ZC3HAV1	Q7Z2W4	ZDHHHC2	Q9UIJ5
ZDHHHC20	Q5W0Z9	ZDHHHC3	Q9NYG2	ZDHHHC5	Q9C0B5
ZDHHHC7	Q9NXF8	ZEB2	O60315	ZFHx3	Q15911
ZFP2	Q6ZN57, Q9NQZ8	ZFP36	P26651	ZFP36L2	P47974
ZFP37	P17032, Q9Y6Q3	ZFP69B	Q9UJL9	ZFP90	Q8TF47
ZFPM1	Q8IX07	ZFPM2	Q8WW38	ZFYVE16	Q7Z3T8
ZIC2	O95409	ZIK1	Q3SY52, Q9BWM5	ZKSCAN1	P17029
ZKSCAN3	Q9BRR0	ZKSCAN7	Q6ZNG0, Q9P0L1	ZMAT5	Q9UDW3
ZNF10	P21506	ZNF101	Q8IZC7	ZNF114	Q8NC26
ZNF138	P52744	ZNF141	Q15928	ZNF154	A6NNF4, Q13106
ZNF17	P17021	ZNF19	P17023	ZNF197	O14709
ZNF20	P17024	ZNF200	P98182	ZNF211	Q13398
ZNF213	O14771	ZNF215	Q8NE65, Q9UL58	ZNF227	Q86WZ6
ZNF23	P17027	ZNF233	Q9NZL3	ZNF256	Q9Y2P7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ZNF260	Q8TAQ5	ZNF264	O43296	ZNF268	Q96BR6
ZNF273	Q14593	ZNF274	Q96GC6	ZNF300	Q96RE9
ZNF304	Q9GZX5	ZNF311	Q5JNZ3	ZNF320	A2RRD8
ZNF324	O75467	ZNF331	Q9NQX6	ZNF333	Q96JL9
ZNF334	Q9HCZ1	ZNF337	Q9Y3M9	ZNF345	Q8NA42
ZNF382	Q96SR6	ZNF385A	Q96PM9	ZNF397	Q96MU6
ZNF417	Q8TAU3	ZNF418	Q8TF45	ZNF423	Q2M1K9
ZNF43	P17038	ZNF436	Q9C0F3	ZNF440	Q8IY18
ZNF441	Q8N8Z8	ZNF442	Q9H7R0	ZNF443	Q9Y2A4
ZNF445	P59923	ZNF454	Q8N9F8	ZNF461	Q8TAF7
ZNF467	Q7Z7K2	ZNF468	Q5VIY5	ZNF480	Q8WV37
ZNF483	Q8TF39	ZNF484	Q5JVG2, Q8N8J6	ZNF490	Q9ULM2
ZNF512B	Q96KM6	ZNF519	Q8TB69	ZNF521	Q96K83
ZNF528	Q3MIS6	ZNF530	Q6P9A1	ZNF534	Q76KX8
ZNF540	Q8NDQ6	ZNF547	Q8IVP9	ZNF548	Q8NEK5
ZNF549	Q6P9A3	ZNF555	Q96BR6	ZNF561	Q8N587
ZNF562	Q6V9R5	ZNF582	Q96NG8	ZNF583	Q96ND8
ZNF587	Q96SQ5	ZNF589	Q86UQ0	ZNF596	Q8TC21
ZNF598	Q86UK7	ZNF600	Q6ZNG1	ZNF605	Q86T29, Q9Y462
ZNF606	Q8WXB4	ZNF620	Q6ZNG0	ZNF627	Q7L945
ZNF638	Q14966	ZNF641	Q96N77	ZNF670	Q9BS34
ZNF671	Q8TAW3	ZNF691	Q5VV52	ZNF700	Q9H0M5
ZNF703	Q9H7S9	ZNF708	P17019	ZNF711	Q9Y462
ZNF714	Q96N38	ZNF730	Q6ZMV8	ZNF74	Q16587
ZNF747	Q9BV97	ZNF75A	Q96N20	ZNF75D	P51815
ZNF76	Q8TAW3	ZNF761	Q86XN6	ZNF774	Q6NX45
ZNF775	Q96BV0	ZNF776	Q68DI1	ZNF778	Q96MU6
ZNF790	Q6PG37	ZNF84	P51786	ZNF853	Q6IV72
ZNF860	A6NHJ4	ZNF875	P10072	ZYX	Q8WUP2

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
ABCA1	ENSG00000165029	ABCC1	ENSG00000103222	ABCC3	ENSG00000108846
ABCF2	ENSG00000033050	ABCG1	ENSG00000160179	ACACA	ENSG00000278540
ACADM	ENSG00000117054	ACOX1	ENSG00000161533	ACSL1	ENSG00000151726
ACTA2	ENST00000224784	ADAP1	ENSG00000105963	AGPAT1	ENSG00000204310
AGPAT2	ENSG00000169692	AGT	ENSG00000135744	AIFM2	ENSG00000042286
ALDH1A3	ENST00000329841	ALK	ENSG00000171094	ALOX15	ENSG00000161905
ALOX5	ENSG00000012779	AMH	ENSG00000104899	ANGPTL4	ENSG00000167772
ANKRD1	ENSG00000148677	ANXA1	ENST00000257497	ANXA2	ENST00000451270
ANXA3	ENST00000264908	ANXA4	ENST00000394295	APC2	ENSG00000234906
APOE	ENSG00000130203	APP	ENSG00000142192	AQP1	ENST00000311813
ARC	ENSG00000198576	ARF1	ENSG00000143761	ARFGAP1	ENSG00000101199
ARID3A	ENSG00000116017	ARL4C	ENSG00000188042, ENST00000339728	ASAH1	ENSG00000104763
ASNS	ENSG00000070669	ATF3	ENSG00000162772	ATF5	ENSG00000169136, ENST00000423777
ATM	ENSG00000149311, ENST00000278616	ATP6AP2	ENSG00000182220	ATP6V0A1	ENSG00000033627
ATP6V0E1	ENSG00000113732	ATP6V1A	ENSG00000114573	ATP6V1H	ENSG00000047249
AXIN1	ENSG00000103126	AXIN2	ENSG00000168646, ENST00000307078	B2M	ENSG00000166710

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
BAX	ENSG00000087088	BCL2A1	ENSG00000140379	BCL2L1	ENSG00000171552
BCL2L14	ENSG00000121380	BCL6	ENSG00000113916	BDNF	ENSG00000176697
BHLHE40	ENSG00000134107	BID	ENSG00000015475	BMP4	ENSG00000125378
BNIP3L	ENSG00000104765	BOC	ENSG00000144857	BRCA1	ENSG00000012048
BST2	ENSG00000130303	BTG1	ENSG00000133639	BTG2	ENSG00000159388
CA9	ENSG00000107159	CALR	ENSG00000179218	CASP1	ENSG00000137752
CASP6	ENSG00000138794	CAV1	ENSG00000105974	CBFA2T2	ENSG00000078699
CBX5	ENSG00000094916	CCL2	ENSG00000108691	CCN1	ENSG00000145386
CCN2	ENSG00000118523	CCNA1	ENSG00000133101	CCND1	ENSG00000110092
CCNG2	ENSG00000138764	CD24	ENST00000606017	CD274	ENST00000381573.3
CD36	ENSG00000135218	CD44	ENST00000428726	CD74	ENST00000009530
CDC25A	ENSG00000164045	CDC25B	ENSG00000101224	CDC25C	ENSG00000158402
CDC7	ENSG00000097046	CDH1	ENST00000261769	CDH11	ENST00000268603, ENST00000394156
CDH19	ENSG00000071991	CDH2	ENSG00000170558	CDK5R1	ENSG00000176749
CDKN1B	ENSG00000111276	CDKN2A	ENSG00000147889, ENST00000304494, ENST00000579755	CDKN2B	ENSG00000147883
CDT1	ENSG00000167513	CEBPB	ENSG00000172216	CEBPD	ENSG00000221869
CEL	ENST00000372080	CES1	ENSG00000125148	CFTR	ENST00000003084
CGB8	ENSG00000104827	CIITA	ENSG00000179583	CISH	ENSG00000114737
CITED2	ENSG00000164442	CNN2	ENSG00000064666	COL1A1	ENSG00000108821
COL1A2	ENSG00000164692	COL4A6	ENSG00000197565	CPA1	ENST00000011292
CPB1	ENST00000282957	CPEB1	ENST00000563519	CPT1A	ENSG00000110090
CRH	ENSG00000147571	CRY1	ENSG00000008405	CRY2	ENSG00000121671
CSF1	ENSG00000184371	CSF3R	ENSG00000119535	CSPG4	ENST00000308508
CSRP1	ENSG00000159176	CTDSP2	ENSG00000175215	CTSD	ENSG00000117984
CUL7	ENSG00000044090	CXCL1	ENSG00000163739	CXCL10	ENSG00000169245
CXCL12	ENSG00000107562	CXXC1	ENSG00000154832	CXXC5	ENSG00000171604
CYCS	ENSG00000172115	DBP	ENSG00000105516	DCUN1D3	ENST00000324344
DDB2	ENSG00000134574	DDIT3	ENSG00000175197	DDIT4	ENSG00000168209
DDX11	ENSG00000013573	DLL1	ENSG00000198719	DLX5	ENST00000648378
DNAJB11	ENSG00000090520	DNAJB6	ENSG00000105993	DNAJB9	ENSG00000128590
DNAJC3	ENSG00000102580	DPPA4	ENSG00000121570	EBF1	ENSG00000164330
EBI3	ENSG00000105246	EDIL3	ENSG00000164176	EGF	ENSG00000138798
EGFR	ENST00000275493	EGR1	ENSG00000120738	EGR2	ENSG00000122877
EHF	ENST00000257831	EIF2AK2	ENSG00000055332	ELL2	ENST00000237853
ELOVL2	ENSG00000012660	ELOVL6	ENSG00000170522	EMX2	ENSG00000170370
ENPP2	ENST00000427067	EPAS1	ENSG00000116016	EPHA1	ENSG00000146904
EPHA4	ENSG00000116106	ERICH5	ENST00000318528	ETS2	ENSG00000157557
EXTL3	ENSG00000012232	EYA1	ENSG00000104313	EZH2	ENSG00000106462
F3	ENSG00000117525	FABP4	ENSG00000170323	FABP7	ENSG00000164434
FADS1	ENSG00000149485	FANCD2	ENSG00000144554, ENST00000419585	FANCI	ENSG00000140525, ENST00000310775
FAS	ENSG00000026103	FASN	ENSG00000169710	FBXO32	ENSG00000156804
FGF2	ENSG00000138685	FGF9	ENSG00000102678	FHL2	ENSG00000115641
FKBP14	ENSG00000106080	FKBP5	ENSG00000096060	FLT4	ENSG00000037280
FN1	ENSG00000115414	FOS	ENSG00000170345	FOXA2	ENST00000377115
FOXF1	ENSG00000103241	FOXL2	ENSG00000183770	FOXO1	ENSG00000150907
FOXP3	ENSG00000049768	FURIN	ENSG00000140564	GAD1	ENSG00000128683
GAD2	ENSG00000136750	GADD45A	ENSG00000116717	GATA4	ENST00000335135

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
GATA6	ENST00000269216	GATA6-AS1	ENST00000579431	GBP1	ENSG00000117228
GBP2	ENSG00000162645	GBP3	ENSG00000117226	GBP5	ENSG00000154451
GBX2	ENSG00000168505	GCLC	ENSG00000001084	GCLM	ENSG00000023909
GEM	ENSG00000164949	GFAP	ENSG00000131095	GFPT1	ENSG00000198380
GLI1	ENSG00000111087	GLI2	ENSG00000074047	GLIPR1	ENSG00000139278
GLS2	ENSG00000135423	GMPR	ENSG00000137198	GOSR2	ENSG00000108433
GPAM	ENSG00000119927	GPR143	ENSG00000101850	GPRC5A	ENST00000014914
GPRIN1	ENSG00000169258	GRB10	ENSG00000106070	GREB1	ENSG00000196208, ENST00000381486.6
GREM1	ENSG00000166923	GRHL1	ENSG00000134317	GRIA2	ENSG00000120251
GRIN2A	ENSG00000183454	GRIN2B	ENSG00000273079	GSDMD	ENSG00000104518
GSDME	ENSG00000105928	GSR	ENSG00000104687	GXYLT2	ENSG00000172986
H19	ENST00000252900	HAND1	ENSG00000113196	HAND2	ENSG00000164107
HDGF	ENSG00000143321	HES1	ENST00000232424	HEYL	ENSG00000163909
HGF	ENSG00000019991	HHEX	ENST00000282728	HHIP	ENSG00000164161
HLA-A	ENSG00000206503	HLA-B	ENSG00000234745	HLA-C	ENSG00000204525
HLA-DPB1	ENSG00000223865	HLA-DRA	ENSG00000204287	HLA-DRB1	ENSG00000196126
HLA-E	ENSG00000204592	HLA-H	ENSG00000206341	HMGCR	ENSG00000113161
HMGCS1	ENSG00000112972	HNRNPDL	ENSG00000152795	HNRNPF	ENSG00000169813
HOXA1	ENSG00000105991, ENST00000343060	HOXA2	ENSG00000105996, ENST00000222718	HOXA3	ENSG00000105997, ENST00000317201
HOXA4	ENSG00000197576, ENST00000360046	HOXB3	ENSG00000120093, ENST00000311626	HOXB4	ENSG00000182742, ENST00000332503
HOXC4	ENSG00000198353, ENST00000303406	HOXD4	ENSG00000170166, ENST00000306324	HSP90B1	ENSG00000166598
HSPA5	ENSG00000044574	HSPA8	ENSG00000109971	HSPH1	ENSG00000120694
ICAM1	ENSG00000090339	ID1	ENSG00000125968	ID4	ENSG00000172201
IFI35	ENSG00000068079	IFIT1	ENSG00000185745	IFITM3	ENSG00000142089
IGF2	ENST00000337883, ENST00000381406	IGFBP3	ENSG00000146674	IGFBP7	ENSG00000163453
IKBIP	ENSG00000166130	IL10RA	ENSG00000110324	IL12A	ENSG00000168811
IL18	ENSG00000150782	IL1B	ENSG00000125538, ENST00000263341	IL1R1	ENSG00000115594
IL6R	ENSG00000160712	INSM1	ENSG00000173404	IP6K2	ENSG00000068745
IQSEC3	ENSG00000120645	IRAK1	ENSG00000184216	IRF1	ENSG00000125347
IRF7	ENSG00000185507	IRX2	ENSG00000170561	ITGA2	ENSG00000164171
ITGA4	ENSG00000115232	ITGA5	ENSG00000161638	ITGB1	ENST00000302278
ITGBL1	ENSG00000198542	JUNB	ENSG00000171223	KDELR3	ENSG00000100196
KDM4B	ENST00000381759	KDM5B	ENST00000235790	KDM6B	ENSG00000132510
KLF4	ENSG00000136826	KLF5	ENSG00000102554	KRT17	ENST00000311208
KRT18	ENST00000388835	KRT19	ENST00000361566	KRT7	ENST00000331817
KRT8	ENST00000293308	LAMA5	ENSG00000130702	LGALS2	ENST00000215886
LIF	ENSG00000128342	LIFR	ENSG00000113594	LIG1	ENSG00000105486
LMNA	ENSG00000160789	LPIN1	ENSG00000134324	LPL	ENSG00000175445
LRIG1	ENST00000273261	MCL1	ENSG00000143384	MCM2	ENSG00000073111
MDM2	ENSG00000135679	ME1	ENSG00000065833	MECOM	ENSG00000085276
MECP2	ENST00000303391, ENST00000453960	MED12L	ENST00000273432	MEF2C	ENSG00000081189
MET	ENSG00000105976	MID1	ENSG00000101871	MITF	ENSG00000187098
MLH1	ENSG00000076242	MME	ENST00000360490	MMP10	ENSG00000166670
MMP2	ENSG00000087245	MPZ	ENSG00000158887	MSH2	ENSG00000095002

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
MSN	ENSG00000147065	MSX1	ENSG00000163132	MT-ATP6	ENST00000361899
MT-CO1	ENST00000361624	MT-CO2	ENST00000361739	MT-CO3	ENST00000362079
MT-ND2	ENST00000361453	MT-ND3	ENST00000361227	MT-ND4	ENST00000361381
MT-ND4L	ENST00000361335	MT-ND6	ENST00000361681	MT-TQ	ENST00000387372
MTERF1	ENSG00000127989	MX1	ENSG00000157601	MYC	ENST00000613283
MYCN	ENSG00000134323	MYDGF	ENSG00000074842	MYL9	ENSG00000101335
MYLIP	ENSG00000007944	MYO5A	ENSG00000197535	MYOCD	ENSG00000141052
NAB2	ENSG00000166886	NAMPT	ENSG00000105835	NCAM1	ENSG00000149294
NDN	ENSG00000182636	NEUROG3	ENSG00000122859	NFE2L2	ENSG00000116044
NKX2-2	ENSG00000125820	NLK	ENST00000407008	NOCT	ENSG00000151014
NOS3	ENSG00000187556	NOTCH1	ENSG00000148400	NOTCH2	ENSG00000134250, ENST00000256646
NOTCH4	ENSG00000206312, ENST00000383264	NOX4	ENSG00000086991	NPAS2	ENSG00000170485
NPNT	ENSG00000168743	NPPA	ENSG00000175206	NR4A3	ENSG00000119508
NR5A2	ENSG00000116833	OAS3	ENSG00000111331	OCLN	ENSG00000197822
ONECUT1	ENST00000305901	ONECUT3	ENSG00000205922	OPRK1	ENSG00000082556
OR10G4	ENSG00000254737	OR51E1	ENSG00000180785	ORC1	ENSG00000085840
OSR1	ENSG00000143867	PAIP2	ENST00000265192	PAX2	ENSG00000075891
PAX3	ENSG00000135903	PAX6	ENSG00000007372	PAX7	ENSG00000009709
PAX8	ENSG00000125618	PCBP4	ENSG00000090097	PCDH19	ENSG00000165194
PDGFA	ENSG00000197461	PDIA2	ENST00000219406	PDIA5	ENSG00000065485
PDIA6	ENSG00000143870	PDK4	ENSG00000004799	PDPN	ENSG00000162493
PERP	ENSG00000112378	PEX11A	ENSG00000166821	PGD	ENSG00000142657
PGR	ENST00000325455	PIM1	ENSG00000137193	PITX2	ENSG00000164093
PLAGL1	ENSG00000118495	PLK2	ENSG00000145632	PLXNA4	ENSG00000221866
PLXND1	ENSG00000004399	PML	ENSG00000140464	PMP22	ENSG00000109099
POLA1	ENSG00000101868	POLR2D	ENST00000272645	POU2F1	ENSG00000143190
POU3F1	ENSG00000185668	POU3F2	ENST00000328345	POU3F3	ENSG00000198914
PPARA	ENSG00000186951	PPP2R5B	ENSG00000068971	PRDM1	ENSG00000057657
PRKCB	ENSG00000166501	PRX	ENSG00000105227	PSAP	ENSG00000122852, ENSG00000185303
PSMB8	ENSG00000204264	PTGS2	ENSG00000073756	PTPN11	ENSG00000179295
PTPN13	ENSG00000163629	PTPN14	ENSG00000152104	PTPN2	ENSG00000175354
PTPN20	ENSG00000204179	PTPN5	ENSG00000110786	PTPN6	ENSG00000111679
PXDN	ENSG00000130508	PXN	ENSG00000089159	RAB27A	ENSG00000069974
RAP1B	ENSG00000127314	RBBP4	ENSG00000162521	RBBP8	ENSG00000101773
RBFOX1	ENSG00000078328	RBFOX3	ENSG00000167281	RBL1	ENSG00000080839
RET	ENSG00000165731	RGCC	ENSG00000102760	RHOA	ENSG00000116574
RIPPLY2	ENSG00000203877	RLN1	ENSG00000107018	ROBO1	ENSG00000169855
ROBO2	ENSG00000185008	ROBO3	ENSG00000154134	RORA	ENSG00000069667
RPS2	ENST00000343262	RRM2B	ENSG00000048392	RSPO3	ENSG00000146374
RUNX1	ENSG00000159216, ENST00000344691	RUNX2	ENSG00000124813	S100B	ENSG00000160307
S1PR1	ENSG00000170989	SALL1	ENSG00000103449	SALL4	ENSG00000101115
SAMHD1	ENSG00000101347	SCD	ENSG00000099194	SCD5	ENSG00000145284
SDC4	ENST00000372733	SEC31A	ENSG00000138674	SERP1	ENSG00000120742
SERPINB2	ENSG00000197632	SERPINE1	ENSG00000106366	SERPINH1	ENSG00000149257
SESN2	ENSG00000130766	SGK1	ENST00000237305	SHH	ENSG00000164690
SIX1	ENSG00000126778	SIX2	ENSG00000170577	SLC27A1	ENSG00000130304
SLC4A4	ENST00000264485	SLIT2	ENSG00000145147	SLPI	ENST00000338380

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
SMAD6	ENSG00000137834	SMAD7	ENSG00000101665	SNAI1	ENSG00000124216
SNCB	ENSG00000074317	SOCS1	ENSG00000185338	SOCS3	ENSG00000184557
SOCS7	ENSG00000180008	SOX1	ENSG00000182968	SOX2	ENSG00000181449
SOX9	ENST00000245479	SP100	ENSG00000067066	SPARC	ENSG00000113140
SQSTM1	ENSG00000161011	SREBF1	ENSG00000072310	SRF	ENSG00000112658
SRPRA	ENSG00000182934	SRPRB	ENSG00000144867	SSR1	ENSG00000124783
SST	ENSG00000157005	STAT1	ENSG00000115415	STAT3	ENSG00000168610
STAT5A	ENSG00000126561	STEAP3	ENSG00000115107	STT3B	ENSG00000163527
SYT10	ENSG00000110975	TALDO1	ENSG00000177156	TBX2	ENSG00000121068
TEP1	ENSG00000171862	TFAP2C	ENST00000201031	TFB2M	ENSG00000162851
TFF3	ENSG00000160180	TFPI2	ENST00000222543	TFRC	ENST00000360110
TGFB1	ENSG00000105329	THBS1	ENSG00000137801	TIGAR	ENSG00000078237
TIMP1	ENSG00000102265	TJP1	ENSG00000104067	TK1	ENSG00000167900
TLN1	ENSG00000137076	TNFRSF10A	ENSG00000104689	TNFRSF10D	ENSG00000173530
TNFRSF1A	ENSG00000067182	TNFRSF21	ENSG00000146072	TP53	ENSG00000141510
TPH1	ENSG00000129167	TPST2	ENSG00000128294	TRIM14	ENSG00000106785
TRIM2	ENSG00000109654	TRIM21	ENSG00000132109	TRIM22	ENSG00000132274
TRIM25	ENSG00000121060	TRIM3	ENSG00000110171	TRIM38	ENSG00000112343
TRIM48	ENSG00000150244	TRIM5	ENSG00000132256	TRIM6	ENSG00000121236
TRPC3	ENSG00000138741	TSC22D1	ENSG00000102804	TUBB4B	ENST00000340384
TUT4	ENSG00000134744	TWIST1	ENSG00000122691	TXNRD1	ENSG00000198431
TYRP1	ENSG00000107165	UBB	ENSG00000170315	VCAM1	ENSG00000162692
VEGFA	ENSG00000112715	VGf	ENSG00000128564	WFDC2	ENSG00000101443
WFS1	ENSG00000109501	WIPI1	ENSG00000070540	WNT11	ENSG00000085741
WNT4	ENSG00000162552	WNT9B	ENSG00000158955	WWC1	ENSG00000113645
YAP1	ENST00000282441	YIF1A	ENSG00000174851	ZEB2	ENSG00000169554
ZIC2	ENSG00000043355	ZNF385A	ENSG00000161642	ZNF521	ENSG00000198795

Input	ChEBI Id
CIT	16947

Input	miRBase Id	Input	miRBase Id
MIR10B	MI0000267	MIR27B	MI0000440

Interactors (4813)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
A2M	P01023	Q8NBL1	AAMDC	Q9H7C9	P54252
AARD	Q4LEZ3	Q13190	AATF	Q9NY61	P08047
AATK	Q6ZMQ8-2	Q15078	ABCA1	O95477	P02647
ABCB1	P08183	P17931	ABCB10	Q9NRK6	Q71RS6
ABCC1	P33527	P17931	ABCC4	O15439	P17931
ABCD3	P28288	P33897	ABCD4	O14678	P23945
ABCE1	P61221	Q5S007	ABCG4	Q9H172	O76024
ABHD10	Q9NUJ1	Q15637, Q96NC0	ABHD15	Q6UXT9	P21673
ABHD17C	Q6PCB6	O14832	ABHD3	Q8WU67	P04201
ABHD4	Q8TB40	P24593	ABI1	Q8IZP0	Q07889
ABI2	Q9NYB9-2	P01189	ABLIM1	O14639	P62258
ABLIM2	Q6H8Q1-8	P35240	ABLIM3	O94929-2	Q13573

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ACACA	Q13085	Q13085	ACADM	P11310	P0DTC5
ACADS	P16219	Q9UH99	ACAN	P16112	P05067
ACAT1	P35610	O75326	ACBD3	Q9H3P7	Q9UBF8
ACCS	Q96QU6	P14927	ACD	Q96AP0	O43768
ACKR3	P25106	P49407	ACOT11	Q8WXI4-2	O14964
ACOT13	Q9NPJ3	Q92993	ACOT9	Q9Y305	P13671
ACOX2	Q99424	P63172	ACP2	P11117	P21453
ACSF2	Q96CM8	P02654	ACSL3	O95573	P27824
ACSL4	O60488	Q9H2J7	ACTA2	P62736	P15498
ACTL6B	O94805	Q99538	ACTMAP	Q5BKX5-3	Q9NQ94
ACTN1	P12814	P12931	ACTN2	P35609	P29373
ACTN4	O43707	Q07157	ACTR2	P61160	Q14247
ACTR3	P61157	O00401	ACTR3C	Q9C0K3	Q9H668
ACVR1	Q04771	Q9NPF0	ACVR1B	P36896	P62942
ACVR2A	P27037	P43115	ACVR2B	Q13705	P27037
ADAM12	O43184	P12931	ADAM15	Q13444	P12931
ADAM22	Q9P0K1	Q12959	ADAM23	Q9UKQ2	P00749
ADAM33	Q08AM2	Q16585	ADAM8	P78325	P42680
ADAM9	Q13443	Q99962	ADAMTS1	Q9UHI8	P08833
ADAMTSL3	P82987	Q7Z3S9	ADAMTSL4	Q6UY14-3	P09681
ADAMTSL5	Q6ZMM2-2	Q7Z3S9	ADAP1	O75689	Q08379
ADARB1	P78563-4	P19525	ADCY3	O60266	P17931
ADCY4	Q8NFM4	P37231	ADCY6	O43306	Q9Y4Z2
ADCY9	O60503	P15514	ADCYAP1	P18509	Q9Y5J6
ADCYAP1R1	P41586-3	P18509	ADD2	P35612	P35611
ADGRA2	Q96PE1	Q12959	ADGRA3	Q8IWK6	Q12959
ADGRB3	O60242	O60883	ADGRE5	P48960	P08174
ADGRL2	O95490	P21926	ADGRL3	Q9HAR2	Q9NZU0
ADI1	Q9BV57	P50281	ADORA1	P30542	P29274
ADORA2B	P29275	P29274	ADRA1B	P35368	P61073
ADRA1D	P25100	P11532	ADRB2	P07550	P12931
AEBP2	Q6ZN18-2	Q9NRD5	AEN	Q8WTP8-2, Q8WTP8	Q13685
AFF3	P51826	P50750	AFG3L2	Q9Y4W6	P42858
AGAP2	Q99490-2	P42229	AGER	Q15109	Q99836
AGFG1	P52594	P51809	AGMAT	Q9BSE5	O43602
AGO4	Q9HCK5	O15397, Q8NDV7	AGPAT1	Q99943	Q8TDQ0
AGPAT3	Q9NRZ7	Q9H2J7	AGPAT4	Q9NRZ5	Q8NBQ5
AGRN	O00468	O15530	AGT	P01019	P11021
AHCYL1	O43865	Q9UG63	AHCYL2	Q96HN2	P31152
AHNAK	Q09666	P18827	AHNAK2	Q8IVF2-3	P24593
AHR	P35869	P27540	AIFM1	O95831	P10599
AIM2	O14862	Q9ULZ3	AIMP1	Q12904-2	Q96CV9
AIMP2	Q13155	P03508	AJUBA	Q96IF1	O95863
AK1	P00568	P10636-8	AK3	P27144	P13501
AK4	P27144	P13501	AKAP11	Q9UKA4	P53611
AKAP12	Q02952	P13861	AKAP14	Q86UN6	P13861
AKAP5	P24588	P13861	AKAP6	Q13023	Q96CV9
AKAP7	Q9P0M2	P17612	AKAP8	O43823	P13861
AKIRIN2	Q53H80	P28066, P49721, P49720	AKR1C1	Q04828	P42330

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
AKR1C2	P52895	P42330	ALCAM	Q13740	P17931
ALDH1A1	P00352	P14136	ALDH1B1	P30837	Q9UMX1
ALDH2	P05091	P02647	ALDH3A2	P51648	P27824
ALDH3B1	P43353	Q13231	ALDH4A1	P30038	Q09028
ALG1	Q9BT22	P13569	ALG2	O75340	P35968
ALG3	Q92685	Q9H015	ALK	Q9UM73	P29353
ALKAL1	Q6UXT8	Q9UM73	ALMS1	Q8TCU4	O75923
ALOX15	P16050	P30086	ALOX5	P09917	Q9P0N9
ALOX5AP	P20292	O15552	ALX1	Q15699	Q92993
AMBRA1	Q9C0C7, Q9C0C7-3	O75385	AMD1	P17707	P18509
AMDHD2	Q9Y303-2	P55212	AMER3	Q8N944	Q9Y2T1
AMFR	Q9UKV5	P62837	AMHR2	Q16671	Q13451
AMIGO1	Q86WK6	P24593	AMMECR1	Q9Y4X0	Q9Y2T1
AMOT	Q4VCS5	Q8N3R9	AMOTL2	Q9Y2J4	Q8N3R9
AMPD3	Q01432	P08238	AMPH	P49418	Q05193
ANAPC1	Q9H1A4	Q9UM11, Q12834	ANAPC5	Q9UJX4	P51955
ANGPTL4	Q9BY76	Q14790	ANK1	P16157-17	Q8IWU4
ANK2	Q01484	Q9P2E9	ANK3	Q12955	O00533
ANKLE2	Q86XL3	P24394	ANKMY2	Q8IV38	Q15185
ANKRD1	Q9CR42	P19338	ANKRD13C	Q8N6S4	P56539
ANKRD27	Q96NW4	P25963	ANKRD40	Q6AI12	P23526
ANKRD46	Q86W74-2	Q969X2	ANKRD50	Q9ULJ7	Q9UJV9
ANKRD52	Q8NB46	P53350	ANKS1A	Q49AR9	P56945
ANKS1B	Q7Z6G8-3	P61586	ANKS6	Q68DC2	P01031
ANO4	Q32M45	Q96S44	ANO6	Q4KMQ2	P0DTC5
ANO9	A1A5B4	Q12959	ANOS1	P23352	P11362
ANP32A	P39687	O15169	ANTKMT	Q9BQD7	O14964
ANTXR1	Q9H6X2	P11021	ANTXR2	P58335	P13423
ANXA1	P04083	Q13546	ANXA2	EBI-8785901, P07355	Q8NBP7
ANXA3	P12429	P13569	ANXA4	P09525	P55212
ANXA5	P08758	Q96CV9	ANXA6	P08133	P10636-8
ANXA8	P13928	Q00403	AP2B1	P63010	P12236
AP2M1	Q96CW1	Q96D71, Q2M2I8, P42566	AP3D1	O14617	Q96EV8
AP3M1	Q9Y2T2	Q9UBH0	AP5B1	Q2VPB7	Q13309
APBA1	Q02410	O14936	APBB1	O00213	P05067-4
APBB2	Q92870-2	Q8NBJ7	APC	P25054	O14936
APC2	Q9UJX6	Q9Y639	APCDD1	Q8J025	P56704
APEX1	P27695	Q09472	APEX2	Q9UBZ4	Q13257
APH1B	Q8WW43	P05067	APLNR	P35414	O43613
APLP1	P51693	P05067-4	APLP2	Q06481-5	Q13190
APOBEC3C	Q9NRW3	Q01831	APOBEC3F	Q8IUX4	Q86VM9
APOBEC3G	Q9HC16	P69723	APOE	P02649	P09238
APOL2	Q9BQE5	Q96PP8	APOL3	O95236-2	Q16585
APOL6	Q9BWW8	Q9UHA4	APP	P05067	P09958
APPL1	Q9UKG1	O14908	APPL2	Q8NEU8	O14908
APRT	P07741	Q6DKK2	AQP1	P29972	P08151
ARAF	P10398	Q92985	ARAP1	Q96P48	P46108
ARC	O60936	Q14790	ARCN1	P48444	P13569
ARF1	P84077	P60880	ARF4	P18085	Q96PM5

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ARF5	P84085	Q9UNA3	ARFGAP1	Q9NP61	P50454
ARFGAP2	Q8N6H7	P05155	ARFGAP3	Q9NP61	P50454
ARFGEF3	Q5TH69	P15907	ARFIP2	P53365	Q92730
ARGLU1	Q9NWB6	P52298	ARHGAP10	Q5T5U3	P29692
ARHGAP12	Q8IWW6	P60709	ARHGAP19	Q14CB8	P54252
ARHGAP21	Q5T5U3	P29692	ARHGAP22	Q7Z5H3	P48730
ARHGAP23	Q9P227	P61981	ARHGAP29	Q52LW3-2	P42858
ARHGAP31	Q2M1Z3	Q9Y490	ARHGAP32	A7KAX9	P06241
ARHGAP33	O14559	P06241	ARHGAP42	A6NI28	P46108
ARHGAP8	P85298-4	Q13526	ARHGDIA	P52565	P61586
ARHGDIB	P52566	P01100	ARHGDIG	Q99819	Q9Y6A5
ARHGEF1	Q92888	P61586	ARHGEF10	O15013	P11142
ARHGEF10L	Q9HCE6	Q8NHY2	ARHGEF11	O15085	O43157
ARHGEF17	Q96PE2	P61981	ARHGEF28	Q8N1W1-4	P27797
ARHGEF3	Q9NR81	Q9NRD5	ARHGEF35	A5YM69	P13716
ARHGEF5	Q12774	P29353	ARHGEF6	Q15052	Q13177
ARHGEF7	O55043, Q14155	Q13177	ARHGEF9	O43307	Q92993
ARID1A	O14497	Q92925, P51531, O96019, Q15532	ARID3A	Q99856	Q06187
ARID3B	Q8IVW6	P48431	ARID5B	Q14865	Q92769
ARK2C	Q6ZSG1	P12755	ARL1	P40616	P13569
ARL4A	P40617	Q08379	ARL4C	P56559	P52306
ARL4D	P49703	P13284	ARL6IP6	Q8N6S5	O15498
ARL8B	Q9NVJ2	P51149	ARLN	Q8WVX3-2	Q9UBD6
ARMCX1	Q9P291	Q9NRD5	ARMH4	Q86TY3	Q9NPF0
ARNT2	Q9HBZ2	Q16665	ARPC1B	O15143	O00401
ARPC2	O15144	Q16236	ARPC5	O15511	P00533
ARPP19	P56211	P05067	ARRB1	P49407	P12931
ARRB2	P32121	P12931	ARRDC3	Q96B67	Q9NPF0
ARSA	P15289	Q8NBK3	ARSB	P15848	P01185
ARSG	Q96EG1	Q8NBK3	ARSI	Q5FYB1	P11021
ARSJ	Q8N5N6	Q12933, Q13077	ARSK	Q6UWY0	Q03405
ARVCF	O00192	O14907	ARX	Q96QS3	P12814
ASAH1	Q13510, Q13510-2	Q13285	ASAP1	Q9ULH1	P12931
ASAP2	O43150	P19174	ASAP3	Q8TDY4	Q86UT5
ASB3	Q9Y575	Q93034	ASCC2	Q9H1I8	Q99633
ASGR2	P07307	P11021	ASH2L	Q9UBL3	Q09472
ASIC2	Q16515	Q9NRD5	ASIC4	Q96FT7	Q9HD20
ASNSD1	Q9NWL6	P18433	ASPG	Q86U10	P57764
ASPH	Q12797-6	Q9HBE4	ASPHD2	Q6ICH7	O75063
ASPM	Q8IZT6	Q9Y4Z2	ASS1	P00966	Q9Y4Z2
ASTN1	O14525-2	Q14145	ASXL1	P59598	Q15788
ATAD1	Q8NBU5	P04201	ATAD3A	Q9NVI7	Q9NYY8
ATAD3B	Q5T9A4	P21802	ATAD5	Q96QE3	O60885
ATAT1	Q5SQI0	P61981	ATCAY	Q86WG3	Q9UNE7
ATF1	P18846	P18846	ATF3	P18847	P18847
ATF5	Q9Y2D1	P49715	ATF7IP	Q6VMQ6	Q13485
ATG5	A9UGY9	P55072	ATIC	P31939	P00533
ATL1	Q8WXF7	P04156	ATL2	Q8NHH9	P63104
ATL3	Q6DD88	Q92911	ATM	Q13315	P46531

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ATOSA	Q32MH5	Q08379	ATP11B	A0A0C4DG94	A6NI15
ATP13A3	Q9H7F0	Q9NRZ7	ATP1A1	P05023	P12931
ATP1A3	P13637	P21453	ATP1B3	P54709	P21926
ATP2B2	P23634, Q01814	P01258	ATP2B4	P23634	P01258
ATP6AP2	O75787	Q15904	ATP6V0A1	Q93050	Q15904
ATP6V0A4	Q9HBG4	Q15904	ATP6V0E1	O15342	P05090
ATP6V1A	P38606	Q9UI12	ATP6V1G2	O95670	P21281, P36543
ATP6V1H	Q9UI12	P42772	ATP7B	P35670	O00244
ATP9A	O75110	Q15363	ATPAF1	Q5TC12	P06576
ATXN1	P54253	P50454	ATXN3	P54252	Q969S8
ATXN7L2	Q5T6C5	P43897	AUTS2	Q8WXX7	Q09472
AXIN1	O15169, Q14DJ8, O35625	P49841	AXIN2	Q9Y2T1	P49841
AXL	P30530	P19174	B2M	P61769	P59596
B3GALNT1	O75752	Q9BYJ1	B3GALNT2	Q8NCR0	Q9H2A9
B3GALT2	O43825	O94822	B3GLCT	Q6Y288	P13671
B4GALT4	O60513	P11021	BACH2	Q9BYV9	O60675
BAG2	O95816	Q9ULW0	BAG3	O95817	Q9BRX2
BAG6	P46379-2	P10451	BAHCC1	Q9P281	Q13547
BAHD1	Q8TBE0	P50454	BAIAP2L1	Q9UHR4	Q9UQB8
BASP1	P80723	P10636-8	BATF3	Q9NR55	P18847
BAX	Q07812	Q9ULZ3	BAZ1A	Q9NRL2	P61981
BAZ2B	Q9UIF8	Q08379	BBLN	Q9BUW7	Q70UQ0
BBOX1	O75936	Q9UHD2	BCAM	P50895	P21926
BCAN	Q96GW7	P07948	BCAR1	P56945	P12931
BCAS3	Q9H6U6	P08670	BCDIN3D	Q7Z5W3	Q9NPC8
BCL2A1	Q07440	P55957	BCL2L1	Q07817	Q9C000
BCL2L14	Q9BZR8	O14757	BCL3	P20749	O95999
BCL6	P41182	P56524	BCL7A	Q4VC05	P48509
BCL9	O00512	P35222	BCL9L	Q67FY2	P35222
BCOR	Q6W2J9	P31749	BCORL1	Q5H9F3	Q9BYE7
BDNF	P23560-2	P18428	BECN1	Q14457	P12931
BEGAIN	Q9BUH8	P28749	BEND3	Q5T5X7	Q13547
BEND5	Q7L4P6	Q8N8N7	BET1	O15155	P51809
BET1L	Q9NYM9	Q9P2W9	BEX1	Q9HBH7	Q9NRD5
BEX2	Q9BXY8	P19883	BEX5	Q5H9J7	P06396
BHLHE40	O14503	Q16613	BID	P70444, P55957, P70444-PRO_0000223236	Q07812
BIN1	O00499	Q13496	BIRC3	Q13489	Q13546, Q9Y572
BIRC6	Q9NR09	Q9NR28	BLCAP	P62952	O95237
BLM	P54132	P39748	BLMH	Q13867	O95336
BLTP3B	A0JNW5	P61981	BLVRA	P53004	Q9H244
BMERB1	Q96MC5	Q8WZA2	BMP1	P13497-2, P13497	29108, P07585
BMP2K	Q9NSY1	Q96D71	BMP4	P12644	P27797
BMP5	P22003	P68871	BMP7	P18075	Q9BUH8
BMS1	Q14692	Q9H9Z2	BNIP3	Q12983	O60313
BNIP3L	O60238	Q9NQ11	BOC	Q9BWV1	Q14623
BOD1	Q96IK1-2	P78333	BOK	Q9UMX3	P16284
BRAF	P15056	Q13233	BRCA1	P38398	Q86YC2
BRICD5	Q6PL45	P98160	BRINP1	O60477	P09651

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BRIX1	Q8TDN6	Q9H9Z2	BRME1	Q0VDD7	P60900
BRSK2	Q8IWW3-3	Q9Y4K3	BSG	P35613	P20020
BST2	Q10589	Q10589	BTAF1	O14981	Q9Y639
BTBD1	Q9H0C5	P61081	BTBD3	Q9Y2F9	P63165
BTBD9	Q96Q07-2	P55212	BTD	P43251	P01008
BTF3	P20290-2	O60551	BTF3L4	Q96K17	P40222
BTG1	P62324	Q13158	BTG2	P78543	P10275
BTN2A2	Q8WVV5	P21579	BTN3A2	P78410	Q96CV9
BTNL8	Q6UX41-6	O00501	BTNL9	Q6UXG8-3	Q8N2H4
BTRC	Q9Y297	P30291	BYSL	Q13895	Q9H0H5
C10orf55	Q5SWW7	P35680	C11orf24	Q96F05	Q96BA8
C11orf87	Q6NUJ2	Q7Z3S9	C14orf39	Q8N1H7	P80297
C15orf39	Q6ZRI6	Q13257	C1GALT1	Q9NS00	P78423
C1GALT1C1	Q96EU7	Q6ZMH5	C1QBP	O35658	Q00059
C1QL1	O75973	O15498	C1QTNF1	Q9BXJ1	P02751
C1R	P00736	P05155	C1S	P09871	P05155
C1orf122	Q6ZSJ8	P40222	C1orf198	Q9H425	Q9UIV8
C1orf226	A1L170	P53350	C1orf50	Q9BV19	P23526
C1orf54	Q8WWF1	Q8NCH0	C1orf94	Q6P1W5	Q9NWB1
C21orf58	Q0VAL7	O15160	C2CD5	Q86YS7	P43115
C2orf72	A6NCS6	O15431	C3	P01024- PRO_0000005911, P01024- PRO_0000005908, P01024	P08603, P15529, P20023, P05156
C3orf18	Q9UK00	P40855	C3orf52	Q5BVD1	Q15109
C4A	P0C0L4	P46059	C4orf46	Q504U0	Q9NRD5
C5orf24	Q7Z6I8	Q15831	C5orf34	Q96MH7	Q14653
C5orf46	Q6UWT4	Q9BRI3	C6	P13671	Q6Y288
C6orf118	Q5T5N4	Q96ES7	C6orf141	Q5SZD1	Q15149
C7	P10643	P01031	C9orf72	Q96LT7, Q96LT7-2, Q96LT7-1	O75385
CA10	Q9NS85	Q8N2W9	CA14	Q9ULX7	Q96BI1
CA8	P35219	P01024	CA9	Q16790	P21291
CABCOC01	Q8IVU9	P42858	CABLES2	Q9BTV7	P80370
CABP7	Q86V35	Q9H2K0	CACFD1	Q9UGQ2	Q96T60
CACNA1A	O00555	Q8N2S1	CACNA1C	Q13936	Q9NZU7
CACNB1	Q02641	P55040	CACNB4	O00305-2	P21579
CACNG2	Q9Y698	P49674	CACNG3	O60359	P53350
CADM1	Q9BY67	Q9BY67	CADM4	Q8NFZ8	Q9UMR7
CADPS	A2RRN7	Q15436	CALB1	P05937	Q9Y6K9
CALCA	P01258	P01258	CALD1	Q05682	Q13813
CALM1	P62158	O15554	CALM2	P62158	O15554
CALM3	P62158	O15554	CALR	P27797	P12644
CALU	O43852	Q05513	CAMK1G	Q96NX5	O76024
CAMK2A	Q9UQM7	Q13224	CAMK2B	P08413	P60709
CAMK4	Q16566	Q9Y2X7	CAMKV	Q8NCB2	P98160
CAMSAP1	Q5T5Y3	P63104	CAMSAP3	Q9P1Y5	P62258
CAMTA2	O94983-5	Q9Y316	CAP1	Q01518	P60709
CAPG	P40121	P11142	CAPN1	P07384	Q92934
CAPN2	P17655	P07237	CAPN6	Q9Y6Q1	Q6PJG9

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CAPNS1	P04632	P20936	CAPRIN2	Q6IMN6	O75197
CAPSL	Q8WWF8	Q15398	CAPZA2	P47755	Q14247
CARD10	Q9BWT7	O14908	CARD11	Q9BXL7	Q9UDY8
CARD8	Q9Y2G2	Q14192	CARF	Q9NXV6	O94913, Q9H0D6
CARNMT1	Q8N4J0	Q13547	CARTPT	Q16568	P13688
CASKIN1	Q8VHK2, Q8WXD9	O14936	CASP1	P29466	Q9C000
CASP2	P42575-1, P42575	P78560, Q9HB75, P42575	CASP4	P49662	P51878
CASP6	P55212	P41222	CASP7	P55210	Q13177
CASP8	Q14790-2, Q14790	O15519-1	CASP8AP2	Q9UKL3	Q14790
CAST	O15446	O15160, O95602, P19388, P61218	CAV1	P33724	P59632
CAV2	P51636	Q15392	CAVIN1	Q6NZI2	P18827
CAVIN3	Q969G5	Q6NZI2	CBARP	Q8N350-4	O43639
CBFA2T2	O43439-4	Q9BRV8	CBFA2T3	O75081	O43439
CBFB	Q13951	Q13761	CBLB	Q13191	Q13882
CBLC	Q9ULV8	Q9BQP7	CBX2	Q14781-2	Q00403
CBX3	Q13185	Q9UI95	CBX4	O00257	Q16665
CBX5	P45973	Q14191	CBX8	Q9HC52	P11686
CCDC102A	Q96A19	O15392	CCDC102B	Q68D86	P01189
CCDC134	Q9H6E4	O75478	CCDC136	Q96JN2-2	Q96HC4
CCDC14	Q49A88-3	Q96T60	CCDC152	Q4G0S7	O95363
CCDC184	Q52MB2	Q13114	CCDC186	Q7Z3E2	P07947
CCDC198	E9PSE9	P84022	CCDC34	Q96HJ3-2	Q8N661
CCDC40	Q4G0X9	Q9HBG6	CCDC59	Q9P031	P98164
CCDC6	Q16204	Q13164	CCDC77	Q9BR77	P36941
CCDC8	Q9H0W5	Q16539	CCDC82	Q8N4S0	Q08945
CCDC85C	A6NKD9	P01730	CCDC86	Q9H6F5	P06748, P62753
CCDC88A	Q3V6T2	Q13257	CCDC88C	CCDC88C	O76024
CCDC92	Q53HC0	Q9UPV0	CCDC93	Q567U6	P36941
CCL2	P13500	P13501	CCN1	P20248	Q13415
CCN2	P29279	P02751	CCN3	P48745	Q12837
CCNA1	P78396	Q05397	CCNC	P24863	Q15648
CCND1	P24385	Q13761	CCND2	P30279	Q08999
CCND3	P30281	P29373	CCNE2	O96020	Q08999
CCNI	Q14094	Q00535	CCNT2	O60583-1, O60583-2, O60583	P50750
CCP110	O43303	Q9Y6M1	CCS	O14618	P98170
CCT5	P48643	Q13371	CCT6A	P40227	Q8NCC3
CD109	Q6YHK3	P23946	CD151	P48509	O14672
CD164	Q04900	P61073	CD2	P08921	P06239
CD200	P41217	Q8TD46	CD22	P20273	P19174
CD24	P25063	P07948	CD244	Q9BZW8	P19174
CD248	Q9HCU0	P27361	CD274	Q9NZQ7-1, Q9NZQ7	Q15116
CD2AP	Q9Y5K6	Q8WUM4	CD302	Q8IX05	Q13651
CD320	Q9NPF0	P20062	CD36	P16671	Q9UKG9
CD4	P01730	Q9NPF0	CD44	P16070	P09038
CD46	P15529	P78504	CD47	Q08722	P78324, Q9P1W8, O43914
CD55	P08174	P48960	CD58	P19256	P17931
CD59	P13987	Q15363	CD6	P30203	P17931

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CD74	P04233-2	O15155	CD81	P60033	P48509
CD83	Q01151	P40818	CD8A	P01732	P06493
CD9	P21926	P48509	CD93	Q9NPY3	Q9BQ95
CD96	P40200	P15151	CDA	P32320	Q96QG7
CDC14A	Q9UNH5	P61981	CDC16	Q13042, Q8R349	P51955
CDC25A	P30304	O96017	CDC25B	P30305	P62258
CDC25C	P30307	O14757	CDC27	P30260	P51955
CDC42BPG	Q6DT37	Q9UHD2	CDC42EP1	Q00587-2	P21673
CDC42EP3	Q9UKI2	Q5S007	CDC42EP4	Q9H3Q1	P60953
CDC7	O00311	Q08379	CDC73	Q6P1J9	O00267
CDCA5	Q9CPY3, Q96FF9	O60216	CDCA7	Q9BWT1	P42858
CDCP1	Q9H5V8	P12931	CDH1	P12830	P12931
CDH10	Q9Y6N8	P35222	CDH13	P55290	Q9NUQ2
CDH2	P19022	O60716	CDH3	P22223	P35222
CDH5	P33151	Q14118	CDH8	P55286	P35221, O60716
CDIP1	Q9H305	P0CG47, P0CG48	CDK14	O94921	P38936
CDK15	Q96Q40	Q13451	CDK17	Q00537	P42772
CDK18	Q07002	Q08379, P62258	CDK2AP1	O14519	P52294
CDK5R1	Q15078	Q9ULW0	CDK7	P50613	P28715
CDK8	P49336	Q15648	CDKL1	Q00532	Q13451
CDKN1B	P46527	P06493	CDKN2A	P42771	O75496
CDKN2AIP	Q9NXV6	O94913, Q9H0D6	CDKN2B	P42772	Q9UI12
CDKN2D	P55273	P22736	CDON	O35158	Q13635
CDR2	Q01850	P51530	CDS2	O75420	Q86UK7
CDT1	Q9H211	Q99741	CDX1	P18111	P20226
CDYL	Q9Y232	Q96KQ7	CDYL2	Q8N8U2	O14744
CEBPB	P17676	Q13485	CEBPD	P49716	Q9BXW9
CELF1	Q92879	P07910	CELF5	Q8N6W0	O15234
CELSR2	Q9HCU4	Q92838	CENPH	Q9H3R5	P20700
CEP112	Q8N8E3	P62258	CEP135	Q66GS9	Q07157
CEP15	Q9HBI5	Q96ST8	CEP170	Q5SW79	Q92845
CEP170B	Q9Y4F5	P62258	CEP192	Q8TEP8	O14965
CEP295	Q9C0D2	Q9NV70	CEP350	Q5VT06	P35221
CEP44	Q9C0F1	P40222	CEP55	Q53EZ4	O00267
CEP57	Q86XR8	P01106	CEP57L1	Q8IYX8-2, Q6P2R3	P09917
CEP68	Q76N32	P53350	CEP70	Q8NHQ1	P46087
CEP78	Q6IRU7	Q13501	CEP83	Q9Y592-2	P18848
CEP97	Q8IW35	O43303	CERCAM	Q5T4B2	O95231
CERK	Q8TCT0	O75496	CERS2	Q96G23	O15245
CERS6	Q6ZMG9	Q9H244	CES1	P02795	P09429
CES2	P23141-3	O95231	CETN2	Q9R1K9, P41208	Q01831
CETN3	O15182	P22692	CFAP100	Q494V2-2	P07237
CFAP161	Q6P656	P41182	CFAP299	Q6V702	Q16236
CFAP46	Q8IYW2	Q9UJK0	CFAP52	Q8N1V2	P31948
CFB	P00751	P61916	CFH	P08603	P01024
CFHR1	Q03591	P01031	CFI	P05156-PRO_0000027569, P05156-PRO_0000027570, P05156	P08603
CFL2	Q549N0, Q9Y281	P23528	CFTR	P13569	O15554

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CGB8	P0DN86	P01215	CGGBP1	Q9UFW8	Q6IN84
CGN	Q9P2M7	Q07157	CGNL1	Q0VF96	P62258
CHAMP1	Q96JM3	Q9UI95	CHAT	P28329-3	Q15814
CHCHD10	Q8WYQ3	Q8N4J0	CHCHD5	Q9BSY4	P53701
CHD3	Q12873	Q99728	CHD4	Q14839	Q13185
CHD5	O00258	P02652	CHD7	Q9P2D1	P48431
CHD8	Q9HCK8	P52292	CHD9	Q3L8U1	P05067
CHGA	P26339	P00441	CHGB	P05060	Q5S007
CHIC2	Q9UKJ5	O96015	CHLSN	Q9BRJ6	Q9H9Z2
CHM	P24386	Q92696	CHML	P26374	Q92696
CHMP2B	Q9UQN3	Q9Y3E7	CHMP6	Q96FZ7	Q9H444
CHN1	P15882	P28482	CHN2	P52757	O43639
CHODL	Q9H9P2	P24593	CHORDC1	Q9D1P4, Q9UHD1	P07900, P08238
CHPF2	Q9P2E5	P0DTC8	CHPT1	Q8WUD6	Q9Y2B1
CHRD	Q9H2X0	P84022	CHRD12	Q6WN34-2	O75093
CHRM3	P20309	Q92538	CHRM4	P08173	Q99808
CHRNA3	P32297	Q8NC56	CHRNA4	P43681	Q16585
CHST14	Q8NCH0	P27824	CHST15	Q7LFX5	P27824
CHST9	Q7L1S5	O75197	CHSY3	Q70JA7	Q03405
CHTOP	Q9Y3Y2	Q14739	CHUK	O15111	Q9ULZ3
CIART	Q8N365	P55212	CIITA	P33076	Q96EB6
CILK1	Q9UPZ9	Q13451	CILP2	Q8IUL8	Q01523
CIMAP3	Q8TCI5	Q9BT73	CISD3	P0C7P0	Q6PJG9
CISH	Q9NSE2	P00533	CIT	O14578	O14907
CITED2	Q99967	P41235	CKAP4	Q07065	O75326
CKMT1A	P12532	P10636-8	CKMT1B	P12532	P10636-8
CLASP1	Q7Z460	P35613	CLASP2	O75122	P49841
CLCC1	Q96S66	P59632	CLCF1	Q9UBD9, Q9UBD9-PRO_0000015616	P26992, O75462
CLCN3	P51790	P17931	CLCN4	P51793	Q8WTS1
CLDN1	Q9NY35	Q9ULC8	CLDN11	O75508	P26715
CLDN2	P57739	P24593	CLDN6	P56747	P45983
CLEC11A	Q9Y240	Q14703	CLEC18A	A5D8T8	P11021
CLEC19A	Q6UXS0	O94888	CLIC3	O95833	Q9NUX5
CLIC4	Q9Y696	O15247	CLINT1	Q14677	O00443
CLIP1	P30622	P13569	CLIP2	Q9UDT6	P10636-8
CLIP3	Q96DZ5	Q08379	CLIP4	Q8N3C7	Q9NPC8
CLK1	P49759	P61981	CLPTM1L	Q96KA5	Q9H1C4
CLSTN1	O94985	Q8IV16	CLSTN3	Q8IUW6	P02649
CLTCL1	P53675	Q96FJ0	CMIP	Q8IY22	P27986
CMKLR1	Q99788	Q9BUV8	CMPK1	P30085	Q6DKK2
CMTM4	Q8IZR5-2	P49675	CMTM7	Q96FZ5	Q15109
CMTR2	Q8IYT2	P62306, P14678, P62304	CMYA5	Q8N3K9	O75923
CNDP2	Q96KP4	P10636-8	CNGA3	Q16281	Q86XL3
CNIH1	O95406	Q6IN84	CNIH3	Q8TBE1	O15554
CNKSR2	Q8WXI2, Q9Z1T4	Q96NW7	CNKSR3	Q6P9H4	P42772
CNMD	O75829-2	Q8WXH5	CNN1	P51911	Q8N0S2
CNN2	Q99439	P11142	CNN3	Q15417	P11142
CNNM1	Q9NRU3	Q12974	CNNM2	Q9H8M5	Q12974

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CNOT11	Q9UKZ1	P09237	CNOT9	Q92600	P09237
CNR1	P21554	P14416	CNRIP1	Q96F85	O95070
CNTFR	P26992	P26441, O75462	CNTN1	Q12860	Q92823
CNTNAP1	P78357	Q9H2A9	CNTNAP2	Q9UHC6	Q12860
COASY	Q13057	P23443	COBL	O75128	P13196
COBLL1	Q53SF7	P62258	COG4	Q9H9E3	Q9Y639
COG6	Q9Y2V7	Q9Y639	COG8	Q96MW5	Q9Y639
COL12A1	Q99715	P08253	COL13A1	Q5TAT6	Q6ISS4
COL14A1	Q05707	P32119	COL18A1	P39060- PRO_0000005794	P19338
COL1A1	P02452	P30101	COL1A2	P08123	Q7Z624
COL26A1	Q96A83-2	Q8WVX9	COL2A1	P02458-1	P06396
COL3A1	P02461-1	P09486	COL4A5	P29400-2	Q8TB40
COL4A6	Q14031	P05067	COL5A1	P20908	P14780
COL5A2	P05997	Q6ISS4	COL6A1	P12109	P14780
COL6A2	P12110	P14780	COL6A3	P12111	Q6ISS4
COL8A1	P27658	Q12837	COL8A2	Q4VAQ0	Q00059
COL9A3	Q14050	P00533	COLEC12	Q5KU26	P08253
COLGALT2	Q8IYK4	Q9BT09	COMMD8	Q9NX08	P0CG47
COMP	P49747	O00244	COMT	P21964	P30518
COPA	P53621	Q8WUJ3	COPB1	P53618	P56945
COPB2	P35606	P19838	COPG1	Q9Y678	Q9NTJ5
COPG2	Q9UBF2	P27824	COPS5	Q92905	P19838
CORO1A	P31146	P40763	CORO6	Q6QEF8	Q9UL45
COTL1	Q14019	P09917	COX5B	P10606	P11182
COX6A1	P12074	P42858	COX7A2L	O14548	P14927, O14949
CPA3	Q9UI42	P01160	CPA4	Q9UI42	P01160
CPB1	P15086	P05067	CPEB1	Q9BZB8	Q92797
CPLX2	Q6PUV4	P43897	CPNE2	Q96FN4	Q8TBC4
CPNE5	Q9HCH3	O60216	CPNE7	Q9UBL6-2	Q9NRD5
CPNE8	Q86YQ8-2	P04271	CPSF2	Q9P2I0	Q92797, Q9C0J8, Q6UN15
CPVL	Q9H3G5	P47756	CRABP1	P29762	P13569
CRABP2	P29373	Q9HAW8	CRACD	Q6ZU35	P47756
CRACDL	Q6NV74	P47756	CRACR2B	Q8N4Y2-3	Q9UBH0
CRB1	P82279	Q8N3R9	CREB3L1	Q96BA8	Q96AA3
CREB3L2	Q70SY1	Q96BA8	CREB3L4	Q8TEY5	O43889
CREB5	Q02930-3	P19883	CREG1	O75629	O96006
CREM	EBI-10048213	P08047	CRH	P06850	P0DPK4
CRIP2	P52943	P82979	CRLF1	O75462- PRO_0000011039, O75462	P26992, Q9UBD9
CRMP1	Q14194	O15169	CROCCP2	Q86T23	Q8IY31
CRTAP	O75718	P02452	CRTC3	Q6UUV7	P62258
CRY1	Q16526	P49674	CRY2	Q49AN0	P49674
CRYAB	P02511	P12830	CRYM	Q14894	Q96CV9
CSF1	P09603	P09603, P07333	CSF2RB	P32927	P08700
CSF3R	Q99062	P40763	CSGALNACT2	Q8N6G5	Q8NBL1
CSMD1	Q96PZ7	P84022	CSNK1G1	Q9HCP0	P78368
CSPG4	Q6UVK1	P17931	CSPG5	O95196	P54198
CSRNP1	Q96S65	P36873	CSRP1	P21291	P00533

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CSRP2	Q16527	P14373	CST3	P01034	P05067
CTBS	Q01459	Q01201	CTDNEP1	O95476	O60602
CTDP1	Q9Y5B0	O14744	CTDSP1	Q9GZU7	O00560
CTDSP2	O14595	P10275	CTH	Q96IQ7	Q15058
CTNNA1	P35221	P25963	CTNNA3	Q9UI47-2	O75582
CTNNAL1	Q9UBT7	P11532	CTNNBIP1	Q9NSA3	Q9BRT9
CTNND2	Q9UQB3	O14907	CTSA	P10619	Q9UM11
CTSB	P07858	P40692	CTSD	P07339	Q8NBJ7
CTSH	P09668	P07711	CTSL	P07711	P43235
CTSS	P25774	P14316	CTSV	O60911	P01160
CTSZ	Q9UBR2	Q7Z3S9, P0DPK4	CTTN	Q14247	P32121
CTTNBP2	Q8WZ74	Q9GZV8	CUBN	O60494	P27352
CUEDC1	Q9NWM3	Q9UM54	CUL1	Q13616	Q15843
CUL4A	Q13619	P19838	CUL4B	Q13620	Q15843
CUL7	Q14999	P04004	CUX1	P39880	P06400
CX3CL1	P78423	P05556	CXCL1	P09341	Q12933
CXCL10	P02778	P13501, P48061	CXCL11	O14625	P13501, P48061
CXCL12	P48061	P78556, P13501, O15444, Q99616	CXCL14	O95715	P13501, P48061
CXCL5	Q6I9S7	Q7Z3S9	CXCL6	P80162	P13501, P48061
CXXC1	Q9P0U4	Q09472	CXXC4	Q9EQC9	O14640
CYB561	P49447	Q12981	CYB561A3	Q8NBI2	Q9NUQ2
CYB5R2	Q6BCY4, Q6BCY4-2	Q08379	CYBRD1	Q53TN4	P09601
CYCS	P99999	Q9Y4K3	CYFIP2	Q96F07	Q9UQB8
CYP21A2	Q08AG9	Q9NRD5	CYSTM1	Q9H1C7	Q6ZMH5
CYTH3	O43739-2	P53365	DAAM1	Q9Y4D1	Q9Y4D1
DAB1	O75553-4	P05067	DAB2	P98078, P98082	Q9UM54
DACH1	Q9UI36	Q13485	DACT1	Q9NYF0	P49841
DAG1	Q14118	Q14118	DAPK1	P53355	P14618-1
DAPK3	O43293	P28908	DAPL1	A0PJW8	Q15915
DAPP1	Q9UN19	Q8WY64	DARS1	P14868	P05412
DAZAP1	Q96EP5	P13569	DAZAP2	Q15038	Q9HAU4
DBN1	Q16643	P60709	DBP	Q10586	P35638
DCAF11	Q8TEB1	O15228	DCAF15	Q66K64	O15431
DCAF4	Q8WV16	Q9Y4Z2	DCAF4L1	Q3SXM0	P36776
DCAF5	Q96JK2	Q9BQB4	DCAF7	P61962	Q13233
DCBLD1	Q8N8Z6	P32971	DCC	P43146	O95631
DCDC2	Q9UHG0	P01178	DCLK1	O15075	O43602
DCLK3	Q9C098	Q00535	DCN	P07585	O75063
DCTD	P32321	O00560	DCTN3	O75935	Q9Y244
DCTN4	Q9UJW0	Q08379	DCTN5	Q9BTE1	P47756
DCTN6	O00399	Q9Y5N6	DCUN1D1	Q96GG9	Q93034
DCX	O43602	Q9BSE5	DDAH1	O94760	O75832
DDB2	Q92466	Q9NYZ3	DDC	P20711	P10275
DDIT3	P35638	P17676	DDIT4	Q9NX09	P16284
DDOST	P39656	Q14165	DDR1	Q08345-1	P29353
DDR2	Q16832	Q16288	DDRKG1	Q96HY6	P55017
DDX10	Q13206	Q9H9Z2	DDX19A	Q9NUU7	P13569
DDX19B	Q9UMR2	P37173	DDX21	Q9NR30	Q86U42
DDX24	Q9GZR7	P01106	DDX25	Q9UHL0	Q9UBB9

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DDX31	Q9H8H2	P20061	DDX3X	O00571	P04608
DDX5	P17844	Q09472	DDX54	Q8TDD1	Q9H9Z2
DEGS1	O15121	Q68D85	DENND11	A4D1U4	Q9Y337
DENR	O43583	Q96HC4	DEPDC5	O75140-2	Q6W0C5
DES	P31001, P17661	Q13496	DESI2	Q9BSY9	Q12982
DEXI	O95424	Q9Y5Y5	DFFA	O00273	O76075
DGCR2	P98153	Q8TC07	DGKA	P23743	P12931
DGKD	Q16760	O75923	DHRS11	Q6UWP2-2	Q92796
DHRS13	Q6UX07	P32971	DHRS3	O75911	O60602
DHRS9	Q9BPW9	Q8TF71	DHX16	O60231	Q92620
DHX37	Q8IY37	P06748	DHX57	Q6P158	Q9Y4Z2
DIAPH3	Q9NSV4	Q6UWR7	DIMT1	Q9UNQ2	Q9H9Z2
DIP2A	Q14689	P19883	DIP2B	Q9P265	Q8TDQ0
DIPK1A	Q5T7M9	Q9UNA3	DIPK1B	Q5VUD6	P21673
DIRAS1	O95057	P52306	DIS3	Q9Y2L1	P05120
DISC1	Q9NRI5	Q8WY54	DISP1	Q96F81	O76024
DKC1	O60832	P06276	DKK3	Q9UBP4	Q8NFX5
DLC1	P63167	P13861	DLG1	Q12959	P78536
DLG2	Q15700	O60469	DLG3	Q92796	P04004
DLG4	P78352	P27037	DLGAP1	O14490, P97836-5	P78352
DLGAP2	Q9P1A6-3	P54252	DLGAP3	O95886	Q08379
DLK1	P80370	Q9HD20	DMAP1	Q9NPF5	P20700
DMKN	Q6E0U4	Q05329	DMRT2	Q9Y5R5	Q9Y5V3
DMRTA1	Q5VZB9	O76024	DMTN	Q08495	Q08379, P62258
DMXL2	Q8TDJ6	O75695	DNAAF1	Q8NEP3	O76003
DNAAF10	Q96MX6	P19388	DNAAF5	Q86Y56	Q9Y639
DNAAF8	Q8IYS4	Q8NFX5	DNAJA3	Q96EY1-2, Q96EY1	P08581
DNAJB11	Q9UBS4	P11021	DNAJB12	Q9NXW2	P13569
DNAJB2	P25686	P13569	DNAJB4	Q9UDY4	Q9Y266
DNAJB5	O75953	P25685	DNAJB6	O75190-2	Q9BRX5
DNAJB9	Q9UBS3	Q03405	DNAJC1	Q96KC8	Q9H1C4
DNAJC10	Q8IXB1	Q9UBX5	DNAJC11	Q9NVH1	Q16891, Q9NX63
DNAJC12	Q9UKB3	Q96LA8	DNAJC15	Q9Y5T4	Q8IUQ4
DNAJC18	Q9H819	P43119	DNAJC3	Q91YW3	P11021
DNAJC6	O75061	P10636-8	DNASE1L1	P49184	P50454
DNASE2	O00115	P27824	DND1	Q8IYX4	Q96JH7
DNM1	Q05193	Q05193	DNM2	P50570	Q05193
DNM3	Q9UQ16	Q05193	DNMBP	Q6XZF7, Q6XZF7-1	P50570, O00401
DNMT3B	Q9UBC3	Q9Y6K1	DNPH1	O43598	O43598
DNTTIP2	Q5QJE6	Q6NZI2	DOC2A	Q14183	Q7Z4L5
DOCK10	Q96BY6	P42357	DOCK11	Q5JSL3	P61981
DOCK2	Q92608-2	P10636	DOCK3	Q8IZD9	Q92556
DOCK4	Q8N1I0	P62993	DOCK5	Q9H7D0	P46109
DOCK7	Q96N67	Q9UM54	DOCK9	Q9BZ29	P84022, Q15796
DOHH	Q9BU89	Q9NQZ5	DOK3	Q7L591-3	Q9NQ94
DOK7	Q18PE1	Q9UKG1	DOT1L	Q8TEK3	P35222
DPEP1	P16444	P17302	DPF1	Q92782	Q8NFD5, P51531
DPF2	Q92785	Q92925, Q8NFD5, P51531, O96019, O94805, O75177, Q15532	DPH2	Q9BQC3	P11142

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DPP3	Q9NY33	Q14145	DPP4	P27487	P27487
DPPA4	Q7L190	Q9BYJ1	DPY19L1	Q2PZI1	O75326
DPYSL2	Q16555	Q8NFD5	DPYSL4	O14531	P19174
DPYSL5	Q9BPU6	P06396	DRAM1	Q8N682	Q9NPL8
DRAXIN	Q8NBI3	O95631	DRC3	Q8TE68	Q9UQB8
DRC7	Q8IY82	P27797	DRD2	P14416	P14416
DSC1	P53805	P16298	DSC2	Q02487	Q14126
DSC3	Q02487	Q14126	DSE	Q9UL01	Q9NRD5
DSG2	Q14126	P13569	DSN1	Q9H410	P51587
DSP	P15924	P07332	DSTN	P60981	P60709
DTL	Q9NZJ0	Q8N4J0	DTNA	Q9Y4J8-9	P40763
DTNB	O60941-5	Q9BUH8	DTNBP1	Q96EV8	A1L3X0
DTWD2	Q8NBA8	P07437	DTX1	Q86Y01	Q96J02
DUSP1	P28562	Q16539	DUSP11	O75319	P17252
DUSP14	O95147	Q9BYZ6	DUSP15	Q9H1R2	P19174
DUSP16	Q9BY84	Q16539	DUSP23	Q9BVJ7	Q8TBC4
DUSP4	Q13115	P28482	DUSP6	Q16828	P28482
DUSP8	Q13202	P45983	DVL3	Q92997	P23025
DYM	Q7RTS9	P21453	DYNLL1	P63167	P13861
DYNLL2	Q96FJ2	O75330	DYNLT1	P63172	P43034
DYNLT3	P51808	P43034	DYSF	O75923	Q96CV9
DZIP1	Q86YF9	P61981	DZIP1L	Q8IYY4	P61981
DZIP3	Q86Y13	P62837	E2F2	Q14209	P06400
E2F4	Q16254	P01189	E2F5	Q15329	Q08999, P06400
E2F8	A0AVK6	P07550	EARS2	Q5JPH6	Q9Y639
EBF3	Q9H4W6-2	Q9HAK2	EBNA1BP2	Q99848	P19525
ECH1	Q13011	Q13114	ECHDC2	Q86YB7	O75380
ECI2	O75521	P13569	ECPAS	Q5VYK3	P08173
ECSCR	Q19T08	O14818	EDA2R	Q9HAV5	Q13114
EDC3	Q96F86	P11309	EDEM2	Q9BV94	Q8TAS1
EDNRA	P25101	P32121, P49407	EEA1	Q15075	Q7Z3T8
EEF1A1	P68104	Q9NYA1	EEF1A2	Q05639	Q15843
EEF1AKMT3	Q96AZ1	Q15915	EEF1B2	P24534	P26641, P68104
EEF1D	P29692-2, P29692	P26641, P29692	EEF2	P13639	Q16539
EEF2K	O00418	Q9H4G0	EFCAB14	O75071	P68400
EFEMP1	Q12805	P24592	EFEMP2	O95967	Q9UBX5, P15502
EFHC2	Q5JST6	Q13882	EFHD1	Q9BUP0	O95409
EFNA1	P20827	P54764	EFNA5	P52803	Q9UBD6
EFNB1	P98172	O15397	EFS	O43281-2, O43281	P51451
EFTUD2	Q15029	P01106	EGF	P01133	P00533
EGFL7	Q9UHF1	Q9UBX5	EGFR	P00533	Q02246
EGLN3	Q9H6Z9	Q13085	EGR1	P18146	Q8N726
EGR2	A0A0S2Z3U4	Q08379	EHBP1L1	Q8N3D4	Q96ES7
EHD1	Q9H4M9	P13569	EHD2	Q9NZN4	Q86UL3
EHD4	Q9H223	Q9H1K0	EHHADH	Q08426	Q16585
EHMT2	Q96KQ7	P48431	EID1	Q9Y6B2	P0CG47
EIF1B	O60739	P62249	EIF2A	Q9BY44	P31371
EIF2AK1	Q9BQI3	Q13451	EIF2AK2	P19525	P55265
EIF2AK4	Q9P2K8	P36873	EIF2S1	P05198	P19525
EIF3A	Q14152	P47813	EIF3B	P55884	Q04637, P06730

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EIF3D	O15371	P04049	EIF3G	O75821	Q14694
EIF3J	O75822	Q04637	EIF3M	Q7L2H7	P13569
EIF4EBP1	Q13541	Q9UKV8	EIF4EBP2	Q13542	P06730
EIF4ENIF1	Q9NRA8	P06730	EIF4G1	Q04637-9	Q9NQ94
EIF4H	Q15056-2	Q9NRD5	EIF5A2	Q9GZV4	O00560
EIF5B	O60841	P14921	ELAVL2	Q12926	O00425, Q9NZI8
ELAVL3	Q14576	Q13627	ELAVL4	P26378-2	Q00994
ELF1	P32519	P08047	ELK3	P41970	O00712, Q12857
ELL	P55199	Q9UHB7, Q96JC9	ELL2	O00472	P07550
ELMO1	Q92556	Q8IZD9	ELMO2	Q96JJ3	Q8IZD9
ELN	P15502	Q9UBX5	ELOVL2	Q9NYP7	P24593
ELOVL4	Q9GZR5	Q12981, O75396	ELOVL6	Q9H5J4	P09601
ELOVL7	A1L3X0	Q96EV8	EMC1	Q8N766	Q15363
EMC2	Q15006	O43868	EMC7	Q9NPA0	Q15363
EMC8	O43402	P27824	EME1	Q96AY2	Q96NY9
EMILIN1	Q9Y6C2	Q8NBN7	EMILIN2	Q9BXX0	Q99435
EMILIN3	Q9NT22	P09238	EML1	O00423	Q9Y266
EML3	Q32P44	P61981	EMP1	P54849	Q9Y5Q5
EMP3	P54852	A1L3X0	EMX2	Q04743	P06730
EN1	Q05925	O00712, Q12857	ENAH	Q03173	Q9UQB8
ENC1	O14682	Q13501	ENG	P17813	P08648
ENKD1	Q9H0I2	P52747	ENKUR	Q8TC29	P37840
ENO2	P09104	P06733	ENOX1	Q8TC92	O00560
ENPP1	P22413	Q9UNQ0	ENTPD2	Q9Y5L3	P18031
ENTPD3	O75355-2	Q96BA8	ENTR1	Q96C92-2	O00560
EOGT	Q5NDL2	Q03405	EPAS1	Q99814	P40818
EPB41	P11171-2, P11171-7	P05067	EPB41L1	Q9H4G0	P61981
EPB41L2	O43491	P62258	EPB41L3	Q9Y2J2	P49841
EPDR1	Q9UM22	P01222	EPHA1	P21709	P57764
EPHA10	Q5JZY3-3	Q8N9N5	EPHA2	P29317	P18827
EPHA3	P29320	Q13485	EPHA4	P54764	P05177
EPHA6	Q9UF33	P04070	EPHA7	Q15375	Q16288
EPHA8	P29322	P11021	EPHB1	P54762	Q14451
EPHB2	P54763	Q13009	EPHB4	P54760	P01588
EPHB6	O15197	Q13451	EPHX1	P07099	Q13077
EPN2	O95208-2	P52888	EPPK1	P58107	P08670
EPS15	P42566	Q9P0V3	EPS8L1	Q8TE68	Q9UQB8
EPS8L2	Q9H6S3	P07196	EPSTI1	Q96J88-3	P17252
ERAP1	Q9NZ08	Q15392	ERBB3	P21860	Q14451
ERBIN	Q96RT1	P35222	ERC2	O15083	P10242
ERCC2	P18074	P00533	ERF	P50548	P28482
ERG	P11308	O00712, Q12857	ERGIC1	Q969X5	P04601
ERGIC2	Q96RQ1	P15907	ERI1	Q8IV48	Q14493
ERICH5	Q6P6B1	Q92845	ERLEC1	Q96DZ1	Q5K4L6
ERMP1	Q7Z2K6	Q9UBD6	ERO1A	Q96HE7	P30101
ERP44	Q9BS26	Q8NBK3	ERVK3-1	Q96A10	O95429
ESRP2	Q9H6T0	P11142	ESYT1	Q9BSJ8	P22309
ESYT2	A0FGR8	P61981	ETF1	P62495	P04798
ETFA	P13804	Q8IWL3	ETFRF1	Q6IPR1	O14561
ETNK2	Q9NVF9	Q16623	ETS1	P14921	P19838

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ETS2	P15036	P50454	ETV1	P50549	Q8NHY2
ETV4	P43268-3	Q13485	ETV5	P41161	Q8NHY2
ETV6	P41212	O75197	ETV7	Q9Y603	O00712, Q12857
EVA1B	Q9NVM1	Q8WVC6	EVA1C	P58658	O14786
EVI5	O60447	P53350	EVL	Q9UI08	P50552
EWSR1	Q01844	Q13114	EXO1	Q9UQ84-1, P39875	P43246
EXOC2	Q96KP1	Q13304	EXT1	Q16394	Q03405
EXTL3	O43909	P61073	EYA1	Q99502	Q15475, Q9NPC8
EYA3	Q99504	P16284	EZH1	Q92800	P15056
EZH2	Q15910	P46100	EZR	P15311	P52565
F11R	Q9Y624	Q9BV20	F2RL1	P55085	P04156
F2RL2	O00254	Q8NI35	F3	P13726	P08709
FABP4	P15090	P08670	FABP5	Q01469	P13569
FABP7	O15540	O76024	FADD	Q13158	P25445
FADS1	O60427	P25103	FADS2	Q8NFF5-2	P29373
FADS3	Q9Y5Q0	Q8N5I4	FAM110C	Q1W6H9	P49674
FAM111A	Q96PZ2	P04637	FAM111B	Q6SJ93	Q01105
FAM114A1	Q8IWE2	Q9UBN6	FAM117A	Q9C073	Q9NRD5
FAM117B	Q6P1L5	P78333	FAM118B	Q9BPY3	Q9Y639
FAM120A	Q9NZB2	Q92900	FAM124A	Q86V42	Q96QF0
FAM131B	Q86XD5	Q99741	FAM133A	Q8N9E0	Q9NRZ7
FAM136A	Q96C01	O00443	FAM13A	O94988	P61981
FAM168A	Q92567	O75177	FAM171A2	A8MVW0	Q16288
FAM171B	Q6P995	P21802	FAM174A	Q8TBP5	Q6UWP7
FAM177A1	Q8N128	O14975	FAM184A	Q8NB25	P06396
FAM199X	Q6PEV8	P49674	FAM200C	Q8IZ13	Q99856
FAM20B	O75063	P07585	FAM20C	Q8IXL6	P10451
FAM210B	Q96KR6	P09601	FAM219A	Q8IW50	O43924
FAM219B	Q5XKK7	Q9NRD5	FAM222A	Q5U5X8	Q9UBE8
FAM222B	Q8WU58	Q9UBE8	FAM234B	A2RU67	O43913
FAM241A	Q8N8J7	Q96FE5	FAM3C	Q92520	Q9UNA3
FAM43A	Q8N2R8	Q9BUR4	FAM53B	Q14153	P62258
FAM53C	Q9NYF3	P53350	FAM78A	Q5JUQ0	P42858
FAM78B	Q5VT40	Q9UI95	FAM83B	Q5T0W9	P49674
FAM83G	A6ND36	P48729	FAM83H	Q6ZRV2	Q13309
FAM91A1	Q658Y4	P08173	FAM98A	Q8NCA5	P59595
FAM98C	Q17RN3	Q00403	FANCA	O15360	Q9Y639
FANCD2	Q9BXW9	O00255	FANCE	Q9HB96	P31749
FANCG	O15287	P08684	FANCI	Q9NVI1	Q9Y5B0
FANCM	Q8IYD8	P49736	FANK1	Q8TC84	P07196
FAR2	Q96K12	P40855	FARP1	Q9Y4F1	P18827
FARP2	O94887	P49674	FAS	P25445	P25445
FASN	P49327	Q16665	FAT1	Q14517	P23946
FAU	P62861	P63244	FAXC	Q5TGI0	Q9H668, Q2NKJ3
FBH1	Q8NFX0	P63208	FBLIM1	Q8WUP2, Q71FD7	P52952
FBLN1	P23142-4	Q96FE5	FBLN2	P98095	P14543
FBLN5	Q9UBX5	Q8N2S1	FBN1	P35555	P15502
FBN2	P35556	Q04721	FBRSL1	Q9HCM7	P68400, P67870
FBXL14	Q8N1E6	Q9UPY3	FBXL16	Q8N461	Q9Y4D1
FBXL17	Q9UF56	O14867	FBXL19	Q6PCT2-2	Q6ZPD8

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FBXL20	Q96IG2	P63208	FBXL3	Q9UKT7	Q9UJ68
FBXL4	Q9UKA2	Q9Y3R4	FBXL8	Q96CD0	O95863
FBXO10	Q9UK96	O43924	FBXO11	Q86XK2	O95863
FBXO16	Q8IX29	P41208	FBXO17	Q96EF6	O60551
FBXO2	Q9UK22	Q8N2S1	FBXO22	Q8NEZ5	O14867
FBXO27	Q8NI29	P63208	FBXO30	Q8TB52	Q9UM54
FBXO33	Q7Z6M2	P04004	FBXO34	Q9NWN3	Q9UM54
FBXO44	Q9H4M3-2	P78537	FBXO46	Q6PJ61	Q92905
FBXO6	Q9NRD1	P98160	FBXO7	Q9Y3I1	O60260
FBXW8	Q8N3Y1	P04637	FBXW9	Q5XUX1	P63208
FCGRT	P55899	P09958	FCHO1	O14526	Q13882
FECH	P22830	P16035	FEM1B	Q9UK73	P49593
FERMT2	Q96AC1	P35222	FES	P07332	P15311
FEZ1	Q9Y250	P30307	FEZ2	Q9UHY8	P54252
FEZF2	Q8TBJ5	P05067	FGD4	Q96M96	P17252
FGD6	Q6ZV73	P62258	FGF10	O15520	P21802
FGF13	Q92913	Q96EB6	FGF14	Q92915-2	P49638
FGF17	O60258	Q9UGY1	FGF19	O95750	P31749
FGF2	P09038	P18827	FGFBP1	Q14512	P09038
FGFR1	P11362, P11362-7	P09038	FGFR1OP2	Q9NVK5	Q9BRV8
FGFRL1	Q8N441	Q15363	FHIP2A	Q5W0V3	Q13304
FHL1	Q13642	P04156	FHL2	Q14192	Q9NYA1
FHL3	Q13643	P01229	FHOD1	Q9Y613	P19474
FHOD3	Q2V2M9, Q2V2M9-4	Q13501	FIGNL1	Q6PIW4	Q14159, Q9NSG2
FIRRM	Q9NSG2	Q13304	FKBP10	Q96AY3	P02452
FKBP14	Q9NWM8	P01222	FKBP5	Q13451	Q9BQI3
FKBP7	Q9Y680	P49638	FKBP9	O95302	Q14118
FLI1	Q01543	P84022	FLII	Q13045	P13569
FLNA	P21333	Q12948	FLNB	O75369	Q13233
FLNC	Q14315	Q92629	FLOT1	P41440	Q03135
FLRT1	Q9NZU1	P42336	FLRT3	Q9NZU0	P05067-4
FLT1	P17948	P46109	FLT4	P35916	P35968
FLVCR1	Q9Y5Y0	P31785	FMNL3	Q6ZPF4	O75044
FMO1	Q01740	Q3KNW5	FMOD	P13605	P08603
FN1	P02751	P29279	FNBP1L	Q5T0N5	Q9Y4D1
FNDC3B	Q53EP0	Q9NPD5	FNDC4	Q9H6D8	Q03135
FOS	P01100	Q9P1W8	FOSB	P53539	Q9BZS1
FOSL1	P15407	P26640	FOSL2	P15408	Q8NFP9
FOXA2	Q9Y261-2	P14136	FOXB1	Q99853	P14373
FOXC1	Q12948	P08651	FOXC2	Q99958	Q99958
FOXF1	Q12946	Q9UPW6	FOXJ1	Q92949	P53350
FOXK1	P85037	P53611	FOXL1	Q12952	O00712
FOXO1	Q12778	P06493	FOXO4	P98177	P98177
FOXP1	Q9H334	Q9BZS1	FOXP3	EBI-10097414	P62805
FOXP4	Q8IVH2	Q9BZS1	FOXRED2	Q8IWF2	Q9UBV2
FOXS1	O43638	P67775	FRAS1	Q86XX4	P08311
FRAT1	Q92837	P49841	FRAT2	O75474	P49841
FRG1	Q14331	Q9HCG8	FRMD3	A2A2Y4	O43169
FRMD4A	Q9P2Q2	O14907	FRMD6	Q96NE9-2	P17252
FRMPD4	Q14CM0	P78352	FRRS1L	Q9P0K9	Q969Z0

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FSTL1	Q12841	P08571	FTCD	O95954	Q8N3R9
FTSJ3	Q8IY81	P19525	FUBP1	Q96AE4	P61981
FUCA1	P04066	Q8NBK3	FURIN	P09958	P50281
FUT8	Q9BYC5	P26572	FUT9	Q9Y231	Q9NWM8
FXYD6	Q9H0Q3	Q9UBD6	FYCO1	Q9BQS8	Q9GZQ8
FYN	P06241	P12931	FYTDD1	Q96QD9	A0AVT1
FZD10	Q9ULW2	P07237	FZD4	Q9ULV1	P42858
FZD7	O75084	P41221	FZD9	Q8TAN2	Q8IUQ4
G3BP1	Q13283	Q14694	G3BP2	Q9UN86-2, Q9UN86	Q14694
GAB1	Q13480	Q06124	GAB2	Q9UQC2-1	P62993-1
GABARAPL2	P60520	Q13685	GABBR1	Q9UBS5	P46459
GABBR2	O75899	P46459	GABRA2	P47869	P00533
GABRE	P78334	O60427	GABRG2	P18507	P27824
GAD1	Q99259	Q13188	GAD2	Q05329	O60906
GADD45A	P24522	P28715	GADD45B	O75293	O14733
GADD45G	O95257	P68104	GADD45GIP1	Q8TAE8	Q14197
GALK1	P51570	Q9NPJ4	GALNS	P34059	Q8NBK3
GALNT10	Q86SR1	Q07817	GALNT18	Q6P9A2	P01185
GALNT2	Q10471	P40855	GALNT6	Q8NCL4	O75063
GALNT7	Q86SF2	Q8IV16	GALT	P07902	P15559
GANAB	Q14697	P01106	GAP43	P17677	Q16822
GAR1	Q9NY12	O14744	GAS2L1	Q99501	Q96B97
GAS7	O60861	Q07666, P23246	GATA4	Q08369	Q09472
GATA5	Q9BWX5	P37231	GATAD2B	Q8WXI9	Q15274
GBP1	P32455	P32455	GBP2	P32456	P32455
GBP3	Q8WXF7	P04156	GBP5	Q96PP8	P32455
GCA	P28676	Q12933, Q13077	GCAT	O75600	O75380
GCFC2	P16383	O43143, O95926, Q9BUQ8, O94906, Q9UBB9	GCLC	P48506	Q6IPR3
GCLM	P48507	P10746	GCN1	Q92616	P01106
GCNT1	Q02742	P62987	GDAP1	Q8TB36	P54252
GDE1	Q9NZC3	Q14973	GDF3	Q9NR23	P35813
GDF5	P43026	P29373	GDI1	P31150	P51149
GEM	P55040	Q9BUH8	GEMIN4	P57678	Q13304
GFAP	P14136	P52747	GFPT1	Q06210	P13569
GFPT2	O94808	Q9NUX5	GFRA1	P56159-2, Q62997	Q92673
GGCX	P38435	P0DTC5	GGH	Q92820	P27824
GGT5	P36269	Q03426	GGT7	Q9UJ14	Q5SRI9
GHR	P10912	P01241	GINM1	Q9NU53	Q14118
GINS2	Q9Y248	O96017	GIPC1	O14908	Q9NZN5
GIPR	P48546	P01275	GJA1	P17302, P08050	Q07157
GJA4	P35212	Q8N661	GJB2	P29033	P29033, P08034
GJC1	Q8N144, P28229	Q07157	GKAP1	Q5VSY0	O00560
GLB1	P16278	Q9UM11	GLB1L2	Q8IW92	P11021
GLCCI1	Q86VQ1	Q13627	GLDC	P23378	Q9BQB4
GLI1	P08151	Q96J02	GLI2	P10070	P84022, Q15796
GLI3	P10071	Q9UMX1	GLI4	P10075	P04004
GLIPR1	P48060	Q9BYJ1	GLIS2	Q9BZE0	O75496

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GLIS3	Q8NEA6-2	Q13077	GLP1R	P43220	Q9HD20
GLRA1	P23415	Q9Y5V3	GLRX5	Q86SX6	Q8IWL3
GLS	O94925	Q9NXA8	GLS2	Q9UI32	Q6GPH4
GLT8D1	Q68CQ7	P48960	GLT8D2	Q9H1C3	P19526
GLUL	P15104	Q9BZE9	GM2A	P17900	Q15915
GMD5	O60547	O60547	GMFB	P60983	Q86X55
GMPPA	Q96IJ6	Q92917	GMPPB	Q9Y5P6	Q8IVS8
GNA14	O95837	O96015	GNAI2	P04899-1	P08173
GNAO1	P59215	P17612	GNAQ	P50148	P49407
GNB4	Q9HAV0	P18827	GNG12	Q9UBI6	Q14192
GNG2	P59768	Q99835	GNG3	P63215	Q13371
GNG4	P50150	Q13371	GNL1	P36915	Q9UG63
GNL2	Q13823	Q9H9Z2	GNL3	Q9BVP2	Q9UKV8
GNL3L	Q9NVN8	Q9UET6	GNPNAT1	Q96EK6	P07148
GNS	P15586	Q8NBK3	GOLGA2	Q08379	Q9ULW0
GOLGA4	Q13439	Q96ES7	GOLGA5	Q8TBA6	Q8NCH0
GOLGA7B	Q2TAP0	O14964	GOLIM4	O00461	Q9Y2C2
GOLM1	Q8NBJ4	P11686	GOLPH3	Q9H4A6	Q9H244
GOLPH3L	Q9H4A5	P21453	GOLT1B	Q9Y3E0	O95197
GORASP1	Q9BQQ3	P30622	GORASP2	Q9H8Y8	P49247
GOSR2	O14653	Q13190	GPAM	Q9HCL2	P12236
GPAT3	Q53EU6	Q9H2J7	GPATCH1	Q9BRR8	Q9BW85, Q95926
GPATCH2L	Q9NWQ4-1, Q9NWQ4	Q9GZV8	GPATCH8	Q9UKJ3	Q9H4G0
GPC3	P51654	P27824	GPC4	O75487	O15530
GPC5	P78333	Q13485	GPC6	Q9Y625	P32249
GPD1L	Q8N335	P13501	GPHN	Q9NQX3-2	P25786
GPM6A	P51674	Q09328	GPM6B	P35803	P31645
GPNUMB	Q14956-1, Q14956	P00533	GPR141	Q7Z602	O00186, Q12846
GPR143	P51810	Q16655, P14679	GPR156	Q8NFN8	P41743
GPR17	Q13304	Q7Z5K2	GPR3	P46089	P05067-4
GPR37	O15354	P29274	GPR39	O43194	P08908
GPR50	Q13585	P04731, P02795	GPR55	Q9Y2T6	P17931
GPR61	Q9BZJ8	Q9UKG4	GPRASP2	Q96D09	P08311
GPRC5A	Q8NFJ5	P13569	GPRC5B	Q9NZH0	P48960
GPRC5C	Q9NQ84	Q9H015	GPRIN1	Q7Z2K8	P21453
GPRIN2	O60269	P31947	GPX7	Q96SL4	P19525
GPX8	Q8TED1	Q92730	GRAMD1A	Q96CP6	P68400
GRAMD1B	Q3KR37	P68400	GRAMD1C	Q8IYS0	P48509
GRAMD2A	Q8IUY3	Q12982	GRAMD2B	Q96HH9	Q12981
GRB10	Q13322-4	P12931	GRB7	Q14451	Q92730
GREM1	O60565	P35968	GRHL1	Q9NZI5	Q9NZI5
GRHL3	Q8TE85	O14744	GRIA1	P42261	Q9UIH9
GRIA2	P42262	P46459	GRIA4	P48058-2	Q8WTS1
GRIN1	Q05586-1	Q13224	GRIN2A	Q12879	P78352
GRIN2B	Q13224	P78352	GRIP2	Q9C0E4	Q15223
GRIPAP1	Q4V328	Q13573	GRK2	P25098, P21146	Q13635
GRK3	P35626	P25098	GRM2	Q14416	Q9H2H9
GRM3	Q14832	O43791	GRM5	P31424-1	P29274
GRN	P28799	P98160	GRPEL1	Q9HAV7	P07902
GRSF1	Q12849	O75832	GSDMD	P57764	P29466

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GSDME	O60443	Q86UL3	GSE1	Q14687	Q92769
GSK3B	G1T8E2, P49841	O15169	GSN	P06396	Q8N5V2
GSPT1	P15170	P08173	GSPT2	Q8IYD1	O00425
GSTA4	O15217	O95995	GSTK1	Q9Y2Q3	O00142
GSTT2B	P0CG30	P13569	GTF2A2	P52657	P20226
GTF2E2	P29084	P39748	GTF2H1	P32780	Q01831
GTF2H2	Q13888	P50613	GTF2H2C	Q6P1K8	P50613
GTF2IRD1	Q9UHL9	Q6VMQ6	GTPBP4	Q9BZE4	P19525
GUCY1A2	P33402	P78352	GXYLT1	Q4G148	Q9UBV2
GXYLT2	A0PJZ3	Q9Y4Z2	GYPC	P04921	Q3KNW5
H1-10	Q92522	P06748	H1-2	P16403	P67809
H2BC21	Q16778	Q15843	H2BC9	Q93079	P62993
H3-3A	P84243	P46100	HACD1	B0YJ81	P46059
HACE1	Q8IYU2	Q96CV9	HADHB	P55084	O15530
HAND2	P61296-2	Q96CV9	HAP1	P27695	Q09472
HBEGF	Q99075	P06396	HCCS	P53701	Q96E11
HCLS1	P14317	P07948	HCRTR2	O43614	P62937
HDAC1	Q13547	Q9UJW3	HDAC5	Q9UQL6	Q13761
HDAC7	Q9UKV0	Q06413	HDAC9	Q9UKV0	Q06413
HDDC2	Q7Z4H3	P01189	HDGF	P51858	P78527
HDHD3	Q9BSH5	O75593	HDLBP	Q00341	Q06609
HEATR1	Q9H583	O14975	HEATR5A	Q86XA9	O75688
HEATR6	Q6AI08	P34741	HECTD1	Q9ULT8	O75496
HECTD3	Q5T447	Q6NZI2	HECTD4	Q9Y4D8	P61981
HEG1	Q9ULI3	P17931	HEMK1	Q9Y5R4	Q9Y5V3
HEPH	Q9BQS7-3	O76003	HERC1	Q15751	O95395
HERC5	Q9UII4	Q9P031	HES1	Q14469	Q9Y5J3
HEXB	P07686	P06865	HEXIM1	O94992	P50750, O60563
HEXIM2	Q96MH2	P11309	HFE	Q30201	O14672
HGF	P14210, P14210-6, Q08048	P08581	HHEX	Q03014	P06730
HHIP	Q96QV1	Q15465	HIBADH	P31937	P63027
HIC1	Q14526	P35222	HIC2	Q96JB3	Q96KQ7
HID1	Q8IV36	O15357	HIF1AN	Q9NWT6	O60547
HINFP	Q9BQA5	P04637	HIP1	O00291	P35222
HIP1R	O75146	Q9NZQ7	HIPK1	Q86Z02	P04637
HIPK2	Q9H2X6	Q01196	HIRIP3	Q9BW71	P42858
HIVEP1	P15822	P84022	HIVEP2	P31629	P61981
HIVEP3	Q5T1R4	P61981	HK2	P52789	P19367
HLA-A	P04439	Q16288	HLA-B	P01889	P61769
HLA-C	P10321	P01222	HLA-DPB1	A0A1C3PI11, P04440	Q96CV9
HLA-DRA	P01903	O75843	HLA-DRB1	P01911	Q13370
HLA-E	P13747	Q09328	HLA-H	P01893	P11142
HLTF	Q14527	Q6ZQN7	HLX	Q14774	O75593
HM13	Q8TCT9	P04578	HMBS	P08397	Q9NRD5
HMCES	Q96FZ2	P11142	HMG20A	Q9NP66	O95863
HMGA2	P52927	P06400	HMGB2	P26583	P48729
HMGCR	P04035	P51684	HMGCS1	Q01581	P18827
HMGN2	P05204	Q04446	HMGN5	P82970	Q8WVK2
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Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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HOGA1	Q86XE5	Q9NQZ5	HOMER2	Q9NSB8	Q9H4P4
HOMER3	Q9NSC5	P26367	HOOK1	Q9UJC3	P53611
HOXA1	P09022	Q99697	HOXA10	P31260	P17947
HOXA5	P20719	Q96RT1	HOXA7	P31268	O00560
HOXA9	P31269	O14744	HOXB5	P09067	Q9Y4D7
HOXB7	P09629	P78527	HOXB9	P17482	P04004
HOXC11	O43248	P42858	HOXC13	P31276	P32519
HOXC4	P09017	Q9Y3P9	HOXC5	Q00444	P40424
HOXC8	P31273	Q06710	HOXC9	P31274	Q06710
HOXD10	P28358	O75925	HOXD4	P09016	Q9NW38
HOXD8	P13378	Q5SW96	HPCAL1	P37235	Q99685
HPS1	Q92902	Q9C0C7	HS2ST1	Q7LGA3	P53794
HS6ST1	O60243	Q9Y2A9	HS6ST2	Q96MM7	P27824
HSBP1L1	C9JCN9	Q9UBN6	HSD17B12	Q53GQ0	P04049
HSD3B1	P14060	P27797	HSD3B7	Q9H2F3	Q96FE5
HSDL1	Q3SXM5	Q00169	HSF2	Q03933	Q9Y315
HSP90AB3P	Q58FF7	Q15831	HSP90B1	P14625	P11021
HSPA4L	O95757	P11021	HSPA5	P11021	P01222
HSPA8	P11142	Q9UNU6	HSPB7	Q9UBY9	Q9GZV8
HSPB8	Q9UJY1	P49247	HSPBP1	Q9NZL4	P32519
HSPG2	P98160- PRO_0000391621, P98160- PRO_0000391622, P98160	P35968	HSPH1	Q92598	P11021
HTATIP2	Q9BUP3-3	Q8TB40	HTR1A	P08908	P08908
HTRA1	Q92743	P01185	HTT	P42858	Q8N5V2
HYCC2	Q8IXS8	Q08379	HYI	Q5T013	Q05519
IBTK	Q9P2D0	Q06187	ICA1	Q05084	Q9NRD5
ICAM1	P05362	P15514	ICE1	Q9Y2F5	P55199
ID1	P41134	O14936	ID4	P47928	Q9Y4Z2
IDE	P14735-1	P10997	IDH2	P48735	P46059
IDH3A	P50213	P13569	IER3	P46695	Q9UKG4
IER3IP1	Q9Y5U9	O14653	IER5	Q5VY09	Q00613
IFFO2	Q5TF58	P21802	IFI16	Q16666	P38398
IFI35	P80217-2	Q8N2W9	IFIH1	Q9BYX4	P05161
IFIT1	P09914	Q86WV6	IFITM3	Q01628	Q9NXB9
IFNAR1	P17181	Q13332	IFNAR2	P48551	P23458
IFNGR1	P15260	Q9NPF0	IFNGR2	P38484	P13861
IFNLR1	Q8IU57	P23458	IFRD2	Q12894	Q9Y6K5
IFT22	Q9H7X7	Q9NQC8, A0AVF1, Q9Y547, Q9BW83, Q8IY31, Q9P2H3, Q9NWB7, Q9Y366, Q9UG01, Q96LB3, Q8WYA0, Q13099	IFT46	Q9NQC8	A0AVF1, Q9Y547, Q9BW83, Q8IY31, Q9NWB7, Q9Y366, Q9H7X7, Q8WYA0, Q96AJ1, Q9P2H3, Q9UG01, Q96LB3, Q13099

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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IFTAP	Q86VG3	P53350	IGDCC4	Q8TDY8	P02788
IGF1	P05019	P08833, P22692	IGF1R	P08069	Q9NZN5
IGF2	P01344	P24593, P22692	IGF2BP1	Q9NZI8	O00425, Q9Y6M1
IGF2BP3	O00425	Q9NZI8	IGFBP3	P17936	Q86XT9
IGFBP4	P22692	P05019	IGFBP5	P24593	Q8TB40
IGFBP6	P24592	P53611	IGFBPL1	Q8WX77	P43146
IGFN1	Q86VF2-5	P19474	IGSF3	O75054	P21926
IGSF8	Q969P0	Q969Z0	IGSF9	Q9P2J2-2	O43169
IKBIP	Q70UQ0	O14920	IKZF1	Q13422-7	Q99728
IKZF4	Q9H2S9	P48730	IL10RA	Q13651	P22301
IL12A	P29459	Q9Y624	IL13RA1	P78552	Q8TDR0
IL13RA2	Q14627	Q14118	IL15	P40933	Q13261
IL16	Q14005-2	P52747	IL17RA	Q96F46	Q9NRM6
IL17RD	Q8NFM7	Q9NYV6	IL18	Q14116	Q13478
IL18R1	Q13478	Q14116	IL1R1	P14778	Q9Y4K3
IL1RAP	Q9NPH3	P01584	IL20RA	Q9UHF4	Q99741
IL32	P24001-4	P24158	IL34	Q6ZMJ4-1	P07333
IL4I1	Q96RQ9	Q96CV9	IL6R	P08887-PRO_0000450730, P08887	P05231
IL6ST	P40189	P05231	IL7R	P16871	P13232
ILF3	Q12906	Q9NZI8, Q9Y6M1	ILRUN	Q9H6K1	P11142
IMP4	Q8TCT7	O75063	IMPA1	P29218-3	P42858
INA	Q16352	P08670	INF2	Q27J81	P50548
ING4	Q9UNL4	Q00994	INHBA	P08476	P31749
INHBE	P58166	Q9Y2C2	INO80D	Q53TQ3	Q9Y265
INPP5D	Q92835, Q92835-2	O75525	INPP5F	Q9Y2H2	P63104
INPPL1	O15357	P35813	INSR	P06213	P29353
INSYN1	Q2T9L4	Q9UL45	INSYN2A	Q6ZSG2	O15111
INTS13	Q9NVM9	Q13951	INTS4	Q96HW7	Q15116
INTS6	Q9UL03	P22891	IP6K2	Q9UHH9	P04637
IP6K3	Q96PC2	Q13114	IPO13	O94829	P47813
IPO9	Q96P70	P08173	IQCC	Q4KMZ1	Q14145
IQGAP1	P46940	Q16539	IQSEC2	Q5JU85	P0DP23
IQUB	Q8NA54	Q15669	IRAK1	P51617	Q92985
IRF1	P15314	Q13114	IRF2BPL	Q9H1B7	P14316
IRF7	Q92985	Q9UHD2	IRGQ	Q8WZA9	Q9H2D1
IRS1	P35570	Q06124	IRS2	Q9Y4H2	P27986
ISCA1	Q9BUE6	Q96RU8	ISCU	Q9H1K1	Q8IWL3
ISLR	O14498	Q99741	ISLR2	Q6UXK2, Q5RKR3	P41159
ISYNA1	Q9NPH2	Q9NUX5	ITGA1	P56199	P17931
ITGA2	P17301	Q9H0F6	ITGA3	P26006	P48509
ITGA4	P13612	P02751	ITGA5	P08648	P02751
ITGA7	Q13683	Q14192	ITGA8	P53708	Q6UX41
ITGAV	P06756	P04004	ITGB1	P05556	P48509
ITGB3	P05106	P02751	ITGB3BP	Q13352	Q969Y2

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ITGB4	P16144	P48509	ITGB5	P18084	P04004
ITGB7	P26010, P26010-1	P21333	ITK	Q08881	P48023
ITLN1	Q8WWA0	P11021	ITM2A	O43736	P05067-4
ITM2B	Q9Y287	Q86XT9	ITPKB	P27987	Q8N9R8
ITPR1	P11881	P10415	ITPR2	Q14571	Q9H244
ITPR3	Q14573	P40763	ITPRID2	P28290	P22681
ITPRIPL1	Q6GPH6-2	Q12933	ITPRIPL2	Q3MIP1	Q9BRI3
ITSN1	Q15811	P60880	ITSN2	Q9NZM3	P00533
IVNS1ABP	Q9Y6Y0	P53350	JADE1	Q6IE81-3	Q96CV9
JADE2	Q9NQC1-2	P01100	JAK1	P23458	O60674
JAKMIP2	Q96AA8	Q13526	JARID2	Q92833	Q09028
JAZF1	Q86VZ6	Q9Y4A5, Q9Y265	JDP2	Q8WYK2	P18848
JOSD1	Q15040	Q13077	JOSD2	Q8TAC2	P21741
JPH1	Q9HDC5	Q5SY16	JPH3	Q8WXH2	Q9NR63
JPH4	Q96JJ6	Q6VN20	JRK	O75564	P52294
JUN	P05412	P45983	JUNB	P17275	P18847
JUP	P14923	P01160	KANK1	Q14678	Q9UQB8
KANK2	Q63ZY3	Q15669	KANSL3	Q9P2N6	Q9BVI0
KAT2B	Q92831	Q09472	KAT7	O95251	Q99728
KAZN	Q674X7-2	O75604	KBTBD4	Q9NVX7	P11142
KBTBD6	Q86V97	Q6NZI2	KBTBD7	Q8WVZ9	Q13009
KCNA2	P16389	Q6UWP7	KCNA3	P15384	P78352
KCNA6	P17658	Q9UPQ8	KCNC4	Q03721	P56817
KCNF1	Q9H3M0	Q92504	KCNIP1	Q9NZI2-2	P40189
KCNIP4	Q6PIL6	P08123	KCNJ10	P78508	Q8IUQ4
KCNJ15	Q99712	P63252	KCNJ5-AS1	Q8TAV5	O43559
KCNK1	O00180	P63165	KCNMA1	Q08460	P46109
KCNMB2	Q9Y691	P42858	KCNS2	Q9ULS6	P0DPK4
KCTD13	Q8WZ19	P46108	KCTD20	Q7Z5Y7	P36873
KCTD3	Q9Y597	P26045	KCTD9	Q7L273	O14786
KDELR1	P24390	Q92796	KDM2A	Q9Y2K7	Q13257
KDM2B	Q8NHM5	P63208	KDM3B	Q7LBC6	P09917
KDM4A	O75164	Q15831	KDM6A	O15550	O43474
KDM6B	Q5NCY0	Q9BVI0	KDR	P35968	P98160
KDSR	Q06136	Q9H5K3	KHDRBS2	Q5VWX1	Q13882
KHDRBS3	O75525	P40938	KIAA0040	A0A384DVV8	P42858
KIAA0232	Q92628	Q8N9F0	KIAA0586	Q9BVV6	Q8IW35, O43303
KIAA1217	Q5T5P2-6	Q13882	KIAA1671	Q9BY89	Q9Y3R4
KIF1A	Q12756	O76024	KIF1B	O60333-2	Q8N5V2
KIF1C	O43896	P62258	KIF21A	Q7Z4S6	Q14678
KIF21B	O75037	P61981	KIF22	Q14807	Q8TC07
KIF24	Q5T7B8	P51955	KIF2A	O00139	Q5S007
KIF3A	Q9Y496	Q5T8I9	KIF3B	O15066	Q9Y496, Q92845
KIF3C	O14782	Q9Y496, Q92845	KIF5A	Q12840	Q15811
KIF5C	O60282	P61981	KIF7	Q2M1P5	P10070
KIF9	Q9HAQ2	P08670	KIFC3	Q9BVG8-5	Q9BUH8
KIRREL1	Q96J84	Q07157	KIRREL3	Q8IZU9	P15311
KITLG	P21583	P10721	KIZ	Q2M2Z5	Q96NL6
KLC1	Q07866-2, Q07866	Q12982	KLC3	Q6P597	Q9BTA9
KLF11	O14901	Q14154	KLF12	Q9Y4X4	Q96KQ7

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
KLF13	Q9Y2Y9	P43694	KLF14	Q3SY56	P17612
KLF16	Q9BXX1	Q12857	KLF2	Q9Y5W3	Q9HCE7
KLF4	O43474	O14964	KLF5	Q13887	P0CG48
KLF6	Q99612	P02545	KLHDC4	Q8TBB5-2	P55212
KLHDC8A	Q8IYD2	Q13371	KLHDC9	Q8NEP7	O76003
KLHL12	Q53G59	O95050	KLHL13	Q9P2N7	P01282
KLHL14	Q9P2G3	Q9UF56	KLHL15	Q96M94	P21854
KLHL17	Q6TDP4	P07585	KLHL20	Q9Y2M5	Q9UF56
KLHL21	Q9UJP4	P23025	KLHL23	Q8NBE8	P29279
KLHL25	Q9H0H3	Q9NPA3	KLHL3	Q9UH77	P06396
KLHL32	Q96NJ5	P27361	KLHL42	Q9P2K6	P07237
KLHL7	Q8IXQ5	Q8N4J0	KLHL8	Q9P2G9-2	P11086
KLHL9	Q9P2J3	Q9UF56	KLK10	O43240	Q9HAW8
KLK5	Q9Y337	P11245	KLK6	Q92876	Q13371
KLRG2	A4D1S0	Q9NT99	KMT2A	Q03164	O00255
KMT2E	Q8IZD2	P04637	KMT5B	Q4FZB7	Q9H2G4
KNTC1	P50748	Q9Y639	KPNA3	O00505	Q8WWX8
KPNA4	O00629	Q9BQB4	KRBOX4	Q5JUW0-3	Q92993
KREMEN2	Q8NCW0	P14138	KRI1	Q8N9T8	P40429
KRR1	Q13601	Q9H2U1	KRT10	P13645	P13569
KRT13	P13646	Q8IVS8	KRT17	Q04695	P25054
KRT18	P05783	Q9NUX5	KRT19	P08727	Q9P0N9
KRT7	P08729	P09429	KRT75	O95678	O43592
KRT8	P05787	Q9Y6K9	KRT80	Q6KB66-2	Q9BRX5
KRT81	Q14533	P49639	KRTAP2-3	P0C7H8	Q8WUK0
KSR2	Q6VAB6	Q93008	KTN1	Q86UP2-3	P20645
L1CAM	P32004	O00533	L1TD1	Q5T7N2	P52292
L3HYPDH	Q96EM0	P46939	L3MBTL1	Q9Y468	P33992, P49736
L3MBTL3	Q96JM7-2	O94955	LACC1	Q8IV20	P49327
LACTB	P83111	Q9NQ92	LACTB2	Q53H82	Q8IUQ4
LAMA1	P25391	P17931	LAMA3	Q16787	P13569
LAMA4	Q16363	P04637	LAMA5	O15230	Q6ISS4
LAMB1	P07942	P17931	LAMB2	P55268	Q03405
LAMC2	Q13753	Q70UQ0	LAMP1	P11279	Q15836
LAMP2	P13473-2	P01730	LAMP3	Q9UQV4	Q92696
LAP3	P28838	Q13185	LAPTM4A	Q15012	P80370
LAPTM4B	Q86VI4	P08581	LARGE1	O95461	Q6VN20
LARGE2	Q8N3Y3	Q03405	LARP1	Q6PKG0	P59595
LARP1B	Q659C4-6	Q3MII6	LASP1	Q14847	Q9H2D1
LATS2	Q9NRM7	O95863	LAYN	Q6UX15-2	P24593
LBR	Q14739	Q9Y4Z2	LBX1	P52954	P46940
LCA5	Q86VQ0	P40222	LCA5L	O95447	Q9NQT4
LCAT	P04180	P02647	LCLAT1	Q6UWP7	Q92504
LCOR	Q96JN0	Q08379	LDB2	O43679	O00560
LDLRAD4	O15165	Q96PU5	LDLRAP1	D3ZAR1	P01130
LODC1	O95751	P56524	LEF1	Q9UJU2	Q16236
LEMD3	Q9Y2U8	Q9UM73	LEPROT	O15243	Q9UBD6
LEPROTL1	O95214	Q9Y5Y5	LETM1	O95202	P04798
LETMD1	Q6P1Q0	Q9H5K3	LGALS1	P09382	P48509
LGALS2	P05162	O00560	LGALS3BP	Q08380	P02751

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LGALS4	P56470	Q9HAS3	LGALSL	Q3ZCW2	P50458
LGMN	Q99538	O94805	LHFPL2	Q6ZUX7	Q14162
LHFPL4	Q7Z7J7	P06576	LHX2	P50458	Q99650
LHX5	Q9H2C1	P06396	LHX9	Q9NQ69	Q2TBE0
LIF	P15018	P40189	LIFR	P42702	P26441
LIG1	Q96JA1	P22681	LIG3	P49916	P06746
LIMA1	Q9UHB6	P60953	LIMCH1	Q9UPQ0-1	Q96JA1
LIME1	Q9H400	Q05329	LIMS1	P48059	Q9Y3P9
LIMS2	Q7Z4I7-5	Q96PM5	LIN28A	Q9H9Z2	O00425
LIN28B	Q6ZN17	Q8IZH2	LIN7B	Q9HAP6	O14936
LINGO1	Q96FE5	P08138	LINGO2	Q7L985	Q96FE5
LIPH	Q8WWY8	O75197	LITAF	Q99732	Q9NPF0
LLGL1	Q15334	Q05513	LLGL2	Q6P1M3	Q05513
LMAN1	P49257	P04578	LMNA	P02545-1, P02545-2, P02545	P20700
LMO4	P61968	Q9BRX2	LMO7	Q8WWI1-3	Q96JA1
LMTK2	Q8IWU2	P36873	LMTK3	Q96Q04	P03372
LMX1A	Q8TE12	P62993	LMX1B	O60663-2	Q02962
LNK1	Q8TBB1	P12931	LNK2	Q8N448	O95050
LONRF1	Q17RB8	Q9UQK1	LONRF2	Q1L5Z9	P99999
LOX	P28300	Q9UBX5, P15502	LOXL1	Q08397	P06276
LOXL2	Q9Y4K0	P02751	LOXL4	Q96JB6	P06396
LPAR1	P61793	Q9NZN5	LPAR2	Q9HBW0	P17931
LPAR4	Q99677	Q13315	LPAR6	P43657	P23458
LPCAT1	Q8NF37	O14975	LPCAT2	Q7L5N7	Q15125
LPCAT3	Q6P1A2	P04004	LPGAT1	Q92604	P22732
LPIN1	Q14693	P12931	LPIN3	Q9BQK8	P62258
LPL	P06858	Q8IV16	LPP	Q93052	P05120
LRAT	O95237	Q7Z5P4	LRATD2	Q96KN1	Q06609
LRCH3	Q96II8	Q9UM54	LRFN3	Q9BTN0	P13385
LRFN5	Q96NI6	P23468, Q13332	LRIG1	Q96JA1	P22681
LRIG2	O94898	Q96JA1	LRIG3	Q6UXM1	Q96JA1
LRP1	Q07954	P12236	LRP10	O75096	Q9BQB4
LRP2	P98164	P29279	LRP4	O75096	Q9BQB4
LRP6	O75581	P98082	LRP8	Q14114	P35968
LRPAP1	P30533	P98155	LRRC1	Q9BTT6	Q9BT09
LRRC18	Q8N456	O43559	LRRC20	Q8TCA0	P60709
LRRC32	Q14392	O60906	LRRC41	Q15345	Q93034, Q9UBF6
LRRC45	Q96CN5	P13196	LRRC46	Q96FV0	Q9UM11
LRRC4C	Q9HCJ2	Q9NT99	LRRC57	Q8N9N7	Q9Y2Z0
LRRC59	Q96AG4	P04049	LRRC8B	Q6P9F7	Q13370
LRRC8C	Q8TDW0	P25090	LRRC8E	Q6NSJ5	P43119
LRRK1	Q38SD2	P29353	LRRK2	Q5S007	O00273
LRRN1	O75427	A1L3X0	LRRN2	O75325	P98170
LRRN4	O75427	A1L3X0	LRRTM2	O43300	Q9BT09
LRRTM3	Q86VH5	O95470	LSAMP	Q13449	Q14154
LSM11	P83369	P62306, P62304	LSM12	Q3MHD2	P10636-8
LSM14A	Q8ND56	P09651	LSR	Q86X29	Q96FT7
LTBP1	Q14766	P09238	LTBP2	Q14767	Q9UBX5
LTBP3	Q14767	Q9UBX5	LTBP4	Q8N2S1	Q9UBX5

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LTBR	P36941	Q9NPF0	LTV1	Q96GA3	P63244
LUM	P51884	P50281	LURAP1L	Q8IV03	O00560
LUZP1	Q86V48	P62258	LXN	Q9BS40	Q9UQK1
LYAR	Q9NX58	P19525	LYN	P07948	P12931
LYNX1	P0DP58-2	P06396	LYPD1	Q8N2G4	P41221
LYPLA1	Q6IAQ1, O75608	Q08379	LZTFL1	Q9NQ48	O43684
LZTS2	Q9BRK4	Q9Y316	LZTS3	O60299	Q13627
MAB21L1	Q13394	Q8IUQ4	MAB21L2	Q9Y586	Q08379
MACIR	Q96GV9	P62258	MAD2L2	Q9UI95	Q9UBZ9
MADD	Q8WXG6	Q9NUX5	MAF	O75444	P18848
MAFK	O60675	O14867	MAGED2	Q9UNF1	P52333
MAGED4	Q96JG8	Q15185	MAGEE1	Q9UBF1	P04637
MAGEH1	Q9H213	P17252	MAGEL2	Q9UJ55	Q93009
MAGI1	Q6P9H4	P42772	MAGI2	Q86UL8	P01135
MAGT1	Q9H0U3	Q15363	MAK	P20794	Q9UM11
MAK16	Q9BXY0	Q9HAP6	MALSU1	Q96EH3	Q14197
MALT1	Q9UDY8	Q9UDY8	MAN1A1	P33908	P09238
MAN1A2	O60476	O60602	MAN2A1	Q16706	Q03405
MAN2B2	Q9Y2E5	Q96BA8	MAP2	P11137-4	P02649
MAP2K3	P46734	Q16539	MAP2K4	P45985	Q13233
MAP2K5	Q13163	Q13164	MAP2K6	P52564	Q16539
MAP3K1	Q13233	P32121	MAP3K14	Q99558	Q13114
MAP3K20	Q9NYL2	O75582	MAP3K21	Q5TCX8	O15111
MAP3K3	Q99759	Q13163	MAP3K4	Q9Y6R4	Q06187
MAP3K5	Q99683	P25445	MAP3K6	O95382	P61981
MAP3K7CL	P57077-4, P57077	P22415, Q13227	MAP3K8	P41279	P19838, Q8NFX5
MAP3K9	P80192	P17252	MAP4	P27816	P10636-8
MAP4K3	Q924I2	Q99962	MAP4K5	Q9Y4K4	P46108
MAP7	Q14244	P61981	MAP7D2	Q96T17	P10636-8
MAP7D3	Q8IWC1	P61981	MAPK10	P53779	P49407
MAPK11	Q15759	P49137	MAPK14	Q16539	O75582
MAPK15	Q8TD08	P31749	MAPK8	P45983-4	Q9BY84
MAPK8IP1	Q9UQF2, Q9WVI9-1	P45983	MAPK8IP2	Q13387-4	P05556
MAPKAP1	Q9BPZ7	P31749	MAPRE2	Q15555	P11142
MAPRE3	Q9UPY8	O00560	MAPT	P10636	P04156
MARCHF3	Q86UD3	Q12959	MARCKSL1	P49006	O60551
MARF1	Q9Y4F3	P61964	MARK1	Q9P0L2	P05067, P62258
MARVELD2	Q8N4S9	Q8NG68	MASP1	P48740-2	P17927
MAST1	Q9Y2H9	P11021	MAST2	Q6P0Q8	P27037
MAT2A	P31153	P02795	MATK	P42679	Q14289
MATN2	O00339	Q96ES7	MATN3	O15232	P02787
MB21D2	Q8IYB1	Q6NZI2	MBD1	Q9UIS9	Q15047
MBD2	Q9UBB5	P26358	MBNL1	Q86VM6, Q9NR56	O75553
MBNL3	Q9NUK0	O75553	MBP	P13727	Q12946
MBTD1	Q05BQ5	P62805	MBTPS1	Q14703	Q8IV16
MCAM	P43121	Q12860	MCAT	Q8IVS2	P13196
MCCC2	Q9HCC0	P13569	MCF2L	Q6PDM6	P61586
MCFD2	Q8NI22	P78382	MCL1	Q07820	Q13794
MCM10	Q7L590	Q7L590	MCM2	P49736	P03433

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MCM7	P33993	Q9Y248, Q9BRT9, Q14691, Q9BRX5	MCM8	Q9UJA3	P43246
MCOLN2	Q8IZK6	Q9GZU1	MCOLN3	Q8TDD5	Q86XT9
MCRIP2	Q9BUT9	O15234	MCU	Q8NE86-1, Q8NE86	Q9H4I9, Q9BPX6
MCUB	Q9NWR8	Q15047	MDFI	Q99750	P06850
MDK	P21741	P62979	MDM2	Q00987	Q15648
MDM4	O15151	P06400	MDN1	Q9NU22	Q9Y639
ME1	P48163	P48163	ME2	P23368	P10809
MEAF6	Q9HAF1	P40937	MEAK7	Q6P9B6	O60551
MECOM	Q03112	P01100	MECP2	Q00566	P46100
MED10	Q9BTT4	Q15648	MED12L	Q86YW9	P61024
MED13L	Q71F56	Q15648	MED15	Q96RN5	Q15648
MED21	Q13503	Q15648	MEDAG	Q5VYS4	Q9H0E2
MEF2C	Q06413	P56524	MEGF6	Q75095	P00533
MEIS1	O00470	P40424	MELTF	P08582	P13569
MEOX2	P50222	Q9UBX5	MESD	Q14696	Q969Y2
MEST	Q5EB52	Q05329	MET	P08581	P12931
METAP2	P50579	Q93034	METRNL	Q641Q3	Q01523
METTTL13	Q8N6R0	Q12933	METTTL21A	Q8WXB1	P11142
MFAP2	P55001	P49639	MFAP4	P55083	Q16625
MFHAS1	Q9Y4C4	Q9Y496	MFN1	Q8IWA4	Q16611
MFSD13A	Q14CX5	P0DPK4	MFSD14B	Q5SR56	Q9NRZ7
MGAM	O43451	Q96BA8	MGARP	Q8TDB4	Q8TAF3, O94782
MGAT1	P26572	Q9BYC5	MGAT2	Q10469	P40855
MGAT4C	Q9UBM8	O75197	MGAT5	Q09328	Q03405
MGST1	P10620	P01133	MGST2	Q99735	Q16585
MIA2	Q96PC5	P18846	MICA	Q29983	P17931
MICAL2	O94851	P54252	MICAL3	Q7RTP6, Q7RTP6-1	Q02241
MICALL1	Q8N3F8	P61981	MICALL2	Q8IY33	P12814
MICB	Q29980	P17302	MICU2	Q8IYU8	Q9H4I9, Q9BPX6, Q8NE86
MID1	P29372	P04156	MIDEAS	Q6PJG2	Q16539
MINDY1	Q8N5J2-3	P55212	MISP	Q8IVT2	P07355
MITF	O75030	P19484	MKKS	Q9NPJ1	P26641
MKLN1	Q9UL63	Q13685	MKRN1	Q9UHC7	P51668
MKRN2	Q9H000	P51668	MKX	Q8IYA7	Q9BZK7
MLH1	P40692	P54278	MLLT1	Q03111	P04608
MLLT3	P42568	P21741	MLPH	Q9BV36	P51159
MLXIP	Q9HAP2	P62258	MLYCD	O95822-2	P52756
MMADHC	Q9H3L0	Q9P0N9	MMD	Q15546	Q3SXY8
MMD2	Q8IY49-2	Q96BA8	MME	P08473	A0PJW6
MMP10	P09238	P01033	MMP14	P50281	P09958
MMP2	P08253	P39060	MMP28	Q9H239	P09958
MMRN1	Q13201	P02649	MMS19	Q96T76	P51530
MMS22L	Q6ZRQ5	Q08945	MNDA	P41218	P08581
MOB1A	Q9H8S9	Q13188, Q13043	MOB1B	Q7L9L4	Q13188, Q13043
MOB3C	Q70IA8	Q9NRD5	MOCOS	Q96EN8	Q14653
MOCS2	O96007	O96033	MOGS	Q13724	Q14974
MON2	Q7Z3U7	P21453	MORC4	Q8TE76	P40763
MORF4L2	Q15014	Q86YC2	MORN4	Q8NDC4	Q16143

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MOV10	Q9HCE1	Q9NZI8	MOXD1	Q6UVY6	O76003
MPDZ	O75970	Q8N3R9	MPG	P29372	P04156
MPHOSPH6	Q99547	P05067	MPND	Q8N594	P05067
MPP2	Q08050	Q13416	MPP4	Q96JB8	Q8N3R9
MPP7	Q5T2T1	Q9HAP6	MPPED2	Q15777	Q9Y244
MPST	P25325	O96006	MPZL1	O95297	Q06124
MR1	Q95460-1	P61769	MRAP	Q8TCY5, Q8TCY5-1	Q01718
MRAP2	Q96G30	Q9Y2C2	MRC2	Q9UBG0	P17931
MRM1	Q6IN84	P11182	MRPL10	Q7Z7H8	Q14197
MRPL16	Q9NX20	P67809	MRPL19	P49406	P40429
MRPL3	P09001	Q14197	MRPL32	Q9BYC8	Q9GZV8
MRPL4	Q9BYD3	Q14197	MRPS10	P82664	P62753
MRPS12	O15235	Q04721	MRPS15	P82914	P62753
MRPS18B	Q9Y676	P06400	MRPS34	P82930	O00425
MRPS35	Q9Y2Q9	P67809	MRS2	Q9HD23	Q8N5V2
MSANTD3	Q96H12	P06396	MSH2	P43246	P52701, P20585
MSH5	O43196	P00519	MSI2	Q96DH6	Q9Y5J6
MSL1	Q68DK7	O75496	MSN	P26038	P16070
MSRB2	Q9Y3D2	P02649	MSRB3	Q8IXL7	Q9UKN5
MSX2	P35548	P28482	MT-CO1	P00395	P25103
MT-CO2	P00403	P15954	MT-CO3	P00414	P37840
MT-ND2	P03891	O75380	MT-ND4	P03905	Q9P0J0
MT-ND6	P03923	O75489	MTBP	Q96DY7	P49336
MTCL2	O94964	P62258	MTDH	Q86UE4	Q9UKV8
MTF2	Q9Y483-4	P54252	MTHFD1	P11586	P13569
MTHFD1L	Q6UB35	Q96GW9	MTHFR	P42898	O60906
MTIF3	Q9H2K0	Q07817	MTMR3	Q13615	P61981
MTMR4	Q9NYA4	P09917	MTMR7	Q9Y216	Q9UJK0
MTMR8	Q96EF0-1, Q96EF0, Q96QG7	Q9Y216, Q96QG7	MTMR9	Q96QG7	Q9Y216, Q96QG7
MTNR1A	P48039	P11021	MTPN	P58546	P13569
MTR	Q99707	Q9Y4U1	MTSS2	Q765P7	P60709
MTTP	P55157	P04114	MTURN	Q8N3F0	Q9H4P4
MTUS1	Q9ULD2-3	Q13190	MTUS2	Q5JR59	P56524
MTX1	Q13505	Q05940	MUTYH	Q9UIF7	P31749
MVP	Q14764	Q9H0E2	MX1	P20591	Q13507
MXD1	Q05195	Q96ST3	MXD3	Q9BW11	Q9UBB9
MXI1	P50539-3	Q7Z3S9, P0DPK4	MXRA8	Q9BRK3	Q9NPB3
MYADM	Q96S97	Q9Y624	MYBBP1A	Q9BQG0	P01106
MYC	P01106	P52701	MYCBP2	O75592	P22692
MYCL	P12524-2	P54646	MYCN	P04198	Q8N726
MYD88	Q99836	Q15109	MYDGF	Q969H8	Q14554
MYEOV	Q96EZ4	Q96CV9	MYH10	P35580	Q14164
MYH7	P12883	Q15475	MYH9	P35579	O00255
MYL10	Q9BUA6	P46940	MYL12A	P19105	P35579
MYL12B	O14950	P60709	MYL9	P24844	P35579
MYLIP	Q8WY64	P54252	MYLK2	Q9H1R3	Q9H4I9
MYLK3	Q32MK0	P53778	MYO1B	O43795	Q5S007
MYO1C	Q12965	P50570	MYO1D	O94832	P43251
MYO1E	Q12965	P50570	MYO5A	Q9Y4I1	O43889

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MYO5B	Q9ULV0	P40855	MYO5C	Q9NQX4	O75496
MYO6	Q9UM54	P98082	MYOCD	Q8IZQ8	P11831
MYORG	Q6NSJ0	P11021	MYOZ3	Q8TDC0	Q13882
MYRIP	Q8NFW9	P51159	MYT1	Q99640	P06493
MZT2A	Q6P582	O00571	N4BP3	O15049	Q9H4P4
NAA10	P41227	Q96FF9	NAA15	Q9BXJ9	Q8WVK2
NAA50	Q9GZZ1	P04798	NAALAD2	Q9Y3Q0	O76024
NAB2	Q15742	P40937	NABP1	Q96AH0	Q5VWX1
NACC1	Q96RE7	P41743	NADK	O95544	O00560
NADSYN1	Q6IA69	Q9UNE7	NAF1	Q15025	P24387
NAGLU	P54802	P27824	NAMPT	P43490	P43490
NAP1L2	Q9ULW6	P42858	NAP1L3	Q99457	P14136
NAP1L5	Q96NT1	P18848	NAPB	Q9H115	Q13190
NAPSA	O96009	P11021	NARF	Q8WVD3	P07550
NASP	P49321	P62805	NAT14	Q8WUY8	P40855
NAT8L	Q8N9F0	P40855	NATD1	Q8N6N6	Q9NRD5
NAV2	Q8IVL1	P62258	NBAS	A2RRP1	Q12981, Q9NZ43, Q9P2W9
NBDY	A0A0U1RRE5	Q9NPI6	NBEA	Q8NFP9	Q9BYJ1
NBEAL2	Q6ZNJ1	Q9Y639	NBL1	P41271-2	P05067
NBN	O60934	Q09472	NCALD	P61601	P23352
NCAM1	P13591	P25098	NCAPH2	Q6IBW4-4	P24593
NCCRP1	Q6ZVX7	Q15915	NCDN	Q9UBB6	P26367
NCF2	P19878	Q92845	NCKAP5	O14513	P25054
NCL	P19338	P19338	NCLN	Q969V3	Q5S007
NCOA2	Q15596	O00482-1	NCOA6	Q14686	P06400
NCOA7	Q8NI08	P27540	NCOR2	Q9Y618	Q01094
NCR3LG1	Q68D85	P47813	NDEL1	Q9GZM8	P23025
NDN	P25233	P08138	NDRG4	Q9ULP0-2	O60906
NDST1	P52848	Q03405	NDUFA13	Q9P0J0	P23786
NDUFA6	P56556	Q8IWL3	NDUFAF3	Q9BU61	Q9NPB3
NDUFB11	Q9NX14	P08034	NDUFS3	O75489	Q9P0J0
NDUFS5	O43920	Q9P0J0	NDUFV3	P56181	Q9P0J0
NEBL	O76041	Q08379, P05067	NECAB2	Q7Z6G3	P29274
NECAP1	Q8NC96	Q96CW1	NECAP2	Q9NVZ3	P42566
NECTIN1	Q15223	Q9NQS3, Q15223	NECTIN2	Q92692-2	P07237
NEDD9	Q14511	Q8IZD9	NEFH	P12036	P05067
NEFL	P07196	Q13569	NEFM	P07197	P06400
NEGR1	Q7Z3B1	P61916	NEK11	Q8NG66	P51955
NEK3	P51956	P32971	NEK6	Q9HC98-4	P12830
NEK7	Q8TDX7	P07237	NELFA	Q9H3P2	O00267
NELL2	Q99435	O00411	NEMP1	O14524-2	Q14162
NET1	Q7Z628	P78352	NETO2	Q8NC67-2	P06396
NEURL4	Q96JN8	Q9UJU6	NEUROG3	Q9Y4Z2	O95631
NEXN	Q0ZGT2	P60709	NFATC1	O95644	P18846
NFATC2	Q13469-2	P05412	NFATC2IP	Q8NCF5	Q96HC4
NFE2L1	Q14494	O15525	NFE2L2	Q16236	O60675
NFE2L3	Q9Y4A8	O60675	NFIA	Q12857	Q02962
NFIB	O00712	Q02962	NFIC	P08651	Q12948

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
NFIX	Q14938	O00712	NFKB1	P19838- PRO_0000030311	O00255
NFKB2	Q00653	P41279, P19838	NFKBIA	P25963	Q15813
NFYB	P25208	P04637	NGB	Q9NPG2	P14635
NGEF	Q8N5V2	P41743	NGF	P01138- PRO_0000019600, P01138	P08138
NGFR	P07174, P08138	P01138	NGLY1	Q96IV0	Q96FT7
NGRN	Q9NPE2	O00425	NHERF1	O14745	P29762
NHERF2	Q15599	Q05066	NHS	Q6T4R5	O43639
NHSL1	Q5SYE7	O43639	NHSL2	Q5HYW2	P53350
NHSL3	Q9P206	Q9UQB8	NIBAN2	Q96TA1	P13569
NID2	Q8IV28	P10745	NIFK	Q9BYG3	Q9H9Z2
NINJ1	Q92982	Q16585	NINL	Q9Y2I6	Q7L590
NIP7	Q9Y221	P46087	NIPAL1	Q6NVV3	Q9UBV7
NIPAL3	Q6P499	P27449	NIPAL4	Q0D2K0	P08034
NIPBL	Q6KC79	P52294	NIPSNAP2	O75323	Q05513
NIPSNAP3A	Q9UFN0	P10809	NKD1	Q969G9	P67775
NKILA	EBI-10826665	P25963	NLGN2	Q8NFZ4	Q8N0Z6
NLGN3	Q9NZ94-2	P00403	NLK	Q9UBE8	Q96PM5
NLN	Q9BYT8	Q5SW96	NLRC5	Q86WI3	O15111, O14920
NLRP1	Q9C000	Q9ULZ3	NLRX1	Q86UT6	P49411
NMB	Q14956-1, Q14956	P00533	NME4	O00746	Q9NPB3
NMI	Q13287	P23434	NMT1	P30419	P04601
NMU	P48645	P12644	NOB1	Q9ULX3	P51530
NOC3L	Q8WTT2	P19525	NOCT	Q9UK39	P11182
NOD1	Q9Y239	Q13153	NOL10	Q9BSC4	P06748, P62753
NOL4	O94818-2	P07237	NOL6	Q9H6R4-4	Q8IUQ4
NOL7	Q9UMY1	Q9H8H0	NOLC1	Q14978	P32121, P49407
NOM1	Q5C9Z4	Q9Y4Z2	NOMO1	Q15155	P25090
NOP14	P78316	O94979	NOP16	Q9Y3C1	P06748
NOP56	O00567	Q9Y4Z2	NOPCHAP1	Q8N5I9	P52294
NOS1AP	O75052	P21741	NOS3	P29474	Q12891
NOTCH1	P46531	P78504	NOTCH2	Q04721	Q8N2S1
NOTCH4	Q99466	P05067	NOX4	Q9NPH5	O00206
NPAS2	Q99743	O43639	NPAS3	Q8IXF0	Q9HNBZ2
NPC2	P61916	P00751	NPHP1	C9J082	P01019
NPLOC4	Q8TAT6	Q9H598	NPM1	P06748	Q8N726
NPM2	Q86SE8, Q86SE8-2	P06748	NPPA	P01160	Q6GTS8
NPPB	P16860	P25788	NR2F1	P10589	Q00403
NR2F2	P24468	P43119	NR2F6	P10588	P24468
NR3C1	P06536	Q15788	NR3C2	P08235	Q9UBK2
NR4A2	P43354	P05067	NR4A3	Q92570	Q9H2F3
NR5A2	O00482	P0DTC5	NRBP1	Q9UHY1	P51530
NRCAM	Q92823	Q12860	NRM	Q8IXM6	P04000
NRN1	Q9NPD7	Q86XT9	NRP2	Q99435	O00411
NRSN2	Q9GZP1	Q8N6M3	NRXN1	Q63373	O14936
NSD2	O96028	P17096	NSD3	Q9BZ95	Q06609
NSF	P46459	P59635	NSG1	P42857	Q96BA8
NSG2	Q9Y328	P11021	NSMAF	Q92636	P60709
NSMF	Q9EPI6	P27361	NSRP1	Q9H0G5	Q8WUA2

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
NSUN2	Q08J23	O14786	NTF4	P34130	P23560
NTN1	O95631	P43146	NTN4	Q9HB63	P16284
NTRK2	Q16620	Q16620	NTRK3	Q16288	Q16620
NUAK1	O60285	Q93008	NUCB1	Q02818	P18509
NUCB2	P80303	P17275	NUDC	Q9Y266	Q13451
NUDCD1	Q96RS6	Q92620	NUDCD2	Q8WVJ2	P43034
NUDT16	Q96DE0	P27930	NUDT16L1	Q9BRJ7	Q9NPB3
NUDT18	Q6ZVK8	Q15633	NUDT2	P50583	Q14566
NUMA1	Q14980	Q9UIL1	NUMBL	Q9Y6R0	Q9NWB1
NUP210	Q8TEM1	Q9H1C4	NUP214	P35658	Q9UBU9
NUP62CL	Q9H1M0	O14641	NUP93	Q8N1F7	O60733
NUP98	P52948	Q9UBP0	NUPR1	O60356	O15534
NUS1	Q96E22	P11142	NVL	O15381	P40429
NXN	Q6DKJ4	O14641	NXPH2	O95156	O75197
NXPH4	O95158	Q8IV16	NYAP1	Q6PFX7	P27986
NYNRIN	Q9P2P1	Q15185	OARD1	Q9Y530	P09874
OAS3	Q9Y6K5	Q7Z434	OAT	P04181	P05067
OBSCN	Q5VST9-3	P16157	OCLN	Q16625	P18827
OCRL	Q01968	Q9NYZ3	ODC1	P11926	Q92993
OGDHL	Q9ULD0	P25963	OGFOD1	Q8N543	P62266
OGFRL1	Q5TC84	P11802	OGN	P20774	Q9BT09
OLFM1	Q99784	Q5SRI9	OLFM3	Q96PB7, Q96PB7-3	Q9UNE7
OLFML2A	Q68BL7	Q15848	OLFML2B	Q68BL8	Q13217
OLR1	P78380	P17987	ONECUT3	O60422	O43639
OPRK1	P41145	P35414	OPTN	Q96CV9	P52888
ORC1	Q13415	O43929, Q9UBD5, Q13416, O43913	ORC2	Q13416	Q13415
ORC4	O43929	Q13415	ORMDL2	Q53FV1	Q9Y6L6
OS9	Q13438	O00468	OSBPL11	Q9BXB4	P09917
OSBPL5	Q9H0X9	P54802	OSR1	O95747	Q8WVK2
OST4	P0C6T2	A6NI15	OSTC	Q9NRP0	P80370
OSTF1	Q92882	O75695	OSTM1	Q86WC4	P98170
OTP	Q5XKR4	O43924	OTUB2	Q96DC9	Q9Y4K3
OTUD4	Q01804	P04004	OTUD7A	Q8TE49	Q13485
OTUD7B	Q6GQQ9	Q8NFZ5	P2RX4	Q99571	Q10472
P2RY6	Q15077	Q14974	P3H1	Q32P28	Q70UQ0
P3H2	Q8IVL5	O95076	P3H3	Q8IVL6	Q03405
P3H4	Q92791	Q9H2A9	P4HA1	P13674	Q6UN15
P4HA2	O15460	P39060	P4HA3	Q7Z4N8	Q14493
P4HB	P07237	O43521	PABIR3	Q6P4D5-2	P27797
PABPC1	P11940	Q86U42	PABPC4	Q13310	P59595
PABPC4L	P0CB38	P16104	PABPN1	Q86U42	P11940
PACS2	Q86VP3	P61981	PACSIN3	Q9UKS6	O14672
PADI1	Q9ULC6	Q9UGK3	PAFAH1B3	Q15102	P43034
PAG1	Q9NWQ8	P41240	PAIP1	Q9H074	O00425
PAIP2	Q9BPZ3	P11940	PAIP2B	Q9ULR5	O15234
PAK1	Q13153	P53667	PAK1IP1	Q9NWT1	P62753
PAK3	O75914	O14907	PAK4	O96013	O00401
PAK5	Q9P286	O14965	PAK6	Q9NQU5	P62258
PALD1	Q9ULE6	Q92985	PALLD	Q8WX93	P13569

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PALM	O75781	P53611	PALM2AKAP2	Q9Y2D5	P17612
PALM3	A6NDB9	Q6IA17	PALMD	Q9NP74	P42858
PALS1	Q8N3R9, Q8N3R9-1	P59637	PALS2	Q9NZW5	Q9HAP6
PAN2	Q9HBH5	P25103	PANX2	Q495U3	P06396
PAPSS2	O95340	Q96CV9	PAQR5	Q9NXX6	P26951
PAQR8	Q8TEZ7	Q08426	PARD3	Q8TEW0	P59637
PARD6B	Q9BYG5	Q05513	PARP1	P09874	P06746
PARP11	Q9NR21-5	P07550	PARP12	Q9H0J9	P48507
PARP14	Q460N5	O43914	PARVA	Q9NVD7	Q96EN8
PARVB	Q9HBI1	Q15404, Q9NVD7, P48059	PASK	Q96RG2	P08311
PATZ1	Q9HBE1	P40429	PAWR	Q96IZ0, Q62627	P11021
PAX2	Q02962	O00712, Q12857	PAX3	P24610	Q08999, P28749
PAX6	P26367	O00712, Q12857	PAX7	P23759	O00712, P08651, Q12857
PAX8	Q06710	O00712, Q12857	PBX2	P40425	Q9UHB7
PBXIP1	Q96AQ6	O15554	PCBD1	P61457	P35680
PCBP4	P57723	P11142	PCDH10	Q9P2E7	P43119
PCDH7	O60245	P34925	PCDHB16	Q9NRJ7	O60906
PCF11	O94913	Q92989	PCGF3	Q3KNV8	P68400, P67870
PCGF5	Q86SE9	Q12933	PCM1	Q15154	Q8IW35, O43303
PCNP	Q8WW12	O75533	PCNX2	A6NKB5-5	Q9BRI3
PCOLCE	Q15113	P07996	PCSK1N	Q9UHG2	P09238
PCSK5	Q92824	Q03405	PCSK6	P29122	P0DTC8
PCSK9	Q8NBP7	P07355	PCYOX1	Q9UHG3	O94955
PCYT1B	Q9Y5K3-3	Q8N0U8	PDCD10	Q9BUL8	Q9BRV8
PDCD1LG2	Q9BQ51	Q9NZQ7	PDCD6IP	Q8WUM4	P12931
PDE1A	P54750-4	P42858	PDE1B	Q01064	Q9NUX5
PDE2A	O00408	O00170	PDE3B	Q13370	O95398
PDE4A	P27815-4	P55212	PDE4D	Q08499-2	Q9HBX9
PDE6D	O43924	P36404	PDE7A	Q13946	P17612
PDGFA	P04085	P16234	PDGFB	P01127	P16234
PDGFC	Q9NRA1	P16234	PDGFD	Q9GZP0	O95714
PDGFRA	P16234	P19174	PDGFRB	P09619	P40818
PDGFRL	Q15198	Q9UBW7	PDHB	P11177	P10515
PDHX	O00330	Q8IWL3	PDIA3	P30101	P27797
PDIA4	P13667	P02452	PDIA5	Q14554	Q14508
PDIA6	Q15084	P11021	PDIK1L	Q8N165	Q13451
PDK1	O15530	P09038	PDK3	Q15120	P10515
PDK4	Q16654	Q15119	PDLIM1	O00151	P43487
PDLIM4	P50479	P12814	PDLIM7	Q9NR12	Q96HC4
PDP1	Q9P0J1	P12644	PDS5A	Q29RF7	Q7Z5K2
PDXDC1	Q6P996	P13569	PDYN	P01213	Q709F0
PDZD2	O15018	P59637	PDZD4	Q17RL8	Q9UJW3
PDZD7	Q9H5P4	P60484	PDZD8	Q8NEN9	P23945
PDZRN3	Q9UPQ7	P31040	PDZRN4	Q6ZMN7	P23634
PEA15	Q15121	P28482	PECAM1	P16284	P12931
PECR	Q9BY49	Q5T8I9	PEG10	Q86TG7	Q8IUQ4
PELO	Q9BRX2	P17252	PERP	Q96FX8	P14136
PES1	O00541	P06748, Q08050	PEX1	O43933	P19429

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PEX11B	O96011	P40855	PEX16	Q9Y5Y5	P56589, P40855
PEX2	P28328	P40855	PEX3	P56589	Q9Y5Y5, P40855
PFDN1	O60925	Q13164	PFKFB1	P16118	P35557
PFKFB3	Q16875	P61981	PFKFB4	Q16877	P61981
PFKP	Q01813	Q5S007	PGAM5	Q96HS1	Q9Y572
PGAP2	Q9UHIJ9-5	Q9Y624	PGAP4	Q9BRR3	Q6ZMH5
PGBD5	Q8N414	P53611	PGD	P52209	P13569
PGF	P49763	P17948	PGK1	P00558	P12004
PGPEP1	Q9NXJ5	P68871	PGR	P06401-1	Q13569
PGRMC1	O00264	P05177	PGRMC2	O15173	O60906
PHACTR2	O75167	P60709	PHACTR3	Q96KR7	Q9Y5Z9
PHACTR4	Q8IZ21	P62258	PHF1	O43189	P04637
PHF10	Q8WUB8	Q92925, O96019	PHF11	Q9UIL8	O00560
PHF12	Q96QT6	Q13547	PHF21A	Q96BD5	Q92769
PHF21B	Q96EK2	Q13547	PHF24	Q9UPV7	P07204
PHF3	Q92576	P28482	PHLDA1	A2BDE7	P0DPK4
PHLDB1	Q86UU1	P62258	PHLDB2	Q86SQ0	P62258
PHLPP1	O60346	Q8TAF3, O94782	PHYH	O14832	O75925
PHYHD1	Q5SRE7	Q9NUX5	PHYHIP	Q92561	Q9NS87
PHYKPL	Q8IUZ5	Q9NUX5	PI4K2B	Q8TCG2	Q03405
PI4KA	P42356	P21462	PIBF1	Q4G0R1	P98170
PICALM	Q13492	O00443, P42566	PICK1	Q9NRD5	P23975
PIEZO1	Q92508	P61073	PIGN	O95427	P43119
PIGS	Q96S52	P13569	PIGW	Q7Z7B1	Q9H8X2
PIK3C2A	O00443	O95630, Q96FJ0	PIK3C2B	O00750	P62258
PIK3CB	P42338	P43115	PIK3CD	O00329, O00329-2	P27986
PIK3R3	Q92569	P12931	PIM1	P11309-2	Q9UNQ0
PIP4K2B	P78356	Q15831	PIP4K2C	Q8TBX8	Q15831
PIP4P1	Q86T03	Q92993	PIP4P2	Q8N4L2	P00740
PIP5K1A	Q99755	Q9H9Z2	PIPOX	Q9P0Z9	Q9NUX5
PITPNB	P48739	P55212	PITRM1	Q5JRX3	P0DTC5
PITX1	P78337	P05121	PITX2	Q99697-3	Q12948
PJA1	Q8NG27	Q93009	PJA2	O43164	P13569
PKD1	Q15139	Q9NPC8	PKD2	Q13563	P21453
PKDCC	Q504Y2	Q9H5K3	PKIA	P61925	P17612
PKIB	Q9C010-2	P17612	PKIG	Q9Y2B9	P17612
PKN3	Q6P5Z2	A1A4S6	PKNOX2	Q96KN3	P48552
PKP2	Q99959	P03431	PKP3	Q9Y446	Q9Y496
PKP4	Q99569	Q8NI35	PLA2G12A	Q9BZM1	O76024
PLAAT5	Q96KN8-3	Q08379	PLAC1	Q9HBJ0	Q92830
PLAT	P00750	Q04721	PLAU	P00749	P05121
PLAUR	Q03405	Q09328	PLB1	Q6P1J6-2	P17252
PLBD1	Q6P4A8	P15498	PLCD1	P51178	Q14974
PLCD3	Q8N3E9	Q9Y4X5	PLCH1	Q4KWH8	P62258
PLCL2	Q9UPR0	P49914	PLD1	Q13393	P23528
PLD2	O14939	P23528	PLD3	Q8IV08	P19838
PLD6	Q8N2A8	P40855	PLEC	Q15149	P10636-8
PLEKHA2	Q9HB19	O14908	PLEKHA7	Q6IQ23-2	Q15669
PLEKHG3	A1L390	P61981	PLEKHG4B	Q96PX9	O76003
PLEKHH1	Q9ULM0	Q7Z5R6	PLIN3	O60664	Q8TB40

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PLK2	Q9NYY3	Q13177	PLN	P26678	Q8N661
PLOD1	Q02809	P55317	PLOD2	O00469	Q9Y230
PLOD3	O60568	P39060	PLP1	P60201-2	P40126
PLP2	Q04941	Q96QE2	PLPP4	Q5VZY2	P17302
PLPPR1	Q8TBJ4	P50454	PLPPR2	Q96GM1	Q9BVC3
PLS1	Q14651	P24387	PLS3	Q9NRY6	Q15915
PLSCR1	O15162	Q9UBR2	PLSCR4	Q9NRQ2	P10745
PLXDC2	Q6UX71	P08833	PLXNA1	Q9UIW2	Q01523
PLXNA4	Q9HCM2	P05067	PLXNB1	O43157	P52198
PLXNB2	O15031	P52198	PLXNB3	Q9ULL4	Q92854
PLXNC1	O60486	O75326	PLXND1	Q9Y4D7	O15041, Q9H3S1
PM20D2	Q8IYS1	O75695	PML	P29590	P25445
PMM1	Q92871	P20340	PMP22	Q01453	O15554
PMPCA	Q10713	P01160	PNKD	Q8N490-3	Q13153
PNMA1	Q8ND90	P56524	PNMA2	Q9UL42	Q7L590
PNMA5	Q96PV4	P41208	PN01	Q9NRX1	Q9NRD5
PNP	P00491	P17612	PNPT1	Q8TCS8	Q8IWL3
PODXL	O00592	P51684	PODXL2	Q9NZ53	P19320
POFUT1	Q9H488	P0DTC8	POGLUT2	Q6UW63	P0DTC8
POGLUT3	Q7Z4H8	P0DTC8	POGZ	Q7Z3K3	P14859, P52747
POLA1	P09884	Q9H668	POLD2	P49005	O60673
POLE	Q07864	P01106	POLE2	P56282	P52594
POLI	Q9UNA4	Q15843	POLR1A	O95602	Q9NYV6
POLR1C	O15160	P23025	POLR1D	P0DPB6	P06396
POLR1E	Q9GZS1	Q9NYV6	POLR1F	Q3B726	Q9NYV6
POLR1G	O15446	O15160, O95602, P19388, P61218	POLR2A	P24928	Q9Y5B0
POLR2D	O15514	Q15648	POLR2G	P62487	Q9Y5B0
POLR2M	P0CAP2	P19388, P62875, P52434	POLR3A	O14802	O15160, P19388, P61218, P62875
POLR3D	P05423	P02753	POLR3E	Q9NVU0	O15160, P53803, P19388, P61218, P62875
POLR3G	O15318	P52292	POLR3H	Q9Y535	P61081
POMK	Q9H5K3	O75197	PON2	Q15165	Q12891
POP1	Q8WXC3	Q9ULZ3	POU2F1	P14859	P14859
POU2F2	P09086-4	P35548	POU3F2	P20265	P56693
POU4F2	Q12837	P24592	POU6F2	P78424	P35680
PPARA	Q07869	P55072	PPFIA1	Q13136	P40222
PPFIA3	O75145	Q9UBB9	PPFIBP1	Q86W92	P62258
PPIB	P23869	P37659	PPIC	P45877	Q8N9N5
PPIF	P30405	P04637	PPIL3	Q9H2H8	O95391
PPIL6	Q8IXY8	O95429	PPL	O60437	Q96QF0
PPM1D	O15297	Q14974	PPM1E	Q8WY54	P61289
PPM1H	Q9ULR3	P62258	PPM1K	Q8N3J5	P11182
PPM1L	Q8BHN0	Q9H3P7	PPP1CB	P62140	Q9UQK1
PPP1R12A	O14974	P01100	PPP1R12B	O60237-2	P06396
PPP1R13L	Q8WUF5	P19838	PPP1R14A	Q96A00	P48729
PPP1R16B	Q96T49	P40937	PPP1R1A	Q13522	Q96LA8
PPP1R2	P41236	P49841	PPP1R26	Q5T8A7	Q08379
PPP1R37	O75864	O75593	PPP1R3B	Q86XI6	P36873
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PPP1R9B	Q96SB3	P38398	PPP2CB	P62714	Q9BRV8
PPP2R1A	P30153	P56211, O43768	PPP2R2A	P63151	P46531
PPP2R2B	Q00005	O15530	PPP2R2C	Q9Y2T4	P17987
PPP2R3A	Q06190	Q9NYZ3	PPP2R5A	Q15172	P56704
PPP2R5B	Q15173	O96017	PPP3R1	P63098	P24592
PPP4R2	Q9NY27	O75533	PPTC7	Q8NI37	P11182
PRAF2	O60831	Q8NBQ5	PRAG1	Q86YV5	P46109
PRCP	P42785	O60883	PRDM1	O75626	Q12857
PRDM6	Q9NQX0	Q13882	PRDX2	P32119	P40763
PRDX3	P30048	O60313	PRDX4	Q13162	P21731
PRDX6	P30041	Q7KZN9	PRELID1	Q9Y255	O43715
PRELID3A	Q96N28	O43715	PREX1	Q8TCU6	P08581
PREX2	Q70Z35	P15056	PRICKLE1	Q96MT3	Q9HAQ2
PRICKLE3	O43900	Q9UJK0	PRIM1	P49642	P09884
PRKAA1	Q13131	O75385	PRKAA2	P54646	P08151
PRKAB1	Q9Y478	O75385	PRKACB	P22694	P13861
PRKAG2	Q9UGJ0	Q13501	PRKCA	P17252	O94844
PRKCB	P05771	P31749	PRKCD	Q05655	Q8N5V2
PRKCE	Q02156	P17252	PRKCI	P41743	Q8N5V2
PRKD1	Q15139	Q9NPC8	PRKD3	O94806	P63027
PRMT3	O60678	P15880	PRMT7	Q582G4	P62805
PRMT8	Q9NR22	Q9Y2W1	PRNP	P04156	P29372
PROCA1	Q8NCQ7-2	P55212	PROCR	Q9UNN8	P25116
PROS1	P07225	P17252	PROSER2	Q86WR7	Q9Y316
PROSER3	Q2NL68	Q96ES7	PRP4K	Q13523	P32121
PRPF3	O43395	Q8WW38	PRPF38A	Q8NAV1	Q96CW1
PRPF38B	Q5VTL8	P61981	PRPF40A	O75400-2	Q9BV20
PRPF6	O94906	P01106	PRPH	P41219	Q13562
PRPSAP2	O60256	P14635	PRR13	Q9NZ81	P30626
PRR14	Q9BWN1	Q13185	PRR16	Q569H4	P78352
PRR3	P79522	Q9NYY8	PRR5	P85299	Q12959
PRRC1	Q96M27	Q15436	PRRG4	Q9BZD6	P46937
PRRT2	Q7Z6L0	Q6UWP7	PRSS23	O95084	P60709
PRSS8	Q16651	P15056	PRXL2C	Q7RTV5	Q9NRD5
PSAP	Q8IWL2	Q9UGM3	PSD3	Q9NYI0	P63104
PSD4	Q8NDX1-2	Q5SW96	PSIP1	O75475	Q03164
PSMB5	P28074	Q9Y4K3	PSMB8	P28062-2	Q13114
PSMB9	P28065	Q15788	PSMD5	Q16401	O00487
PSMD6	Q15008	Q9Y4K3	PSME1	Q06323	P60896
PSME3	P61289	Q8WY54	PSME3IP1	Q9GZU8	Q12933
PSPC1	Q8WXF1	O60563	PSTPIP2	Q9H939	Q05209
PTBP1	P26599	Q9NZI8	PTBP2	Q9UKA9	P0DTD2
PTBP3	O95758	Q92900	PTGER3	P43115	P27037
PTGES2	Q9H7Z7	Q9UG63	PTGFRN	Q9P2B2	O14672
PTGIR	P43119	Q7L8W6	PTGS1	P23219	P48645
PTK2	P34152, Q05397	P12931	PTK7	Q13308	P35222
PTN	P21246	P50454	PTOV1	Q86YD1	P40855
PTP4A3	O75365	Q12974	PTPMT1	Q8WUK0	Q8IUQ4
PTPN11	Q06124	Q7Z6A9	PTPN13	Q12923	P25445
PTPN14	Q15678	P46937	PTPN2	P17706	Q6ZMH5

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PTPN21	Q16825	P51813	PTPN5	P54829	P28482
PTPN6	P29350	P35968	PTPRA	P18433	P12931
PTPRD	P23468	O14965	PTPRF	P10586	Q6PJG9, Q9P244
PTPRG	P23470	Q6VY07	PTPRH	Q9HD43	P00533
PTPRJ	Q62884	P12931	PTPRK	Q15262	P17931
PTPRN	Q16849	Q13332	PTPRO	Q92729	Q13685
PTPRQ	Q15256	Q16539	PTPRS	Q13332	Q16288
PTPRT	O14522	O60469	PTPRU	Q92729	Q13685
PTPRZ1	P23471	Q12860	PTRHD1	Q6GMV3	Q9H244
PTTG1IP	P53801	Q16585	PTX3	P26022	P09038
PUM3	Q15397	P40429	PUS7L	Q9H0K6	Q9Y4X5
PUSL1	Q8N0Z8	P84103	PVR	P15151	P11686
PWP1	Q13610	P48431	PXDN	Q92626	P01282
PXN	P49023	P12931	PYCARD	Q9ULZ3	Q9C000
PYCR1	P32322	Q99497	PYCR3	Q53H96	P42772
PYGL	P06737	Q13451	QPCT	Q16769	Q9Y5Y2
QPRT	Q15274	O00443	R3HDM1	Q15032	P62258
RAB11A	P62491	Q13352	RAB11B	Q15907	P12931
RAB11FIP1	Q6WKZ4	P62258	RAB11FIP2	Q7L804	P62258
RAB11FIP4	Q86YS3	Q8NFZ5	RAB11FIP5	Q9BXF6	P63104
RAB12	Q6IQ22	Q5S007	RAB15	P59190	P52306
RAB17	Q9H0T7	P24386	RAB1A	P62820	Q92696
RAB1B	Q9H0U4	Q92696	RAB20	Q9NX57	P52306
RAB27A	P51159	Q92696	RAB27B	O00194	P51159, Q9HCH5
RAB29	O14966	Q92696	RAB2A	P61019	P51149
RAB32	Q13637	Q92696	RAB33A	Q14088	Q8IUQ4
RAB38	P57729	Q5S007	RAB39A	Q14964	Q08379
RAB39B	Q96DA2	P46059	RAB3A	P20336	Q9HD47
RAB3C	Q96E17	Q15078	RAB3GAP2	Q9H2M9	P13569
RAB40C	Q96S21	P53611	RAB4A	P20338	Q96JA1
RAB6B	Q9NRW1	Q92696, P53611	RAB8B	Q92930	P24386
RAB9B	Q9NP90	Q92696	RABEP2	Q9H5N1	Q9NWB7
RABEPK	Q7Z6M1	Q15569	RABGAP1	Q9Y3P9	P40763
RABGAP1L	Q5R372-2	P06400	RABGGTB	P53611	Q9UBV7
RABL2B	Q9UNT1-2	P06396	RABL6	Q3YEC7-3	P17252
RAC2	P15153	P52306	RAD23A	P54725	Q15843
RAD23B	P54727	Q92681	RAD54L2	Q9Y4B4	Q13185
RADX	Q6NSI4	Q13451	RAI14	Q9P0K7	P04004
RALB	P11234	P11182	RALBP1	Q15311	P43897
RALGDS	Q12967	P15056	RALYL	Q86SE5, Q86SE5-3	P07910
RAMP3	O60896	Q99943	RANBP6	O60518	O43808, O14975
RANBP9	Q96S59	P01275	RAP1A	P62834	P52306
RAP1B	P61224	Q12967	RAP1GAP	P47736	Q684P5
RAP2A	P10114	P53611	RAP2C	Q9Y3L5	Q13371
RAPGEF1	Q13905	P00519	RAPGEF2	Q9Y4G8	P46937
RAPGEF3	O95398	Q13370	RAPGEF4	Q8WZA2	P05455
RAPGEF5	Q92565	Q12955	RAPGEF6	Q8TEU7	P62258
RAPH1	Q70E73	Q99962	RARA	P10276	Q15648
RARB	P10826-2	Q9UBK2	RARG	P13631	Q08999
RARS1	P54136	Q96K83	RASAL2	Q9UJF2	P49674

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RASD2	Q96D21	O75197	RASGRP3	Q8IV61	Q8NBQ5
RASGRP4	Q8TDF6-2	Q12982	RASL10B	Q96S79	Q9H3L0
RASSF10	A6NK89	O96014	RASSF3	Q86WH2	Q13188, Q13043
RASSF5	Q8WWW0	Q8WY64	RASSF7	Q02833	O43639
RASSF8	Q8NHQ8	Q13485	RAVER1	Q8IY67	O75791
RBBP4	Q09028	P07237	RBBP8	Q99708	Q9UQ84
RBFOX1	Q9NWB1-5	O95231	RBFOX2	O43251	Q9NWB1
RBFOX3	A6NFN3	P22732	RBKS	Q9H477	Q9H477
RBL1	P28749	Q9BUH8	RBM10	P98175	Q9GZV8
RBM11	P57052	Q8N2W9	RBM12	Q9NTZ6	P04798
RBM17	Q96125	P21673	RBM19	Q9Y4C8	Q9H9Z2
RBM3	P98179	Q9Y6M1	RBM33	Q96EV2	O75152
RBM47	A0AV96	P0DTC9	RBM4B	Q9BQ04	Q8NB12
RBM6	P78332	P09651	RBM7	Q9Y580	P52298
RBM8A	Q9Y5S9	Q92900	RBMS1	P29558	P84103
RBMS2	Q15434	O00425	RBMX	P38159	P10636-8
RBMX2	Q9Y388	Q9NR30	RBP1	P09455	Q9UBN6
RBP7	Q96R05	Q8WXH5	RBPMS	Q93062	Q53HV7
RBSN	Q9H1K0	Q9NRW7	RCAN1	P53805	P16298
RCC1L	Q96I51	O00425	RCL1	Q9Y2P8	Q08379
RCN1	Q15293	Q9H4P4	RCN3	Q96D15	Q9UMX1
RCOR1	Q9UKL0	O95863	RCOR2	Q8IZ40	Q92769
RCVRN	P35243	Q53FT3	RDH10	Q8IZV5	Q05329
RDH14	Q9HBH5	P25103	RDH5	Q92781	O43929
REC8	O95072	Q9UNE7	RECK	Q6P9E2	Q96LC9
RECQL4	O94761	Q9UBX5	REEP1	Q9H902	P21291
REEP3	Q6NUK4	P62258	REEP4	Q9H6H4	Q13190
REL	Q04864	P48065	RELA	Q04206	Q13568
RELB	Q01201	P19838, Q8NFFZ5	RELCH	Q9P260	P21709
RELL1	Q8IUW5	P26572	RELL2	Q8NC24	O75396
REPS1	Q96D71	Q2M2I8	RESF1	Q9HCM1	Q9NV70
REST	Q13127	P07900	RET	P07949	P22455
RETREG2	Q8NC44	Q9GZQ8	RETSAT	Q6NUM9	Q8WTR4
REV3L	O60673	Q9UI95	REX1BD	Q96EN9	O00487
REXO5	Q96IC2	Q15185	RFC5	P40937	P06400
RFTN1	Q14699	P18827	RFTN2	Q52LD8	Q70CQ3
RFX2	P48378	O75928-2	RFX3	P48380	Q12933
RGR	P47804-3	P06396	RGS17	Q9UGC6	Q9H2F3
RGS19	P49795	Q12837	RGS2	P41220	Q9P244
RGS20	O76081-6	Q8WUK0	RGS4	P49798	P00533
RGS6	P49758	P08238	RHBDD1	Q8TEB9	O75396
RHBDD2	Q6NTF9-3	P50454	RHBDF1	Q96CC6	Q16539
RHBDF2	Q80WQ6	P78536	RHEB	Q15382- PRO_0000082708	O43924
RHEBL1	Q8TAI7	Q9UKG1	RHEX	Q6ZWK4	Q05329
RHOB	P62745	Q92888	RHOBTB1	O94844	P61081
RHOC	P08134	P52565	RHOD	O00212	Q9Y4D1
RHOF	Q9HBH0	P52306	RHOG	P84095	P52565
RHOJ	Q9H4E5	Q13177	RHOJ	Q7L0Q8	Q14289
RHPN2	Q8IUC4	P61981	RIC1	Q4ADV7	P20340

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RILPL1	Q5EBL4-3	Q86T03	RIMKLA	Q8IXN7	Q96EN8
RIMS1	Q86UR5	P06241	RIMS3	Q9UJD0	Q9NRD5
RIMS4	Q9H426	Q96FA3	RIN3	Q8TB24	P19174
RINT1	Q6NUQ1	Q9Y639	RIOK1	Q9BRS2	O14744
RIOK3	O14730	Q92851	RIOX2	Q8IUF8	P06748
RIPK1	Q13546	Q13546, Q9Y572	RIPK2	O43353	Q13114
RIPK4	P57078	P98170	RIPOR2	Q9Y4F9, Q9Y4F9-2	P61586
RIPOR3	Q96MK2	P61586	RIPPLY2	Q5TAB7	P42858
RIPPLY3	P57055	Q13416	RMC1	Q96DM3	P07196
RMDN2	Q96LZ7	Q9Y4Z2	RMI1	Q9H9A7	Q14159
RMI2	Q96E14	Q14159	RNASEH1	O60930	P17152
RND1	Q92730	O43157	RND2	P52198	O43157
RND3	P61587	O43157	RNF11	Q9Y3C5	Q8N5V2
RNF112	Q9ULX5	O14832	RNF122	Q9H9V4	O95674
RNF125	Q96EQ8	P55072	RNF130	Q86XS8	Q6UX41
RNF135	Q8IUD6	P40222	RNF139	Q8WU17	Q8TCT9
RNF144A	P50876	Q96AA3	RNF144B	Q7Z419	Q07812
RNF145	Q96MT1	P0DTC5	RNF146	Q9NTX7	Q86WJ1
RNF149	Q8NC42	Q30201	RNF150	Q9ULK6-3	P17252
RNF152	Q8N8N0	Q15125	RNF167	Q9H6Y7	Q9UNA3
RNF182	Q8N6D2	P51668	RNF185	Q96GF1	P62837
RNF19A	Q9NV58	P00441	RNF20	Q5VTR2	O96017
RNF208	Q9H0X6	P49247	RNF213	Q63HN8	Q13304
RNF216	Q9NWF9	O14933	RNF217	Q8TC41	P62736
RNF38	Q9H0F5-2	P42858	RNF41	Q9H4P4	P09681
RNF43	Q68DV7	P62837	RNF44	Q7L0R7	Q8TDS5
RNF5	Q99942	P62837	RNF6	Q9Y252	P09917
RNF7	Q9UBF6	Q8WXH5	RNFT1	Q5M7Z0	Q9NRM6
RNH1	P13489	Q12834	RNMT	O43148	P06730
ROBO1	Q9Y6N7	O94813	ROBO3	Q96HH0	Q9NRD5
ROCK1	Q13464	P61586	ROPN1L	Q96C74	P24588
ROR1	Q01973	P12931	ROR2	Q01974	O95231
RORA	P35398-4	Q9BZS1	RORB	Q92753-1	Q96GM5
RPA2	P15927	P23025	RPAIN	Q86UA6	O15315
RPE65	Q16518	Q9UBW7	RPGR	Q92834-2, Q92834	O43924
RPH3A	Q9Y2J0	O14936	RPL12	P30050	P27824
RPL23A	P62750	Q01105	RPL36A	P83881	P63244
RPL7L1	Q6DKI1	P40429	RPL8	P62917	Q15843
RPLP1	P05386	O43809	RPLP2	P05387	Q9H9Z2
RPN1	P04843	Q14165	RPN2	P04844	Q14165
RPP38	P78345	P06748	RPS13	P62277	Q99558
RPS16	P62249	Q15843	RPS2	P15880	P05455
RPS27A	P62979	P08311	RPS27L	Q71UM5	Q00987
RPS6KA2	Q15349	Q12849	RPS6KA4	O75676	Q16539
RPS6KA5	O75582	Q16539	RPS6KB1	P23443-2	P08151
RPS6KB2	Q9UBS0	P67870	RRAGA	Q7L523	P61916
RRAS	P10301	Q9UBH0	RRAS2	P62070	Q92696, P53611
RRBP1	Q9P2E9	P13569	RREB1	Q92766	Q92769
RRM2B	Q7LG56	O43929	RRP1	P56182	P04004

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
RRP12	Q5JTH9	Q9H9Z2	RRP15	Q9Y3B9	Q9H9Z2
RRP1B	Q14684-1	P19338	RRS1	Q15050	P19338
RSBN1	Q5VWQ0	Q9BYW2	RSF1	Q96T23	P09429
RSKR	Q96LW2	O95231	RSL1D1	O76021	P78563
RSPH9	Q9H1X1	Q9UPT8	RSPO2	Q6UXX9-2	Q92504
RSPO4	Q2I0M5	O14964	RSU1	Q15404	Q9NVD7, Q9HBI1, P48059
RTL8B	Q17RB0	O00560	RTL8C	A6ZKI3	P42772
RTN1	Q16799-3	O00560	RTN3	O95197-3	Q9NTJ5
RTN4RL1	Q86UN2	Q7Z3S9, P0DPK4	RTN4RL2	Q86UN3	P17931
RTP5	Q14D33	Q86U10	RUBCNL	Q9H714	Q9P2Y5
RUNDC3A	Q59EK9	Q96CW1	RUNDC3B	Q96NL0	P78368
RUNX1	Q01196	Q9H2X6	RUNX2	Q13950	P17480
RUSC1	Q9BVN2	P42772	RUVBL1	Q9Y265	O96019, O94805
RWDD2B	P57060	Q96HC4	RXYLT1	Q9Y2B1	O75072
RYK	P34925	Q68D85	RYR1	P11716	P62942
S100A10	P60903	P07355	S100A11	P31949	P04637
S100A2	P29034	P35579	S100B	P04271	Q15109
S100PBP	Q96BU1	O43809	S1PR1	P21453	Q7Z2K8
S1PR2	O95136	P17677	S1PR3	Q99500	P21964
SALL1	Q9NSC2	Q13547	SALL2	Q9Y467	P08138
SAMD11	Q96NU1	Q12933, Q13077	SAMD3	Q8N6K7-2	P04156
SAMD4A	Q9UPU9	Q08379, P62258	SAMD4B	Q5PRF9	P62258
SAMHD1	Q9Y3Z3	O75923	SAP130	Q9H0E3	P48431
SAP30	O75446	Q00403	SAP30BP	Q9UHR5	Q08379
SAR1A	Q9NR31	P78382	SARM1	Q6SZW1	Q99836
SASH1	O94885	P61981	SAT2	Q96F10	P21673
SAV1	Q9H4B6	P18509	SBDS	Q9Y3A5	Q15149
SCAI	Q8N9R8	Q13526	SCAMP2	O15127	P13569
SCAMP4	Q969E2	Q8WTR4	SCAMP5	Q8TAC9	P49638
SCAP	Q9NP31	O14543	SCAPER	Q9BY12	P20248
SCARA5	Q6ZMJ2-2	Q969S0	SCARB1	Q8WTV0	P11686
SCARB2	Q14108	Q9UBD6	SCARF2	Q96GP6	P06241
SCD	O00767	P08034	SCFD1	Q8WVM8	P16234
SCG2	O00255-2, O00255	P35579	SCG3	Q8WXD2	O00560
SCHIP1	P0DPB3	Q9UIL1	SCLT1	Q96NL6	A1L190
SCML2	Q9UQR0-1	P38936	SCN2B	O60939	P23468
SCN3B	Q9NY72	Q96L58, O75063	SCNN1A	P37088	P46934
SCO1	O75880	Q14061	SCUBE3	Q8IX30	P37173
SCYL1	Q96KG9-4	O15169	SDAD1	Q9NVU7	P06748, P62753
SDC2	P34741	P09038	SDC3	A0A0S2Z4U3	P63252
SDC4	P31431	O15530	SDCBP	O00560	Q5VTY9
SDF2L1	Q9HCN8	Q9H5K3	SDF4	Q9BRK5	P50548
SDHB	P21912	Q4G176	SDK2	Q58EX2	P17931
SEC11C	Q9BY50	P09601	SEC13	P55735	P49674
SEC14L1	SEC14L1	Q07955	SEC22B	O75396	O95249, Q13190
SEC23A	Q15436	P50570	SEC23IP	Q9Y6Y8	A1L4H1
SEC24A	O95486	Q15436	SEC24B	O95487	Q15436
SEC24D	O94855, O94855-2	Q15436	SEC31A	O94979	Q15436
SEC61A1	P61619	Q16288	SEC61A2	Q9H9S3	P04049

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SEC61B	P60468	P51572	SEC62	Q99442	O43914
SEC63	Q9UGP8	Q8N6T3	SEH1L	Q96EE3	Q15274
SEL1L3	Q68CR1	Q8IV16	SELENOF	O60613	Q8TC12
SELENOM	Q8WWX9	P61604	SELENOS	Q9BQE4	Q6Q788
SELENOT	P62341	P40855	SEMA3B	Q13214-2	Q9NRD5
SEMA3C	Q99985	P18827	SEMA3E	O15041, P70275	Q9Y4D7
SEMA4A	Q9H3S1	Q9Y4D7	SENP1	Q9P0U3	P46060
SENP7	Q9BQF6	Q13185	SEPSECS	Q9HD40	P04591
SEPTIN3	Q9UH03	Q15019	SEPTIN4	O43236	P05067
SEPTIN7	Q16181	Q15019	SEPTIN8	Q92599	Q15019
SEPTIN9	Q9UHD8	Q15019	SERBP1	Q8NC51	P50454
SERINC1	Q9NRX5	Q06710	SERP1	Q9Y6X1	Q12846
SERP2	Q8N6R1	Q9H2H9	SERPINA1	P01009	P08246
SERPINA5	P05154	P09958	SERPINB1	P30740	P48431
SERPINB2	P05120	Q9NRY4	SERPINB7	O75635	Q9NYY3
SERPINB8	P50452	Q9Y324	SERPINB9	P50453	Q11203
SERPINE1	P05121	P04004	SERPING1	P05155	P00750
SERPINH1	P50454	P12830	SERTAD2	Q14140	O60733
SERTAD4	Q9NUC0	P54652	SESN2	P58004	Q13501
SESTD1	Q86VW0	Q15561	SETD7	Q8WTS6	P26358
SETDB1	Q15047-2	O94844	SEZ6L2	Q6UXD5	P17931
SF1	Q13285	P40937	SF3A2	Q15428	Q09161
SFI1	A8K8P3	P41208	SFMBT1	Q9UHJ3	P62805
SFRP4	Q6FHJ7	Q8NBQ5	SFT2D2	O95562	Q8NBQ5
SFXN5	Q8TD22	P46059	SGCB	Q16585	Q01453
SGCG	Q13326	O75923	SGK1	O00141	P49815
SGK3	Q96BR1	Q96J02	SGPP1	Q9BX95	Q8WTR4
SGSM1	Q2NKG1-4, Q2NKG1	P41208	SH2B2	O14492-2	Q9UBH0
SH2D3A	Q9BRG2	P00533	SH2D4A	Q9H788, Q9H788-2	O14936, Q9UJU6
SH3BGRL	O75368	P00533	SH3BP2	P78314	Q9UJU6
SH3BP4	Q9P0V3	O14908	SH3BP5	O60239	Q06187
SH3D19	Q5HYK7	P50570	SH3GL1	Q99961	O14672
SH3GL2	Q99962	Q99962	SH3GLB1	Q9Y371	P01730
SH3RF1	Q7Z6J0	P62258	SH3RF2	Q08AM8, Q8TEC5	Q08379
SH3RF3	Q8TEJ3	Q13177	SHANK1	Q9Y566	P46937
SHANK2	Q9UPX8	P19174	SHARPIN	Q9H0F6	P08648
SHB	Q15464	P00519	SHC3	Q92529	Q86U10
SHC4	Q6S5L8	P00533	SHD	Q96IW2	O43639
SHH	Q62226, Q15465	Q4KMG0, Q96QV1	SHISA5	Q8N114	Q9NZC7
SHISA6	Q6ZSJ9	P38936	SHISAL1	Q3SXP7	P29033
SHLD2	Q86V20-2	Q12888	SHOX2	O60902-3	Q14957
SHPRH	Q149N8-1	P61088	SHQ1	Q6PI26	Q9Y265
SHROOM2	Q13796	P61981	SHROOM3	Q8TF72	P62258
SIGMAR1	Q99720	P14416	SIK2	Q9H0K1	P55072
SIL1	Q9H173	P11021	SIM2	Q14190	Q15185
SIMC1	Q8NDZ2	Q8IWZ6	SIN3A	Q96ST3	O95863
SIN3B	O75182, Q62141	P01106	SINHCAF	Q9NP50	Q9Y6N5
SIPA1L3	O60292	P62258	SIRPA	P78324	Q08722
SIRT6	Q8N6T7	P01106	SIX1	Q15475	Q99502
SIX2	Q9NPC8	Q7Z5W3	SKAP2	O75563	Q9Y2R2

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SKI	P12755	P36941	SKIC2	Q15477	P06748
SKIC3	Q6PGP7	P06748	SKIDA1	Q1XH10	Q15022
SKIL	P12757	P23025	SKP2	Q13309	Q13415
SLAIN1	Q8ND83	P09917	SLAIN2	Q9P270	P61981
SLBP	Q14493	Q14493	SLC10A7	Q0GE19	P04114
SLC12A4	Q9UP95	P20340	SLC12A7	Q9Y666	P43119
SLC12A8	A0AV02	P26367	SLC15A3	F5H1C8	Q14974
SLC16A1	P53985	P35613	SLC16A13	Q7RTY0	Q96BA8
SLC16A2	P36021	P21926	SLC16A3	O15427	P35613
SLC16A6	O15403	O76024	SLC18A1	P54219	Q7L8W6
SLC19A1	P41440	Q03135	SLC1A3	P43003	Q8WVV5
SLC20A2	Q08357	P43088	SLC22A15	Q8IZD6	Q86TM6
SLC22A17	Q8WUG5	P07237	SLC22A5	O76082	Q9H015
SLC24A3	Q9HC58	Q86TM6	SLC25A16	P16260	Q15116
SLC25A22	Q9H936	Q5S007	SLC25A23	Q9BV35	Q8N695
SLC25A24	Q6NUK1	P13569	SLC25A28	Q96A46	P46059
SLC25A30	Q5SVS4	Q07954	SLC25A40	Q8TBP6	P02452
SLC25A43	Q8WUT9	P10809	SLC25A46	Q96AG3	Q8N695
SLC25A48	Q6ZT89	Q13077	SLC26A8	Q96RN1	Q9H0H5
SLC26A9	Q7LBE3-1	P13569	SLC27A1	Q6PCB7	P40855
SLC27A3	Q5K4L6	P06576	SLC27A6	Q9Y2P4	Q9Y2C2
SLC30A1	Q9Y6M5	P43119	SLC30A10	Q6XR72	Q9BRI3
SLC30A5	Q8TAD4	P21217	SLC30A7	Q8NEW0	Q9Y2C2
SLC31A1	O15431	Q9H9P8	SLC32A1	Q9H598	Q9P0P0
SLC34A2	O95436	Q15904	SLC35A2	P78381-1, P78381-2	Q9Y2D2
SLC35B4	Q969S0	O43914	SLC35C1	Q96A29	P05090
SLC35E3	Q7Z769	P49638	SLC35F1	Q5T1Q4	P21964
SLC35F2	Q8IXU6	Q9H5K3	SLC35F5	Q8WV83	P61073
SLC35F6	Q8N357	P38606	SLC37A3	Q8NCC5	Q9Y4D1
SLC38A10	Q9HBR0	Q30201	SLC38A2	Q96QD8	P43657
SLC39A12	Q504Y0	Q9BPX6	SLC39A13	Q96H72	O15173
SLC39A14	Q15043	P48960	SLC39A7	Q92504	Q9UBV7, O75063
SLC39A8	Q9C0K1	Q09328	SLC40A1	Q9JHI9	O60674
SLC41A2	Q96JW4	P24593	SLC43A2	Q8N370	P0DPK4
SLC44A1	Q8WWI5	O14672	SLC44A2	Q8IWA5	P0DTC4
SLC44A3	Q8N4M1-3	Q8WTS1	SLC44A5	Q8NCS7	P50454
SLC50A1	Q9BRV3	O00560	SLC52A3	Q9NQ40	P54252
SLC66A2	Q8N2U9	Q9NR28	SLC6A1	P30531	Q9H3Z4
SLC6A13	Q9NSD5-3	P04792	SLC6A5	P23975	Q9NRD5
SLC6A8	P48029	P43119	SLC7A14	Q8TBB6	P15291
SLC7A2	P52569	P30825	SLC7A3	Q8WY07	P22732
SLC7A6	Q92536	Q9UPQ8	SLC7A8	Q9UHI5	O60906
SLC9A6	Q92581	Q8NAT1	SLC9B2	Q86UD5	O15173
SLCO1C1	Q9NYB5	P78352	SLCO4C1	Q6ZQN7	Q14527
SLF1	Q9BQI6	P19838	SLF2	Q8IX21	P25786
SLIT1	O75093	P49639	SLIT2	O94813	O94813
SLITRK3	O94933	O76024	SLITRK4	Q8IW52	Q9BT09
SLITRK5	O94991	Q6PJG9	SLMAP	Q14BN4	Q9BRV8
SLPI	P03973	P49639	SLX4	Q8IY92	P43246
SLX4IP	Q5VYV7	Q16539	SMAD2	Q15796	P98082

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SMAD3	P84022	P98082	SMAD6	O43541-2, O35182	O95630
SMAD7	O15105	P62942	SMAD9	O15198, O15198-2	Q13485
SMAP2	Q9H0E9	O96019	SMARCC2	Q8TAQ2	Q92925, Q8NFD5, P51531, O96019
SMARCD1	Q96GM5	Q9H8M2	SMARCD2	Q92925	O96019, Q15532
SMC5	Q8IY18	Q9H668	SMCO4	Q9NRQ5	P20645
SMG5	Q9UPR3	P04637	SMG6	Q86US8	P55212
SMG7	Q92540	Q92900	SMIM1	B2RUZ4	Q86WA9
SMIM5	Q71RC9	Q08426	SMOC1	Q9H4F8	Q04721
SMPD2	O60906	P49675	SMPD3	Q9NY59	P04114
SMTN	P53814-5	O76024	SMTNL2	Q2TAL5	P20248
SMYD3	Q9H7B4	Q15915	SNAI1	O95863	P49841
SNAP23	O00161	Q16623, Q13277	SNAP25	P60880-2	P63027
SNAP47	Q5SQN1	P51809	SNAP91	O60641	P38398
SNAPC4	Q5SXM2	Q9UPP1	SNCA	P37840	Q16143
SNCAIP	Q9Y6H5, Q9Y6H5-2	P37840	SNCB	Q16143	P05067
SND1	Q7KZF4	P11362	SNPH	O15079	P23025
SNRNP200	O75643	P01106	SNRPD3	P62318	Q13573
SNTA1	Q13424	P60484	SNTB1	Q13884	P60484
SNTB2	Q13425	O14936	SNTG1	Q9NSN8	Q9NT99
SNX10	Q9Y5X0	P17252	SNX17	Q15036	P19174
SNX18	Q96RF0	O14672	SNX24	Q9Y343, Q9Y343-2	P49674
SNX27	Q96L92	P27037	SNX29	Q8TEQ0	Q13077
SNX3	O60493	P56539	SNX30	Q8WV41	Q05193
SNX31	Q8N9S9-2	P63000	SNX4	O95219	Q6ZPD8
SNX5	Q9Y5X3	Q9Y639	SNX6	Q9UNH7	Q5T8I9
SNX7	Q9UNH6	O14964	SNX9	Q9Y5X1	Q05193
SOBP	A7XYQ1	Q13363	SOCS1	O15524	P46109
SOCS3	O14543	Q15369, Q15370	SOCS5	O75159	Q93034, Q9UBF6
SOCS6	O14544	P46937	SOCS7	Q8WXH5	Q8N3R9
SORBS1	Q9BX66	P00519	SORBS2	O94875	Q13177
SORBS3	O60504-2	P00519	SORL1	Q92673	Q15669
SORT1	Q99523	P01138	SOS2	Q07890	P19174
SOX10	P56693	O00712, Q12857	SOX12	O60248	O00712, Q12857
SOX13	Q9UN79	Q12933	SOX2	P48431	P30740
SOX21	Q9Y651	P48431	SOX3	P41225	Q12933
SOX6	P35712	O00712, P08651, Q12857	SOX9	P48436	Q12857
SP100	P23497	Q9Y6K1	SP3	Q02447	P03372
SP6	Q3SY56	P17612	SPACA6	W5XKT8	Q8N9N5
SPAG8	Q99932-2	P10745	SPAG9	O60271	P19838
SPANXB1	Q9NS25	Q9BZL1	SPART	Q8N0X7	P19338
SPATA13	Q96N96-6	P51570	SPATA2	Q9UM82	Q9UI12
SPATA20	Q8TB22	P47985	SPATA2L	Q8IUW3	P02649
SPATA6	Q9NWH7	Q92466	SPATS2L	Q9NUQ6	Q9H9Z2
SPC24	Q8NBT2	Q96ES7	SPCS3	P61009	P27824
SPDL1	Q96EA4	P25963	SPECC1L	Q69YQ0	P25963
SPEF1	Q9Y4P9	O96006	SPEG	Q15772-4	Q9NRD5
SPG21	Q9NZD8	P42772	SPHK1	Q9NYA1-2	Q7Z4W1
SPIN1	Q9Y657	Q86U90	SPIN4	Q56A73	Q96LA8

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SPINT1	O43278-2	P02787	SPINT2	O43291	Q99640
SPIRE1	Q08AE8	P60709	SPIRE2	Q8WWL2-2	O14964
SPOCK1	Q08629	Q96BA8	SPOPL	Q6IQ16	Q99836
SPPL2A	Q8TCT8	O76024	SPPL3	Q8TCT6	Q9H015
SPRED1	Q7Z699	Q9UBX5	SPRED2	Q7Z698	P27797
SPRY1	O43609	O43597	SPRY2	O43597	P22681
SPRY4	Q9C004	P22681	SPRYD4	Q8WW59	P24534
SPRYD7	Q5W111	Q86UT5	SPSB1	Q96BD6	Q13153
SPSB4	Q96A44	Q15369, Q15370, Q93034	SPTA1	P02549	Q9NQ4T
SPTB	P11277	P05067	SPTBN1	Q01082	Q13813
SPTLC1	O15269-2	Q8WXH5	SPTY2D1	Q68D10	P78563
SPX	Q9BT56	Q6IN84	SQSTM1	Q13501	Q9UDY8
SRCIN1	Q9QWI6, Q9C0H9	P56945	SREBF1	P36956-PRO_0000314029	P20700, P02545-1
SREBF2	Q12772	P08047	SRF	P11831	Q969V6
SRGAP2	O75044	O75044	SRI	P30626	P30626
SRM	P19623	P19320	SRP54	P61011	P11142
SRP68	Q9UHB9	P46087	SRP72	O76094	P48431
SRPRA	P08240	Q6UWB1	SRPRB	Q9Y5M8	Q99650
SRPX	P78539	Q9BRX2	SRR	Q9GZT4	Q9H4P4
SRRM1	Q8IYB3	P40429	SRRM4	A7MD48	P61964
SRSF11	Q05519	Q92844	SRSF2	Q01130	Q9HD47
SRSF3	P84103	P09429	SS18	Q15532	Q92925
SSB	P05455	Q8WZA2	SSBP2	P81877	Q9UBH0
SSBP4	Q9BWG4	Q9UBH0	SSC5D	A1L4H1	O00560
SSH1	Q8WYL5	P53667	SSH2	Q76I76	P62258
SSH3	Q8TE77	O43639	SSR1	P43307	P27824
SSR3	Q9UNL2	P27824	SSR4	P51571	Q5S007
SST	P61278	P05067	SSTR2	P30875, P30874	P27986
SSU72	Q9NP77	O60216	SSUH2	Q9Y2M2-2	Q969R5
ST3GAL1	Q11201	Q9ULF5	ST6GALNAC1	Q9NSC7	P13569
ST8SIA1	Q92185	Q96CV9	ST8SIA3	O43173	P62987
ST8SIA4	Q92187	Q92854	STAC	Q99469	Q02548
STAMPB	O95630	O00443	STAMPBPL1	Q96FJ0	O43166
STAP2	Q9UGK3	Q99836	STAR	P49675	O60906
STARD3	Q14849	P46059	STAT1	P42224	Q05397
STAT2	P52630	O94782	STAT3	P40763	P12931
STAT5A	P42229	P10515	STAT5B	P51692	P00519
STAT6	P42226	Q9UHD2, P42226	STING1	Q86WV6	Q96N66
STK17B	O94768	P08173	STK26	Q9P289	Q9BRV8
STK3	Q13188	P18509	STK32C	Q86UX6	O14964
STK36	Q9NRP7	P08151	STK38	Q15208	Q9Y5B0
STK40	Q8N2I9	Q8NHY2	STMN2	Q93045	P40222
STMN3	Q9NZ72	O14964	STMN4	Q9H169-2	Q9NR28
STOM	P27105	Q96N66	STOX1	Q6ZVD7	Q01974
STOX2	Q9P2F5	P49674	STRA6	Q9BX79-6	O76003
STRADB	Q9C0K7	Q15831	STRIP1	Q5VSL9	Q9BRV8
STRIP2	Q9ULQ0	Q9BRV8	STT3A	P46977	Q9H5K3
STT3B	Q8TCJ2	Q14165	STX11	O75558	P60880

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TBL2	Q9Y4P3	O15530	TBRG4	Q969Z0	Q9Y639
TBX15	Q96SF7	O94955	TBX18	O95935	Q13077
TBX2	Q13207	Q6DKK2	TBX3	O15119	Q08117
TBXA2R	Q0VAB0	P29033	TC2N	Q8N9U0	Q96JCB
TCAF1	Q9Y4C2	P01275	TCAP	O15273	Q13153
TCEA3	O75764	Q8WVC0	TCEAL1	Q15170	O14757
TCEAL8	Q8IYN2	P04792	TCEAL9	Q9UHQ7	Q6VN20, Q96S59
TCEANC2	Q96MN5	Q9NRD5	TCF12	Q99081	P30626
TCF20	Q9UGU0	Q92547	TCF21	O43680	O00560
TCF7	P36402	P35222	TCHH	Q07283	P24385
TCIRG1	Q13488	Q15904	TCN2	P20062	Q9NPF0
TCP11L2	Q8N4U5	O14964	TCTA	P57738	P78382
TDG	Q13569	Q9GZX7	TDRD7	Q8NHU6	P42771
TEAD3	Q99594	P46937	TEAD4	Q15561	P46937
TEDC2	Q7L2K0	Q7Z3S9	TEF	Q10587	Q16236
TEK	Q02763	P29353	TEKT1	Q969V4	O00560
TENM2	Q9NT68	P21926	TENM3	Q9P273	P21926
TENM4	Q6N022	Q92608	TENT5A	Q96IP4	P43405
TENT5B	Q96A09	O95947	TEP1	P60484	Q8NI35
TERF2IP	Q9NYB0	O75347	TES	Q9UGI8	P13569
TET1	Q8NFU7	O15294	TET3	Q8BG87, O43151	O15294
TEX264	Q9Y6I9	Q13370	TF	P02787	O00501
TFAP2B	Q92481	Q92993	TFAP2C	Q92754	P63279
TFB2M	Q9H5Q4	O00411	TFCP2	Q12800	P03428
TFDP1	Q14186	Q08999, P06400	TFDP2	Q14188	Q08999, P06400
TFE3	P19532	O75030	TFEB	P19484	P62258
TFF3	Q07654	Q07654	TFG	Q92734	O75593
TFRC	P02786	Q9P0V3	TGFB1	P01137	P07237
TGFB1I1	Q62219, O43294	Q05397	TGFB2	P61812	P05067
TGFB3	P10600	P37173	TGFB2R2	P37173	Q99741
TGFBRAP1	Q8WUH2	Q9UBF6	TGIF1	Q15583	O15169
TGIF2	Q9GZN2	Q96ST3	TGM2	P21980	P02751
TGM5	O43548	P0CG47	THAP10	Q9P2Z0	P61289
THAP4	Q8WY91	P24593	THBD	P07204	P55056
THBS1	P07996-PRO_0000035842	P16671	THBS2	P35442	Q6Y288
THBS3	P49746	Q9P0V3	THNSL1	Q8IYQ7	Q9NPC8
THOC2	Q8NI27	Q13769	THOC6	Q86W42	Q13838
THPO	P40225	P40238	THRB	P10828	P60709
THSD7A	Q9UPZ6	P21964	THSD7B	Q9C0I4	Q9H2K0
THYN1	Q9P016	Q9UJH6	TIAL1	Q01085-2	Q9Y5V3
TIAM1	Q13009	P18827	TICAM1	Q8IUC6	Q9UHD2
TIE1	P35590	P04629	TIFA	Q96CG3	O60443
TIGAR	Q9NQ88	P55212	TIGD1	Q96MW7	P42858
TIGD4	Q8IY51	Q12933	TIGD7	Q6NT04	Q9BZF9
TIGIT	Q495A1	P15151, Q9NQS3	TIMM23	O14925	Q9BVV7
TIMMDC1	Q9NPL8	O75063	TIMP1	P01033	P14780, P09238
TIMP2	P16035	P50281, P08253	TIMP3	P35625	P78536
TINAGL1	Q9GZM7	O75386	TJP1	Q07157	P59637
TK1	P04183	Q9UM11	TLCD4	Q96MV1	P24593

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TLE2	Q04725	P11142	TLE3	Q04726-4	Q15475, Q9NPC8
TLE4	Q04727	O95231	TLK1	Q9UKI8	P26367
TLN1	Q9Y490	O60331	TLR2	O60603	Q99836
TLR3	O15455	P12931	TLR6	Q9Y2C9	O60603
TLX1	P31314	P14921	TM2D1	Q9BX74	P05067
TM2D2	Q9BX73	P29033	TM2D3	Q9BRN9	P0DTC8
TM6SF1	Q9BZW5-2	P06396	TM9SF3	Q9HD45	Q9H244
TMA16	Q96EY4	Q96CW1	TMBIM1	Q969X1	Q8NCR0
TMBIM6	P55061	Q9H5K3	TMC7	Q7Z402	P50747
TMCO3	Q6UWJ1	P00533	TMED1	Q13445	Q15363
TMED10	P49755	Q9Y3A6	TMED2	Q15363	Q9Y3A6
TMED3	Q9Y3Q3	Q15363	TMED5	Q9Y3A6	Q15363
TMED7	Q9Y3B3	P31371	TMED8	Q6PL24	Q12893
TMED9	Q9BVK6	Q9Y3A6	TMEFF1	Q8IYR6	Q13627
TMEFF2	Q9UIK5	Q9Y5Z9	TMEM102	Q8N9M5	Q96QF0
TMEM106A	Q96A25	Q16585	TMEM106C	Q9BVX2	P11686
TMEM107	Q6UX40	Q8N661	TMEM108	Q6UXF1	Q9NYV6
TMEM109	Q9BVC6	Q9H2K0	TMEM120A	Q9BXJ8	P16410
TMEM130	Q8N3G9	P22732	TMEM131	Q92545	P0DTC3
TMEM132A	Q24JP5	Q01523	TMEM132D	Q14C87	P62136
TMEM143	Q96AN5	Q6ZPD8	TMEM144	Q7Z5S9	Q9NPL8
TMEM14A	Q9Y6G1	P49682	TMEM14C	Q9P0S9	A1L3X0
TMEM150A	Q86TG1	O60563	TMEM154	Q6P9G4	Q15052
TMEM163	Q8TC26	Q9Y6M5, Q9BRI3	TMEM165	Q9HC07	P13569
TMEM169	Q96HH4	P08183	TMEM176A	Q96HP8	Q9H2K0
TMEM176B	Q3YBM2	P17252	TMEM184A	Q6ZMB5	Q4G176
TMEM184B	Q9Y519	O15357	TMEM190	Q8WZ59	O00501
TMEM192	Q8IY95	Q9UBV7	TMEM201	Q5SNT2-2	Q8N183
TMEM205	Q6UW68	O14653	TMEM214	Q6NUQ4	O75326
TMEM218	A2RU14	Q9NXB9	TMEM229B	Q8NBD8	O75396
TMEM230	Q96A57	P27824	TMEM231	Q9H6L2	Q7Z3S9
TMEM243	Q9BU79	Q6IN84	TMEM248	Q9NWD8	Q8N661
TMEM255A	Q5JRV8	Q96J02	TMEM255B	Q8WV15	Q6ZPD8
TMEM258	P61165	P0DTC4	TMEM266	Q2M3C6	O60341
TMEM30A	Q8VEK0	P05067-4	TMEM35A	Q53FP2	O15155, O14653
TMEM38B	Q9NVV0	Q8WTS1	TMEM39B	Q9GZU3	O75326
TMEM43	Q9BTV4	Q6Q788	TMEM45A	Q9NWC5	P24593
TMEM47	Q9BQJ4	P42858	TMEM51	Q9NW97	Q96J02
TMEM52B	Q4KMG9	P24593	TMEM59	Q9BXS4	Q13332
TMEM63A	O94886	P43119	TMEM63B	Q5T3F8	P43119
TMEM65	Q6PI78	P15151	TMEM79	Q9BSE2	O00631
TMEM86A	Q8N2M4	O43914	TMEM88	Q6PEY1	P17152
TMEM92	Q6UXU6	Q96J02	TMEM94	Q12767	P80370
TMEM98	Q9Y2Y6	P08034	TMF1	P82094	P53350
TMOD2	Q9NZR1	Q9UM54	TMTC1	Q8IUR5-4	P06396
TMTC3	Q6ZXV5	Q6PIU2	TMTC4	Q5T4D3	Q9Y4Z2
TNFAIP1	Q13829	P12931	TNFAIP3	P21580	Q14790
TNFAIP8	O95379	O00560	TNFRSF10A	O00220	Q14790
TNFRSF10D	Q9UBN6	Q05329	TNFRSF11B	O00300	Q9Y6Q6
TNFRSF12A	Q9NP84	Q12933	TNFRSF19	Q9NS68	P52565

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TNFRSF1A	P19438	Q13546	TNFRSF21	O75509	P08138
TNFRSF8	P28908	Q99741	TNFRSF9	Q07011	Q9ULF5
TNFSF13	O75888-3	P05067	TNFSF14	O43557	Q92956
TNFSF15	O95150	P42858	TNFSF4	P23510	P43489
TNFSF9	P41273	P09238	TNIP2	Q8NFZ5	P41279, P19838
TNIP3	Q96KP6	P21580	TNK1	Q13470	P62258
TNKS	O95271	Q86WJ1	TNKS1BP1	Q9C0C2	P09237
TNKS2	Q9H2K2	O60547	TNMD	Q9H2S6-2	P27797
TNNC1	P63316	P19429	TNNI1	P19237	Q9NPB3
TNNT1	P13805-3, P13805	Q16236	TNNT2	P45379-11	O60341
TNR	A1L306	O43639	TNRC6B	Q9UPQ9	P11940
TNRC6C	Q9HCJ0, Q9HCJ0-1	P11940	TNS2	Q63HR2	Q9Y5Y2
TNS3	Q68CZ2	P12931	TOM1L1	O75674	P56539
TOM1L2	Q6ZVM7	Q9UM54	TOMM34	Q15785	P07900
TOP2B	Q02880	P46531	TOP3A	Q13472	P23025
TOR1A	O14656-2	Q14154	TOR1AIP1	Q5JTV8	P10599
TOR1AIP2	Q8NFQ8	Q8NE86	TOR2A	Q5JU69	Q13363
TP53	P04637	O00255	TP53BP1	Q12888	P38398
TP53RK	Q96S44	Q9BTK6	TP73	O15350, O15350-1	P46937
TPD52	P55327	Q86T03	TPD52L1	Q16890	P63104
TPH1	P17752	O95619	TPM1	P09493-10, P09493, P09493-5	Q08379
TPM2	P07951	P13569	TPM4	P67936	Q9NRD5
TPMT	P51580	P61981	TPPP	O94811	P37840
TPRG1	Q6ZUI0	P55056, P02654	TPST1	O60507	P61769
TPST2	O60704	Q9BUV8	TPT1	P13693	P11940
TRABD2B	A6NFA1	P41221	TRADD	Q15628	Q13546
TRAF3	Q13114	Q13546	TRAF3IP2	O43734	Q9Y4K3
TRAF5	O00463	Q92956	TRAF7	Q6Q0C0	Q9NWB7
TRAIP	Q9BWF2	O15160	TRAK1	Q9UPV9	P49841
TRAM1	Q9Y6Q9	Q09472	TRAPPC2	P0DI81	Q96QF0
TRAPPC2L	Q9UL33	Q96QF0	TRAPPC3	O43617	Q96QF0
TRAPPC4	Q9Y296	P28482	TRAPPC5	Q8IUR0	Q9P2M4
TRAPPC6A	O75865-2	Q9BRX5	TRAPPC9	Q96Q05	Q96QF0
TRDN	Q13061-3	O14543	TRERF1	Q96PN7	P46937
TRIB2	Q92519	Q8NHY2	TRIL	Q7L0X0	O15455
TRIM14	Q14142	P61024	TRIM21	P19474	P19474
TRIM22	Q8IYM9	Q9HC29	TRIM23	P36406	Q8NFF5
TRIM24	O15164	P04637	TRIM25	Q14258	P59595
TRIM3	O75382	P51668	TRIM36	Q9NQ86	Q14451
TRIM38	O00635	Q13546	TRIM41	Q8WV44	P03466
TRIM47	Q96LD4	Q9UPW6, Q01826	TRIM5	Q9C035-3	P07947
TRIM56	Q9BRZ2	P05388	TRIM65	Q6PJ69	P09917
TRIM71	Q2Q1W2	Q9H9Z2	TRIM9	Q9C026	P48730
TRIO	O75962	Q9NV70	TRIP10	Q15642-2	Q9Y4D1
TRIP6	Q15654	Q9Y316	TRMT6	Q9UJA5	Q13451
TRMT61A	Q96FX7	Q14192	TRNT1	Q96Q11-3	O76024
TRO	Q12816	Q05655	TROAP	Q12815	Q12933
TRPC4AP	Q8TEL6	P01160	TRPM4	Q8TD43	Q9NWB7
TRPM7	Q923J1	O00571	TRPV4	Q9EPK8	Q96J02

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TRUB1	Q8WWH5	Q96DP5	TRUB2	O95900	O00425
TSC1	Q92574	P49815	TSC22D1	Q15714	Q15047
TSGA10IP	Q3SY00	Q6NZI2	TSHR	P16473	Q9NPA3
TSHZ2	Q9NRE2	Q08379	TSHZ3	Q63HK5	Q08379
TSNARE1	A0AVG3	Q8N661	TSPAN14	Q8NG11	P48509
TSPAN17	Q96FV3	Q16585	TSPAN18	Q8WUV1	P48509
TSPAN2	O60636	O00501	TSPAN3	O60637	Q99741
TSPAN4	O14817	O43559	TSPAN5	P62079	P48509
TSPAN7	P41732	Q8TDQ0	TSPAN9	O75954	P21926
TSP0	P30536	P29762	TSPYL4	Q9UJ04	O00273
TSPYL5	Q86VY4	Q93009	TSR1	Q2NL82	Q14974
TSSC4	Q9Y5U2	Q08379	TTBK2	Q6IQ55	Q9UPV0
TTC19	Q6DKK2	Q16774	TTC23	Q5W5X9-3	A8MTQ0
TTC28	Q96AY4	O43639	TTF2	O00358	P19419
TTI1	O43156	Q9NZQ7	TTL	Q8NG68	P01189
TTLL12	Q14166	P68104	TTYH1	Q9H313	O15354
TTYH3	Q9C0H2	P21453	TUBA1A	Q71U36	Q9UBX3
TUBA1C	Q9BQE3	P36507	TUBA3D	Q13748	Q13287
TUBB2A	Q13885	P36404	TUBB2B	Q9BVA1	P36404
TUBB4A	P04350	P36404	TUBB4B	P68371	Q9BTW9, P36404
TUBB6	Q9BUF5	Q5S007	TUFT1	Q9NNX1	Q92993
TULP3	O75386	P98160	TUSC3	Q13454	Q14165
TUT4	Q5TAX3	Q9H9Z2	TVP23B	Q9NYZ1	Q8NBQ5
TWF1	Q12792	P13569	TWIST1	Q15672	O60885
TXK	P42681	O14543	TXLNB	Q8N3L3	Q9BRV8
TXNDC11	Q6PKC3	Q99633	TXNDC5	Q8NBS9	P0DTC3
TXNDC9	O14530	O00628	TXNIP	Q9H3M7	P10599
TXNRD1	Q16881	O75695	TXNRD3	Q86VQ6	Q14192
TYK2	P29597	O75496	TYMP	P19971	Q9BW85
TYRP1	P17643	O14908	TYW3	Q6IPR3	P01130
U2AF1	Q01081	Q8N4J0	UACA	Q9BZF9	P04004
UBA1	P22314	P61088	UBA3	Q8TBC4	Q15843
UBA6	A0AVT1	Q8WVK2	UBA7	P41226	P05161
UBAC2	Q8NBM4	Q8TC07, Q9HA65	UBAP1	Q9NZ09	Q99816
UBAP2L	Q14157	P10636-8	UBB	P0CG47	Q9NZ52
UBE2A	P49459	Q8WZA2	UBE2D3	P61077	Q9Y4K3
UBE2E1	P51965	Q6ZNA4	UBE2E2	Q96LR5	Q96EQ8
UBE2E3	Q969T4	Q96NR8	UBE2G1	P62253	P06307
UBE2G2	P60604	P0CG48	UBE2H	P62256	Q08379
UBE2J1	Q9Y385	Q9UBV2	UBE2K	P61086	Q15843
UBE2L6	O14933	Q9NWF9, Q96EQ8	UBE2O	Q9C0C9	P12644
UBE2Q1	Q7Z7E8	Q9UNE7	UBE2Q2	Q8WVN8	P04792
UBE3B	Q7Z3V4	P80370	UBE3C	Q15386	P60896
UBL4A	P11441	O15055	UBQLN1	Q9UMX0	P23352
UBR4	Q5T4S7	P0DP23	UBR5	O95071	P49841
UBTD1	Q9HAC8	P62837	UBXN2A	P68543	P55072
UBXN4	Q92575	P21802	UBXN6	Q9BZV1, Q9BZV1-1, Q9BZV1-2	P55072
UBXN7	O94888	Q15843	UCN	P55089	P06850
UGCG	Q16739	P43119	UGDH	O60701	P13569

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UGGT2	Q9NYU1	P11021	UGP2	Q16851	O76003
UGT8	Q16880	O75396	UHMK1	Q8TAS1-2	Q9BU61
UIMC1	Q96RL1	P38398	ULK3	Q6PHR2	Q13451
UNC119	Q13432	P36404	UNC45A	Q9H3U1	P48729
UNC93B1	Q8VCW4	Q96N66	UPF3B	Q9BZI7, Q9BZI7-2	Q92900
UPK3B	Q9BT76	O75843	UQCC2	Q9BRT2	P62993
URB1	O60287	Q13304	URB2	Q14146	P40429
URI1	O94763	P19388	UROD	P06132	P00533
USO1	O60763	Q9Y6L6	USP11	P51784	Q86YC2
USP14	P54578	P60896	USP15	Q9Y4E8	P60896
USP18	Q9UMW8	P05161	USP20	Q9Y2K6	Q99814
USP22	Q9UPT9-2, Q9UPT9	Q9NRN7	USP25	Q9UHP3	Q9UKG9
USP27X	A6NNY8	Q96B97	USP30	Q70CQ3	Q9H5K3
USP33	Q8TEY7	P00533	USP34	Q70CQ2	P21709
USP37	Q86T82	Q9UM11	USP43	Q70EL4	P62258
USP44	Q9H0E7	P41208	USP49	Q70CQ1	P41208
USP53	Q70EK8	P46109	USP54	Q70EL1-9	P35680
USP8	P40818	Q16665, Q99814	USP9X	Q93008	P29353
UST	Q9Y2C2	Q8WVC6	UTP14A	Q9BVJ6	P05067
UTP15	Q8TED0	Q8TC07	UTP20	O75691	Q9Y639
UTP23	Q9BRU9	Q08379	UTP25	Q68CQ4	P32121
UTP3	Q9NQZ2	P02787	UTRN	P46939	Q14118
UXS1	Q8NBZ7	P40855	VAMP2	P63027	Q12846
VAMP3	Q15836	Q9Y6L6	VAMP5	O95183	Q13190, Q9NYM9
VAMP8	Q9BV40	Q9P2W9	VANGL1	Q8TAA9	P18827
VANGL2	Q9ULK5	P11142	VAPB	O95292	Q9NZN5
VARS1	P26640	P26641, P68104	VARS2	Q5ST30	O95861
VASN	Q6EMK4	P78504	VASP	P50552	P22736
VAT1L	Q9HCJ6	P42858	VAV3	Q9UKW4	P62993
VAX1	Q5SQQ9-2	Q9NRD5	VAX2	Q9UIW0	Q9Y3E7
VCAM1	P19320	P30626	VCL	P18206	Q9Y490
VCP	P55072	P60880	VDR	P11473	Q15648
VEGFA	P15692-4, Q9MYV3-3, P15692	P35968, P17948	VEGFB	P49765	P15692
VEGFD	O43915	Q15836	VEZF1	Q14119	Q96KQ7
VGF	O15240	P01160	VGLL2	Q8N8G2	Q99594
VGLL3	A8MV65-2	Q15561, Q99594	VIPR2	P41587	P53794
VKORC1L1	Q8N0U8	P01019	VMA22	Q96NT0	O15342
VMP1	Q96GC9	Q07157	VOPP1	Q96AW1	Q9NZC7
VPS26A	O75436	O60493	VPS41	P49754	P0DTC3
VPS45	P38932	Q13190	VPS8	Q8N3P4	P21462
VRK2	Q86Y07-2, Q86Y07-1	Q13469	VSIR	Q9H7M9	Q969S0
VWA2	Q5GFL6	Q12933	VWA5B2	Q8N398	P20336
VWCE	Q96DN2	P11021	VWF	P04275	Q76LX8
WASF1	Q92558	P00519, Q9UQB8	WASF2	Q9Y6W5	P00519, Q9UQB8
WASF3	Q9UPY6	Q13485	WBP1L	Q9NX94	P06396
WDCP	Q9H6R7	P53350	WDFY2	Q96P53	Q9Y266
WDHD1	O75717	Q9BQP7	WDR1	O75083	P60709
WDR3	Q9UNX4	P06748	WDR33	Q9C0J8	P04004
WDR37	Q9Y2I8	Q6VY07	WDR43	Q15061	Q13901, Q9H8H0

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WDR45B	Q5MNZ6	P51813	WDR46	O15213	P06748
WDR48	Q8TAF3	Q96B01	WDR54	Q9H977	Q9Y316
WDR55	Q9H6Y2	P06748	WDR82	Q6UXN9	P61964
WDR91	A4D1P6	P42858	WDTC1	Q8N5D0	Q13371
WFDC2	Q14508	P07355	WFS1	O76024	Q9UBX5
WIF1	Q9Y5W5	P78560	WIPF2	Q8TF74	O00401
WIPF3	A6NGB9	O00401	WIP11	Q5MNZ9	P63244
WLS	Q5T9L3	P41221	WNK4	Q80UE6	O00141
WNT16	Q9UBV4	O96014	WNT2	P09544	P11021
WNT2B	Q93097	Q70UQ0	WNT3	P56703	P56704
WNT3A	P56704	Q5T9L3	WNT4	P56705	Q5T9L3
WNT5A	P41221	Q5T9L3	WNT7A	O00755	Q5T9L3
WNT7B	P56706	P56705	WT1	P19544-6	O15397
WT1-AS	Q06250	P14373	WTAP	Q15007-2	P54252
WWC1	Q8IX03	Q05513	WWC2	Q6AWC2	P62258
WWTR1	Q9EPK5	Q07157	XPNPEP1	Q9NQW7-3	Q15669
XPO1	O14980	Q9UKV8	XPO4	Q9C0E2	Q9H244
XRCC1	P18887	P06746	XRCC2	O43543	Q5TEU4
XRN2	Q9H0D6	Q9H0D6	XYLT2	Q9H1B5	Q9Y2C2
YAE1	Q9NRH1	P08670	YAP1	P46937	Q8N3R9
YARS2	Q9Y2Z4	Q12834	YBX3	P16989	P59595
YIF1A	O95070	Q9NQ11	YIPF1	Q9Y548	O00501
YIPF2	Q9BWQ6	Q86WV6	YIPF5	Q969M3	O95070
YIPF6	Q96EC8	Q9UBD6	YKT6	O15498	Q92696, P53611
YMEIL1	Q96TA2	Q12824	YRDC	Q86U90	Q5VWX1
YTHDC1	Q96MU7	Q9NRD5	YTHDF1	Q9BYJ9	P08311
ZBTB1	Q9Y2K1	P11309	ZBTB12	Q9Y330	O43639
ZBTB16	Q05516	P05231	ZBTB18	Q99592	P11021
ZBTB2	Q8N680	P07237	ZBTB24	O43167	Q9NRD5
ZBTB3	Q9H5J0	Q8NB12	ZBTB39	O15060	P55854, P61956
ZBTB4	Q9P1Z0	Q08379	ZBTB42	B2RXF5	O60733
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ZDHHC20	Q5W0Z9-4	O76024	ZDHHC5	Q9C0B5	P0DTC2
ZDHHC7	Q9NXF8-2	O43561-2	ZEB2	O60315	Q96KQ7
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ZMAT5	Q9UDW3	P40692	ZMPSTE24	O75844	P02545-1
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ZNF121	P58317	P01106	ZNF134	P52741	P55081
ZNF138	P52744	Q96EB6	ZNF148	Q9UQR1	P52294
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ZNF211	Q13398	Q13263	ZNF213	O14771	Q9UMX1
ZNF227	Q86WZ6	P05067	ZNF23	P17027	Q13077
ZNF233	Q9NZL3	P98164	ZNF24	P17028	P31749
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ZNF84	P51523	Q13263	ZNFX1	Q9P2E3	Q9Y3F4
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GOLPH3	Q9H4A6	17526	GRB7	Q14451	16152
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LRRK2	Q5S007	17552	MMADHC	Q9H3L0, Q99LS1	16856
NPC2	P61916	16113	PAK1	Q16512, Q13153	15996, P05067
RAB1A	P62820	15996	RAB32	Q13637	15996
RAB38	P57729	15996	RAB4A	P20338	15996
RHOG	P84095	16152	RILPL2	Q969X0	15996
SNCA	P37840	17552	SYT1	P21707	29036
THBS1	P07996	28815	TLL1	O43897	29108

7. Identifiers not found

These 2061 identifiers were not found neither mapped to any entity in Reactome.

ABHD13	ABHD2	ABTB2	ACAP3	ACTA2-AS1	ACTG1P17	ACTG1P2
ACTR3B	ACTRT3	ACYP1	ADAM7-AS1	ADAMTS16-DT	ADARB2	ADD3-AS
ADGRA1-AS1	ADGRB1	ADGRB2	ADGRB3-DT	ADGRD1	ADGRG2	ADGRL1
ADIRF-AS1	ADTRP	AEBP1	AFAP1	AFAP1L1	AFAP1L2	AGAP1
AJAP1	AJM1	AKIRIN1	AKTIP	ALDH1A3-AS1	ALDH1L1-AS2	ALPK2
AMER2	AMZ1	AMZ2P1.1	ANGPTL1	ANKRD12	ANKRD13B	ANKRD18
ANKRD36BP2.1	ANKRD36C	ANKRD42	ANKRD63	ANXA11	ANXA13	ANXA2P
ANXA8.1	ANXA8L1	AOPEP	APIAR	AP1G2-AS1	AP3B2	AP3M2
APOL4	APOLD1	APPAT	ARHGAP19-SLIT1	ARHGAP36	ARHGAP42-AS1	ARHGEF17-
ARHGEF35-AS1	ARHGEF7-AS2	ARMC2	ARMC3	ARMCX4	ARMH3	ARPP21
ASPHD1	ASTN2	ASXL3	ATAD2B	ATF7IP2	ATOH1	ATOH8
ATRAID	ATRN	ATXN1-AS1	BAALC	BAALC-AS2	BACE1-AS	BAIAP2-D
BANCR	BBOF1	BEND6	BEX4	BHLHE40-AS1	BICC1	BLACAT
BMP3	BMP6	BMS1P14.1	BNAT1	BNC1	BNC2	BNC2-AS
BRSK1	BTBD17	BTBD19	BTBD7	BTBD8	BTG1-DT	BTNL12P
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C1QTNF4	C1QTNF6	C1RL	C1RL-AS1	C1orf115	C1orf162	C1orf21
C20orf204	C22orf42	C2CD2	C3orf80	C4orf50	C5orf15	C5orf58
C7orf57	C8orf34	C8orf88	CAAP1	CALB2	CALHM2	CAMK2N
CAMTA1	CAPN1-AS1	CAPS2	CARD14	CARD16	CARD8-AS1	CARINH
CARMN	CASD1	CASP4LP.1	CASZ1	CBLN1	CBLN2	CBLN3
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CCNT2-AS1	CCNYL1	CCSER1	CD276	CD44-DT	CD99.1	CDH13-AS
CDK6-AS1	CDKN2B-AS1	CDV3	CECR2	CELF2	CELF3	CELF4
CELSR3	CENPBD1P	CENPVL3	CERNA2	CEROX1	CES4A	CFAP12P
CFAP157	CFAP221	CFAP251	CFAP276	CFAP43	CFAP45	CFAP54
CFAP74	CFAP77	CFAP90	CFAP92	CFAP97	CFAP99	CFDP1
CGREF1	CHASERR	CHI3L2	CHRA1	CHROMR	CIMAP1B	CIMIP5
CLEC18B	CLEC18C	CLEC4GP1	CLMAT3	CLN8	CLSTN2	CLXN
CNN3-DT	CNTNAP3B	CNTNAP5	CNTNAP5-DT	COL6A4P2	COPB2-DT	COPG2IT
COSMOC	CPAMD8	CPB2-AS1	CPEB2	CPEB3	CPNE4	CPQ
CPXM2	CRB2	CRIM1	CRIM1-DT	CRMA	CROCCP2.1	CRYZL2P
CSMD2	CSRNP3	CT75.1	CTAGE4	CTAGE8	CTB-30L5.1	CTBP2P8
CTIF	CTNNA2-AS1	CTNS-AS1	CTXN1	CTXN2	CTXND1	CYCSP34
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DPY19L1P1	DRC1	DRC11	DRGX	DSP-AS1	DTD2	DUBR
DUSP18	DYNLRB2-AS1	EBPL	ECHDC1	ECHDC3	EDIL3-DT	EEF1DP2
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ELAPOR2	ELF4	ELFN1	ELK4	ELMOD2	ELN-AS1	ELOVL2-A
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FAM131A	FAM151B	FAM171A1	FAM174A-DT	FAM174B	FAM180B	FAM181A
FAM200B	FAM201A	FAM227B	FAM234A	FAM43B	FAM53A	FAM66D
FAM76A	FAM81B	FAM89A	FAR2P1.1	FAR2P2.1	FAT3	FAT4
FBLN7	FERMT1	FGD5-AS1	FGF14-AS2	FHAD1	FHDC1	FHIP1A
FIGNL2	FILIP1L	FIRRE	FJX1	FLJ16779	FNDC1	FNDC1-AS1
FOXD1	FOXL2NB	FREM1	FREM2	FRMD4B	FRMD8	FTH1P2
GAPDHP21	GAS1RR	GAS6-AS1	GAS6-DT	GASK1B	GASK1B-AS1	GBGT1
GCC2-AS1	GCNA	GCNT2	GDAP1L1	GDF10	GDF11	GDF6
GDPD2	GIN1	GJD3-AS1	GLDN	GLIPR2	GLMP	GLOD4
GLYCTK-AS1	GNAO1-DT	GOLM2	GPATCH4	GPC1-AS1	GPR12	GPR137
GPR137C	GPR139	GPR153	GPR157	GPR180	GPR19	GPR26
GPR87	GPRASP1	GPRC5D-AS1	GPRIN3	GPX1P1	GREB1L	GREP1
GS1-24F4.2	GSAP	GSDMB	GSX1	GSX2	GTF2IP14	GULP1
GZMA	H1-10-AS1	H6PD	HAND2-AS1	HAPSTR2	HARBI1	HCFC1R
HCG23	HCP5	HECA	HECW2-AS1	HEIH	HEPACAM	HES3
HHIPL1	HIVEP2-DT	HK2-DT	HLA-F-AS1	HMGA2-AS1	HMGB3P10	HMGNI1
HMGXB3	HMSD	HOATZ	HOMER3-AS1	HOTAIR	HOXA-AS2	HOXA-AS1
HOXA11-AS	HOXA13	HOXB6	HOXB8	HOXC10	HOXC6	HOXD-AS1
HPAT5	HRK	HROB	HSPD1P11	HYDIN	HYMAI	IAH1
IGDCC3	IGF2-AS	IGFL2	IGLON5	IGSF10	IGSF11	ILDR2
IMMP2L	IMPG2	INCR1	INIP	INSYN1-AS1	INSYN2B	INTS6L
IQANK1	IQCJ-SCHIP1	IRX2-DT	IRX3	IRX5	ISL1-DT	ITGB1P1
ITPRID1	JAKMIP1-DT.1	JAKMIP2-AS1	JCAD	JHY	JMJD8	KANSL1-AS1
KBTBD11	KBTBD11-AS1	KC6	KCNJ13	KCNQ1OT1	KCTD11	KCTD13-DT
KCTD2	KCTD8	KHDC1-AS1	KIAA0513	KIAA1210	KIAA1549L	KIAA1751
KIF25-AS1	KLF3-AS1	KLHDC7A	KLHL1	KLHL29	KLHL31	KLHL4
KLLN	KLRK1-AS1	KNCN	KRBA2	KRCC1	KRT18P15	KRT8P10
KRT8P36	KRT8P45	KRTAP5-AS1	LAD1	LAMP5	LASP1NB	LBX1-AS1
LCTL	LDAH	LDLRAD2	LENG9	LETM2	LGSN	LHFPL3
LHX8-AS1	LIF-AS2	LIFR-AS1	LINC00052	LINC00174	LINC00240	LINC00321
LINC00342	LINC00391	LINC00404	LINC00463	LINC00467	LINC00472	LINC00477
LINC00511	LINC00605	LINC00607	LINC00632	LINC00639	LINC00641	LINC00651
LINC00663	LINC00665	LINC00682	LINC00683	LINC00702	LINC00703	LINC00707
LINC00847	LINC00857	LINC00858	LINC00862	LINC00877	LINC00880	LINC00888
LINC00928	LINC00957	LINC00960	LINC00968	LINC01012	LINC01013	LINC01031
LINC01091	LINC01102	LINC01111	LINC01133	LINC01138	LINC01141	LINC01151
LINC01173	LINC01186	LINC01201	LINC01224	LINC01239	LINC01257	LINC01261
LINC01278	LINC01307	LINC01322	LINC01342	LINC01410	LINC01414	LINC01481
LINC01503	LINC01508	LINC01546	LINC01558	LINC01572	LINC01579	LINC01591
LINC01619	LINC01629	LINC01630	LINC01632	LINC01638	LINC01717	LINC01719
LINC01801	LINC01806	LINC01836	LINC01852	LINC01896	LINC01902	LINC01911
LINC01933	LINC01963	LINC01965	LINC01993	LINC02073	LINC02082	LINC02088
LINC02128	LINC02175	LINC02200	LINC02210	LINC02223	LINC02293	LINC02331
LINC02350	LINC02352	LINC02381	LINC02388	LINC02418	LINC02428	LINC02451
LINC02461	LINC02487	LINC02525	LINC02538	LINC02550	LINC02568	LINC02591
LINC02615	LINC02643	LINC02714	LINC02717	LINC02725	LINC02742	LINC02791
LINC02851	LINC02883	LINC02885	LINC02889	LINC02891	LINC02893	LINC02911
LINC02930	LINC02966	LINC02980	LINC02981	LINC02982	LINC02985	LINC03011

LINC03021	LINC03053	LINC03057	LINC03077	LINC03113	LINC03126	LIPT2-AS1
LIX1	LIX1L	LMBR1L	LMLN	LMNB1-DT	LMNTD2-AS1	LMX1B-DT
LNCOC1	LNCOG	LNCTAM34A	LNP1	LOC100128079	LOC100129434	LOC100130079
LOC100240735	LOC100288175	LOC100289230	LOC100294145	LOC100310756	LOC100506076	LOC100506076
LOC100507053	LOC101927480	LOC101928188	LOC101928416	LOC101928499	LOC101929494	LOC101929494
LOC102723475	LOC102723670	LOC102724019	LOC102724220	LOC102724434	LOC102724680	LOC103344079
LOC105369591	LOC105370941	LOC105371401	LOC105371735	LOC105373289	LOC105374338	LOC105375079
LOC105376654	LOC105376944	LOC105377259	LOC105377329	LOC105377378	LOC105377503	LOC105378079
LOC105379199	LOC107984262	LOC107984270	LOC107985177	LOC107985206	LOC107985433	LOC107985433
LOC107986524	LOC107986952	LOC116435278	LOC121725015	LOC124900217	LOC124900275	LOC124900275
LOC124902301	LOC124902479	LOC124902546	LOC124902752	LOC124903135	LOC124903670	LOC124904079
LOC124907878	LOC124909496	LOC283299	LOC283731	LOC343052	LOC400499	LOC61326079
LOC728743	LOC729732	LOC729998	LOC730101	LOC730668	LOC93429	LOXL1-AS1
LRP3	LRRC10B	LRRC15	LRRC17	LRRC2	LRRC23	LRRC24
LRRC37B	LRRC3B	LRRC4	LRRC42	LRRC43	LRRC55	LRRC74H
LRRC75B	LRRC77P	LRRC1	LRRIQ1	LRRN4CL	LSG1	LSP1P5
LY6G5C	LYPD6	LYSMD3	LYSMD4	MAGI2-AS3	MAGOH2P	MAMDC3
MAMSTR	MANEAL	MAP3K10	MAP3K14-AS1	MAP3K19	MAP3K9-DT	MAP6
MARCHF1	MARCHF9	MARVELD1	MARVELD3	MAST4	MATCAP1	MBNL1-AS1
MCPH1-AS1	MCTP1	MCTP2	MDFIC	MEG3	MEGF10	MEGF9
METTL24	METTL25	METTL2A	METTL8	MEX3A	MFSD1	MFSD12-AS1
MGP	MIAT	MICA-AS1	MINDY4	MIR100HG	MIR124-1HG	MIR124-2HG
MIR137HG	MIR17HG	MIR1915HG	MIR193BHG	MIR210HG	MIR219A2HG	MIR222HG
MIR23AHG	MIR3125	MIR31HG	MIR325HG	MIR34AHG	MIR3681HG	MIR4435-2HG
MIR600HG	MIR9-2HG	MIR9-3HG	MIR99AHG	MIRLET7A1HG	MIRLET7BHG	MITA1
MMP23B	MMP24OS	MN1	MOAP1	MOSPD1	MPV17L	MPV17L
MROCK1	MRPL20-AS1	MRPL9P1	MRT04	MS4A8	MSANTD1	MSANTD1
MSC	MSC-AS1	MXRA5	MXRA5Y	MXRA7	MYCBP	MYOF
MYRF	MZF1	NACC2	NAGA	NALF1	NALF2	NAT16
NDNF	NDRG3	NDUFA4L2	NEAT1	NECTIN3-AS1	NEK5	NELL1
NETO1-DT	NEUROD2	NEUROG2	NEXMIF	NFIA-AS2	NFIB-AS1	NFKBIZ
NIPAL4-DT	NIT1	NLRP6	NMD3	NME9	NNT-AS1	NOMO2
NOVA1-DT	NOVA2	NPAP1	NPIP15	NPIP15	NPR3	NPTN-IT1
NR2F1-AS1	NRAV	NREP	NRK	NRSN1	NSUN7	NT5DC2
NUP153-AS1	NUTM2B-AS1	NUTM2D	NWD2	NXPH1	OBSCN-AS1	OCSTAM
OGFOD1P1	OGFR	OLFML1	OLFML3	OLMALINC	ONECUT2	OPA3
OSER1-DT	OSGIN2	OTOR	OTUD6B-AS1	OVCH2	PABPC1L2B	PABPC5
PAMR1	PAPLN	PAQR4	PARM1	PARPBP	PART1	PAXBP1-AS1
PCAT1	PCAT7	PCDH1	PCDH10-DT	PCDH11X	PCDH12	PCDH17
PCDH9	PCDHA10	PCDHB10	PCDHB2	PCDHB3	PCDHB9	PCDHGB
PCOLCE-AS1	PCP4L1	PDCD6IP-DT	PDCL3P4	PDE4DIPP6.1	PDE6B-AS1	PDE7B-AS1
PDLIM3	PEG13	PENK-AS1	PEPD	PEX5L	PFN1P11	PGCKA1
PGLS-DT	PGM5	PGM5P2.1	PHACTR1	PHLDA2	PHLDA3	PHOX2B
PHYHIPL	PI15	PI4KAP2	PICART1	PIEZO2	PIK3CD-AS2	PIM3
PKHD1L1	PLAC9	PLAG1	PLBD2	PLCE1-AS1	PLCL1	PLEK2
PLPP7	PLSCR5	PLXDC1	PMFBP1	PMS2P1.1	PNCK	PNMA3
PNMA6F	PNMA8A	PNMA8B	PNMA8C	PODN	POM121L9P.1	POPCD1
POSTN	POU3F4	PPDPF	PPIEL.1	PPM1M	PPP1CB-DT	PPP1R14
PPP1R1C	PPP1R35	PPP1R3B-DT	PPP4R4	PRCD	PRDM13	PRDX6-AS1
PRICKLE2	PRKCQ-AS1	PRKCZ-AS1	PROM2	PRPF40B	PRR12	PRR15
PRR9	PRRG1	PRRG3	PRRT3	PRRT4	PRRX1	PRSS12

PRSS35	PRTG	PSD2	PSORS1C1	PSRC1	PTER	PTMS
PTPRD-AS1	PUS1-AS1	PVT1	PWRN1	PWRN4	PXDC1	QRICH2
R3HCC1L	R3HDM4	RAB33B-AS1	RAB6C-AS1.1	RAB6D	RALGPS1	RALY-AS
RANBP17	RAPGEFL1	RARA-AS1	RARRES1	RASGEF1C	RASSF9	RBBP9
RBM20	RBMS3	RCBTB1	REM2	RERG	RFLNA	RFLNB
RFPL3S	RFX3-DT	RFX4	RHBDL3	RIMBP2	RIMS2	RLIG1
RN7SKP296	RN7SL381P	RN7SL674P	RNF157	RNF175	RNF207	RNF32-D
RNPEP	RNPEPL1	RNU6ATAC18P	ROM1	RPL17P47	RPL17P50	RPL29P1
RPL36AP54	RPL41P1	RPL7AP64	RPLP0P2	RPRML	RPS20P22	RPS2P14
RPS2P5	RRM2P3	RSPH1	RSPH4A	RTL1	RTL4	RTL5
RTTN	RUFY3	RUNDC3A-AS1	RWDD4P2	SALL3	SAMD14	SAMD5
SAMD9L	SANBR	SATB1-AS1	SATL1	SBK1	SBK2	SBNO2
SCDP1	SCML4	SCOC-AS1	SDR42E2	SEC14L2	SECISBP2L	SECTM1
SEMA3F	SEMA6A-AS2	SEMA6B	SEMA6C	SENP6	SEPSECS-AS1	SEPTIN7-I
SERPINE3	SERPINI1	SERTAD4-AS1	SETD5	SEZ6L	SFMBT2	SFTA1P
SH3TC1	SH3TC2	SHISA2	SHISA3	SHISA7	SIDT1	SIDT2
SILC1	SIPA1L1-AS1	SIX4	SKOR1	SKOR2	SLC10A3	SLC16A1-A
SLC16A5	SLC25A25-AS1	SLC25A33	SLC25A34	SLC25A35	SLC25A5-AS1	SLC25A5
SLC2A4RG	SLC35E2B	SLC35F4	SLC44A3-AS1	SLC45A1	SLC46A3	SLC4A11
SLC66A3	SLC6A17	SLC7A4	SLEAR	SLFN11	SLFN13	SLIT3-AS
SMANTIS	SMIM10	SMIM10L2A	SMIM15	SMIM18	SMIM29	SMIM30
SMIM45	SMIM7	SMOC2	SMPDL3A	SMPDL3B	SMYD4	SNAI1P1
SNED1	SNHG1	SNHG16	SNHG17	SNHG18	SNHG26	SNHG3
SNHG4	SNHG5	SNHG8	SNN	SNX25	SOCS3-DT	SORCS1
SOX1-OT	SOX11	SOX21-AS1	SOX9-AS1	SP140L	SP8	SPA17
SPART-AS1	SPATA7	SPINT1-AS1	SPMIP1	SPMIP5	SPON1-AS1	SRRM3
SSR4P1	SSXP10	ST18	ST20-AS1	ST3GAL1-DT	STC1	STEAP2
STK32B	STK38L	STON1-GTF2A1L	STPG1	STUM	STX16-NPEPL1	STXBP5-A
STYX	SUGCT	SUNO1	SUSD1	SUSD2	SUSD4	SVEP1
SVOP	SWINGN	SWT1	SYNAGE	SYNM	SYPL1P2	SYT13
SYT15B	TAFA2	TAFA3	TAGLN	TAGLN3	TANGO6	TBC1D23
TBCK	TBILA	TBX4	TCEAL2	TCEAL3	TCEAL7	TCF15
TDRKH-AS1	TEKTIP1	TENM2-AS1	TENM3-AS1	TENT2	TEX261	TFIP11-D
THA1P	THAP9-AS1	THBS1-IT1	THCAT155	THUMPD3-AS1	TILAM	TIMP4
TKTL1	TLCD2	TLCD3A	TLCD3B	TLX1NB	TM4SF1	TM7SF3
TMC5	TMCC3	TMCO4	TMEM116	TMEM117	TMEM119	TMEM121
TMEM125	TMEM131L	TMEM132B	TMEM135	TMEM138	TMEM141	TMEM14
TMEM151A	TMEM151B	TMEM161B	TMEM161B-DT	TMEM175	TMEM178B	TMEM17
TMEM185B	TMEM191C	TMEM196	TMEM198	TMEM200B	TMEM200C	TMEM21
TMEM221	TMEM229A	TMEM235	TMEM240	TMEM249	TMEM250	TMEM26
TMEM272	TMEM37	TMEM39A	TMEM59L	TMEM63C	TMEM64	TMEM71
TMEM8B	TMPRSS7	TMSB15A	TMSB15B-AS1	TMSB4XP4	TMT1B	TNS2-AS
TOP1MT	TOP6BL	TOR3A	TOX	TOX-DT	TOX3	TPBG
TPBGL-AS1	TPPP3	TRABD	TRAM2	TRANK1	TRAPPC13P1	TRIDDEN
TRIM66	TRIM67	TRNP1	TRPC5OS	TSC22D2	TSHZ1	TSKU
TSPEAR-AS1	TTBK1	TTC29	TTC38	TTC3P1	TTC7A	TTC9B
TWSG1	TXNDC15	TXNDC16	UAP1L1	UBAP1L	UBE2QL1	UBE2R2-A
UBR7	UEVLD	UFC1	UFM1	UNC5B-AS1	UNC5C-AS1	USF3
USP2-AS1	USP34-DT	USP51	VEPH1	VPS13B-DT	VPS9D1-AS1	VSTM2A
VSTM2L	VWA3A	VWA3B	WAPL-DT	WDPCP	WDR17	WDR38
WDR76	WFDC21P	WFIKKN1	WSCD1	WWC2-AS2	WWC3	XG

XKR7	YBX1P6	YDJC	YJEFN3	YPEL1	ZBED10P	ZBED2
ZBTB37	ZBTB46-AS2	ZCCHC14	ZCCHC2	ZCCHC24	ZDHHHC11B	ZDHHC1
ZDHHHC8BP.1	ZEB2-AS1	ZFHX3-AS1	ZFP92	ZFR2	ZFX	ZFYVE28
ZIC5	ZKSCAN2	ZNF117	ZNF146	ZNF185	ZNF213-AS1	ZNF225-A
ZNF407-AS1	ZNF469	ZNF470-DT	ZNF525	ZNF527	ZNF532	ZNF587E
ZNF674-AS1	ZNF778-DT	ZNF780B	ZNF789	ZNF8-DT	ZNF821	ZNF862
ZNF891	ZNF892	ZNF91	ZNG1E	ZNNT1	ZNRF2P2	ZSCAN2
ZSWIM1	ZSWIM4	ZSWIM5	ZYG11A	gene		